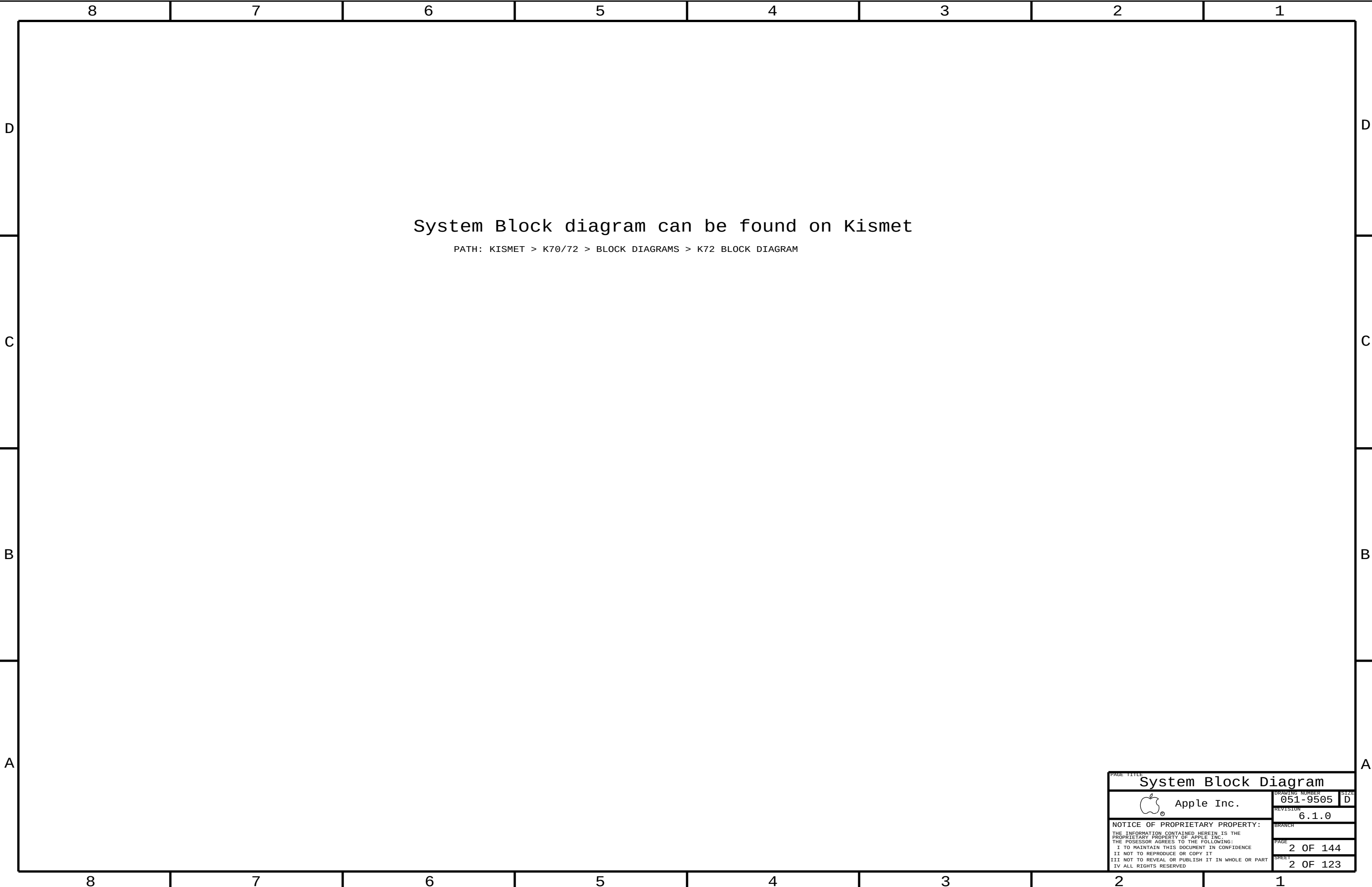
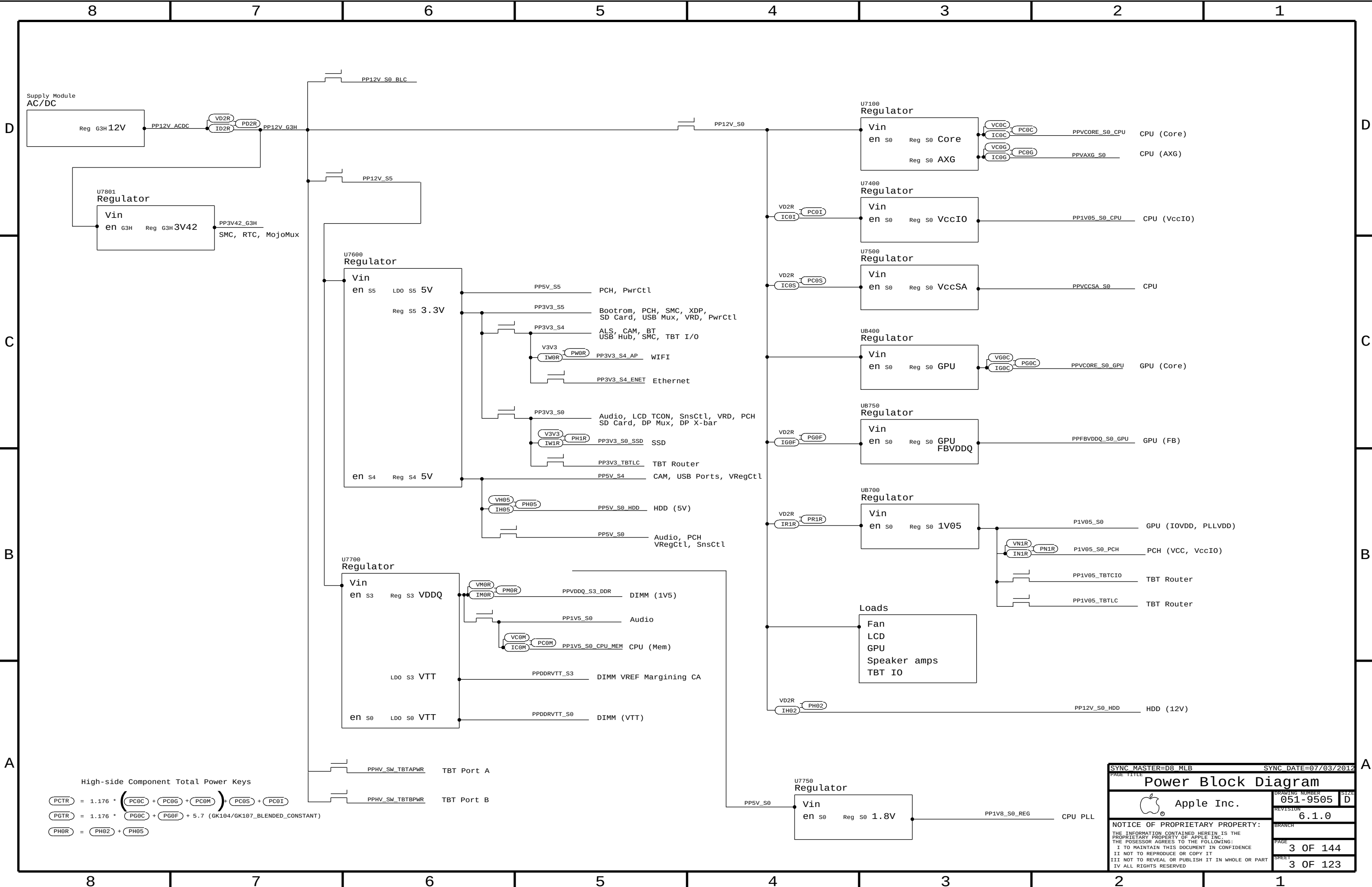




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 Apple Inc.		DRAWING NUMBER	051-9505
		SIZE	D
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SYNC DATE=07/03/2012

Power Block Diagram

 Apple Inc.

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051-9505

REVISION
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BOM Variants

BOM NUMBER	BOM NAME	BOM OPTIONS
639-3662	PCBA, MLB, ULTIMATE, 3. 4G, GTX, SAM, 2GB, D8	D8_COMMON, D8, CPU:4C_3P4GHZ, GPU:104GTX, FB:2G_SAMSUNG, EEEE:F0V5
639-3950	PCBA, MLB, ULTIMATE, 3. 2G, GTX, SAM, 2GB, D8	D8_COMMON, D8, CPU:4C_3P2GHZ, GPU:104GTX, FB:2G_SAMSUNG, EEEE:F49R
639-3560	PCBA, MLB, ULTIMATE, 3. 4G, GT, SAM, 1GB, D8	D8_COMMON, D8, CPU:4C_3P4GHZ, GPU:104GT, FB:1G_SAMSUNG, EEEE:DYN3
639-3949	PCBA, MLB, ULTIMATE, 3. 2G, GT, SAM, 1GB, D8	D8_COMMON, D8, CPU:4C_3P2GHZ, GPU:104GT, FB:1G_SAMSUNG, EEEE:F49P
639-4087	PCBA, MLB, ULTIMATE, 3. 4G, GTX, HYN, 2GB, D8	D8_COMMON, D8, CPU:4C_3P4GHZ, GPU:104GTX, FB:2G_HYNIX, EEEE:F64W
639-4091	PCBA, MLB, ULTIMATE, 3. 2G, GTX, HYN, 2GB, D8	D8_COMMON, D8, CPU:4C_3P2GHZ, GPU:104GTX, FB:2G_HYNIX, EEEE:F652
639-4086	PCBA, MLB, ULTIMATE, 3. 4G, GT, HYN, 1GB, D8	D8_COMMON, D8, CPU:4C_3P4GHZ, GPU:104GT, FB:1G_HYNIX, EEEE:F64V
639-4090	PCBA, MLB, ULTIMATE, 3. 2G, GT, HYN, 1GB, D8	D8_COMMON, D8, CPU:4C_3P2GHZ, GPU:104GT, FB:1G_HYNIX, EEEE:F651
085-4435	PCBA, MLB, DEV, D8. ULTIMATE	DEVELOPMENT, D8_DEVEL

Bar Code Labels / EEEE #'s

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
825-7122	1	MLB LABEL, 48.0X4.8	EEEE_DYW3	CRITICAL	EEEE: DYW3
825-7122	1	MLB LABEL, 48.0X4.8	EEEE_F0V5	CRITICAL	EEEE: F0V5
825-7122	1	MLB LABEL, 48.0X4.8	EEEE_F49P	CRITICAL	EEEE: F49P
825-7122	1	MLB LABEL, 48.0X4.8	EEEE_F49R	CRITICAL	EEEE: F49R
825-7122	1	MLB LABEL, 48.0X4.8	EEEE_F4MW	CRITICAL	EEEE: F4MW
825-7122	1	MLB LABEL, 48.0X4.8	EEEE_F4TY	CRITICAL	EEEE: F4TY
825-7122	1	MLB LABEL, 48.0X4.8	EEEE_F64W	CRITICAL	EEEE: F64W
825-7122	1	MLB LABEL, 48.0X4.8	EEEE_F652	CRITICAL	EEEE: F652
825-7122	1	MLB LABEL, 48.0X4.8	EEEE_F64V	CRITICAL	EEEE: F64V
825-7122	1	MLB LABEL, 48.0X4.8	EEEE_F651	CRITICAL	EEEE: F651

BOM Groups

BOM GROUP	BOM OPTIONS
D8_COMMON	COMMON,ALTERNATE,D8_COMMON1,D8_PROGPARTS
D8_COMMON1	XDP,RSMRST:GATE,SPEAKERID,VREF:CPU,TBTHV:P12V,PRE_BOOST:Y
D8_PROGPARTS	SMC:PROG,BOOTROM:PROG,TBTROM:PROG,CIVROM:PROG,CAMROM:PROG,BLCMCU:PROG
D8_DEVEL	XDP_CONN,LPCPLUS,VREFMRGN:EXT,BKLT_PWM,DEVEL_AUDIO,TEMPSNSDEV

CPUs

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
33754356	1	1VB, SR0TS, PRQ, N3, 3, 2, 77W, 4+1, 1.1, 6M, LGA	CPU	CRITICAL	CPU:4C_3P2GHZ
33754247	1	1VB, SR0PK, PRQ, E1, 3, 4, 77W, 4+2, 1.15, 8M, LG	CPU	CRITICAL	CPU:4C_3P4GHZ

ASICs

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
337S4277	1	IC, PANTHER POINT, C3, SL3C7, PRQ, 8D82277	U1800	CRITICAL	
338S1113	1	IC, TBT, CR-4C, B1, PRQ, 288 FC8GA, 12X12MM	U3600	CRITICAL	
343S0616	1	IC, BCM57766A1, ENET&SD, 8X8	U3900	CRITICAL	
337S4333	1	IC, GPU, NV GK104 7-4-PS-A2	UA000	CRITICAL	GPU:104GT
337S4333	1	IC, GPU, NV GK104 7-4-PS-A2	UA000	CRITICAL	GPU:104GT2
337S4332	1	IC, GPU, NV GK104 8-4-PS-A2,	UA000	CRITICAL	GPU:104GTX

Programmable Parts

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
341S3640	1	IC,EEPROM,CR,V11.1 (B1),D8	U3690	CRITICAL	TBTR0M:PROG
33S0865	1	IC,EEPROM,SERIAL,8KB,MLP8	U3690	CRITICAL	TBTR0M:BLANK
341S3641	1	IC,PROGRMD,EFI ROM,PIB,D7/D8	U5110	CRITICAL	BOOTROM:PROG
33S0867	1	IC,64 MBIT SPI SERIAL FLASH	U5110	CRITICAL	BOOTROM:BLANK
341S3394	1	IC,PROGRMD,SMC,A3,V2.2A32,D8	U4900	CRITICAL	SMC:PROG
33S1098	1	IC,SMC,LX4F51AH5BBCIGA3	U4900	CRITICAL	SMC:BLANK
341S3646	1	IC,CAMERA FLASH,V7226,D7/D8	U4202	CRITICAL	CAMROM:PROG
33S0852	1	IC,FLASH,SPI,1MBIT,3V3	U4202	CRITICAL	CAMROM:BLANK
341S3645	1	IC,ENET 1MBIT, SPI,ROM, V.1.3 D8	U3990	CRITICAL	CIVROM:PROG
33S0862	1	IC,SERIAL FLASH,2MBIT, 2.7V, REF F	U3990	CRITICAL	CIVROM:BLANK
341S3627	1	IC,BLC,MCU, PRPGRAMMED, V0203, D8	U9700	CRITICAL	BLCMCU:PROG
33S3978	1	IC,BLC MCU LPC2132FBD64/01, LQFP64	U9700	CRITICAL	BLCMCU:BLANK

CPU SOCKET

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
511S0073	1	SOCKET.MOLEX, LGA1155,CPU- LF	U1000	CRITICAL	

CPU SOCKET ALTERNATES

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
511S0071	511S0073		ALL	TYCO SOCKET
511S0072	511S0073		ALL	FOXCONN SOCKET

D8 SCHEMATIC / PCB #'S

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
051-9505	1	SCH, MLB, D8, ULTIMATE	SCH1	CRITICAL	D8
820-3299	1	PCBF, MLB, D8, ULTIMATE	PCB1	CRITICAL	D8

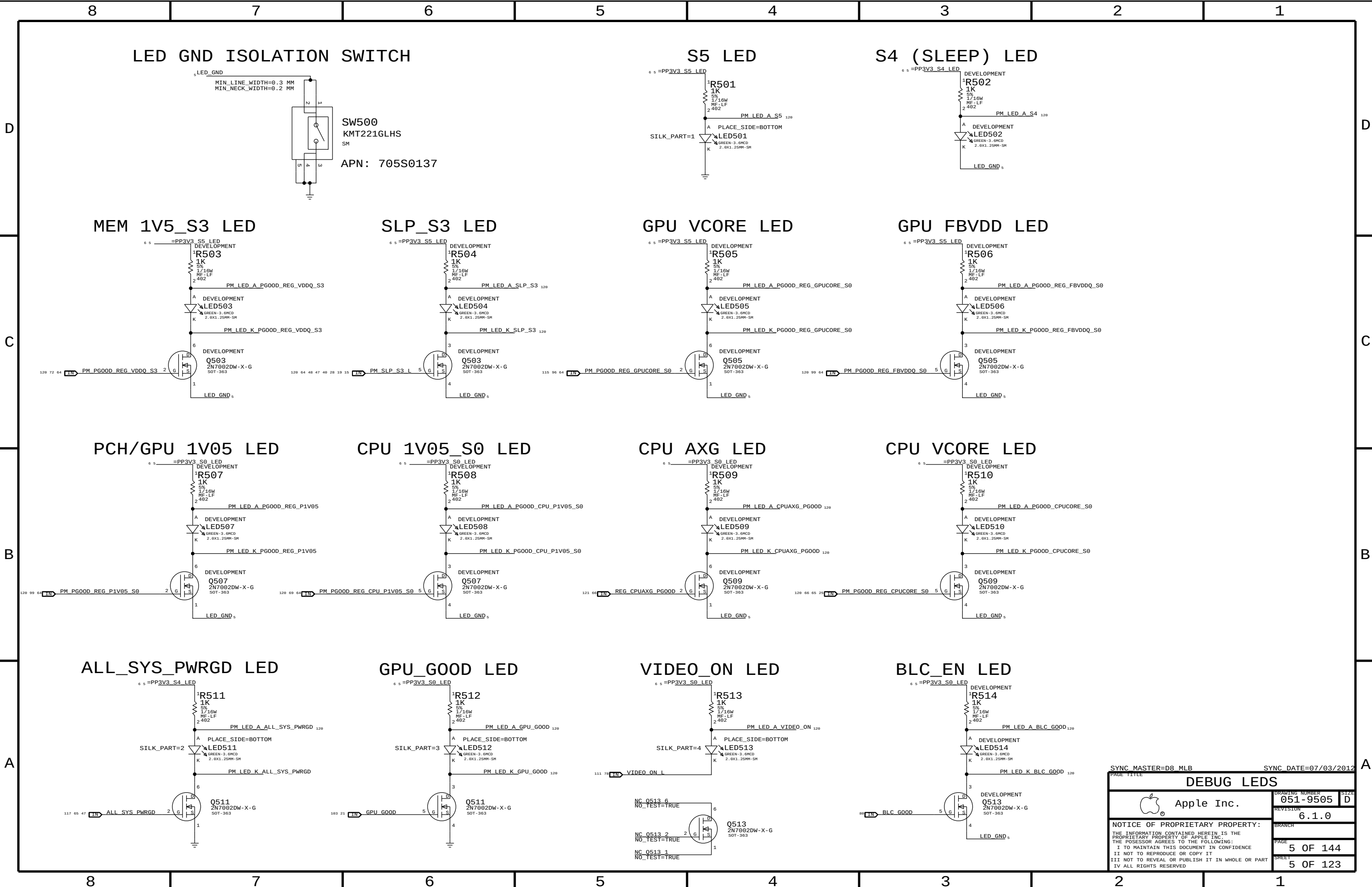
D8 ALTERNATES

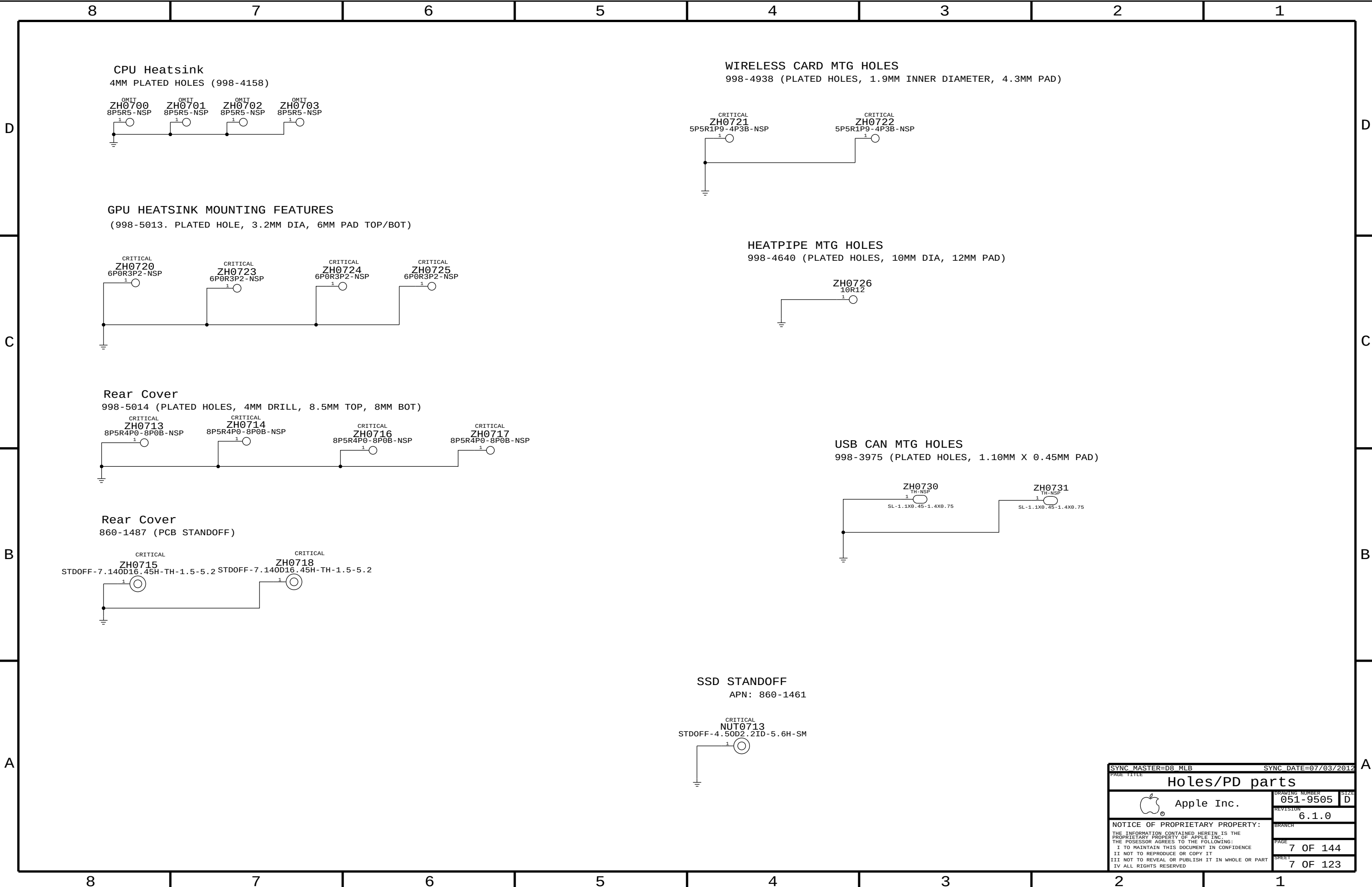
PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
377S0147	377S0126		ALL	USB diodes
157S0084	157S0058		ALL	Enet Magnetics
341S3644	341S3645		U3990	CIVROM
376S0975	376S1081		ALL	P/NCH DUAL FET
152S1668	152S1591		ALL	33UH BLC INDUCTOR
128S0365	128S0368		ALL	150UF CAPS BLK

VRAM Module Parts

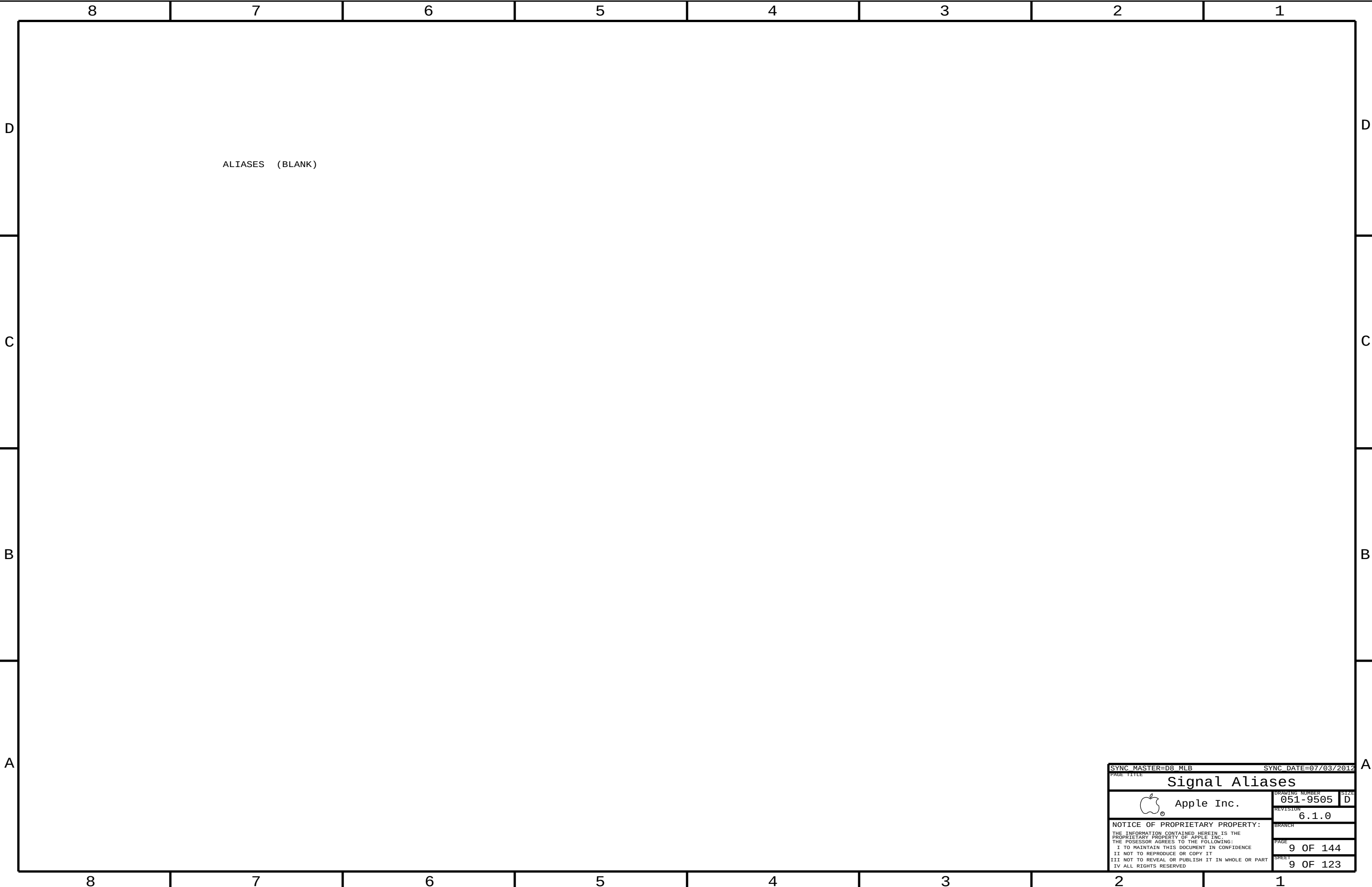
PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
33356619	4	IC, SGRAM, GDDR5, 32MX32, 1.5GHZ, G-DIE, HF	UA300, UA350, UA400, UA450	CRITICAL	FB:1G_SAMSUNG
33356619	4	IC, SGRAM, GDDR5, 32MX32, 1.5GHZ, G-DIE, HF	UA500, UA550, UA600, UA650	CRITICAL	FB:1G_SAMSUNG
33356626	4	IC, GDDR5, 32MX32, 1.5GHZ, VEGA 44NM, B-DIE	UA300, UA350, UA400, UA450	CRITICAL	FB:1G_HYNIX
33356626	4	IC, GDDR5, 32MX32, 1.5GHZ, VEGA 44NM, B-DIE	UA500, UA550, UA600, UA650	CRITICAL	FB:1G_HYNIX
33356631	4	IC, SGRAM, GDDR5, 64MX32, D-DIE	UA300, UA350, UA400, UA450	CRITICAL	FB:2G_SAMSUNG
33356631	4	IC, SGRAM, GDDR5, 64MX32, D-DIE	UA500, UA550, UA600, UA650	CRITICAL	FB:2G_SAMSUNG
33356636	4	IC, GDDR5, 64MX32, A-DIE	UA300, UA350, UA400, UA450	CRITICAL	FB:2G_HYNIX
33356636	4	IC, GDDR5, 64MX32, A-DIE	UA500, UA550, UA600, UA650	CRITICAL	FB:2G_HYNIX

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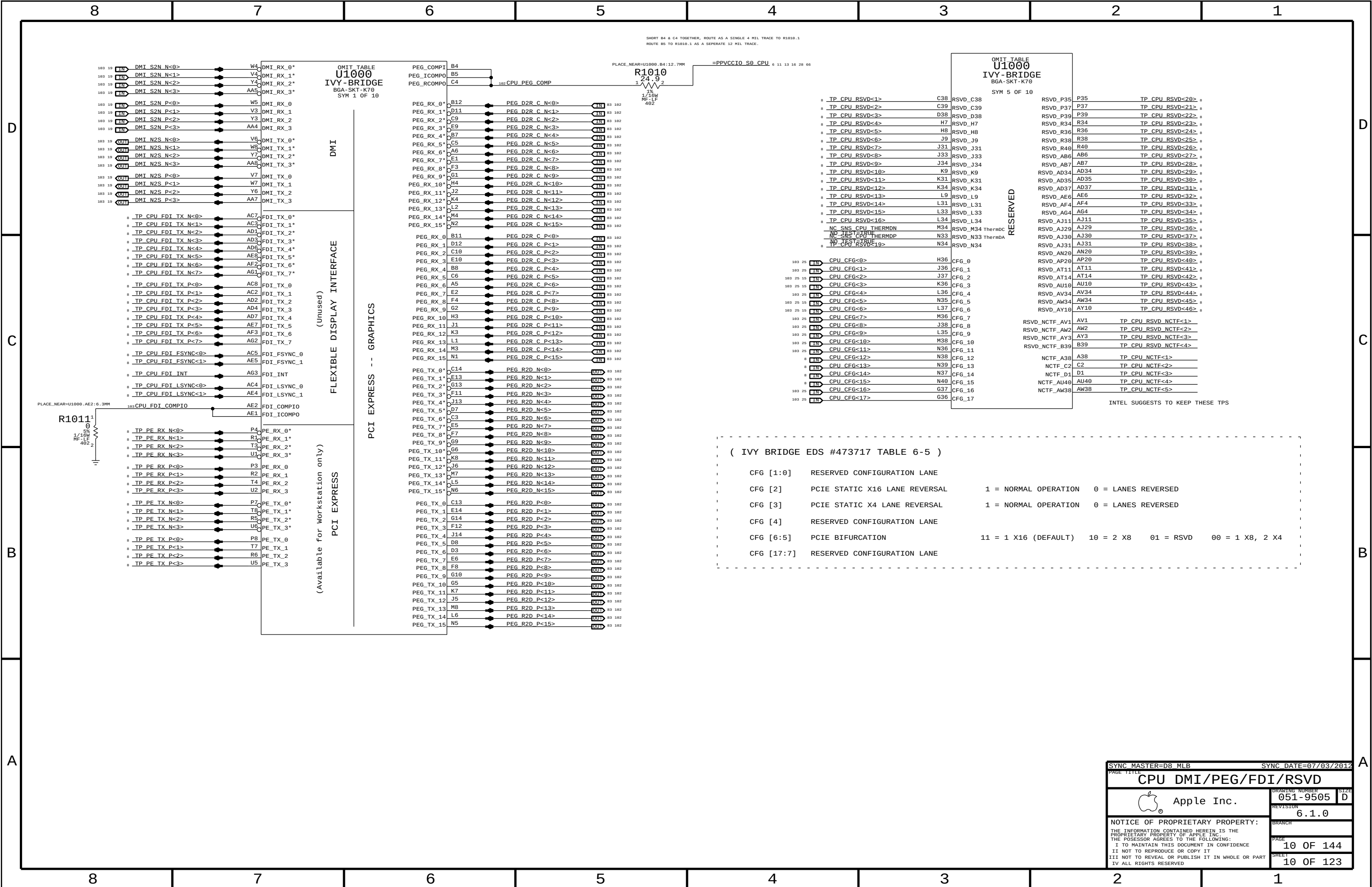


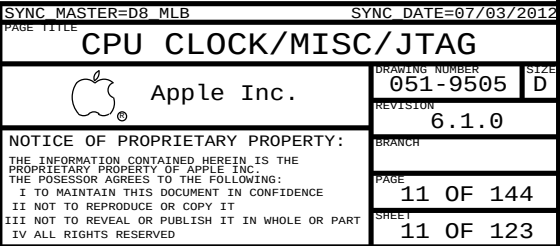


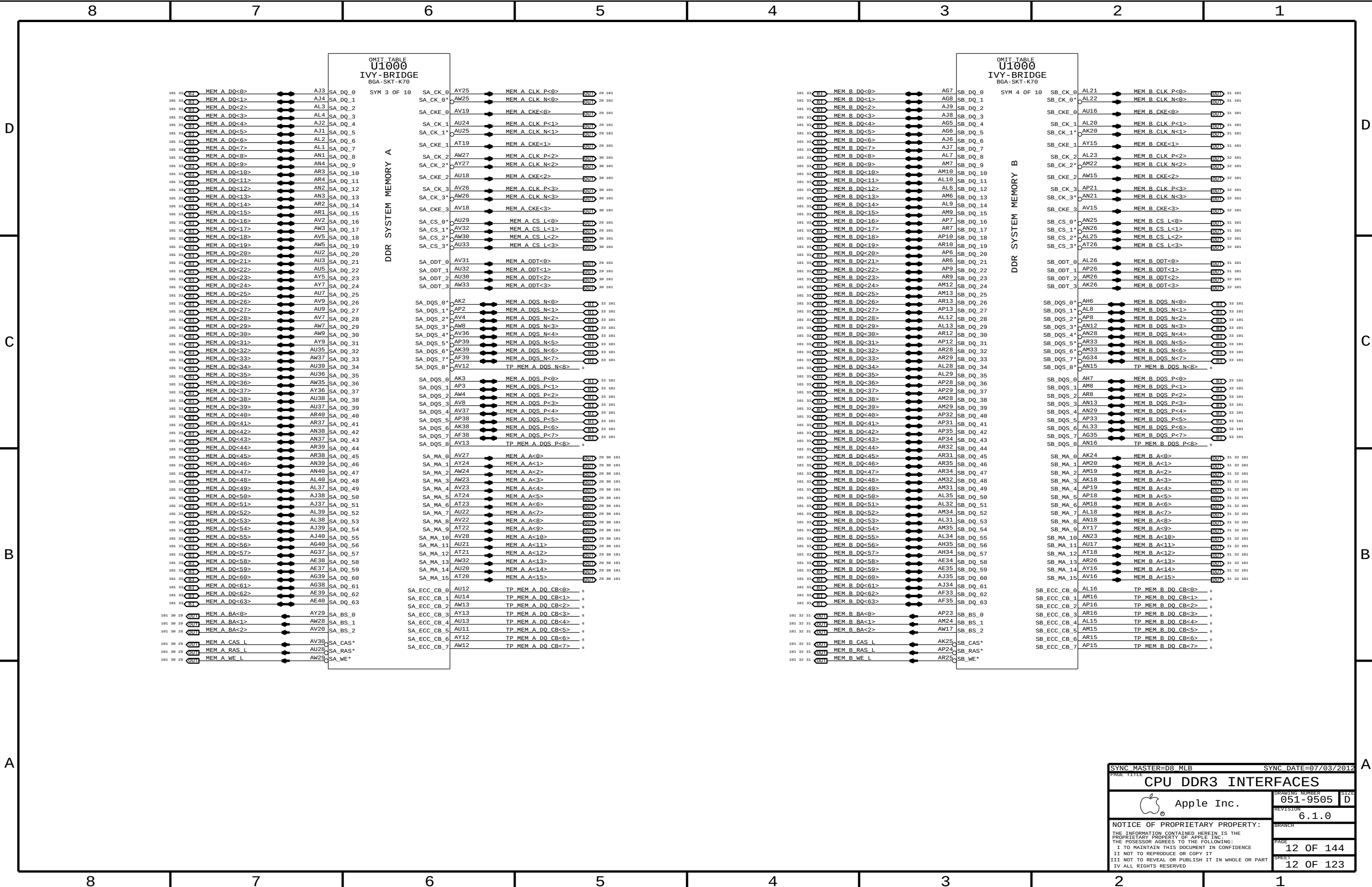
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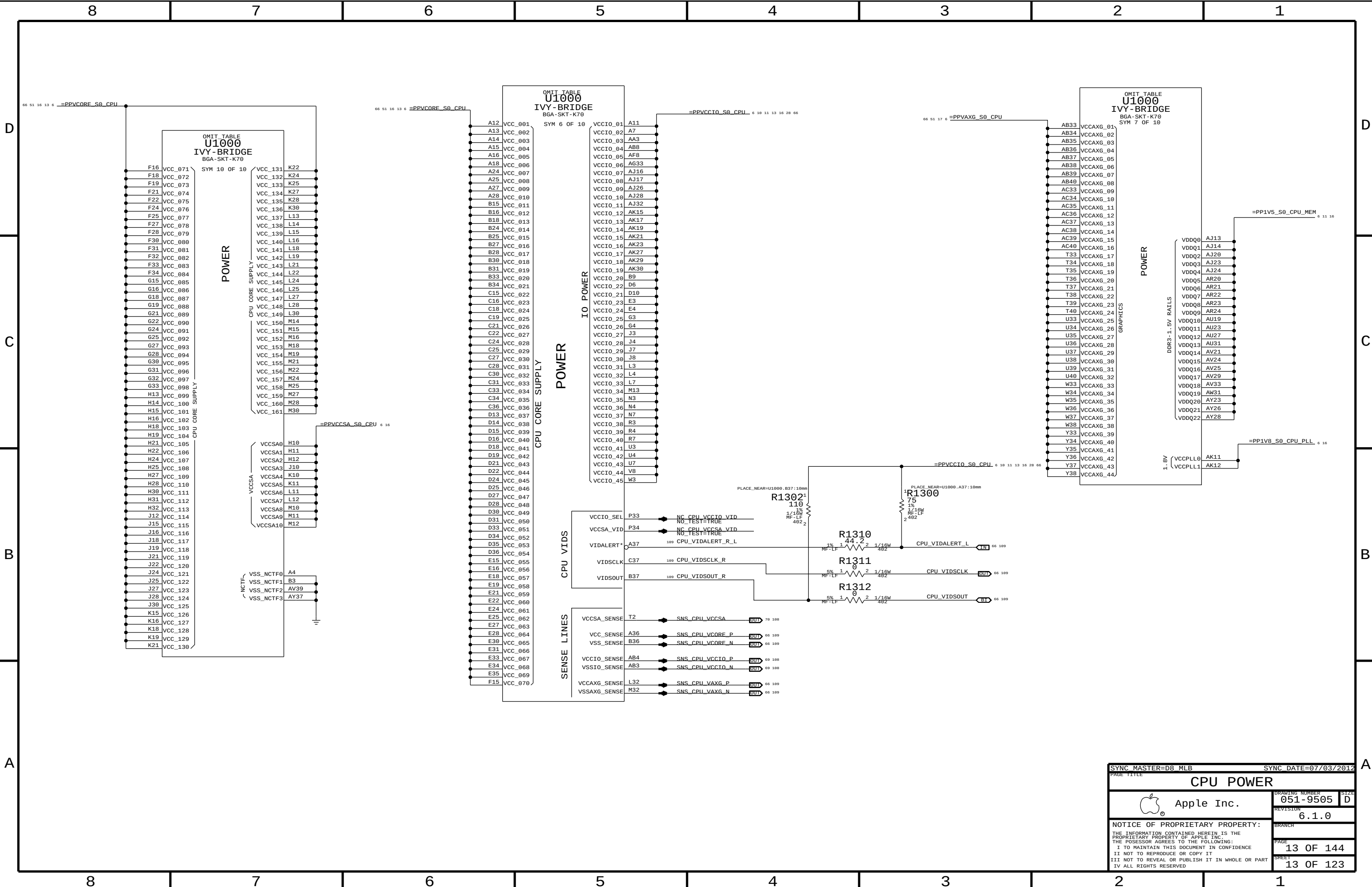


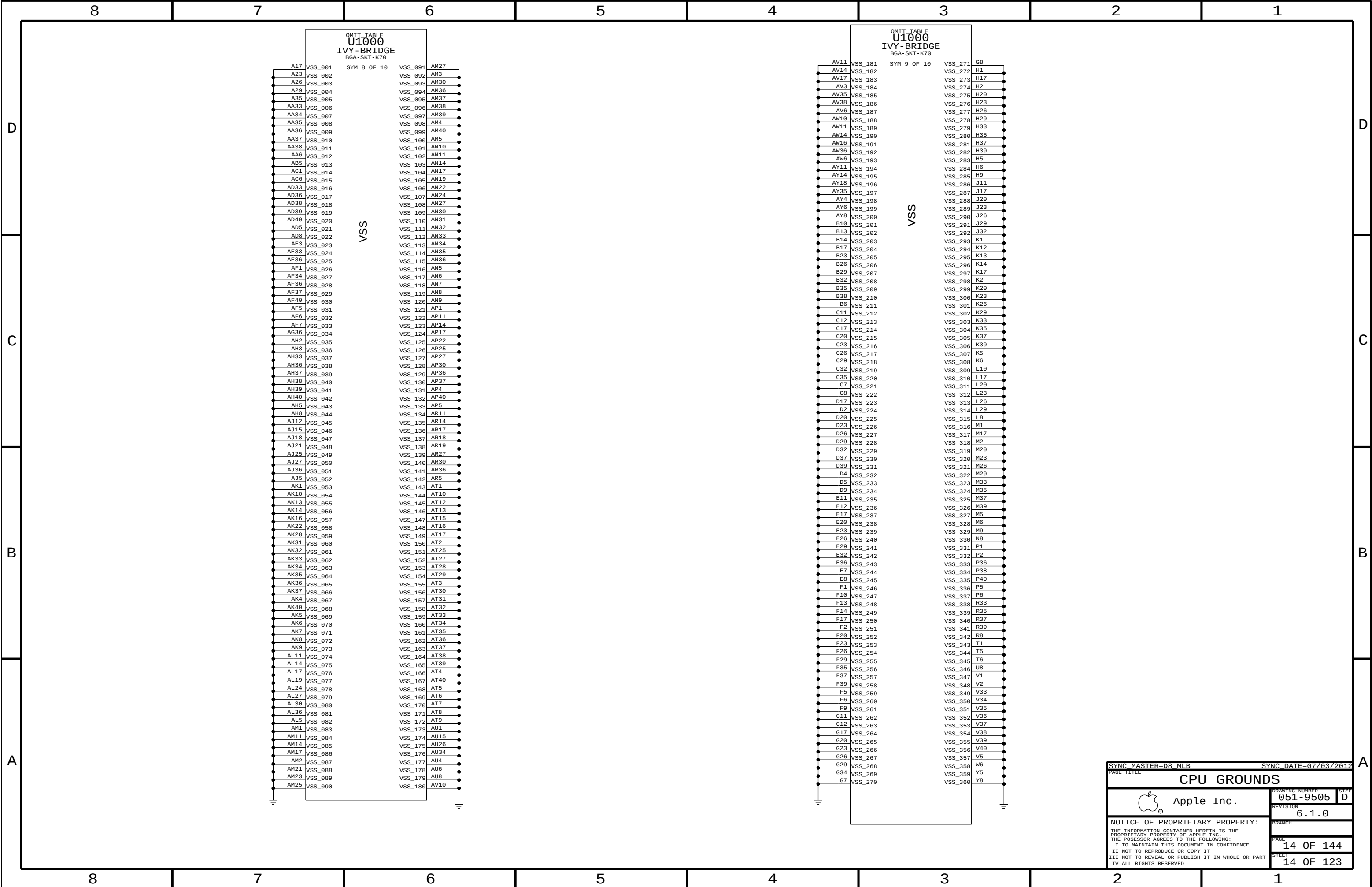
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SYNC DATE=07/03/2012

PAGE TITLE

CPU GROUNDS

 Apple Inc.

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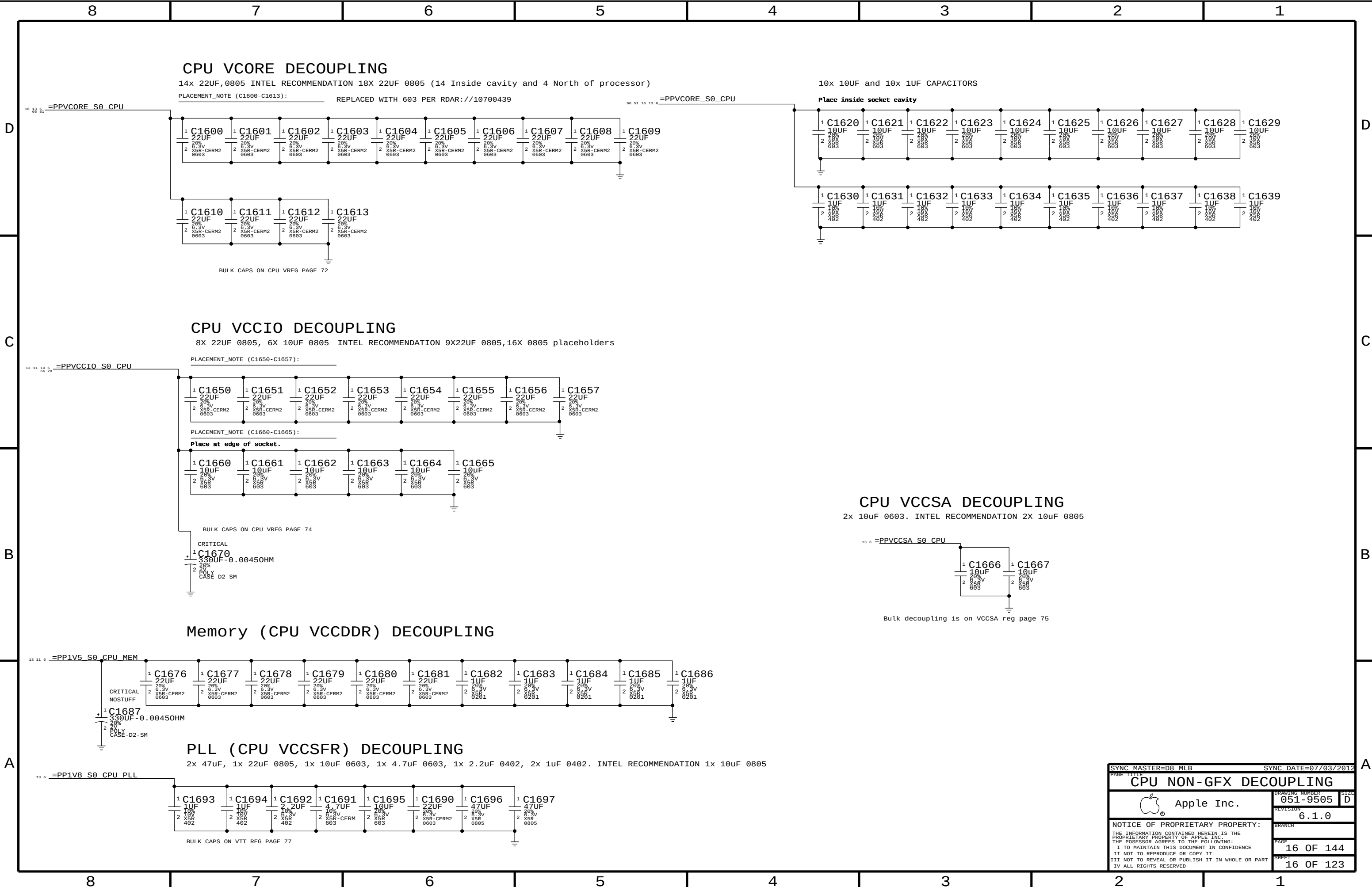
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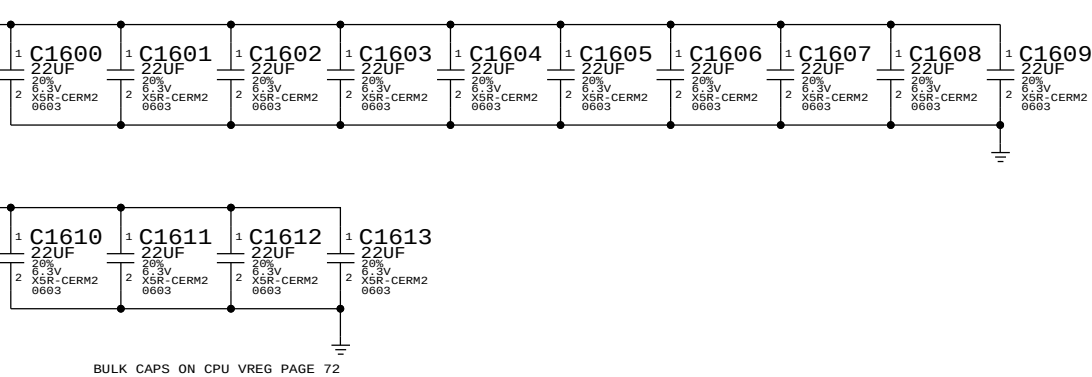
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14 OF 123



CPU Vcore DECOUPLING

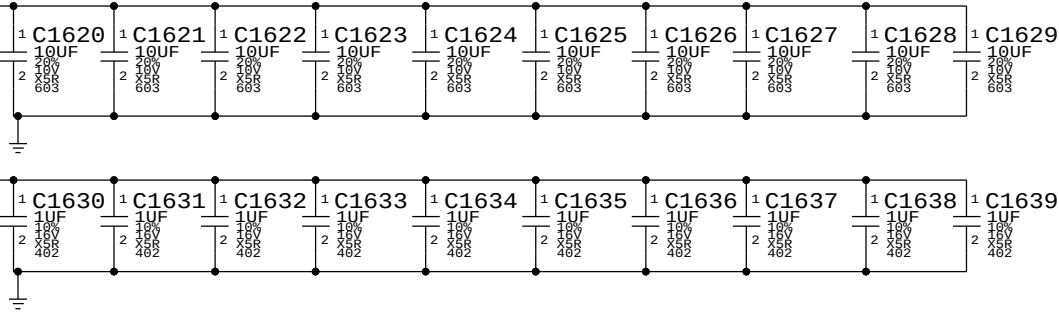
14x 22uF,0805 INTEL RECOMMENDATION 18X 22uF 0805 (14 Inside cavity and 4 North of processor)

PLACEMENT_NOTE (C1600-C1613): REPLACED WITH 603 PER RDAR://10700439



10x 10uF and 10x 1uF CAPACITORS

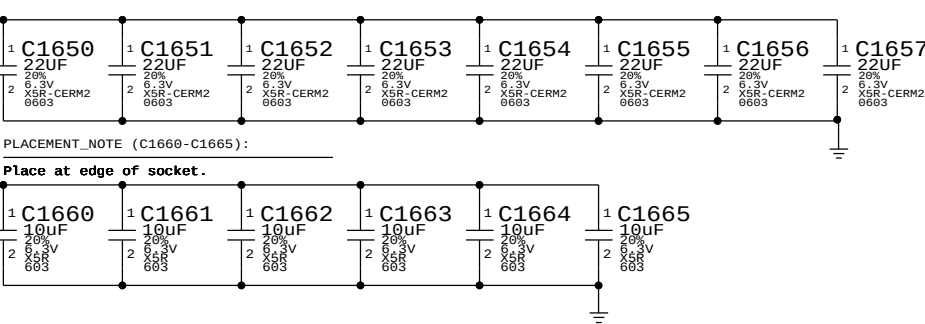
Place inside socket cavity



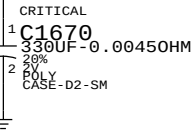
CPU Vccio DECOUPLING

8X 22uF 0805, 6X 10uF 0805 INTEL RECOMMENDATION 9X22uF 0805,16X 0805 placeholders

PLACEMENT_NOTE (C1650-C1657):

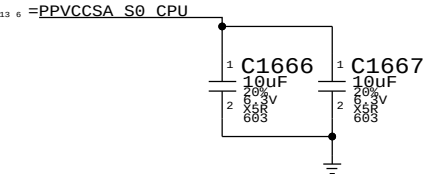


BULK CAPS ON CPU VREG PAGE 74



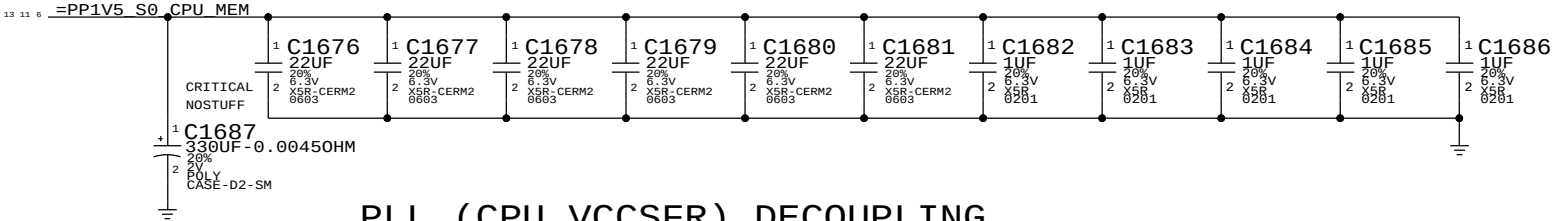
CPU Vccsa DECOUPLING

2x 10uF 0603. INTEL RECOMMENDATION 2X 10uF 0805



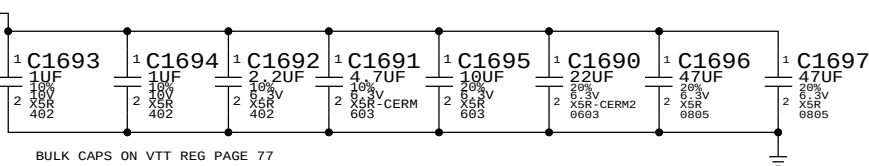
Bulk decoupling is on Vccsa reg page 75

Memory (CPU Vccddr) DECOUPLING

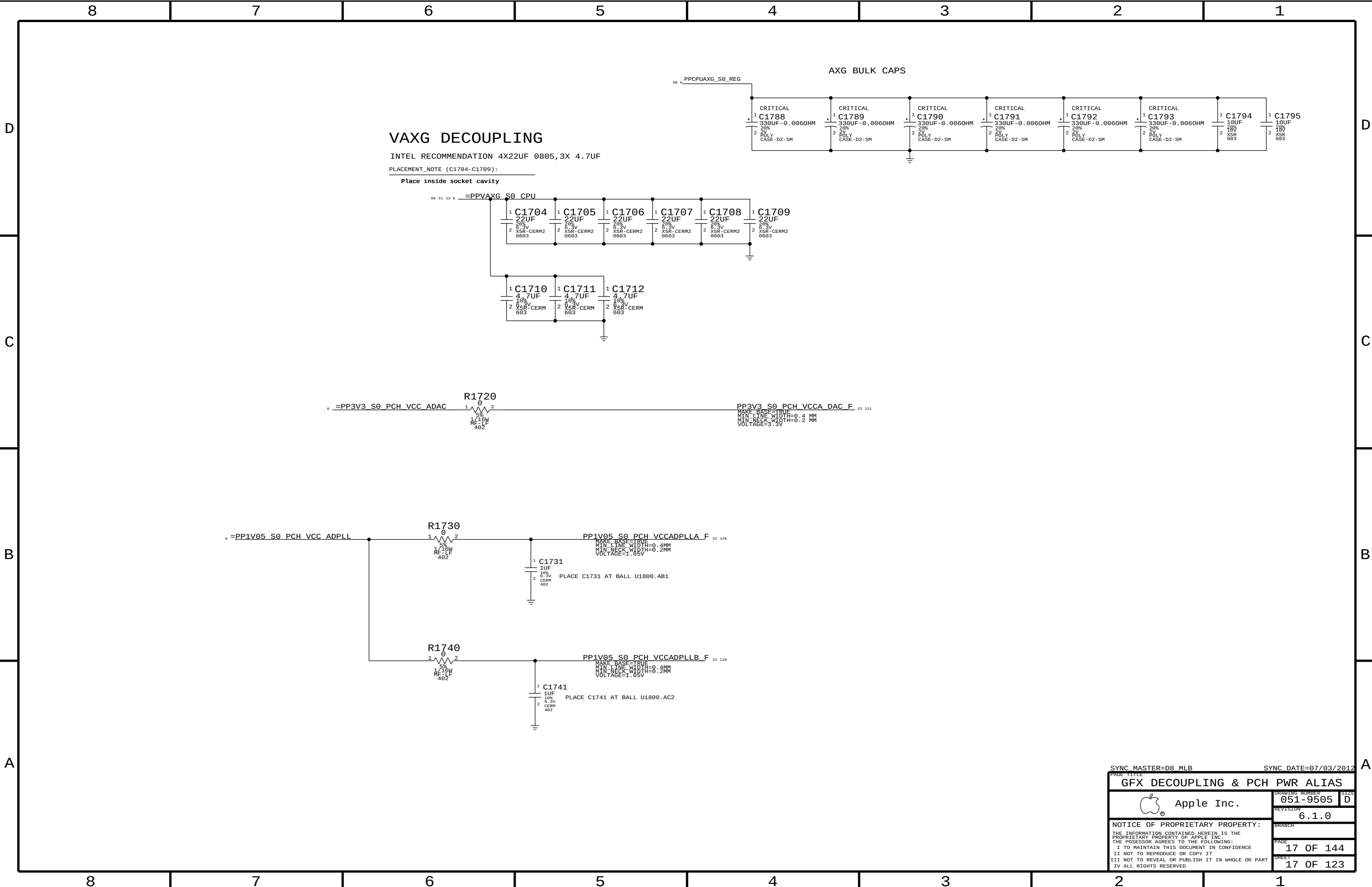


PLL (CPU Vccsfr) DECOUPLING

2x 47uF, 1x 22uF 0805, 1x 10uF 0603, 1x 4.7uF 0603, 1x 2.2uF 0402, 2x 1uF 0402. INTEL RECOMMENDATION 1x 10uF 0805



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CPU NON-GFX DECOUPLING			
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VAXG DECOUPLING

INTEL RECOMMENDATION 4X22UF 0805,3X 4.7UF

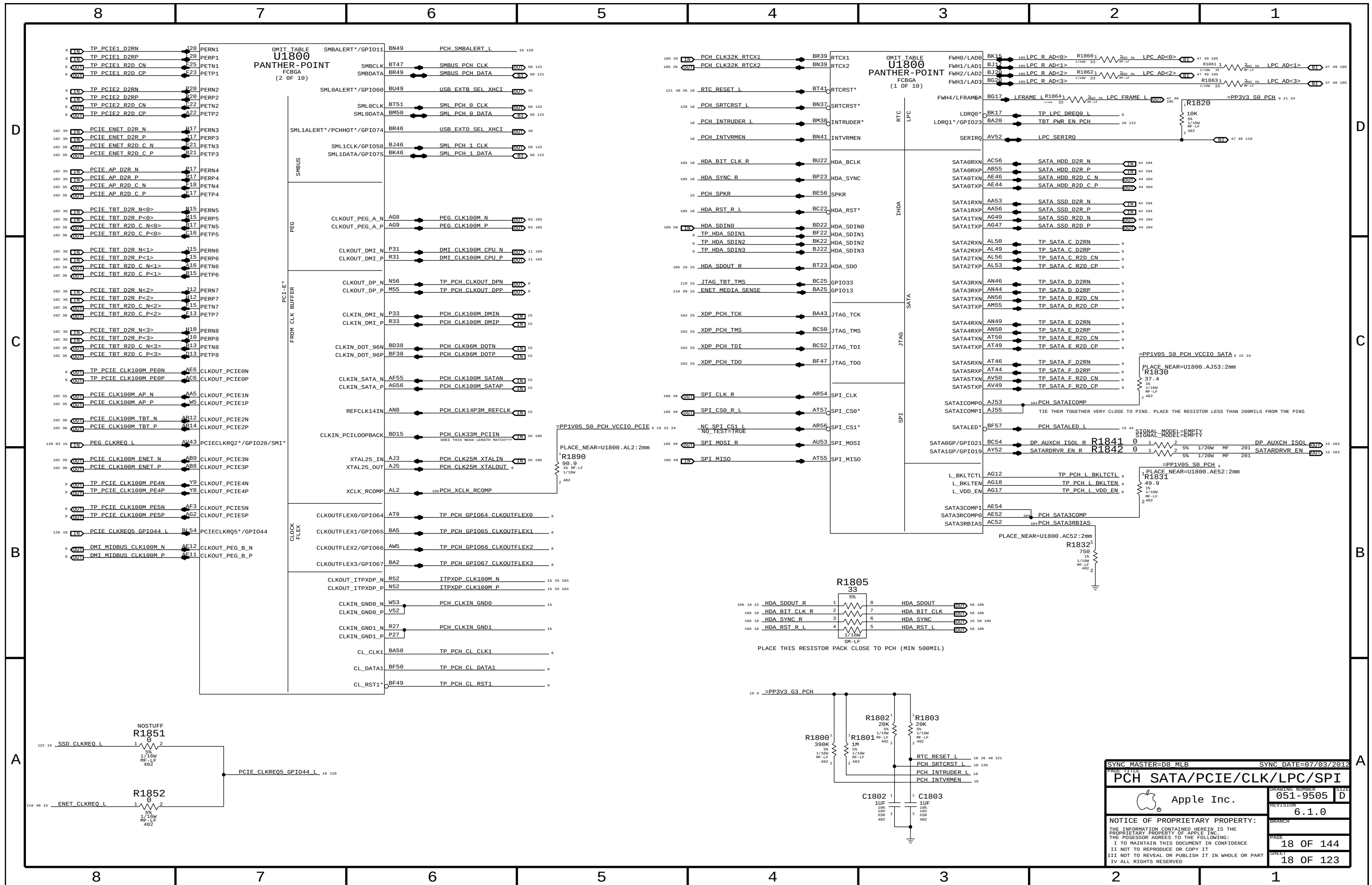
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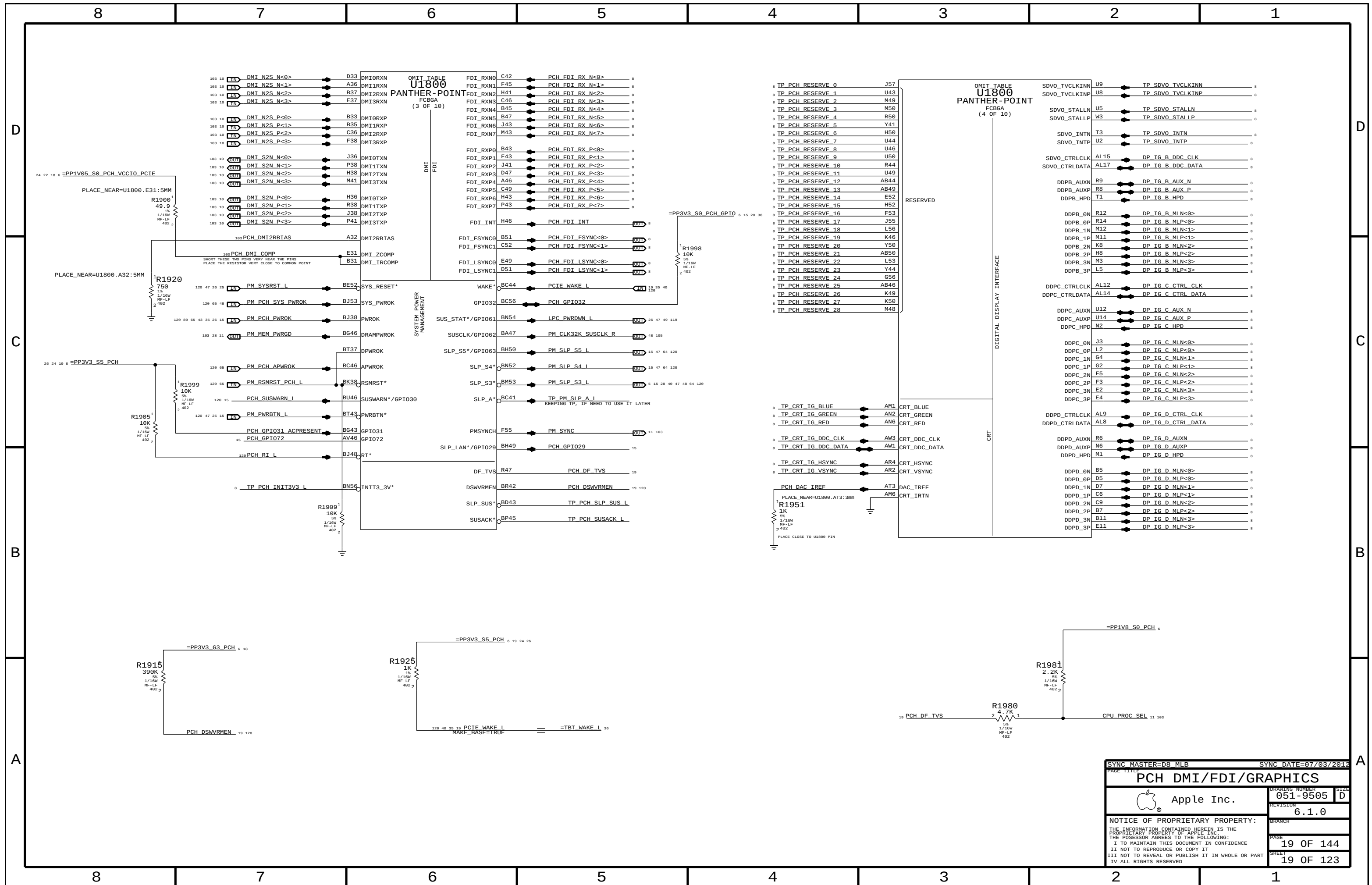
Place inside socket cavity

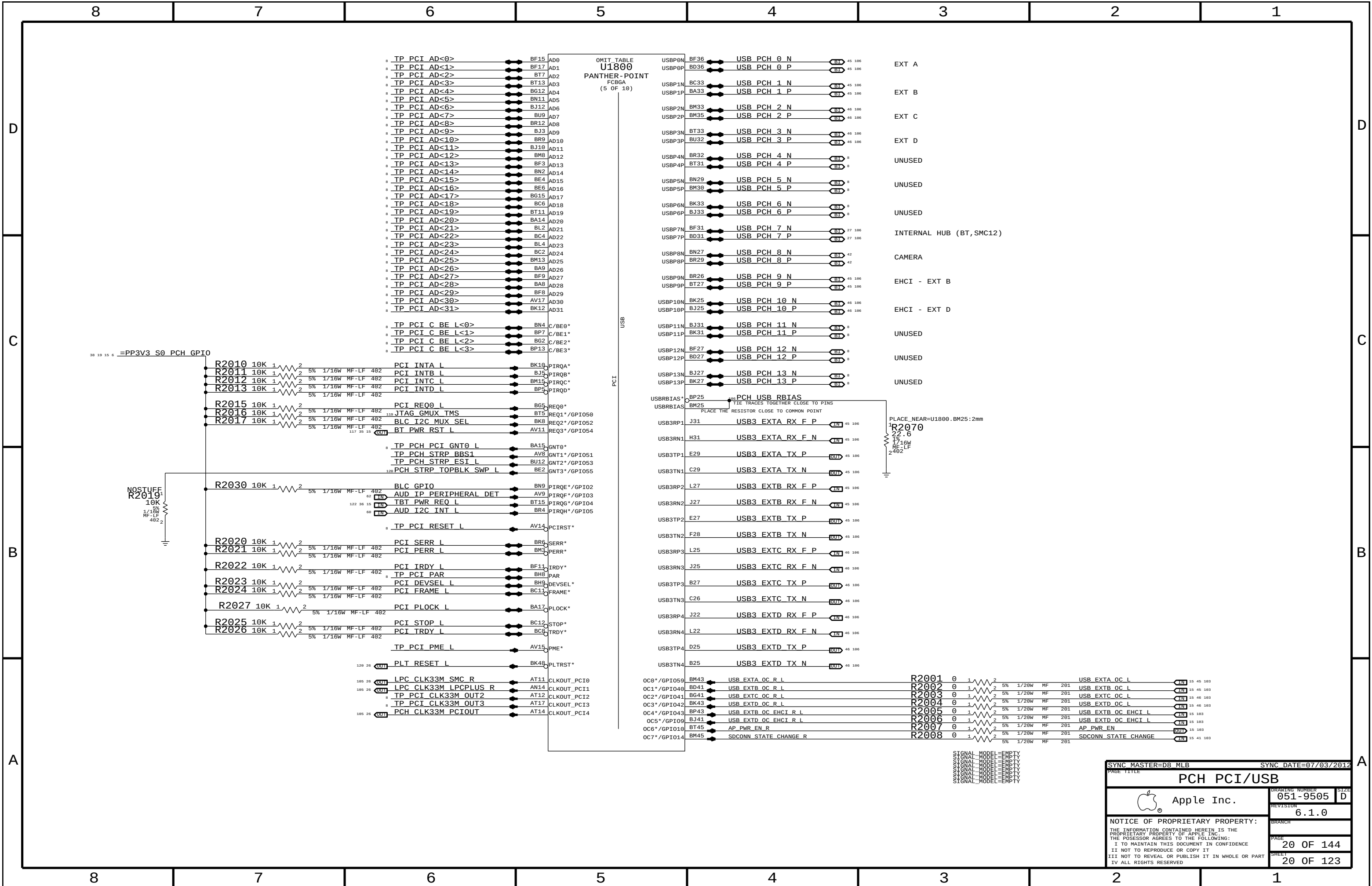
AXG BULK CAPS

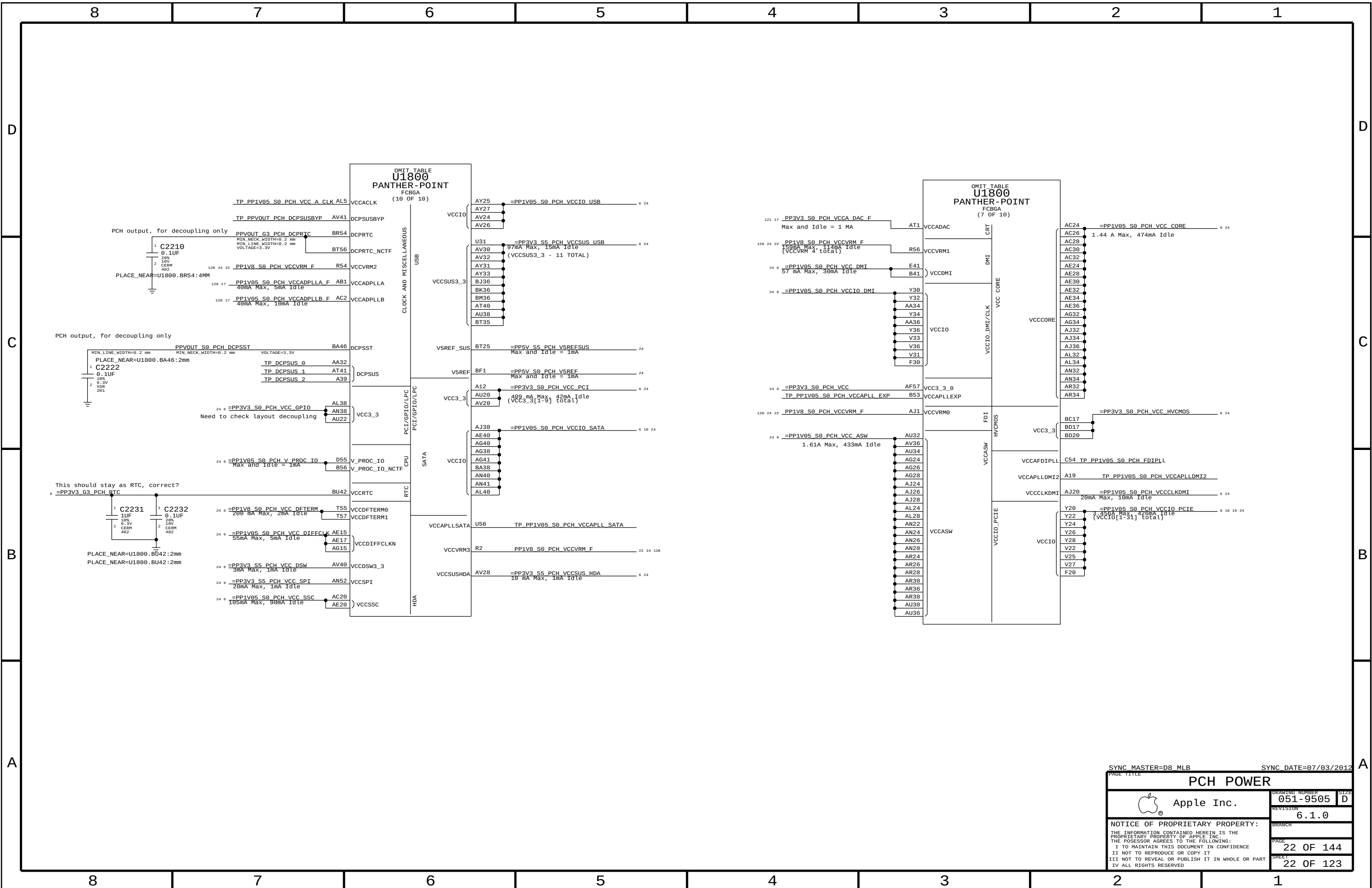
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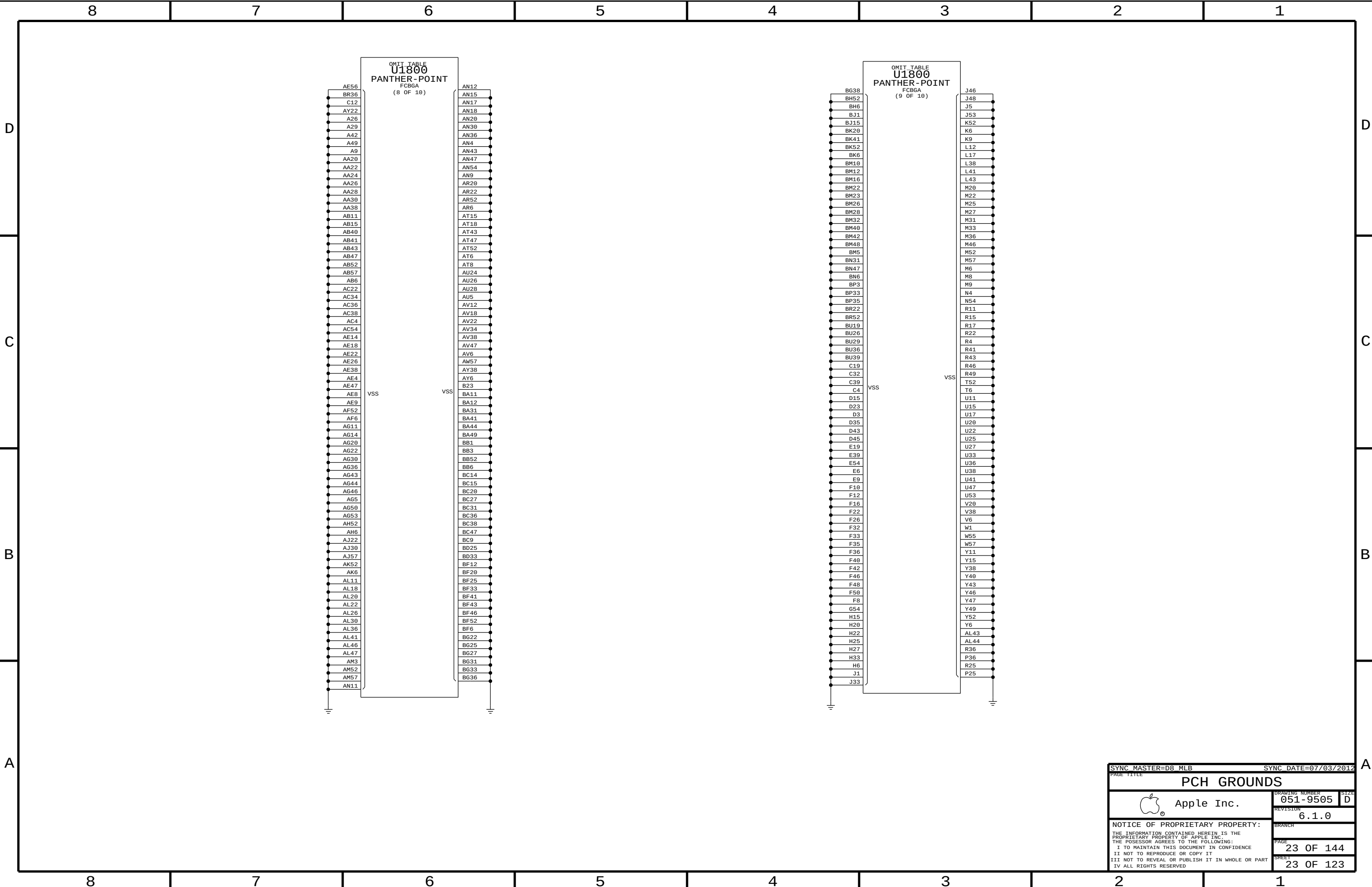
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GFX DECOUPLING & PCH PWR ALIAS		051-9505		D
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		SHEET		17 OF 123







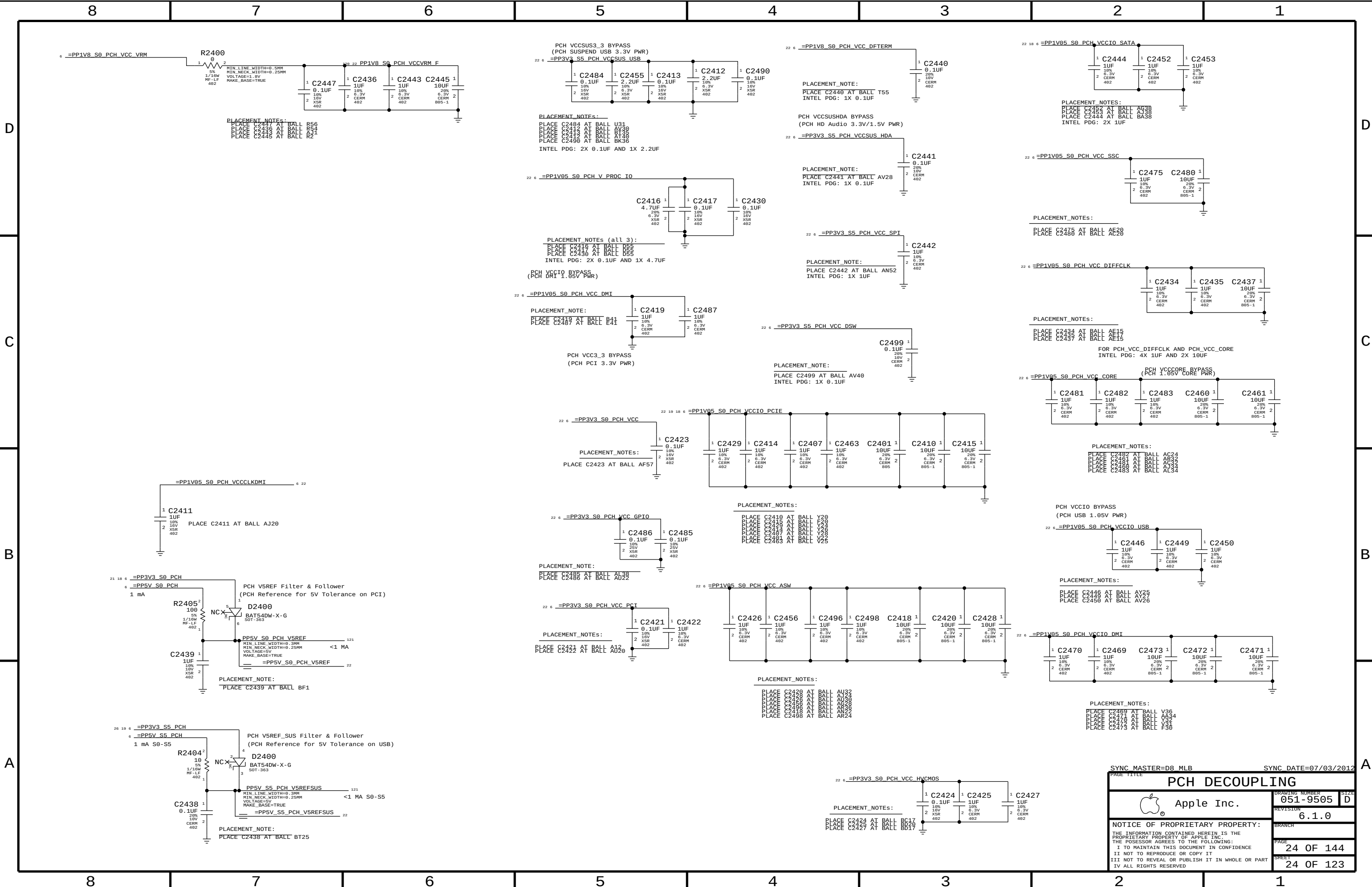




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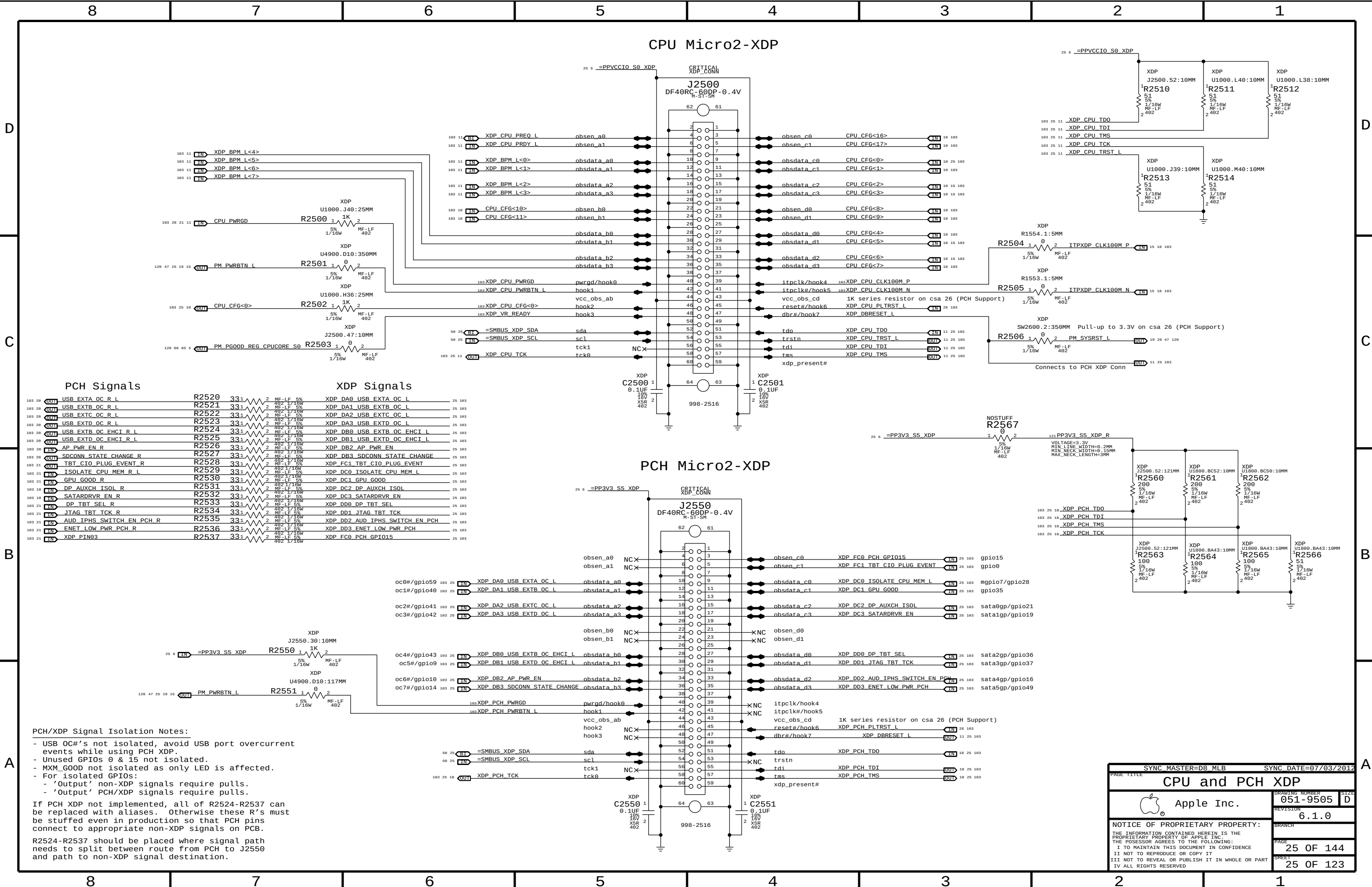
PAGE TITLE		PCH GROUNDS	
 Apple Inc.		DRAWING NUMBER	051-9505
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CPU Micro2-XDP

PCH Micro2-XDP

PCH Signals

XDP Signals

Table mapping PCH signals to XDP signals. Columns include PCH Signals, PCH Pin, XDP Signals, and XDP Pin. Rows include USB signals, power management signals, and various I/O signals.

PCH/XDP Signal Isolation Notes:

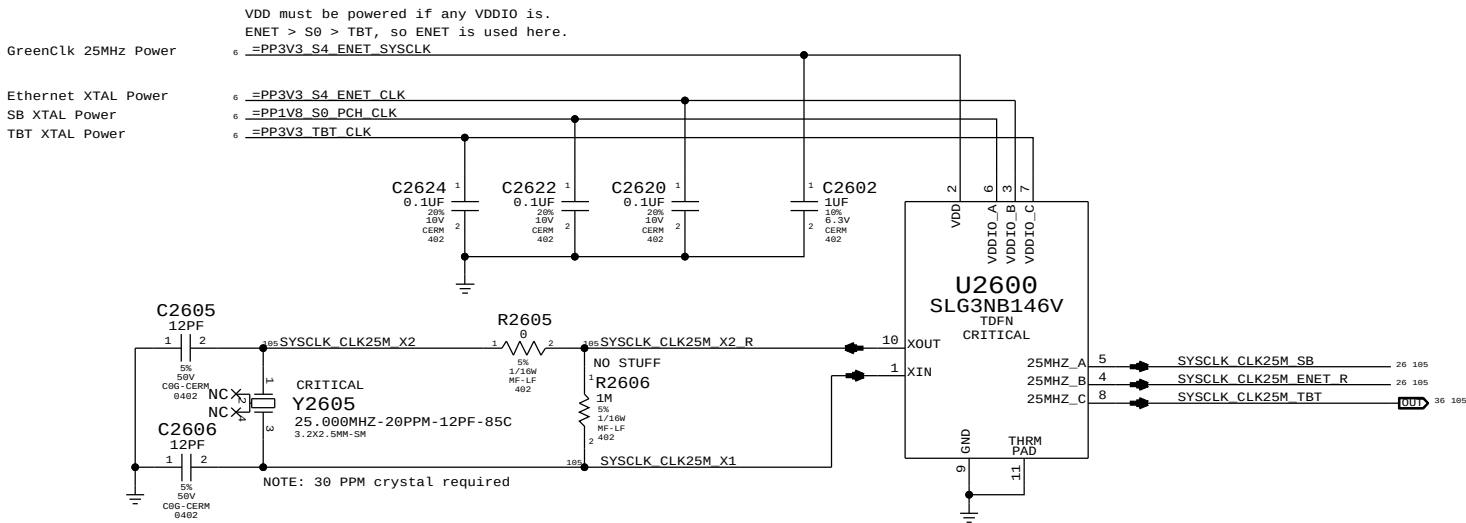
- USB OC#'s not isolated, avoid USB port overcurrent events while using PCH XDP.
- Unused GPIOs 0 & 15 not isolated.
- MXM_GOOD not isolated as only LED is affected.
- For isolated GPIOs:
 - 'Output' non-XDP signals require pulls.
 - 'Output' PCH/XDP signals require pulls.

If PCH XDP not implemented, all of R2524-R2537 can be replaced with aliases. Otherwise these R's must be stuffed even in production so that PCH pins connect to appropriate non-XDP signals on PCB.

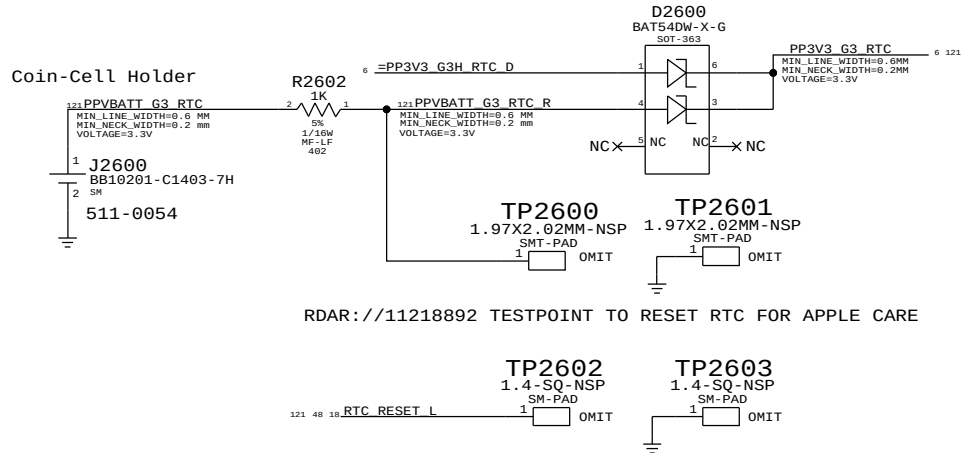
R2524-R2537 should be placed where signal path needs to split between route from PCH to J2550 and path to non-XDP signal destination.

Form containing title block information: SYNC MASTER=D8 MLB, SYNC DATE=07/03/2012, CPU and PCH XDP, Apple Inc. logo, Drawing Number 051-9505, Revision 6.1.0, and a table of page numbers (25 OF 144, 25 OF 123).

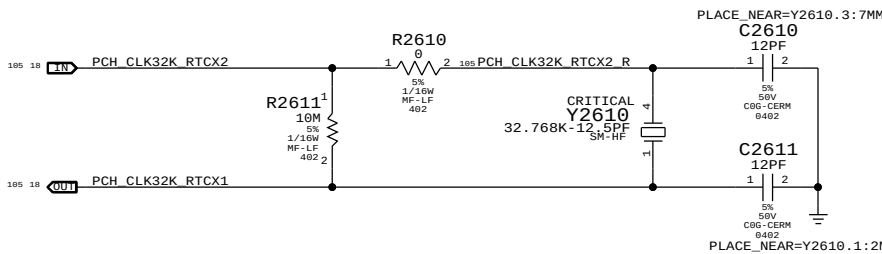
System 25MHz Clock Generator



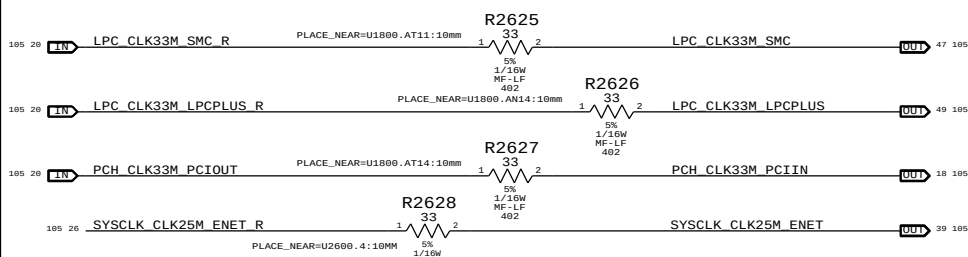
RTC Power Sources



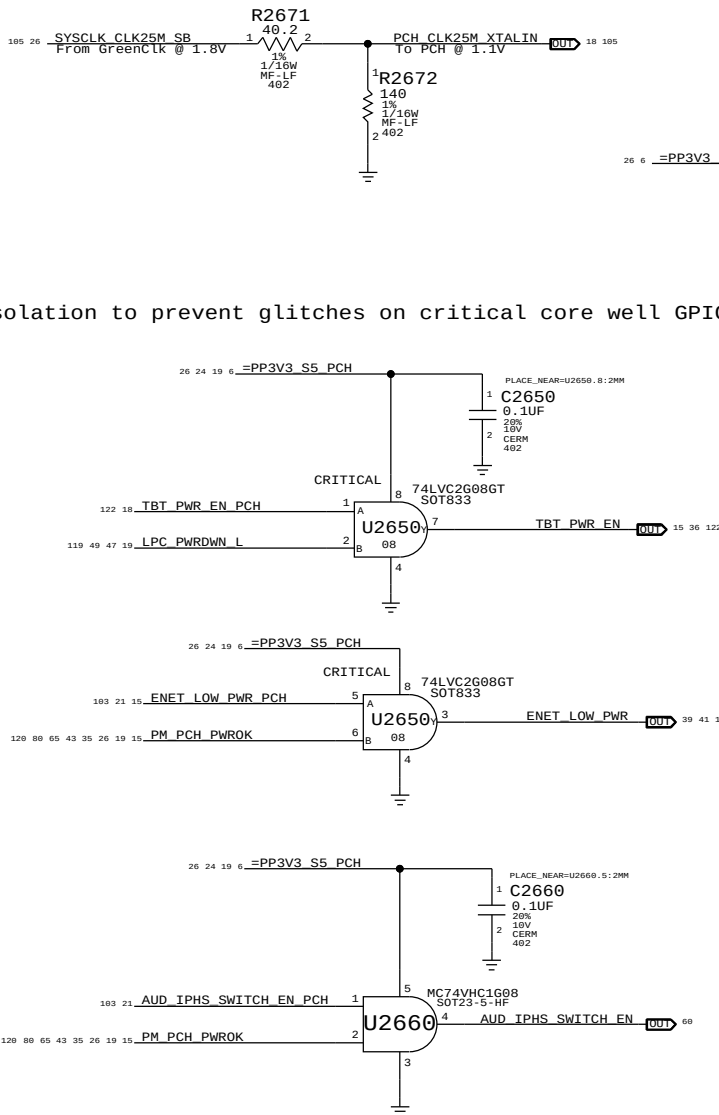
PCH RTC Crystal



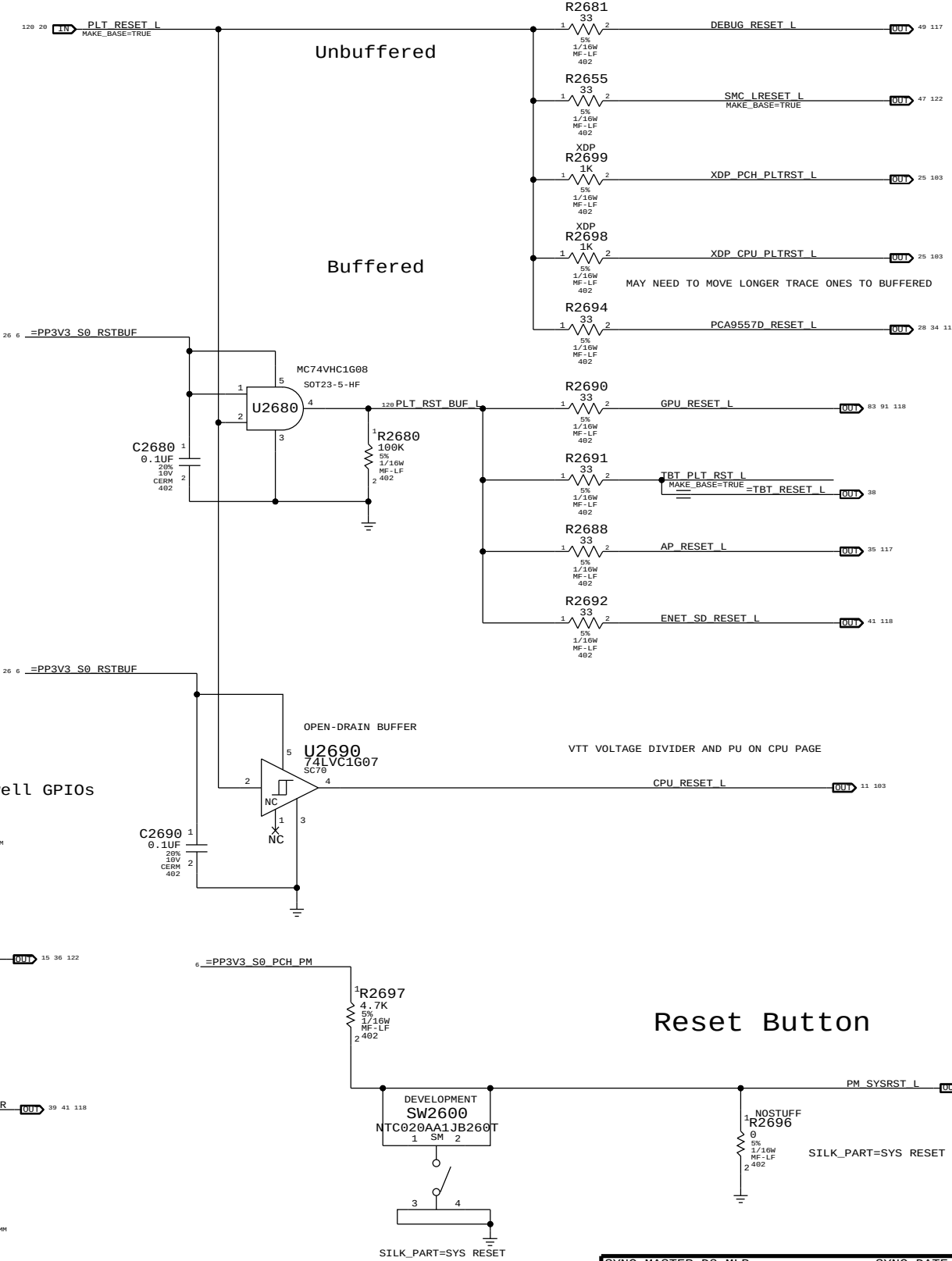
Clock series termination



PCH 25MHZ CLOCK

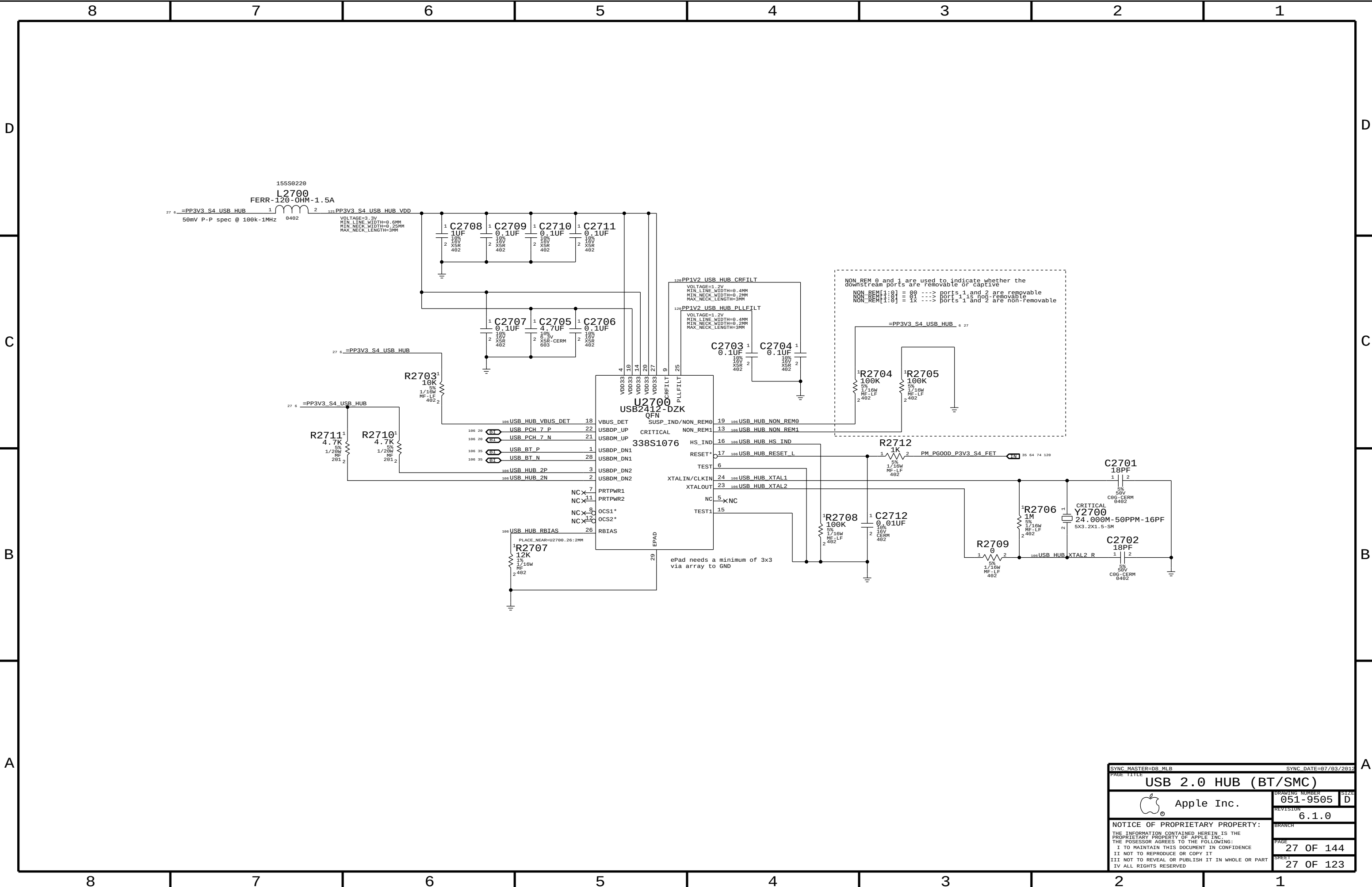


Platform Reset Connections



Reset Button

SYNC MASTER=D8_MLB		SYNC DATE=07/03/2012	
PAGE TITLE		CHIPSET SUPPORT	
Apple Inc.		DRAWING NUMBER	051-9505
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PAGE TITLE			
USB 2.0 HUB (BT/SMC)			
 Apple Inc.		DRAWING NUMBER	051-9505
		REVISION	6.1.0
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MEM_RESET_L Generator

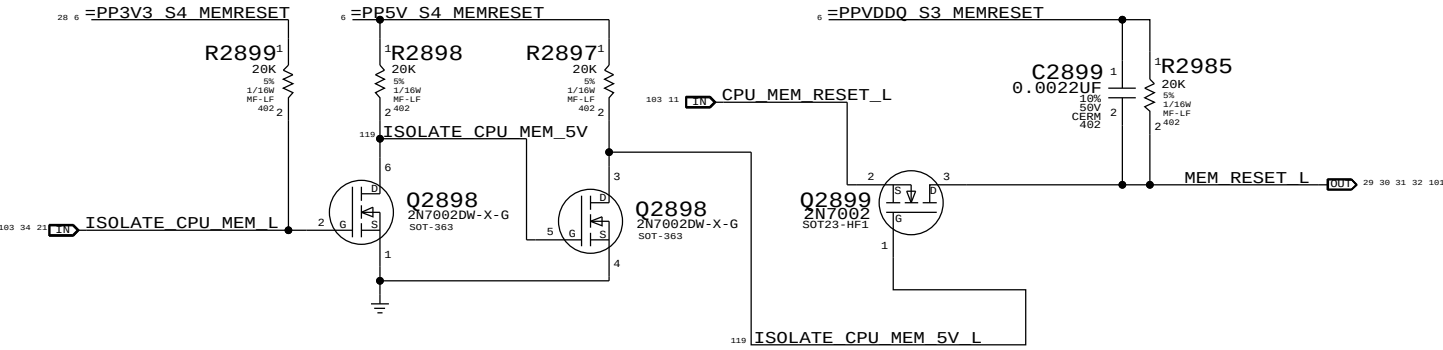
The circuits below handle MEMVTT power during S0->S3->S0 transitions, as well as isolating the CPU's SM_DRAMRST# output from the S0-DIMMs when necessary.

ISOLATE_CPU_MEM_L GPIO state during S3<->S0 transitions determines behaviour of signals.

WHEN HIGH: MEM_RESET_L NOT ISOLATED.

WHEN LOW: MEM_RESET_L IS ISOLATED.

MEM_RESET_L = !ISOLATE_CPU_MEM_L + CPU_MEM_RESET_L (Block CPU from driving MEM_RESET_L in S3)

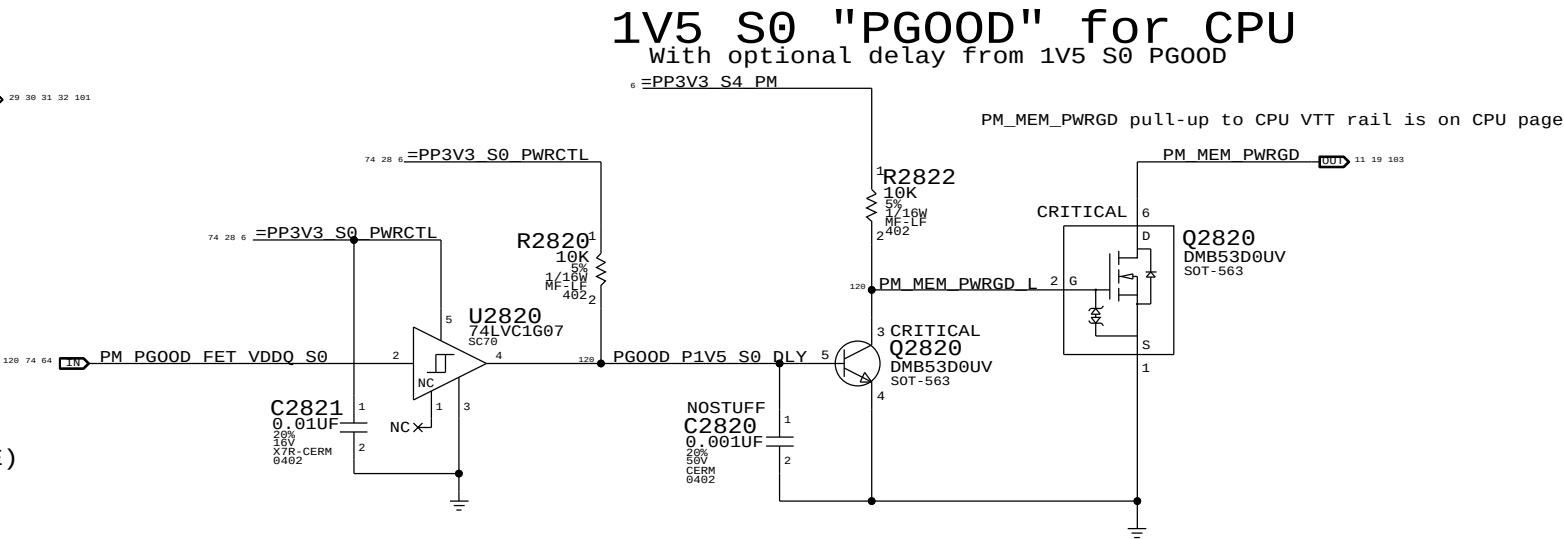


MEMVTT_EN Generator

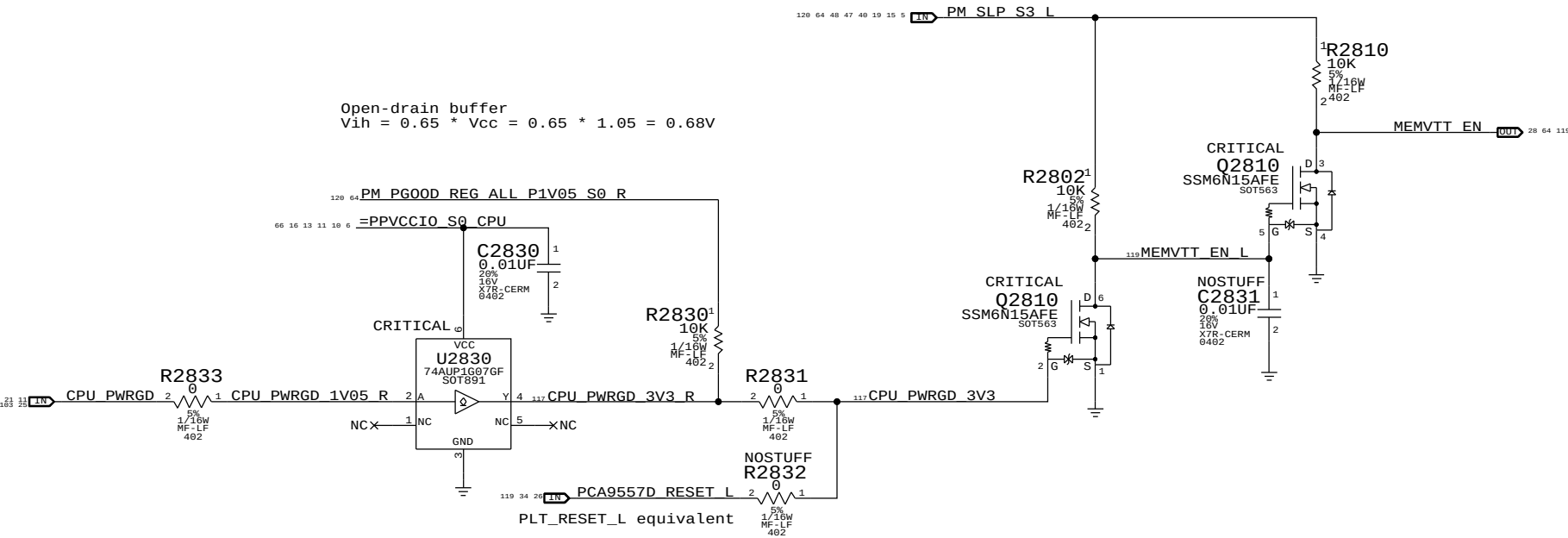
rdar://11117167 Enables MEMVTT when PCH drives CPU PWRGD.

CPU does not drive MEM_CKE until VCCORE activated but CPU 1V5 (VDDQ) leaks into it. Clamping MEMVTT will keep the MEM_CKE low until CPU actively controls it. MEMVTT Clamp actively holds MEMVTT rail low until MEMVTT is enabled.

MEMVTT_EN = CPU_PWRGD * PM_SLP_S3_L (VTT is enabled when PCH tells CPU to enable VCCORE)

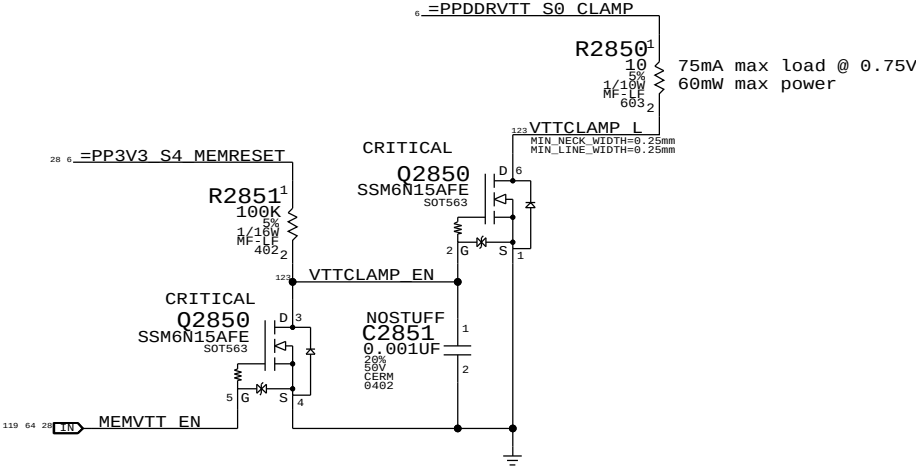


Open-drain buffer
Vih = 0.65 * Vcc = 0.65 * 1.05 = 0.68V



MEMVTT Clamp

Ensures CKE signals are held low in S3 and in S0 before CPU PWRGD



Step	ISOLATE_CPU_MEM_L	PM_SLP_S3_L	CPU_PWRGD	CPU_MEM_RESET_L	MEM_RESET_L	MEMVTT_EN
S0	0	1	1	CPU_MEM_RESET_L	1	1
1	0	1	1	1	1	1
2	0	1	1	1	1	1
3	0	0	0	X	1	0
4	0	0	0	X	1	0
5	0	1	1	0 (*)	1	1
6	0	1	1	1	1	1
S0	1	1	1	CPU_MEM_RESET_L	1	1

(*) CPU_MEM_RESET_L asserts due to loss of PM_MEM_PWRGD, must wait for software to clear before deasserting ISOLATE_CPU_MEM_L GPIO.

NOTE: On a S5->S0 transition, ISOLATE_CPU_MEM_L will default low. Rails will power-up as if from S3, but MEM_RESET_L now needs to be asserted in S0. Software must de-assert ISOLATE_CPU_MEM_L and then generate a valid reset cycle on CPU_MEM_RESET_L.

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CPU Memory S3 Support

Apple Inc.

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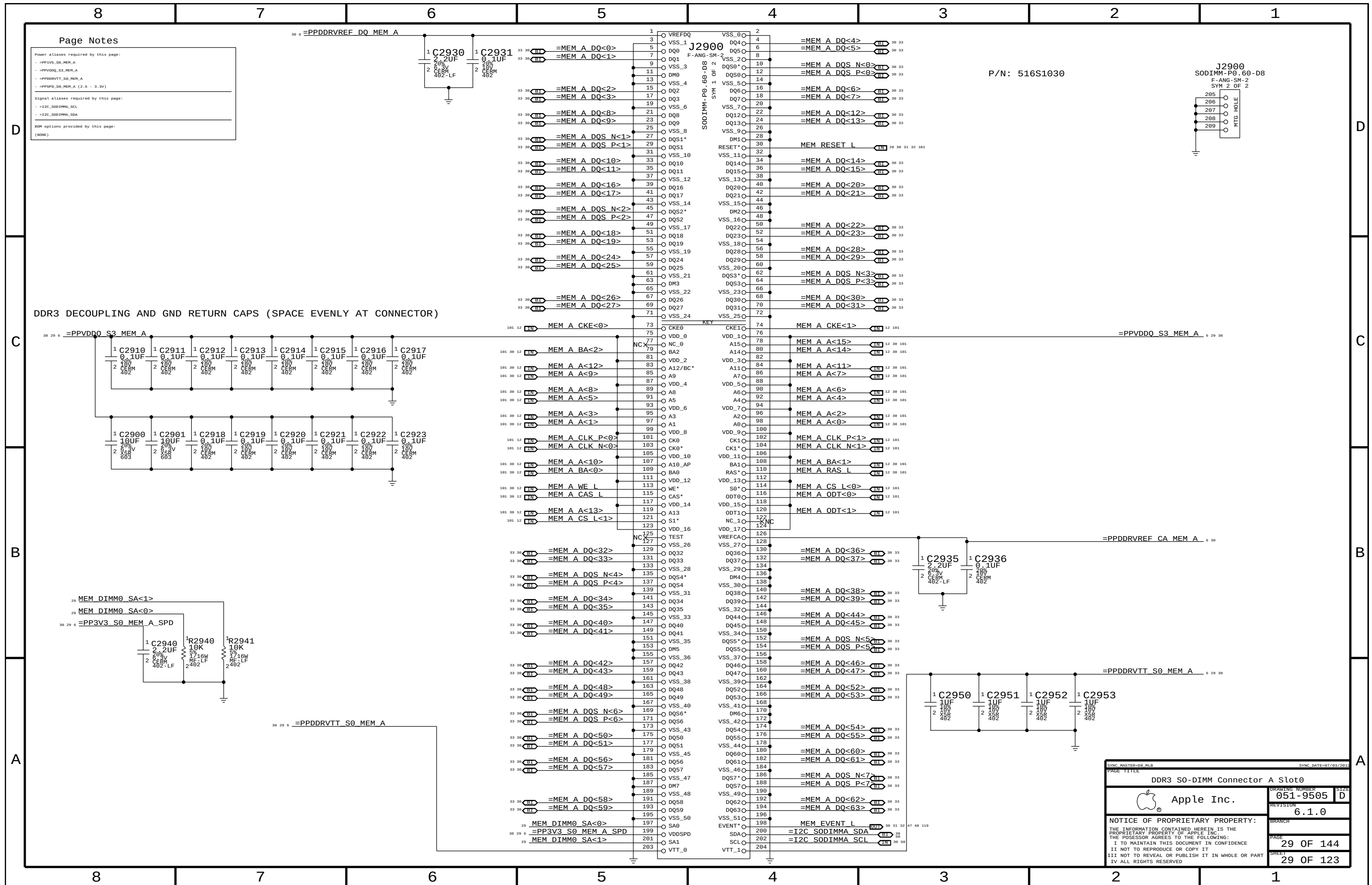
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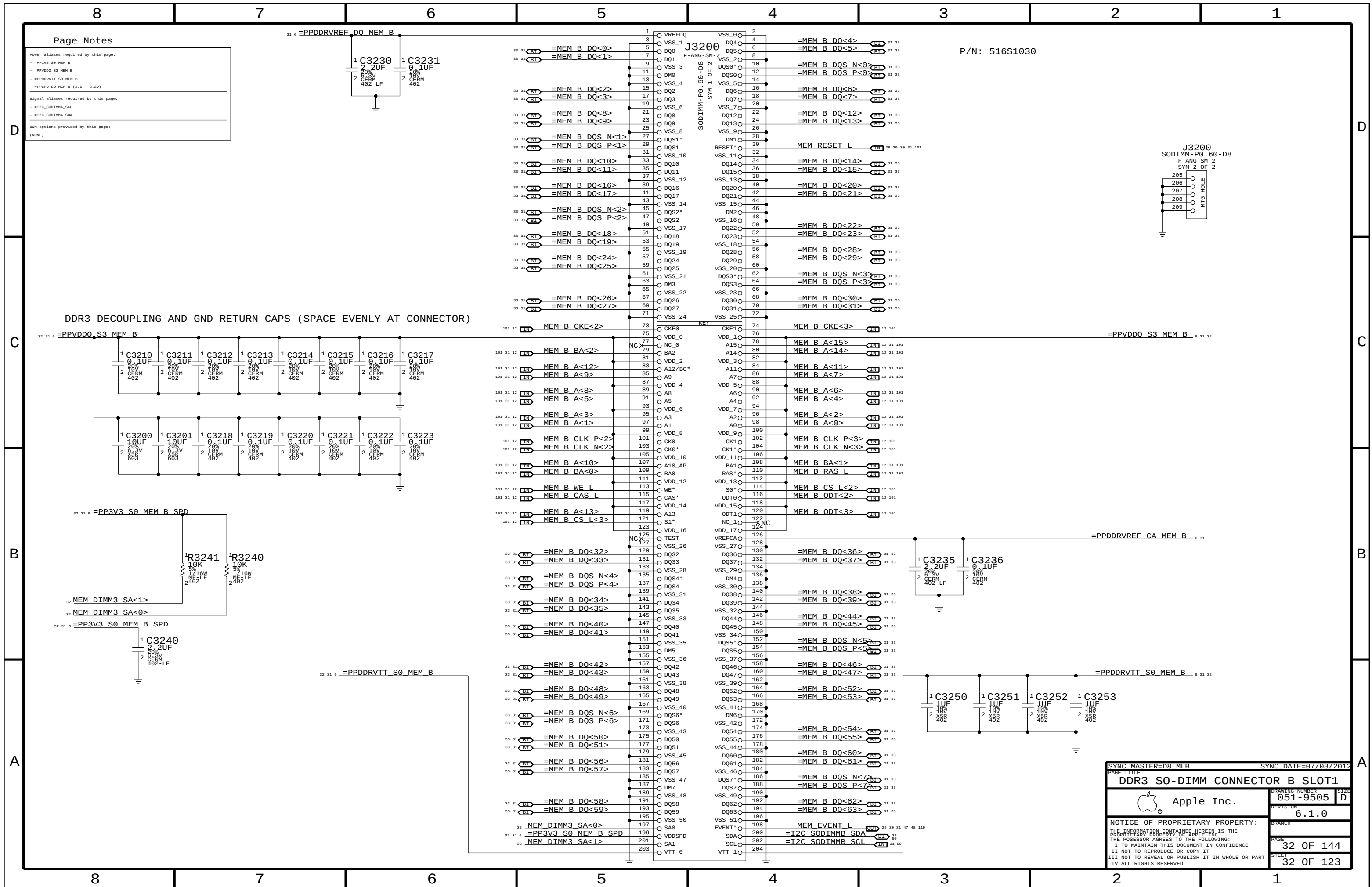
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8		7		6		5		4		3		2		1	
D	101 12	MEM A DQS N<0>	==	=MEM A DQS N<0>	29 30	101 12	MEM B DQS N<0>	==	=MEM B DQS N<0>	31 32	D				
	101 12	MEM A DQS P<0>	==	=MEM A DQS P<0>	29 30	101 12	MEM B DQS P<0>	==	=MEM B DQS P<0>	31 32					
	101 12	MEM A DQ<7>	==	=MEM A DQ<7>	29 30	101 12	MEM B DQ<7>	==	=MEM B DQ<3>	31 32					
	101 12	MEM A DQ<6>	==	=MEM A DQ<6>	29 30	101 12	MEM B DQ<6>	==	=MEM B DQ<6>	31 32					
	101 12	MEM A DQ<5>	==	=MEM A DQ<4>	29 30	101 12	MEM B DQ<5>	==	=MEM B DQ<5>	31 32					
	101 12	MEM A DQ<4>	==	=MEM A DQ<1>	29 30	101 12	MEM B DQ<4>	==	=MEM B DQ<4>	31 32					
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	101 12	MEM A DQ<1>	==	=MEM A DQ<5>	29 30	101 12	MEM B DQ<1>	==	=MEM B DQ<1>	31 32					
	101 12	MEM A DQ<0>	==	=MEM A DQ<0>	29 30	101 12	MEM B DQ<0>	==	=MEM B DQ<0>	31 32					
C	101 12	MEM A DQS N<1>	==	=MEM A DQS N<1>	29 30	101 12	MEM B DQS N<1>	==	=MEM B DQS N<1>	31 32	C				
	101 12	MEM A DQS P<1>	==	=MEM A DQS P<1>	29 30	101 12	MEM B DQS P<1>	==	=MEM B DQS P<1>	31 32					
	101 12	MEM A DQ<15>	==	=MEM A DQ<15>	29 30	101 12	MEM B DQ<15>	==	=MEM B DQ<11>	31 32					
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B	101 12	MEM A DQS N<2>	==	=MEM A DQS N<2>	29 30	101 12	MEM B DQS N<2>	==	=MEM B DQS N<2>	31 32	B				
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A	101 12	MEM A DQS N<3>	==	=MEM A DQS N<3>	29 30	101 12	MEM B DQS N<3>	==	=MEM B DQS N<3>	31 32	A				
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SYNC MASTER=D8_MLB

SYNC DATE=07/03/2012

DDR3 ALIASES AND BITSWAPS

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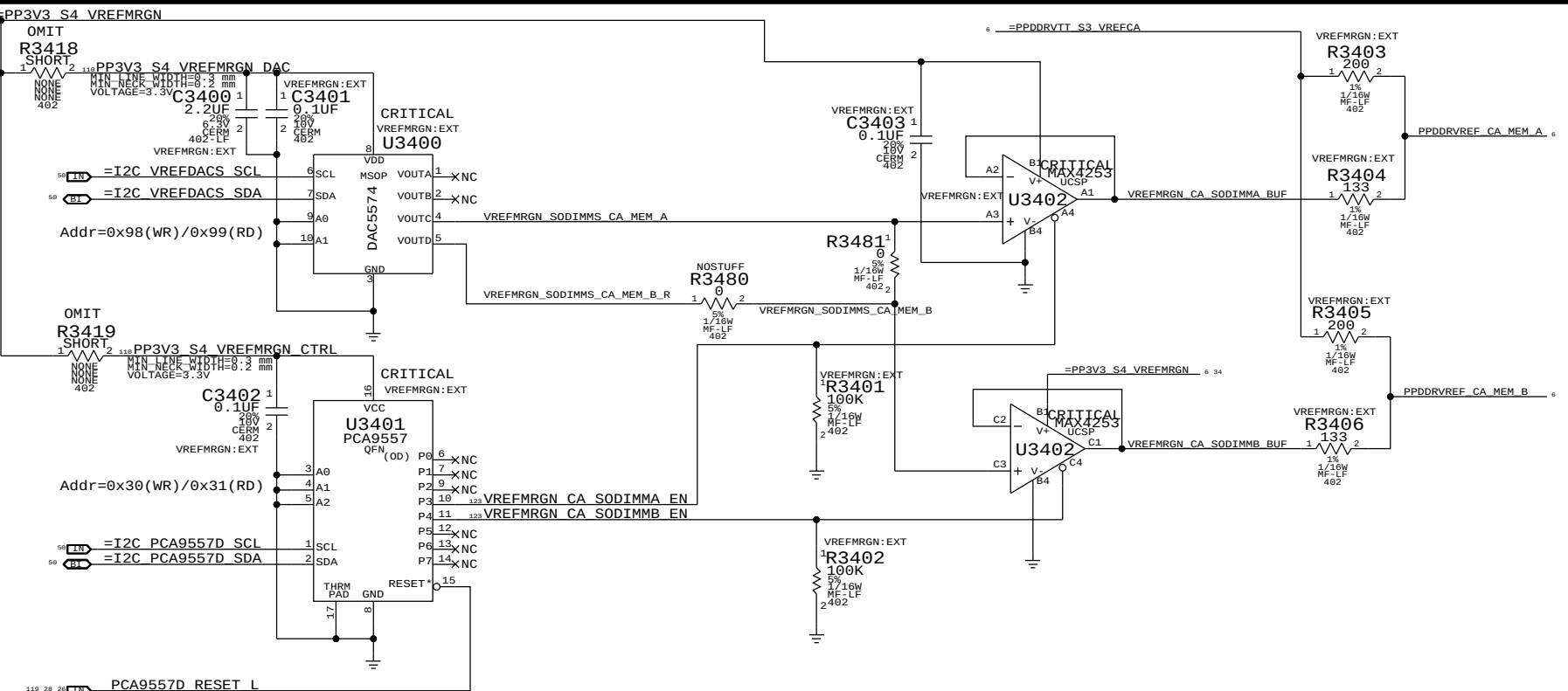
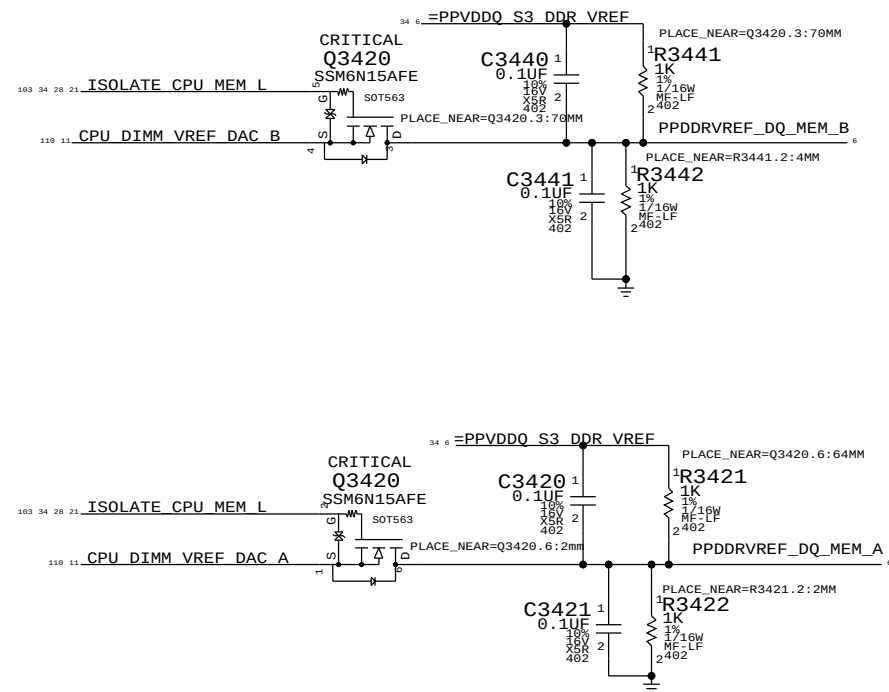
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Driven by CPU

NOTE: CPU DAC output step sizes:
DDR3 (1.5V) 7.70mV per step



RST* on 'platform reset' so that system watchdog will disable margining.

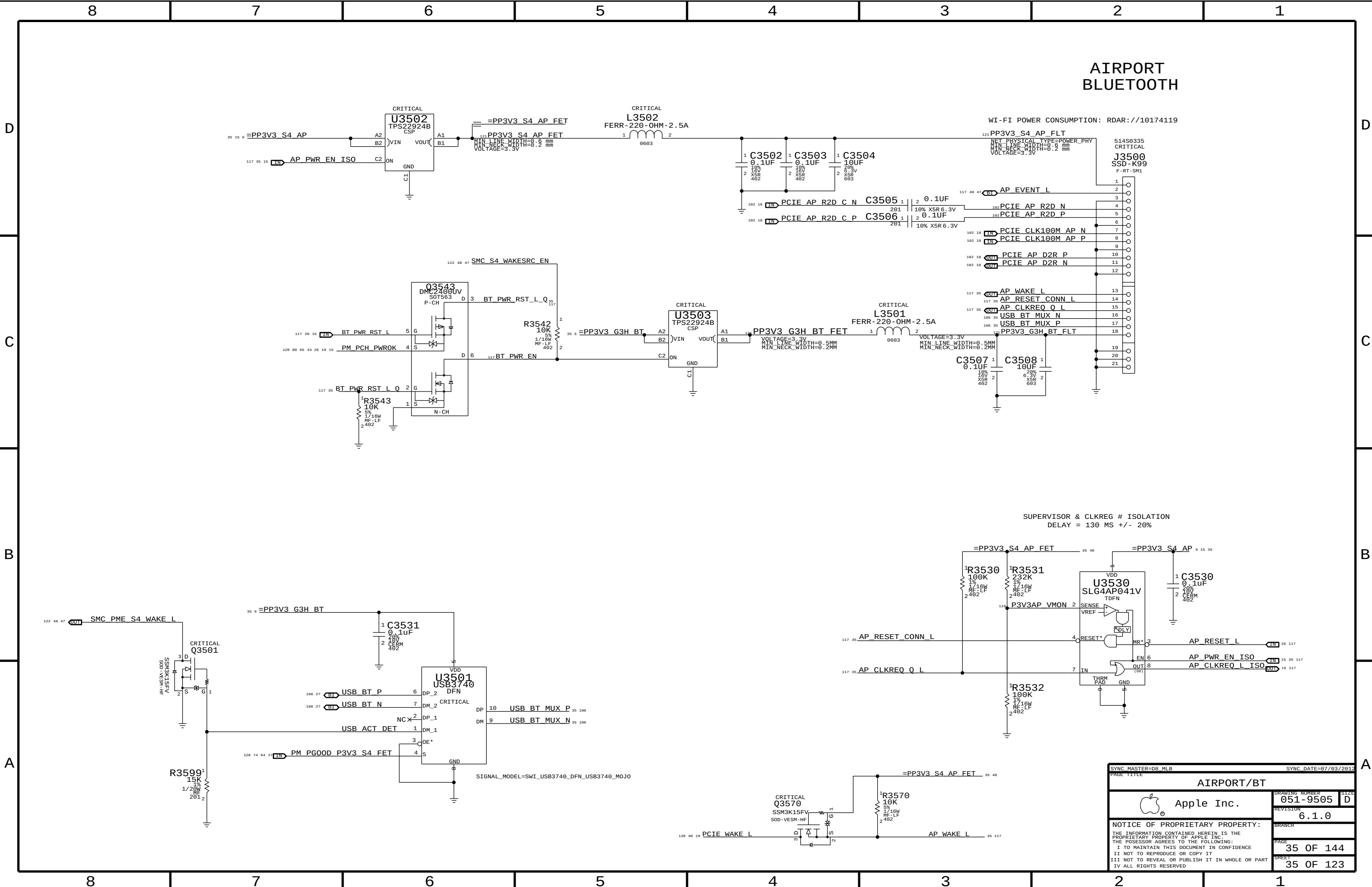
NOTE: MEMVREG and FRAMEBUF share a DAC output, cannot enable both at the same time!

NOTE: Margining will be disabled across all soft-resets and sleep/wake cycles.

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
116S0004	2	RES, MTL FLM, 0, 5%, 402, SM, LF	R3403, R3405		VREFMRGN:N

	MEM A VREF CA	MEM B VREF CA	MEM VREG	GPU Frame Buffer (1.8V, 70% Vref)
DAC Channel:	C	C	D	D
PCA9557D Pin:	3	4	5	6
Nominal value	0.75V (DAC: 0x3A)		1.5V (DAC: 0x3A)	1.267V (DAC: 0x8B)
Margined target:	0.300V - 1.200V (+/- 450mV)		1.000V - 2.000V (+/- 500mV)	1.056V - 1.442V (+/- 180mV)
DAC range:	0.000V - 1.501V (0x00 - 0x74)		0.000V - 3.000V (0x00 - 0x74)	0.000V - 3.300V (0x00 - 0xFF)
VRef current:	+3.4mA - -3.4mA (- = sourced)		+61uA - -61uA (- = sourced)	+6.0mA - -5.0mA (- = sourced)
DAC step size:	7.69mV / step @ output		8.59mV / step @ output	1.51mV / step @ output

SYMC MASTER=D8 MLB		SYMC DATE=07/03/2012	
PAGE TITLE		DRAWING NUMBER	
DDR3/FRAMEBUF VREF MARGINING		051-9505	
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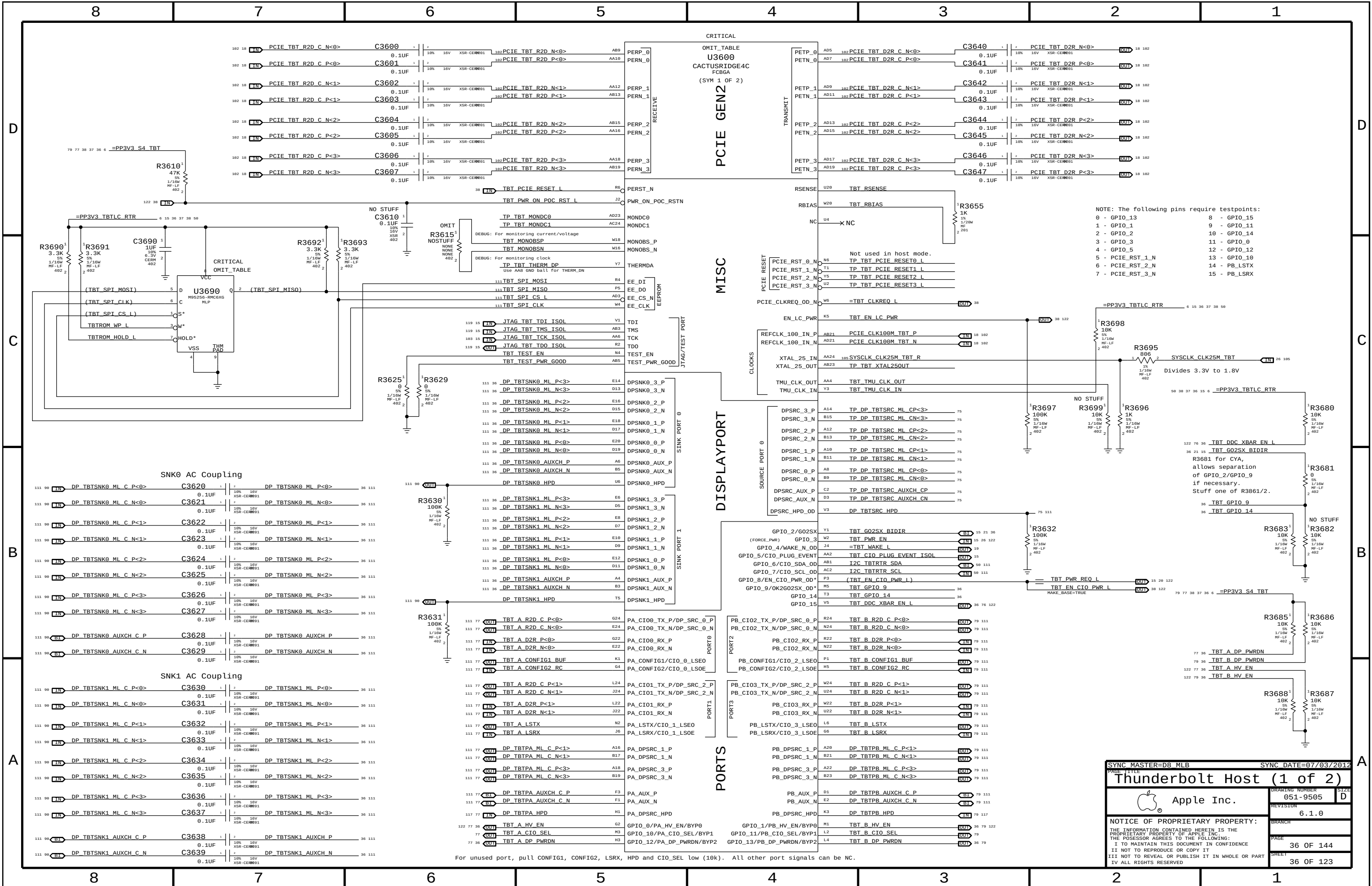


AIRPORT
BLUETOOTH

WI-FI POWER CONSUMPTION: RDAR://10174119

SUPERVISOR & CLKREG # ISOLATION
DELAY = 130 MS +/- 20%

SYNC MASTER=D8_MLB		SYNC DATE=07/03/2012	
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AIRPORT/BT			
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Thunderbolt Host (1 of 2)

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051-9505

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Page Notes

Power aliases required by this page:

- =PPVIN_SW_TBTBST (8-13V Boost Input)
- =PP15V_TBT_REG (15V Boost Output)
- =PP3V3_TBT_P3V3TBTFTET (3.3V FET Input)
- =PP3V3_TBT_FET (3.3V FET Output)
- =PP3V3_S0_TBTTPWRCTL
- =PP1V05_TBT_P1V05TBTFTET (1.05V FET Input)
- =PP1V05_TBT_FET (1.05V FET Output)

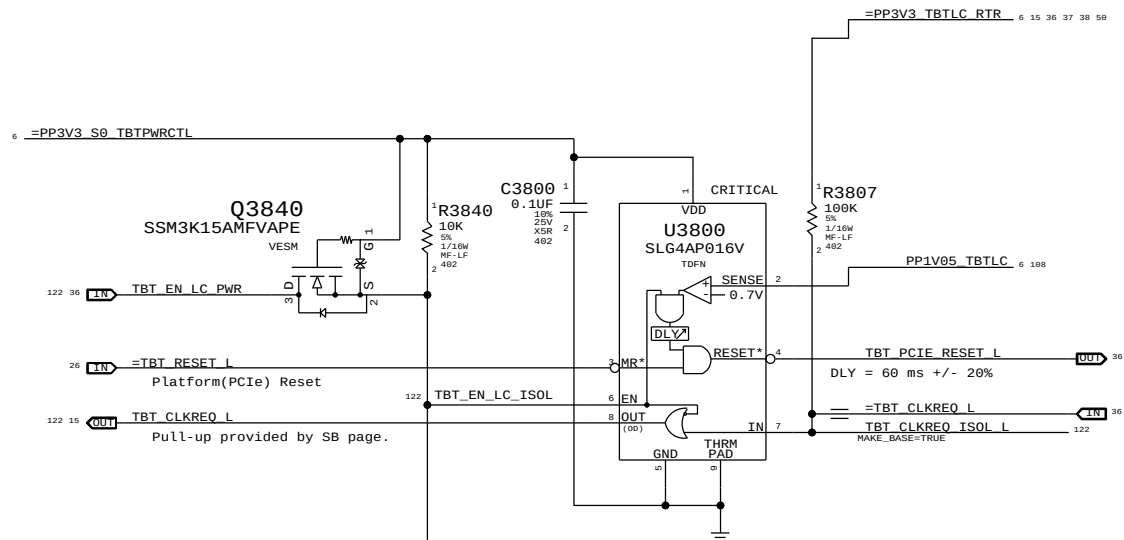
Signal aliases required by this page:

- =TBT_CLKREQ_L
- =TBT_RESET_L

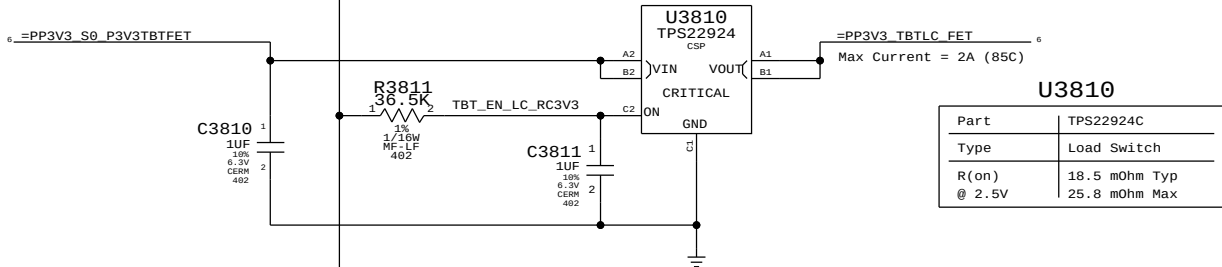
BOM options provided by this page:

TBTBST:Y - Stuffs 15V boost circuitry.

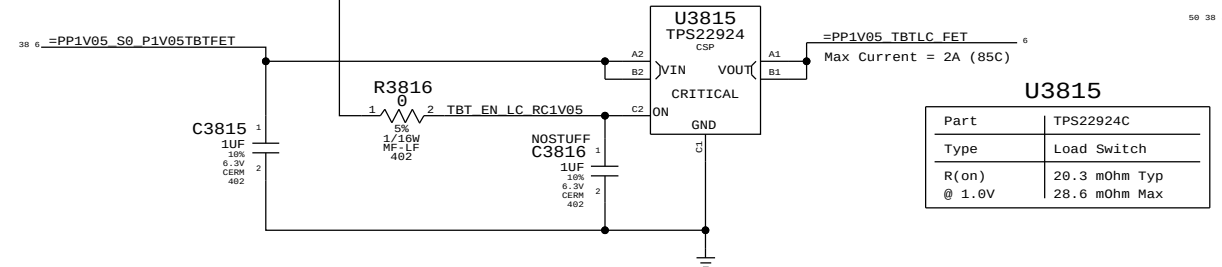
Supervisor & CLKREQ# Isolation



3.3V TBT "LC" Switch

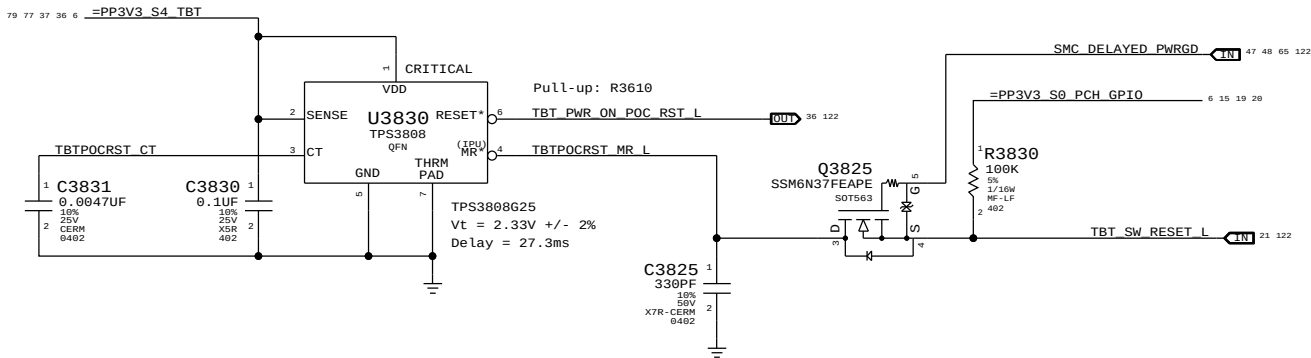


1.05V TBT "LC" Switch

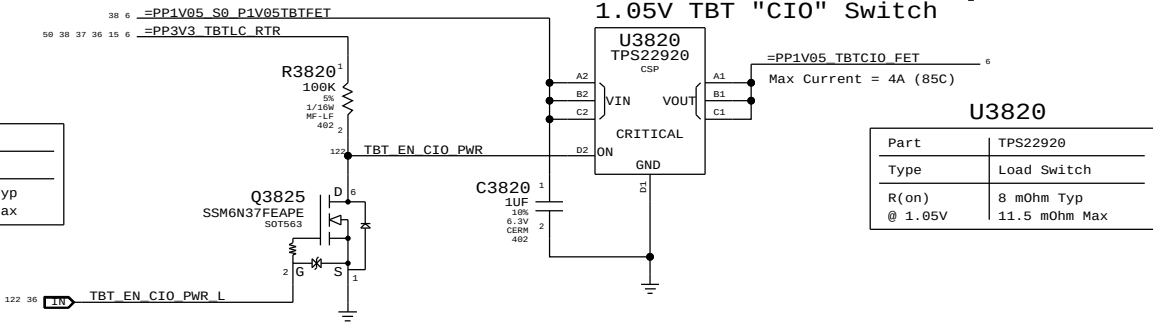


TBT "POC" Power-up Reset

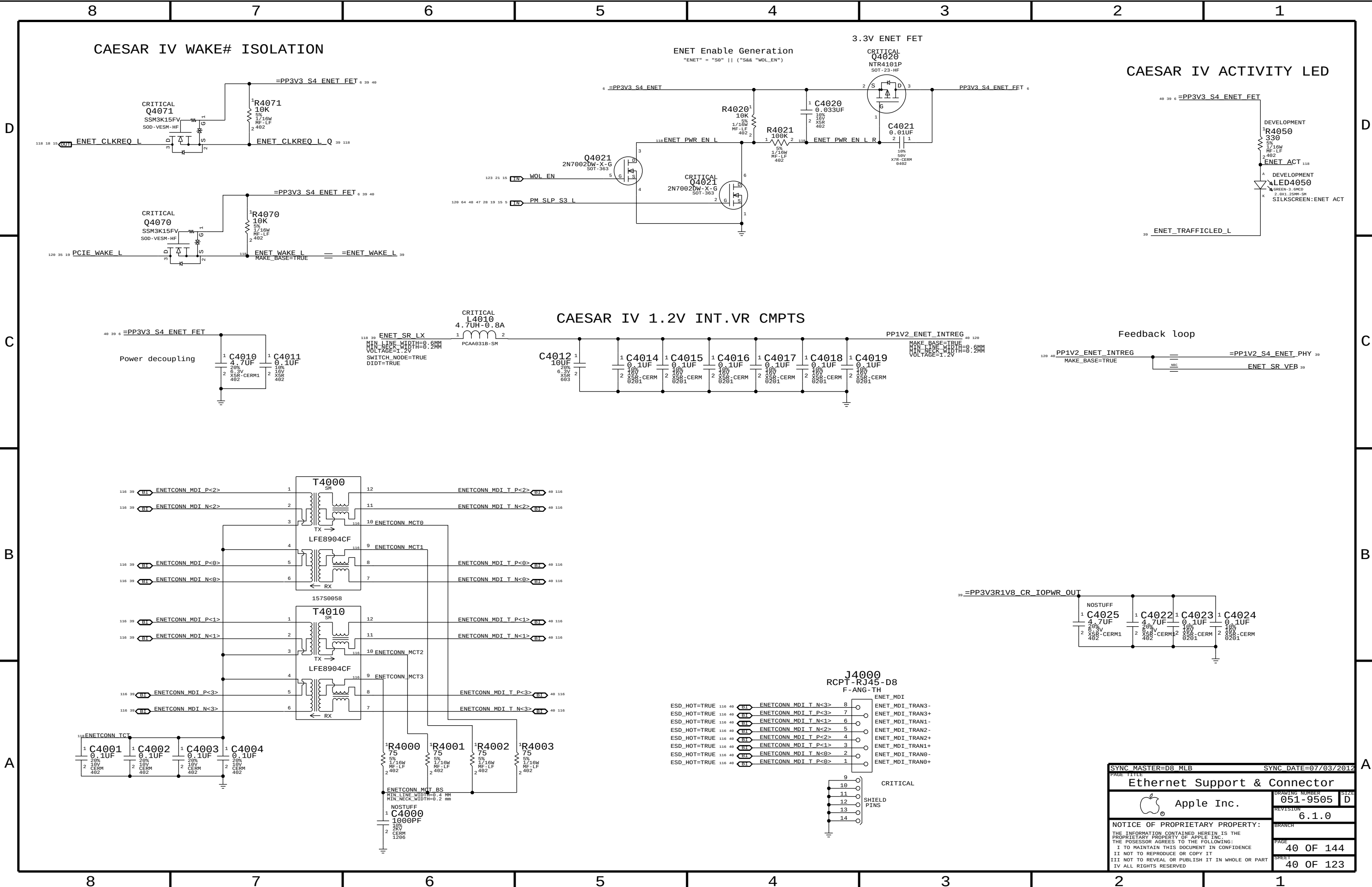
Intel investigating whether RC is sufficient.



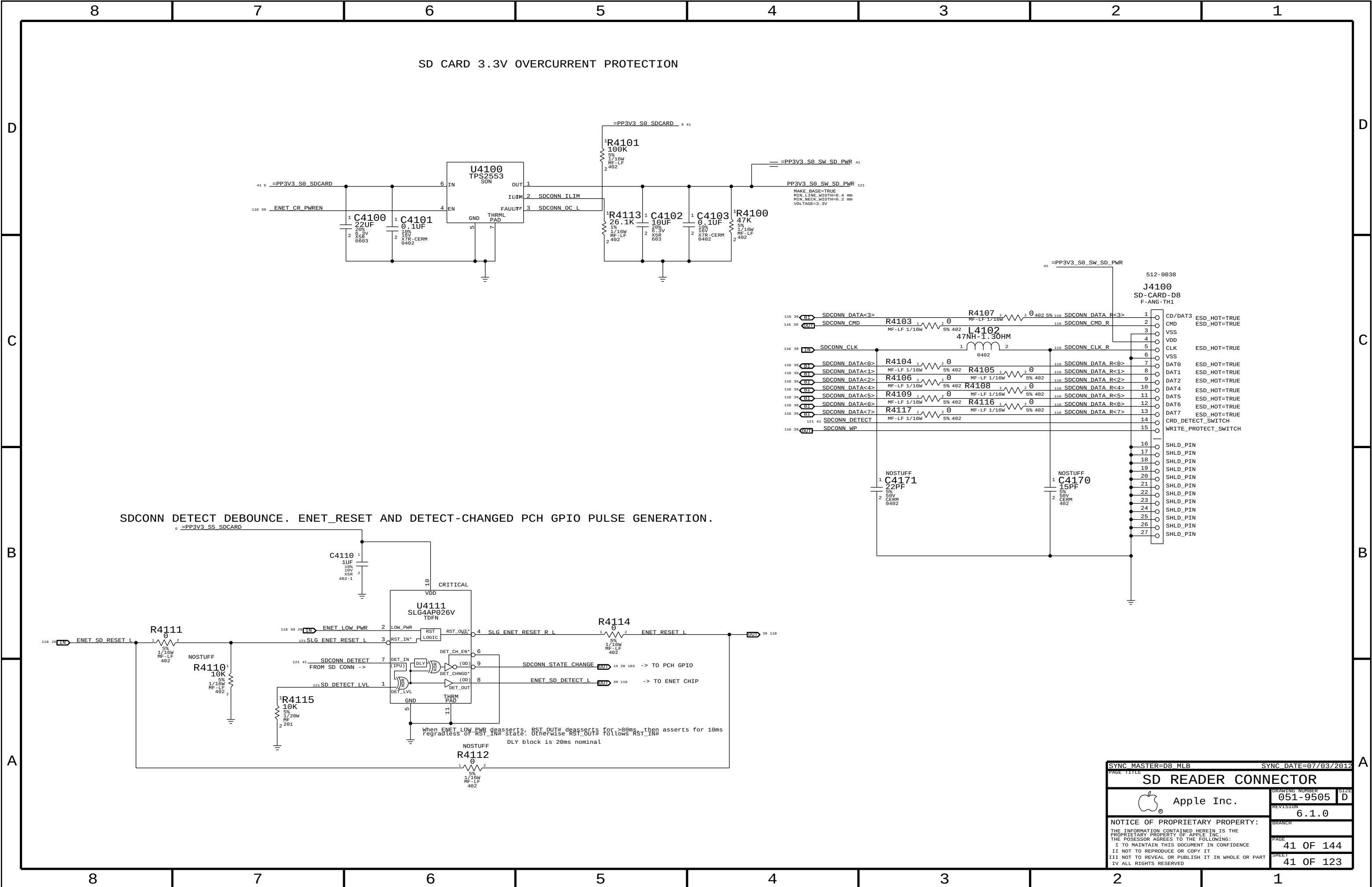
1.05V TBT "CIO" Switch



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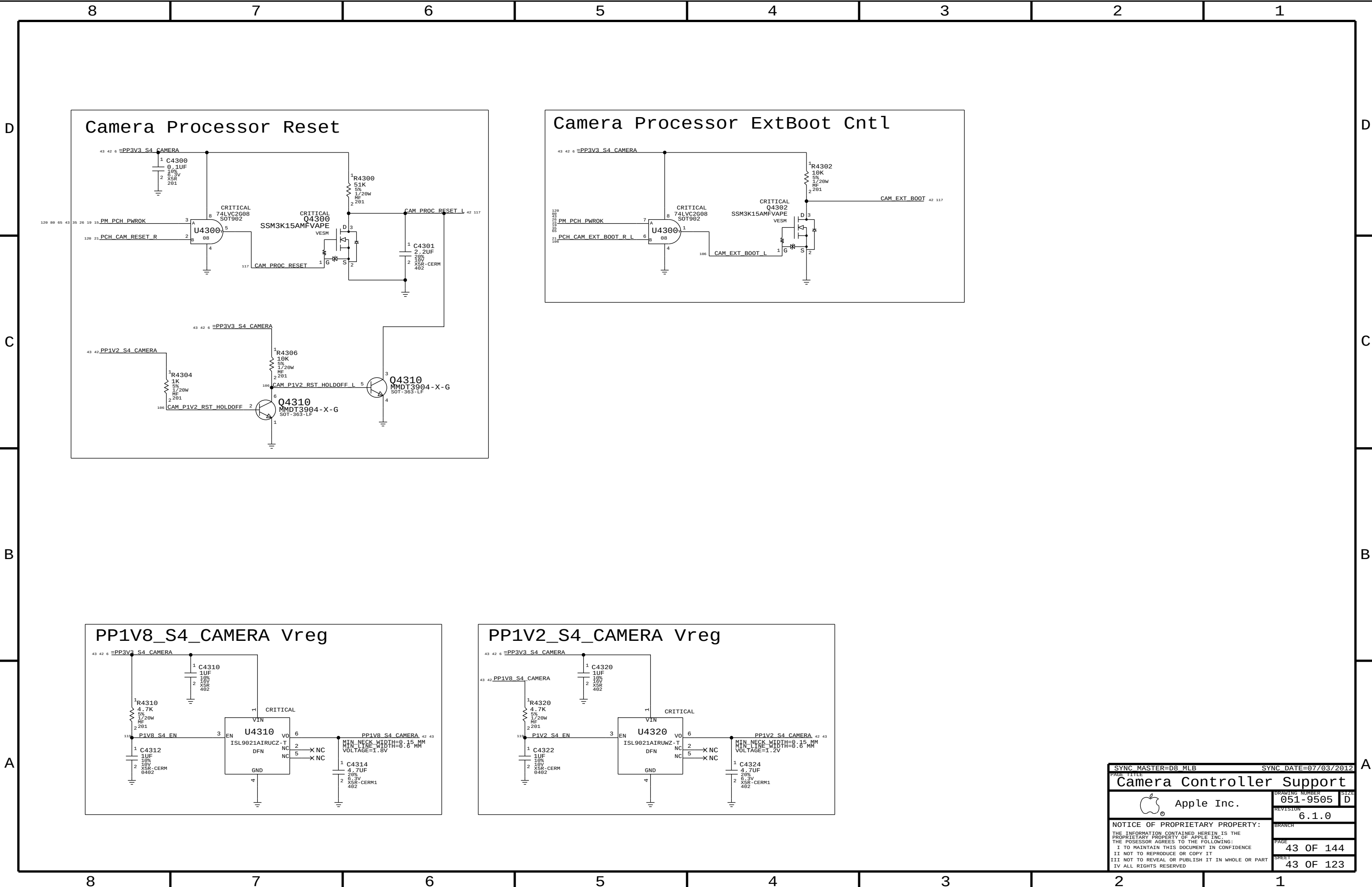


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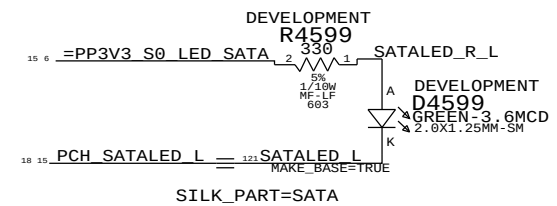
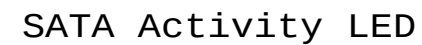
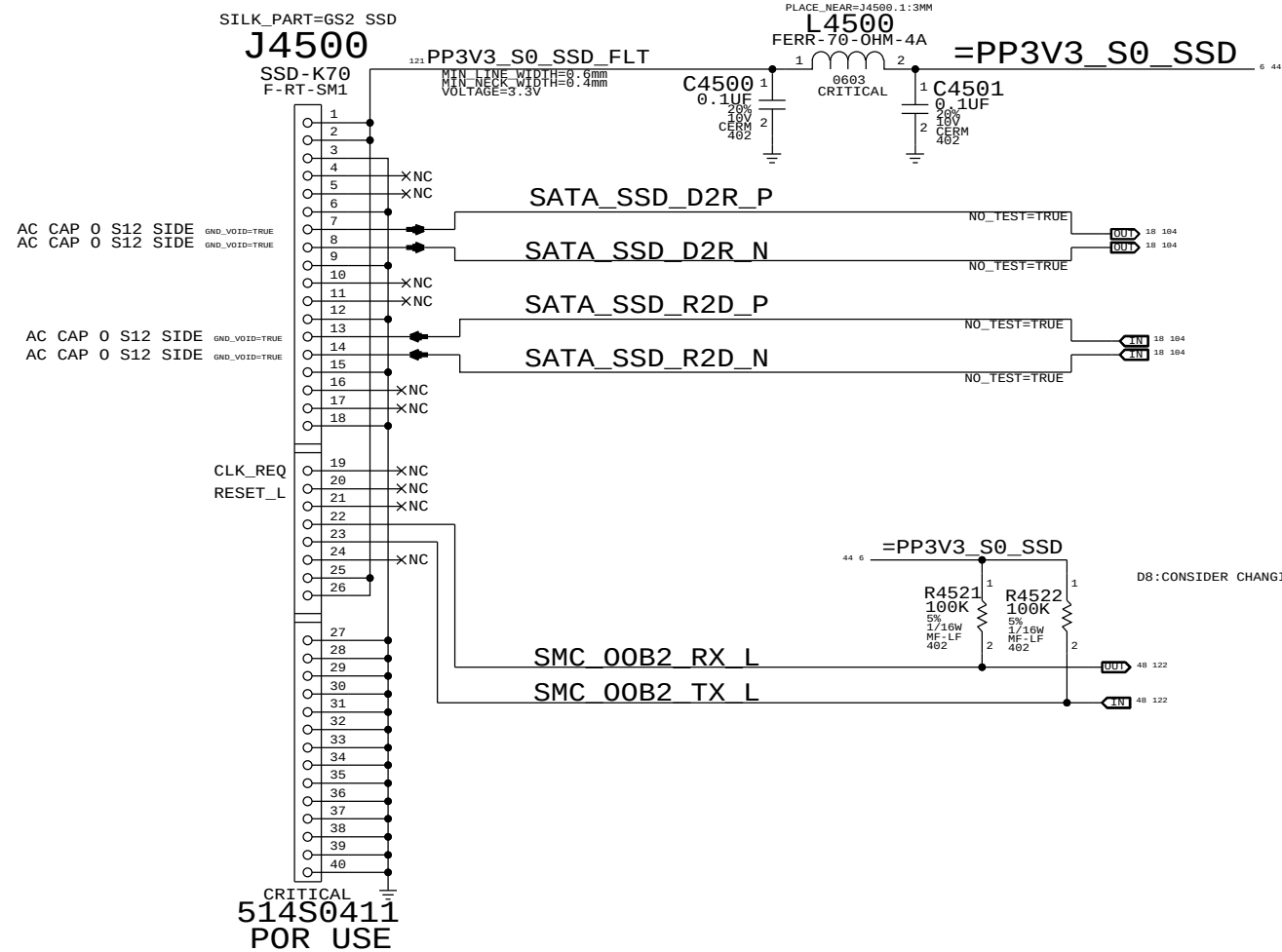
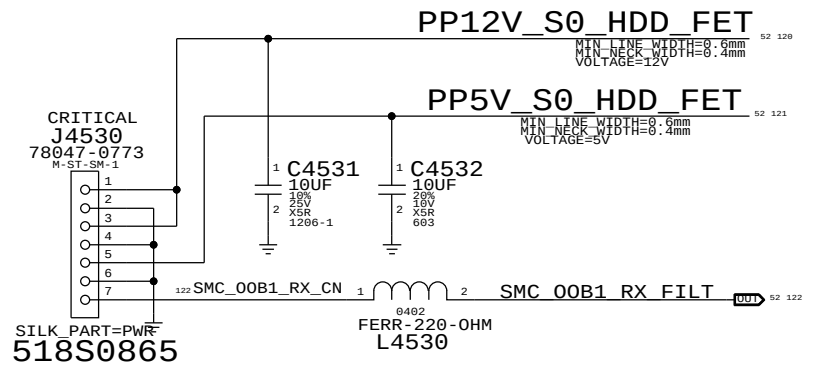
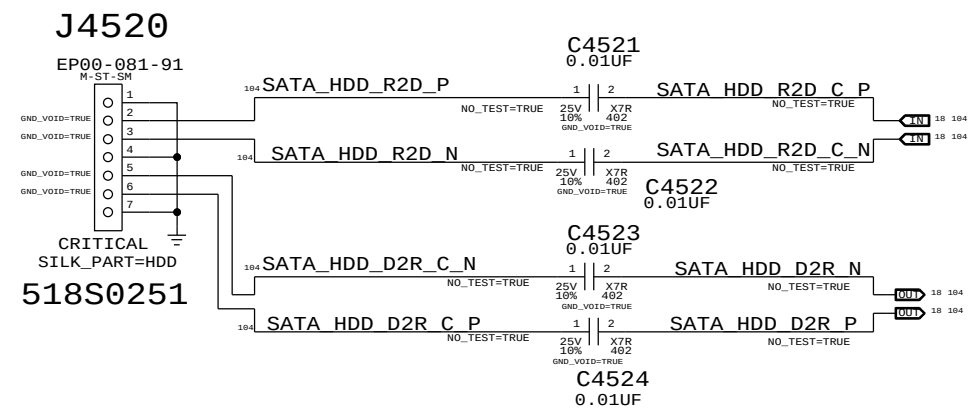


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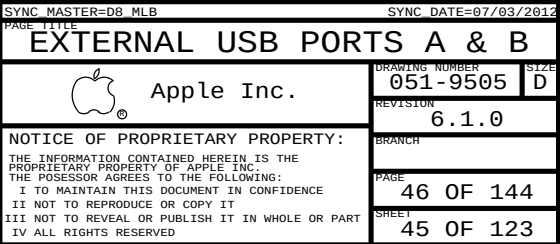
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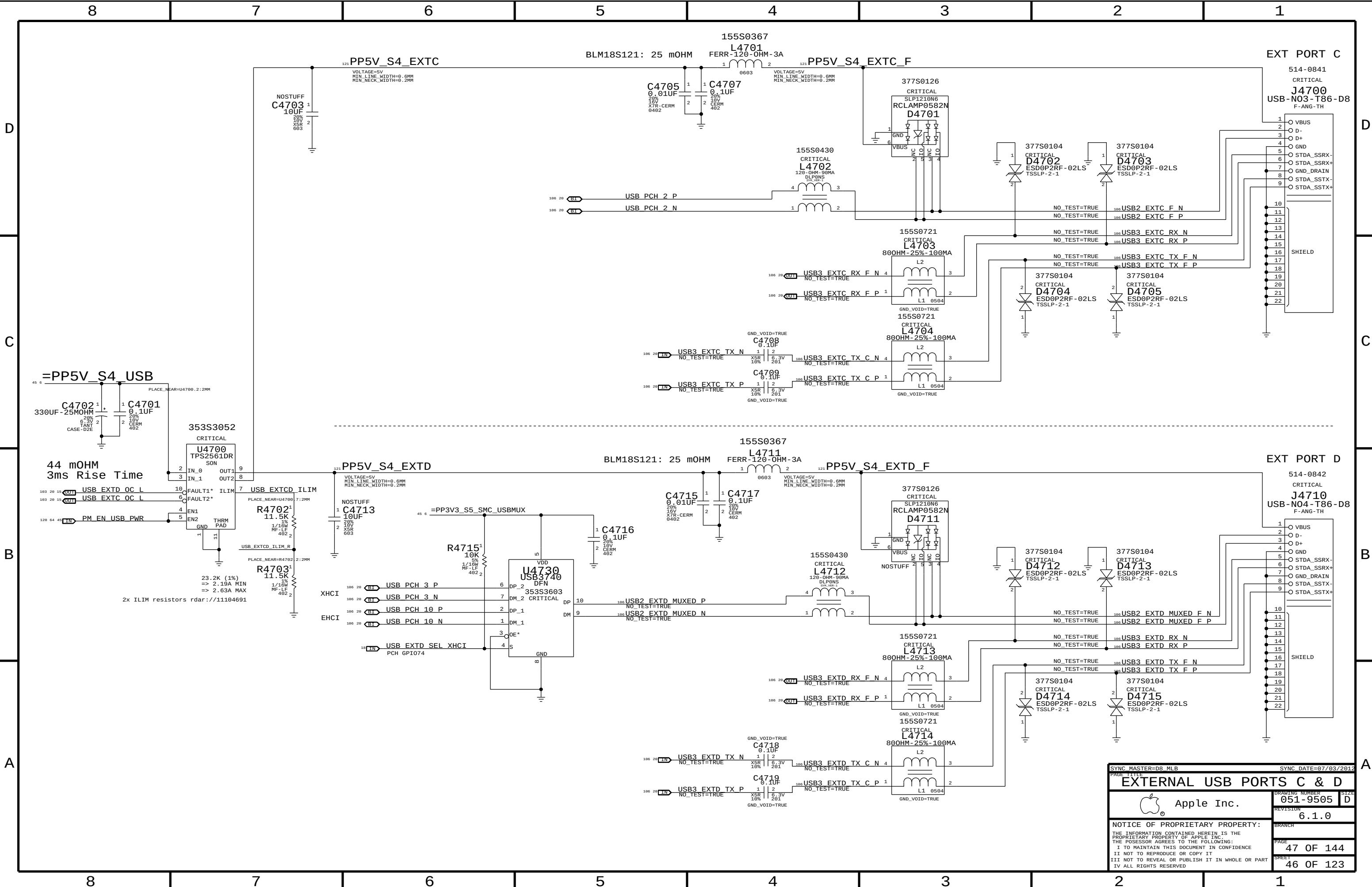


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		SHEET	43 OF 123

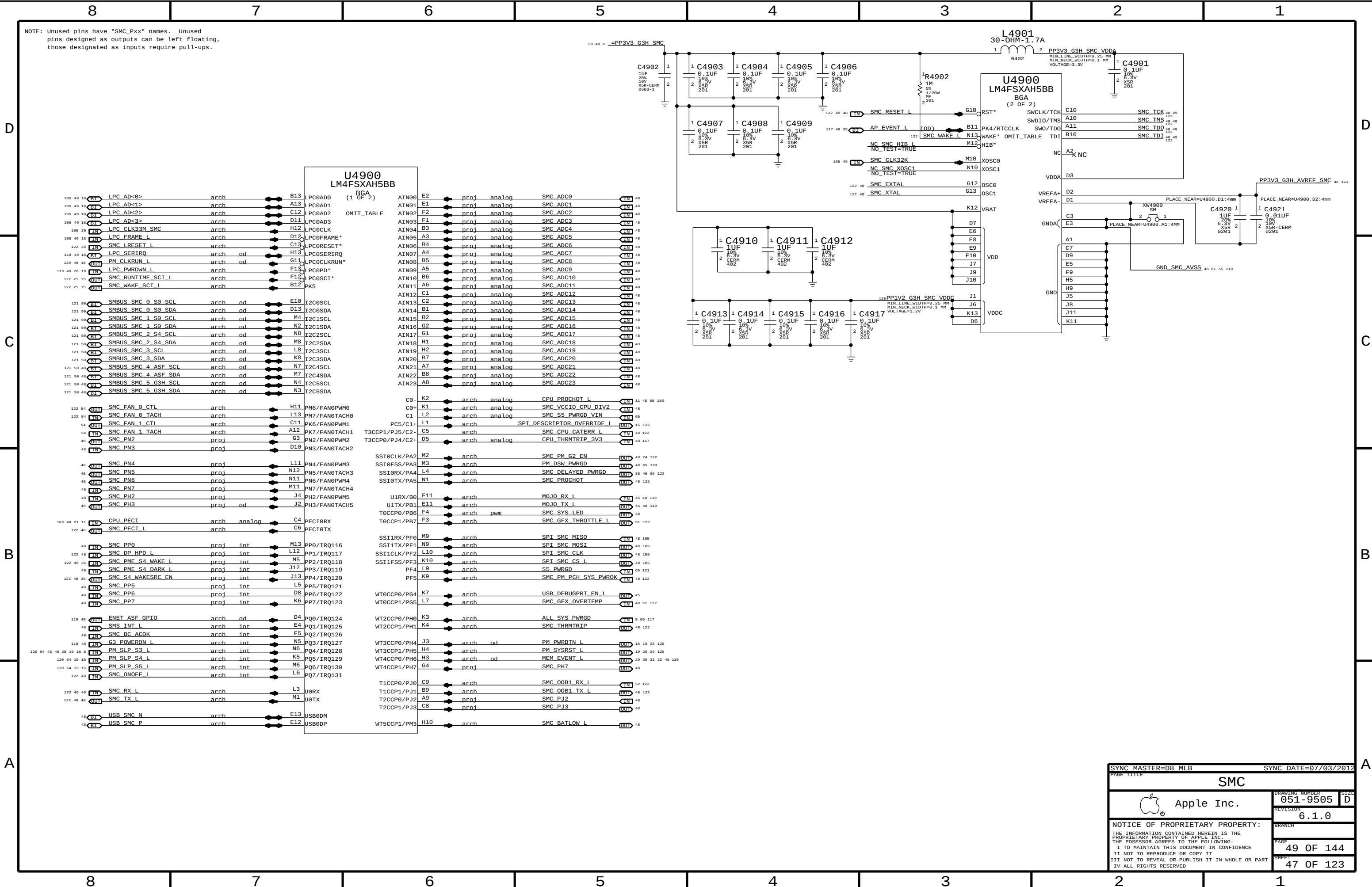


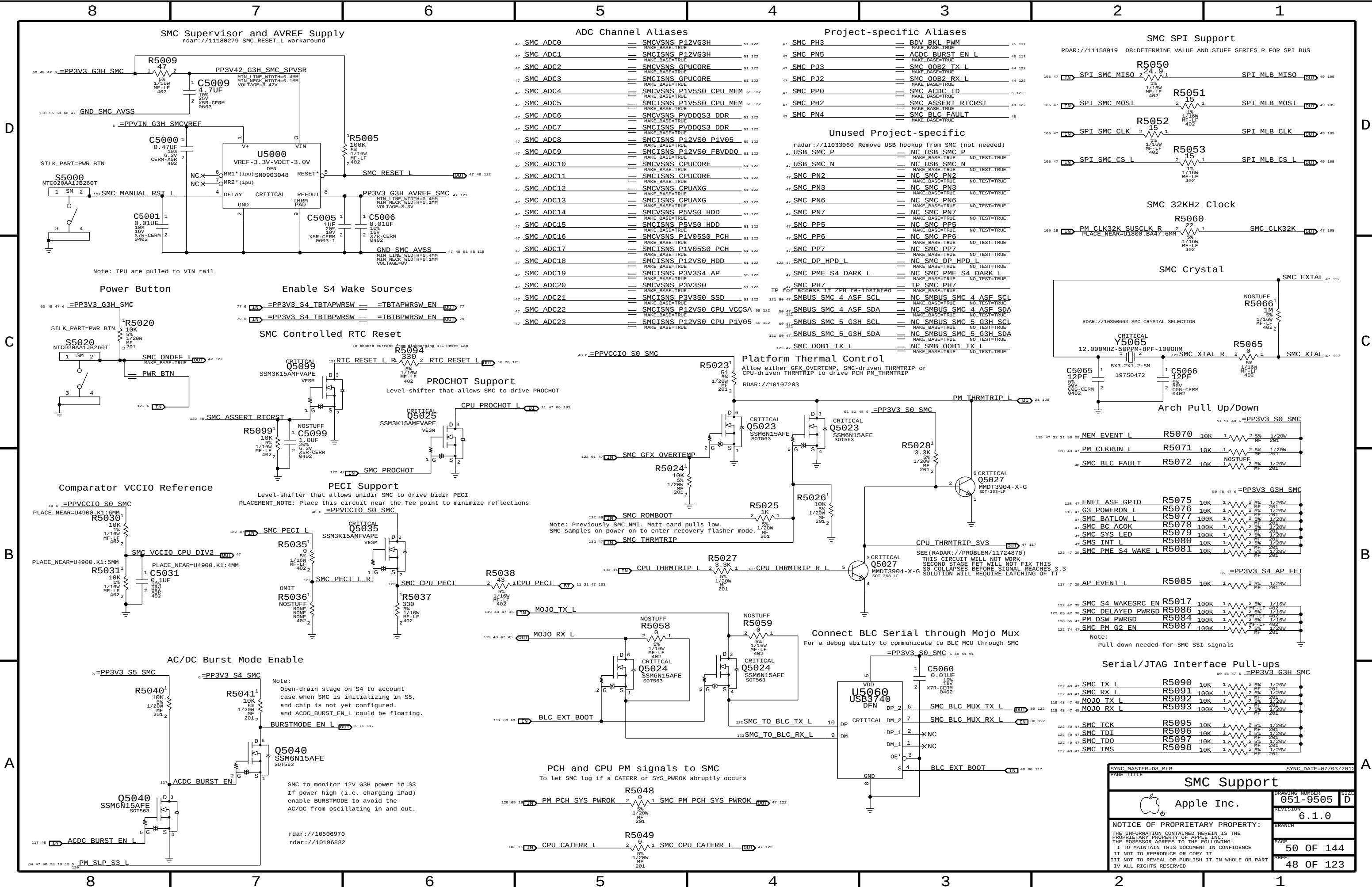
SYNCHMASTER=D8_MLB		SYNCH DATE=07/03/2012	
PAGE TITLE			
SATA Connectors			
 Apple Inc.	DRAWING NUMBER		SIZE
	051-9505		D
	REVISION		
	6.1.0		
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EXTERNAL USB PORTS C & D		051-9505	
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		PAGE	
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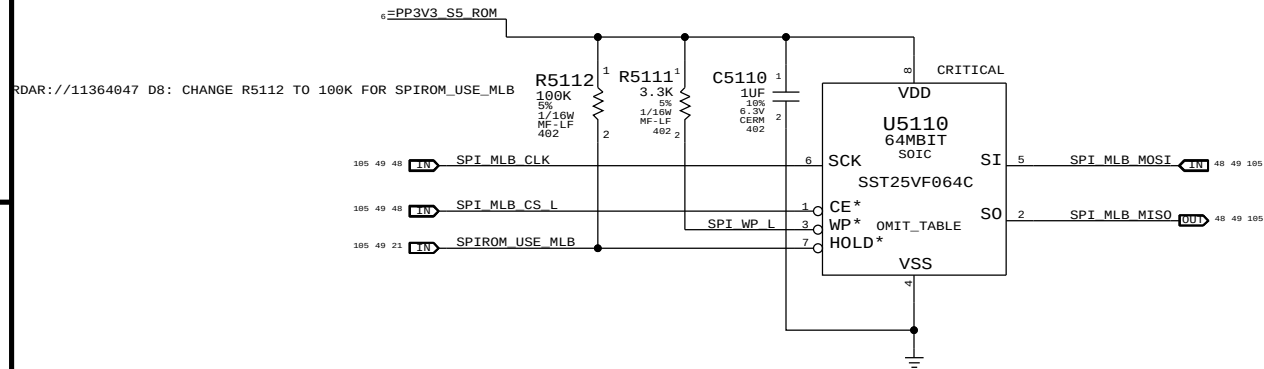
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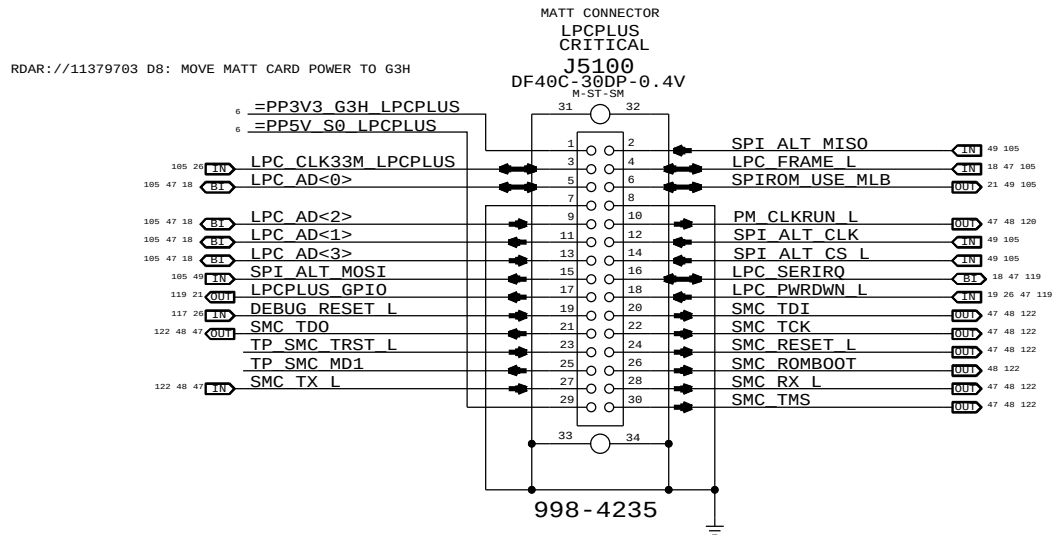
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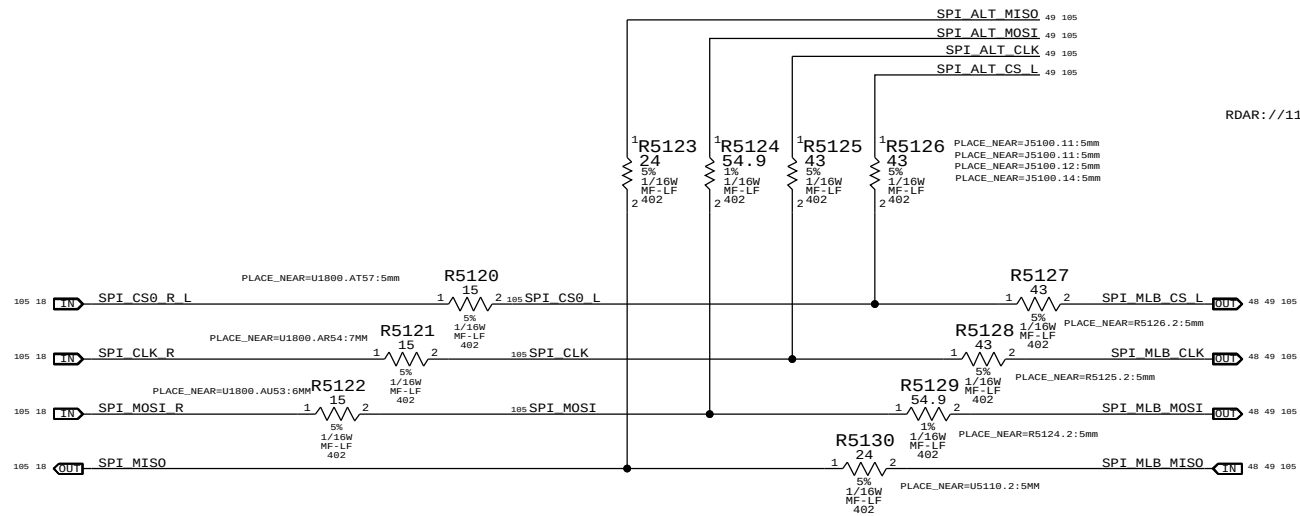
SPI BootROM



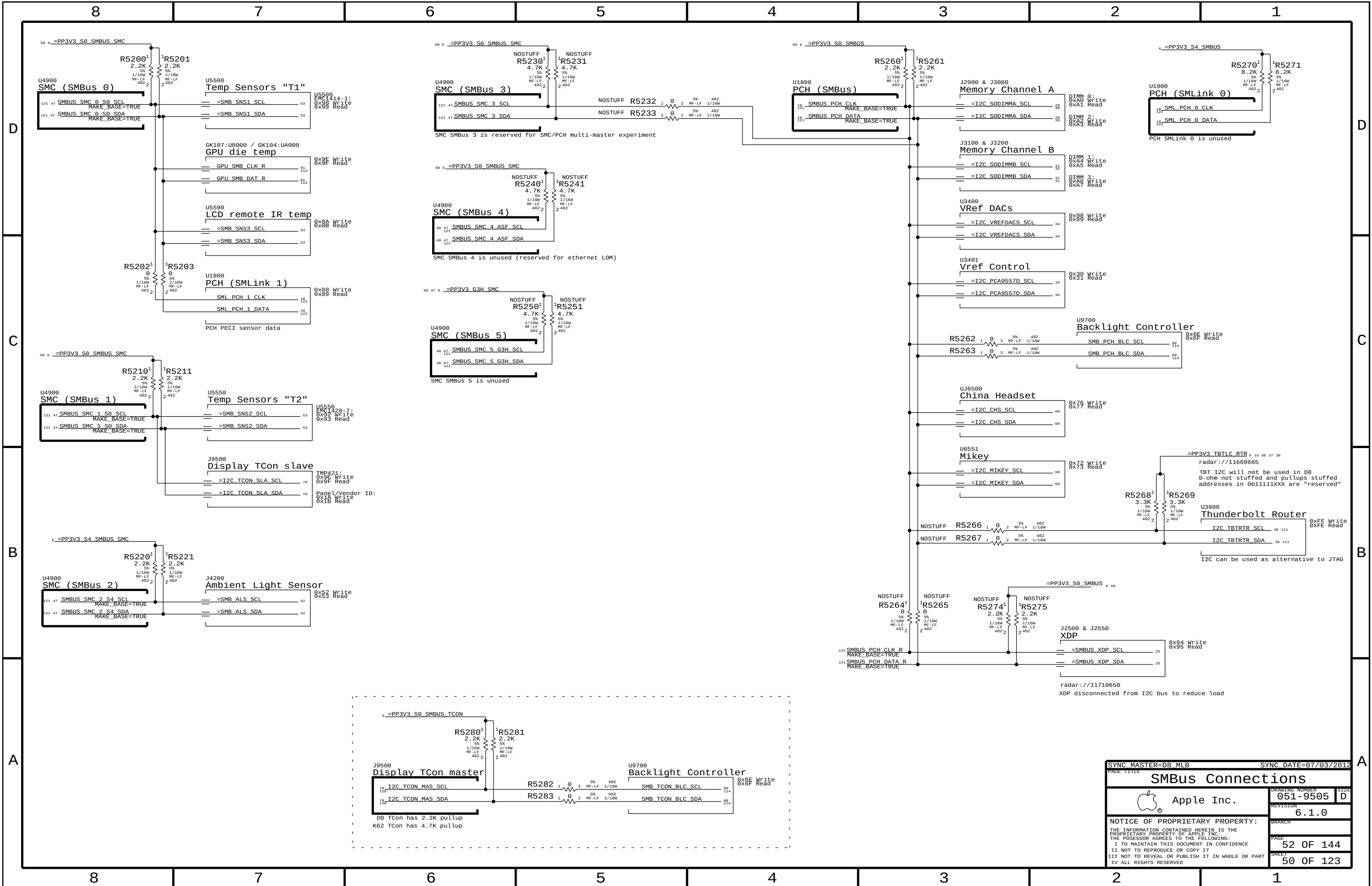
LPC+SPI Connector

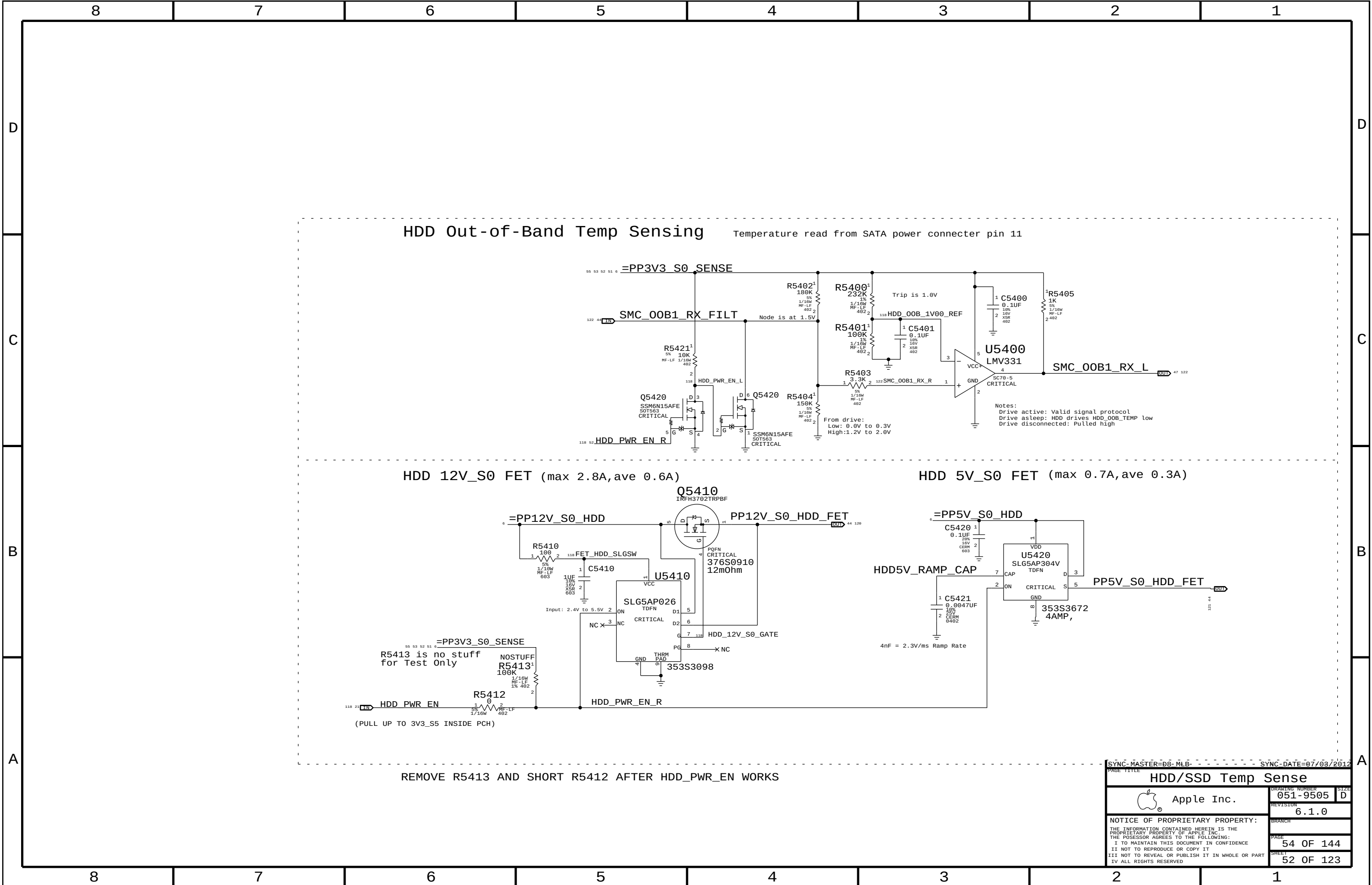


SPI Series Termination



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		REVISION	6.1.0
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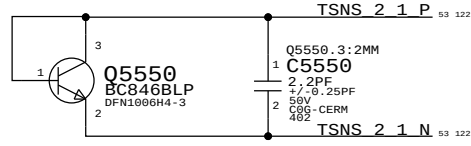


PAGE TITLE		HDD/SSD Temp Sense	
DRAWING NUMBER		051-9505	SIZE
REVISION		6.1.0	
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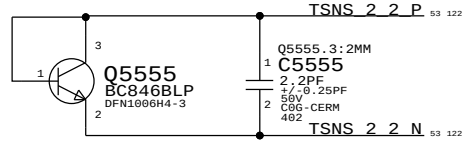
Temperature Sensor T2 EMC1428: Near GPU VR

SNS T2: TEMP SENSOR IC

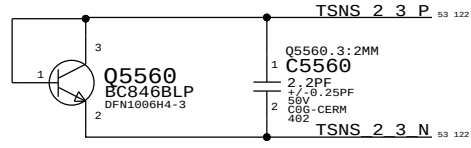
GPU Proximity



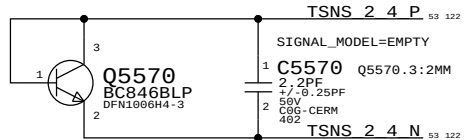
Ambient



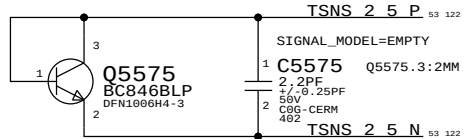
BLC Proximity



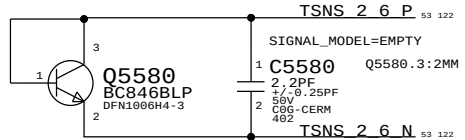
SO-DIMM Proximity 1



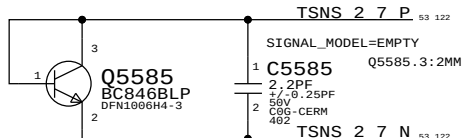
SO-DIMM Proximity 2



SO-DIMM Proximity 3



SO-DIMM Proximity 4



PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
372S0186	372S0185		ALL	Alternate Temp Diode

GPU Prox (T60p)

TSNS_2_1_P

TSNS_2_1_N

Ambient (TA0p)

TSNS_2_2_P

TSNS_2_2_N

SoDIMM Prox 1 (TM0p)

TSNS_2_4_P

TSNS_2_4_N

SoDIMM Prox 3 (TM2p)

TSNS_2_6_P

TSNS_2_6_N

SoDIMM Prox 4 (TM3p)

TSNS_2_7_N

TSNS_2_7_P

SoDIMM Prox 2 (TM1p)

TSNS_2_5_N

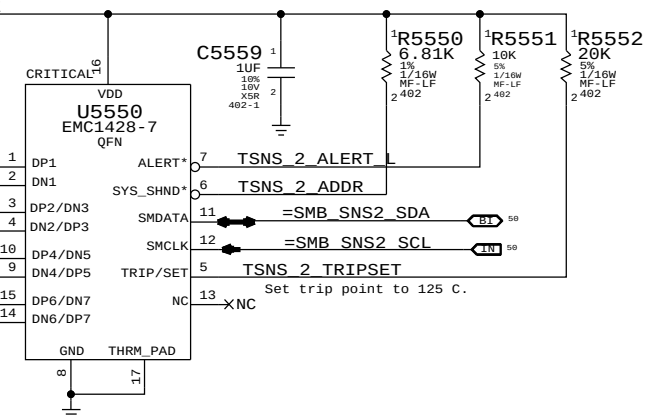
TSNS_2_5_P

BLC PROX (Tb0p)

TSNS_2_3_N

TSNS_2_3_P

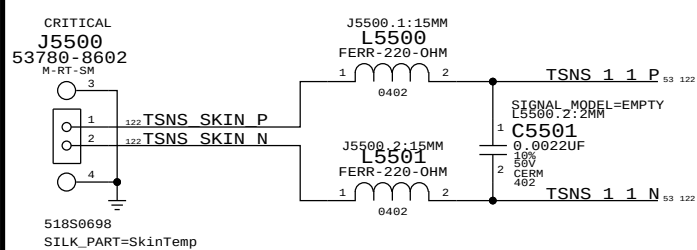
55 53 52 51 6 =PP3V3_S0_SENSE



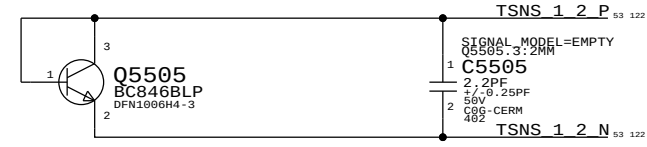
EMC1428-7: 6.8K PULL UP: I2C ADDRESS: WRITE: 0x92, READ: 0x93

Temperature Sensor T1 EMC1414: Near PSU Conn

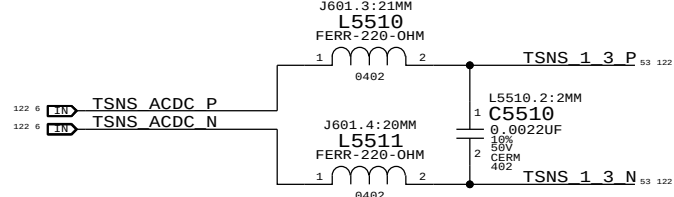
Skin



CPU Proximity



AC/DC via connector to diode inside PSU



Skin Temp (TS0p)

CPU Prox (TC0p)

AC/DC (Tp2p)

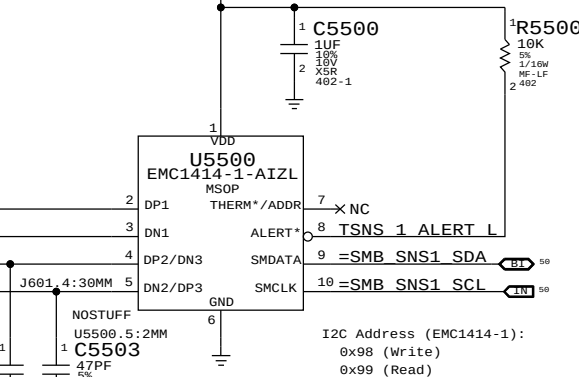
TSNS_1_2_P

TSNS_1_2_N

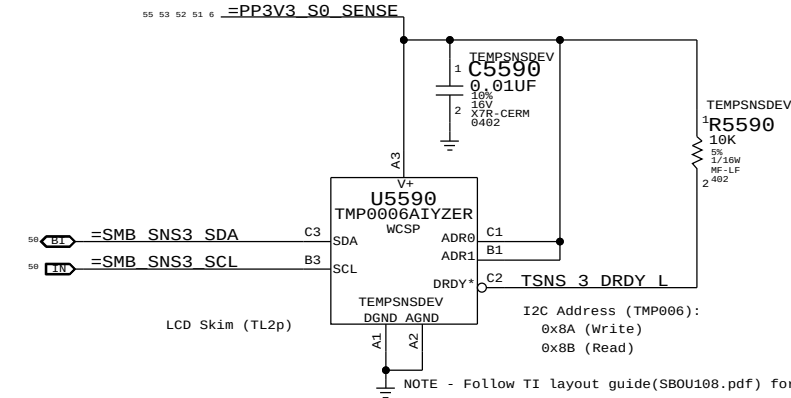
TSNS_1_3_P

TSNS_1_3_N

55 53 52 51 6 =PP3V3_S0_SENSE



Temperature Sensor T3: LCD Remote Sensor (Dev4Now)



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Temperature Sensors			
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		REVISION	6.1.0
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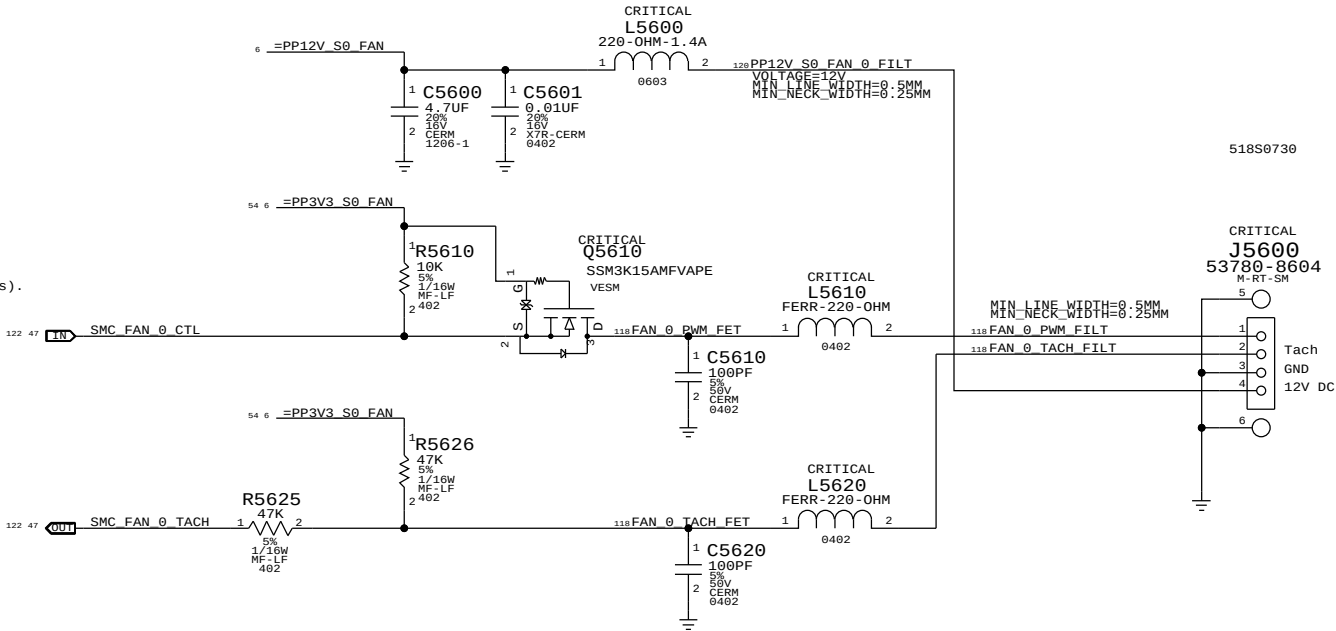
A

SMC Fan 0 (System)

Note:
The circuit for the PWM input to the fan acts as a non-inverting level-shifter to protect the SMC. It is assumed there is a pull-up to 5V/12V inside the fan, otherwise when the SMC PWM goes low and Q5610 turns on, there would be 5V/12V present on the SMC pin! Then by definition, the drain of Q5610 is at common and the SMC sinks current when Q5610 is on.

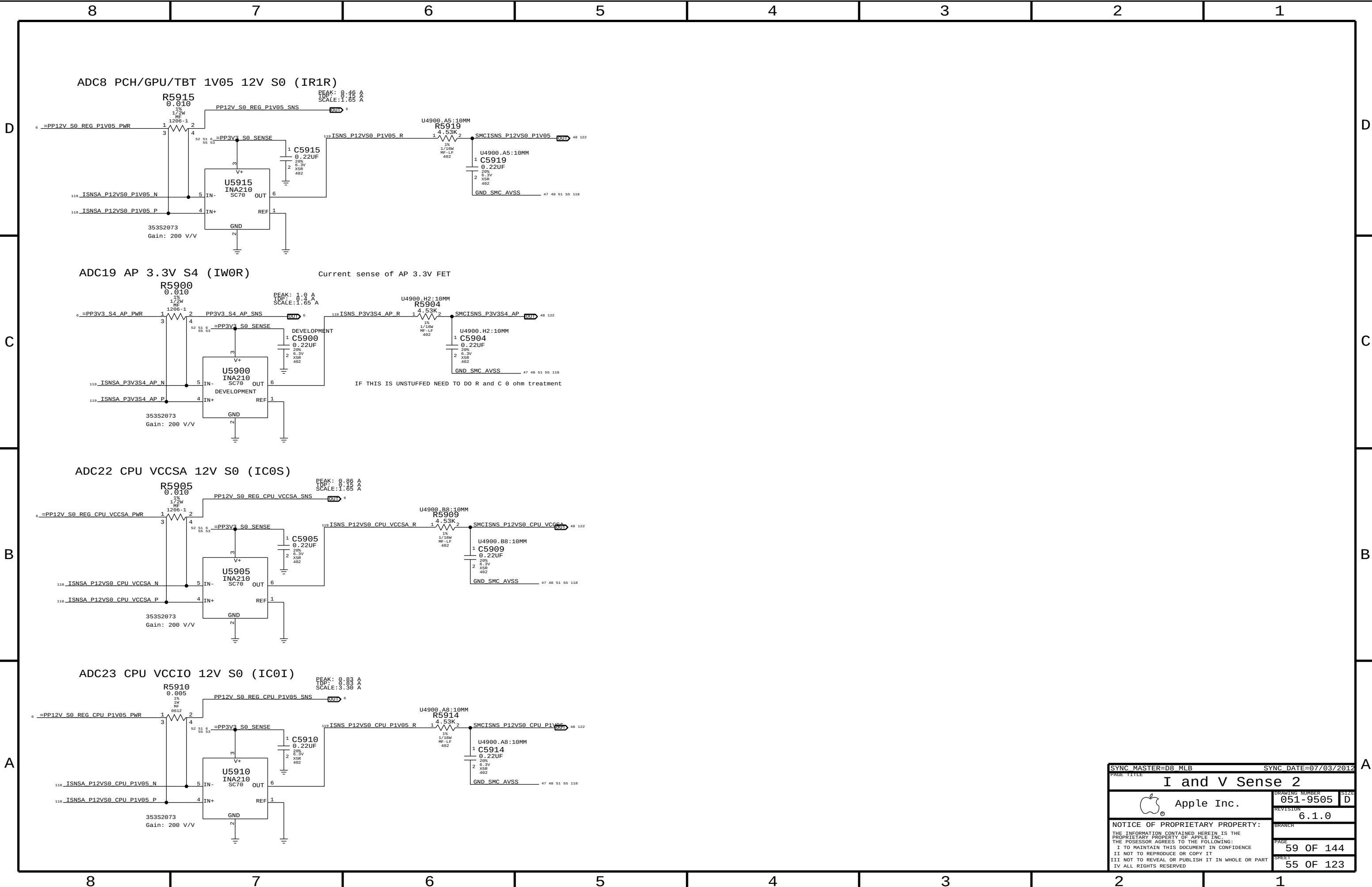
This resembles an open-drain if there is a pull-up, going to a Hi-Z FET input.

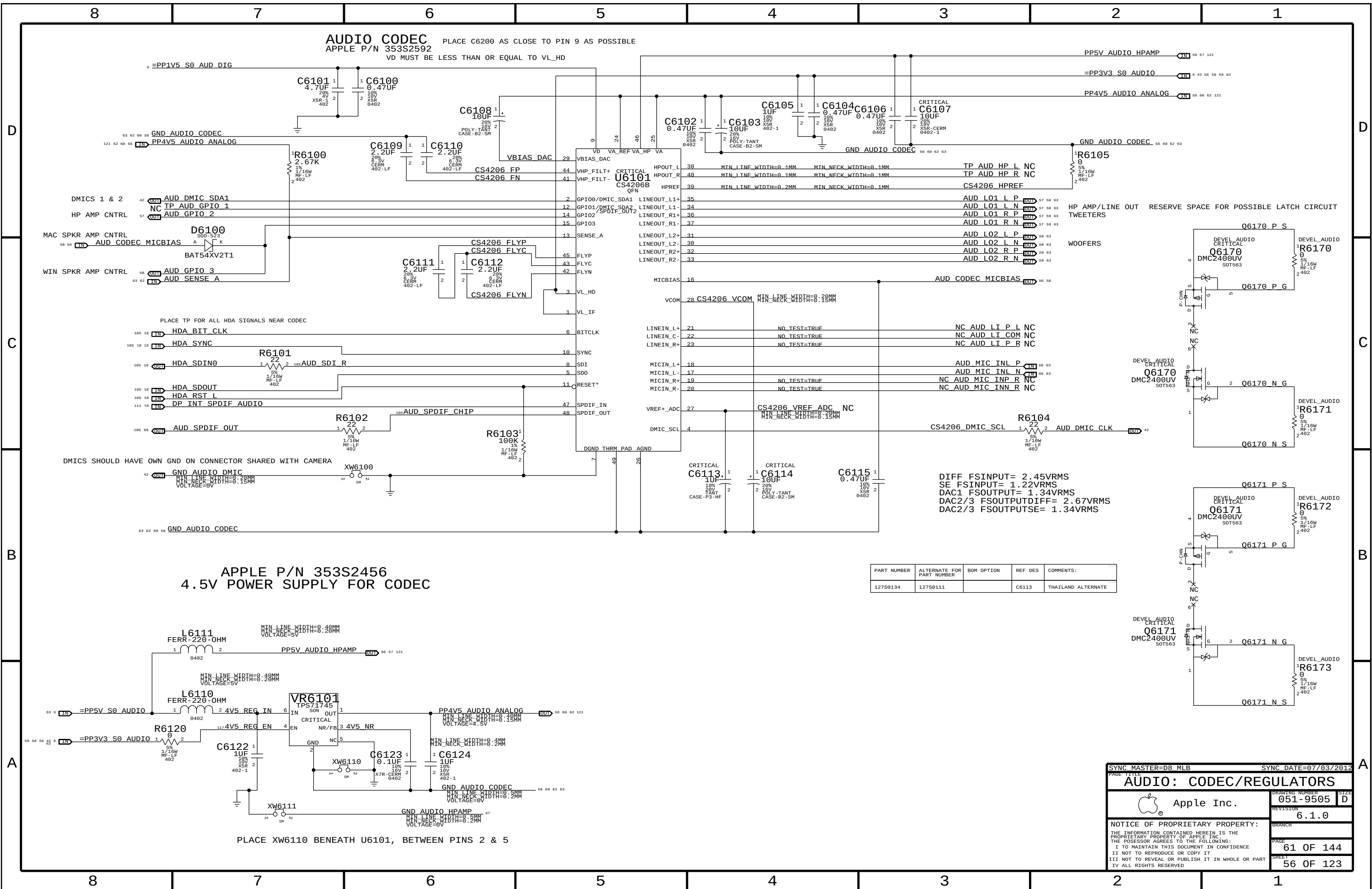
Otherwise, this is simply a pass-FET. See RADAR: 10565825- D7: Need schematic and PCB file of fan(All Vendors).



SMC Fan 1 (Unused)







LEFT CH SPEAKER AMP
APPLE P/N 353S3163

SPEAKER AMP GAIN = +9 DB
SPEAKER AMP RIN = 40K NOMINAL
FC_HPFF, TWEETERS = ~847 HZ (4700 PF)
FC_HPFF, WOOFERS = ~4 HZ (1.0 UF)

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INPUT POLARITY FLIP OK -- TRUE DIFF INPUTS

OUTPUT POLARITY FLIP TO
MAKE LAYOUT MORE LOGICAL

PINS 14 & 15 ARE TEST PINS AND
SHOULD BE TIED TO GND

EDGE RATE
CONTROL
ON
OFF

R6304
0 OHM
NOSTUFF

R6305
0 OHM
NOSTUFF

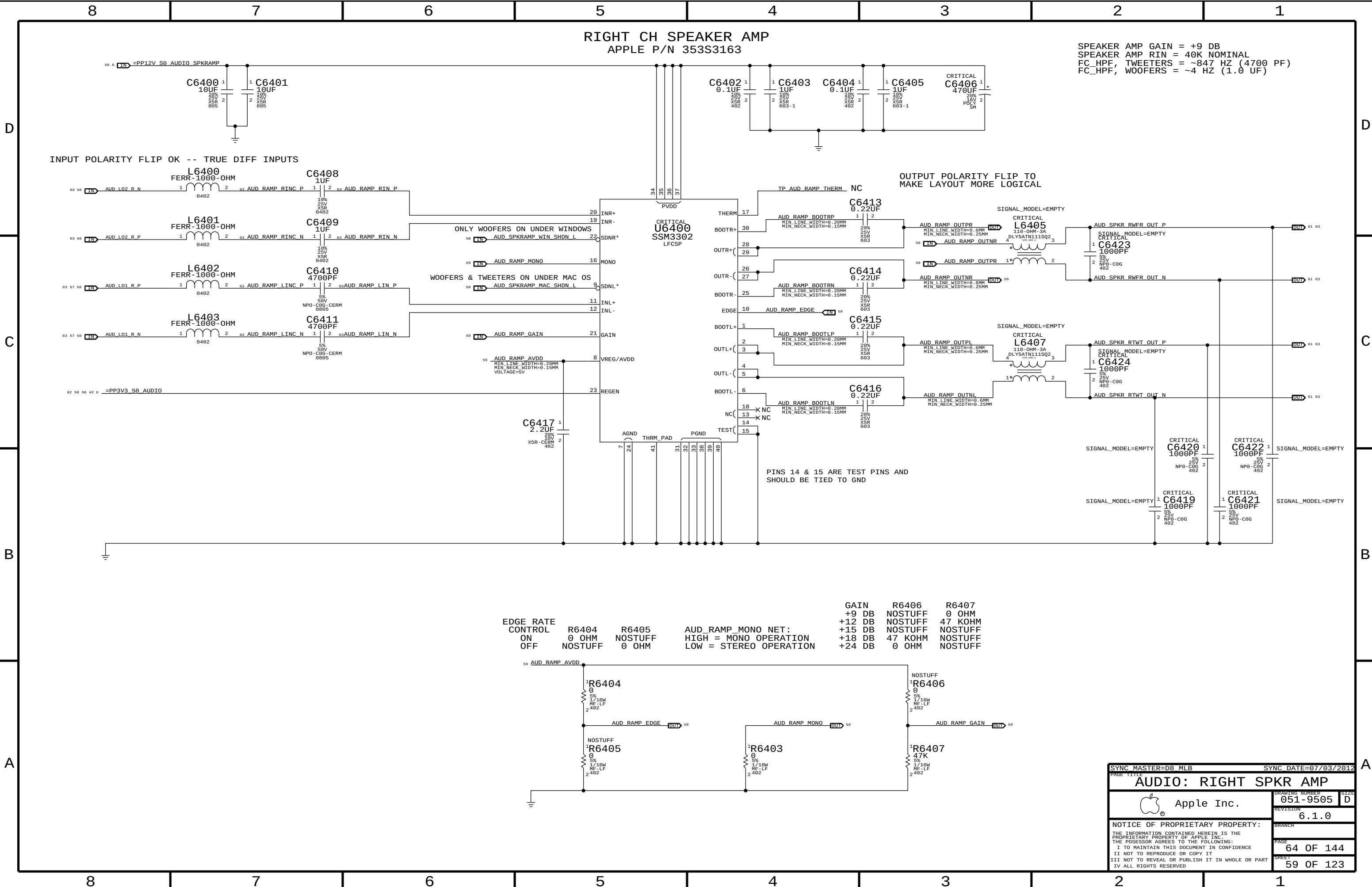
AUD_RAMP_MONO NET:
HIGH = MONO OPERATION
LOW = STEREO OPERATION

GAIN
+9 DB
+12 DB
+15 DB
+18 DB
+24 DB

R6306
NOSTUFF
NOSTUFF
47 KOHM
0 OHM

R6307
0 OHM
NOSTUFF
NOSTUFF
NOSTUFF

SYNC MASTER=D8_MLB		SYNC DATE=07/03/2012	
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MIKEY RECEIVER CKT

WRITE: 0X72 READ: 0X73 APN 353S3231

I2C ADDRESSES

MIKEY	U6551	READ	0111 0011	0X73
MIKEY	U6551	WRITE	0111 0010	0X72

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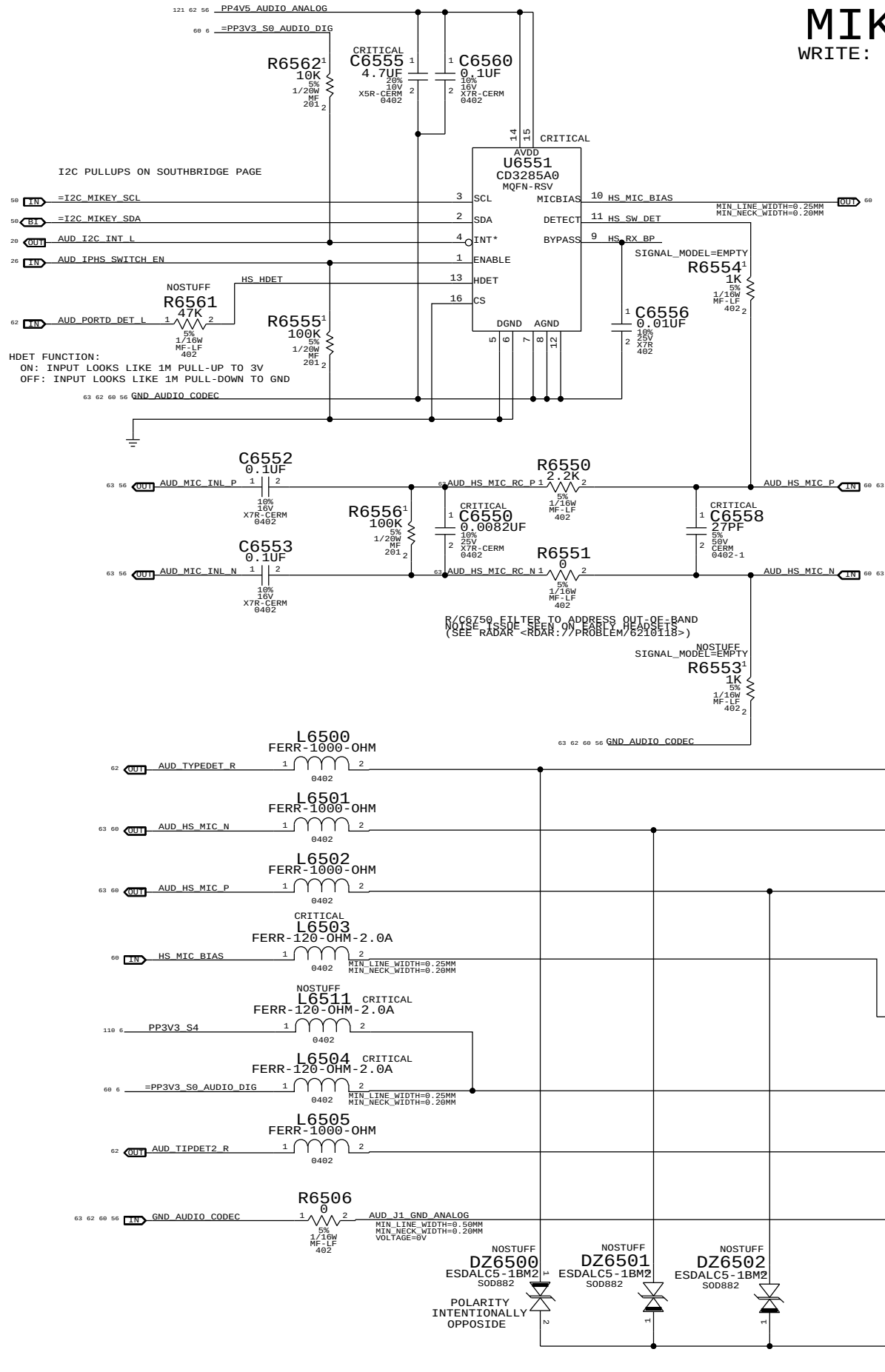
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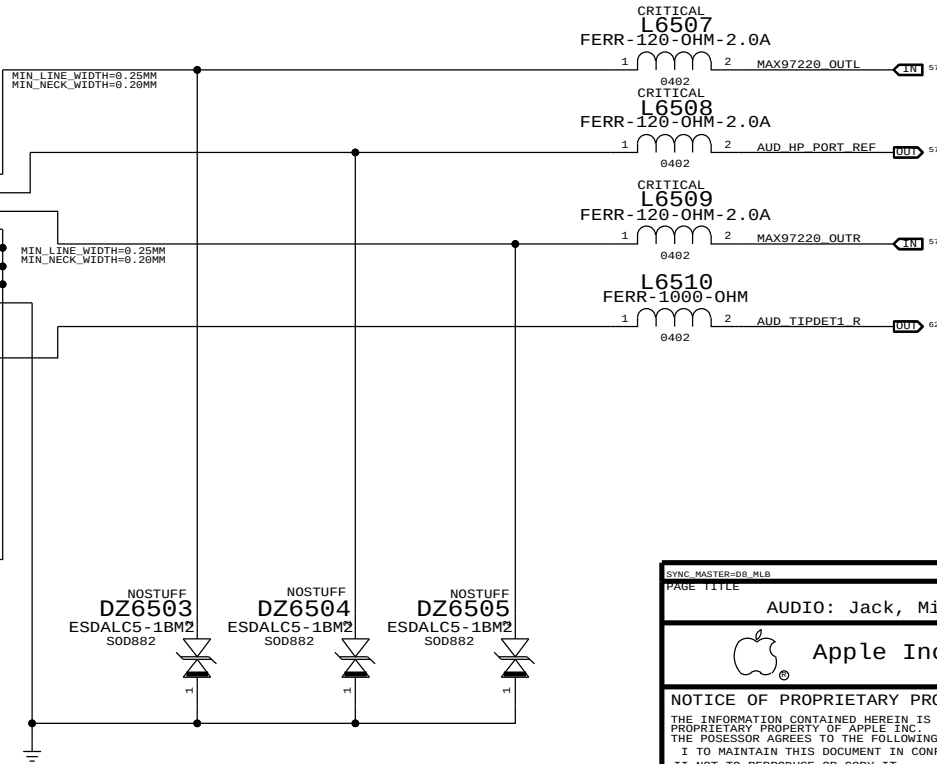
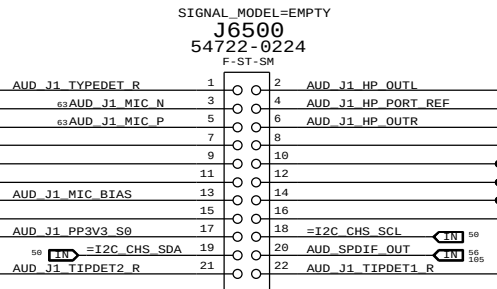
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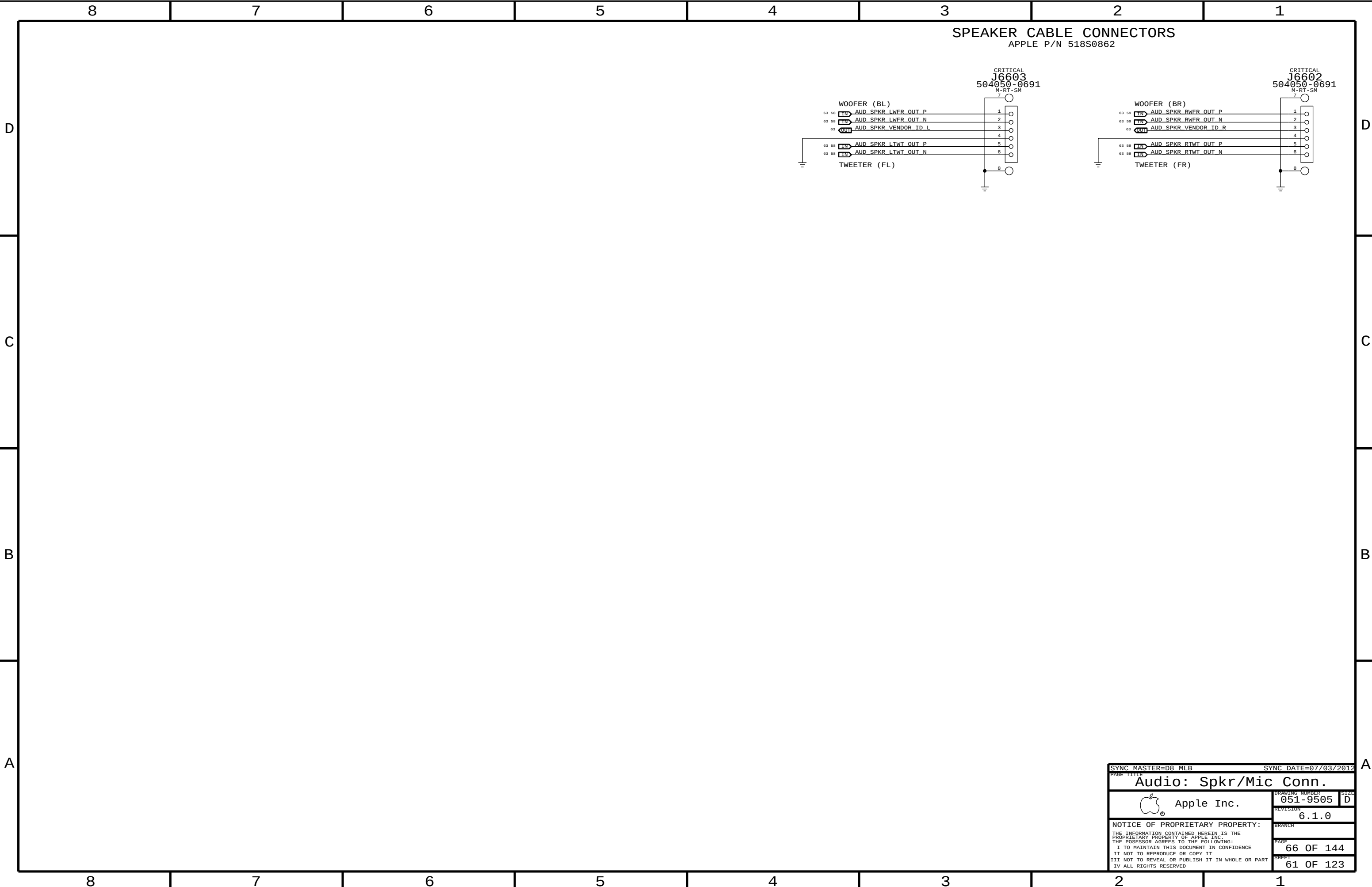
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APPLE P/N 518S0687



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AUDIO: Jack, Mikey, CHS Switch		DRAWING NUMBER	
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IPHS HS Detect Debounce CKT

AUDIO CONNECTOR DETECT STATES

	NOTHING	SPDIF	HEADPHONE
AUD_TYPEDET_R	1	1	0
AUD_TIPDET*_R	0	1	1
AUD_OUTJACK_INSERT	0	1	1
AUD_SENSE_A	1	20K/2.67K RDIV	5.11K/2.67K RDIV

LI Insert Detect (DETECT A)

TBT/DP Audio Enable

PLACE C6799 CLOSE TO Q6700 PIN 4

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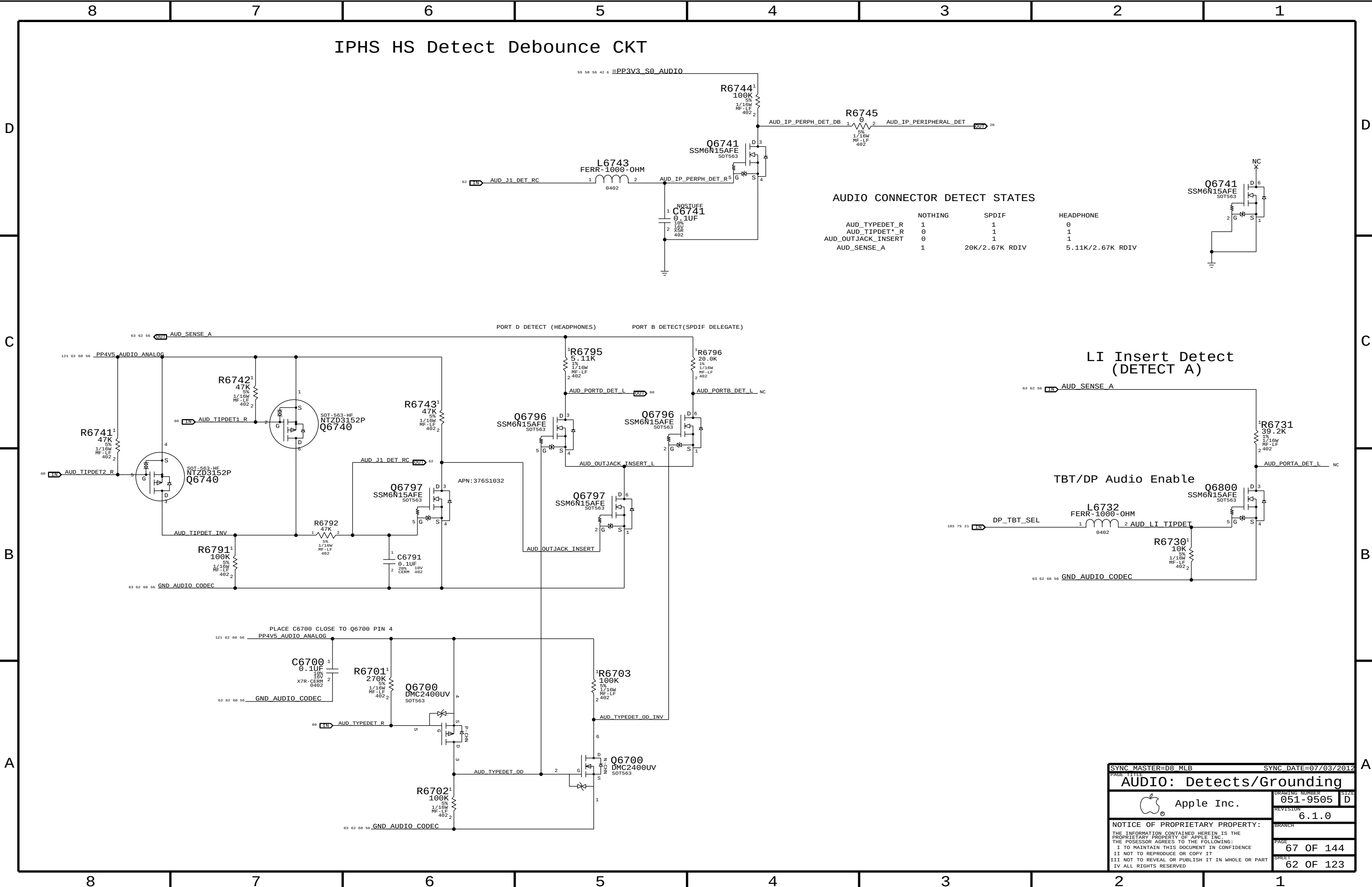
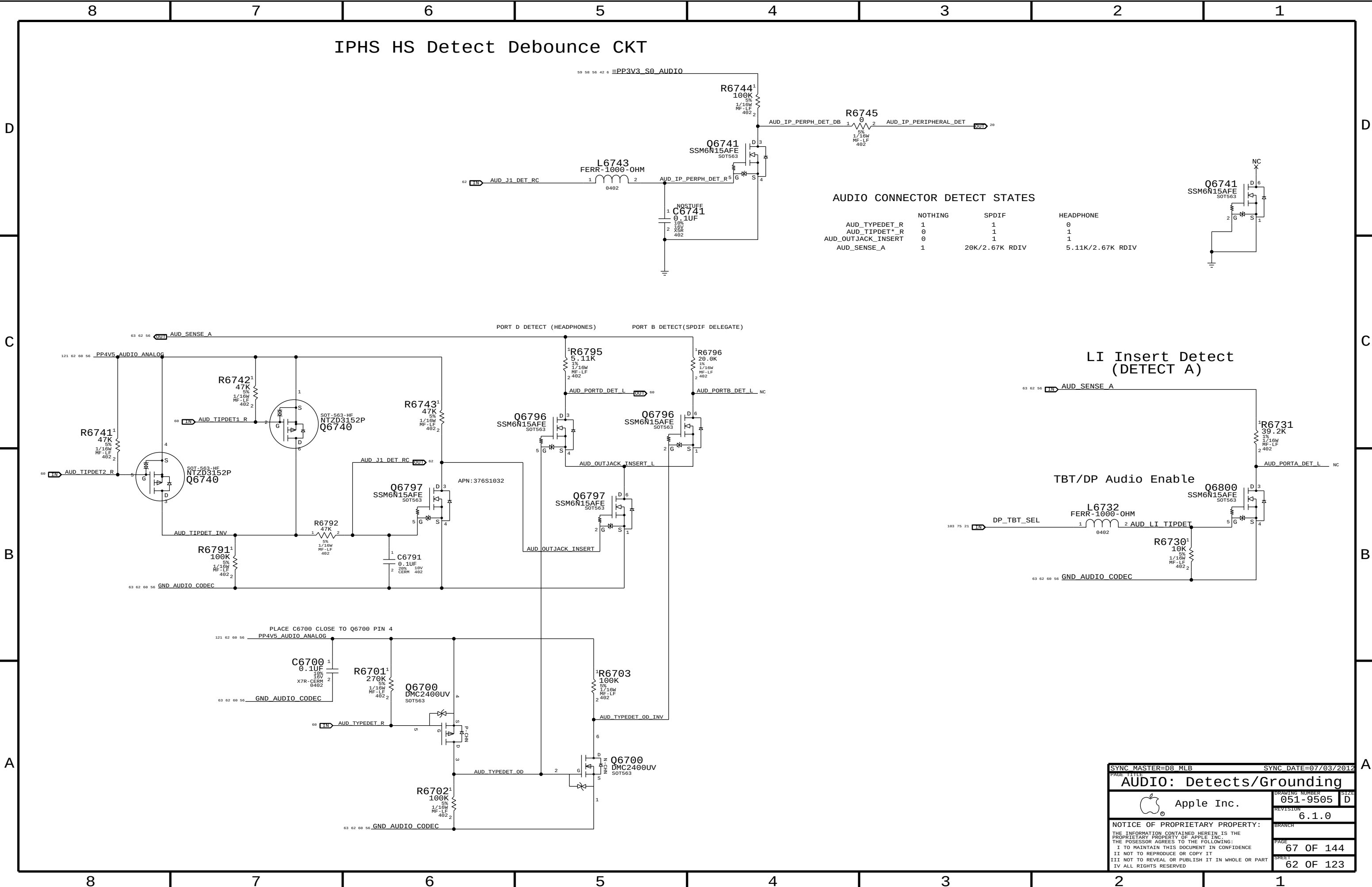
Apple Inc.

051-9505

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[illegible]

IPHS HS Detect Debounce CKT

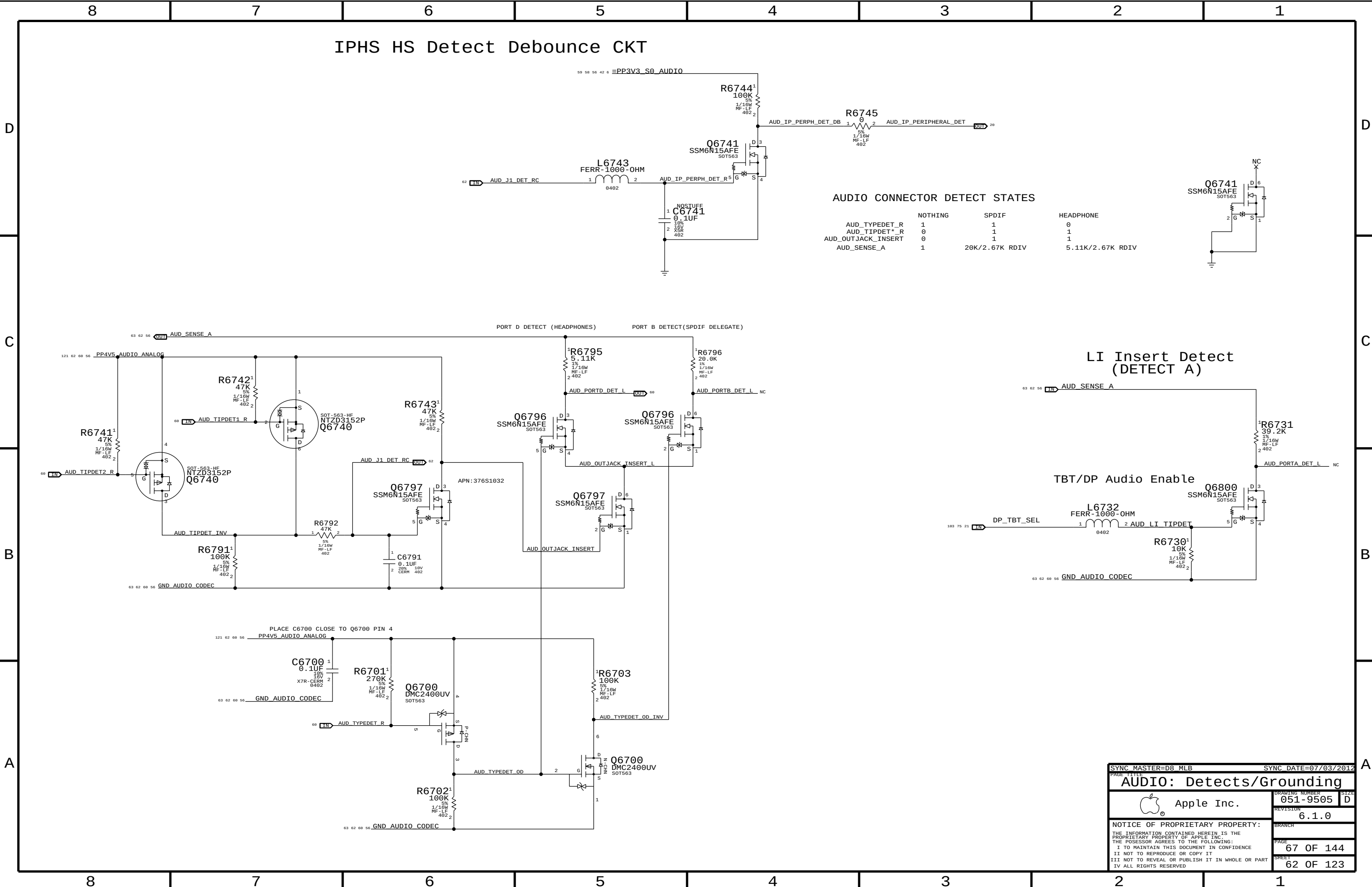
AUDIO CONNECTOR DETECT STATES

	NOTHING	SPDIF	HEADPHONE
AUD_TYPEDET_R	1	1	0
AUD_TIPDET*_R	0	1	1
AUD_OUTJACK_INSERT	0	1	1
AUD_SENSE_A	1	20K/2.67K RDIV	5.11K/2.67K RDIV

LI Insert Detect (DETECT A)

TBT/DP Audio Enable

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IPHS HS Detect Debounce CKT

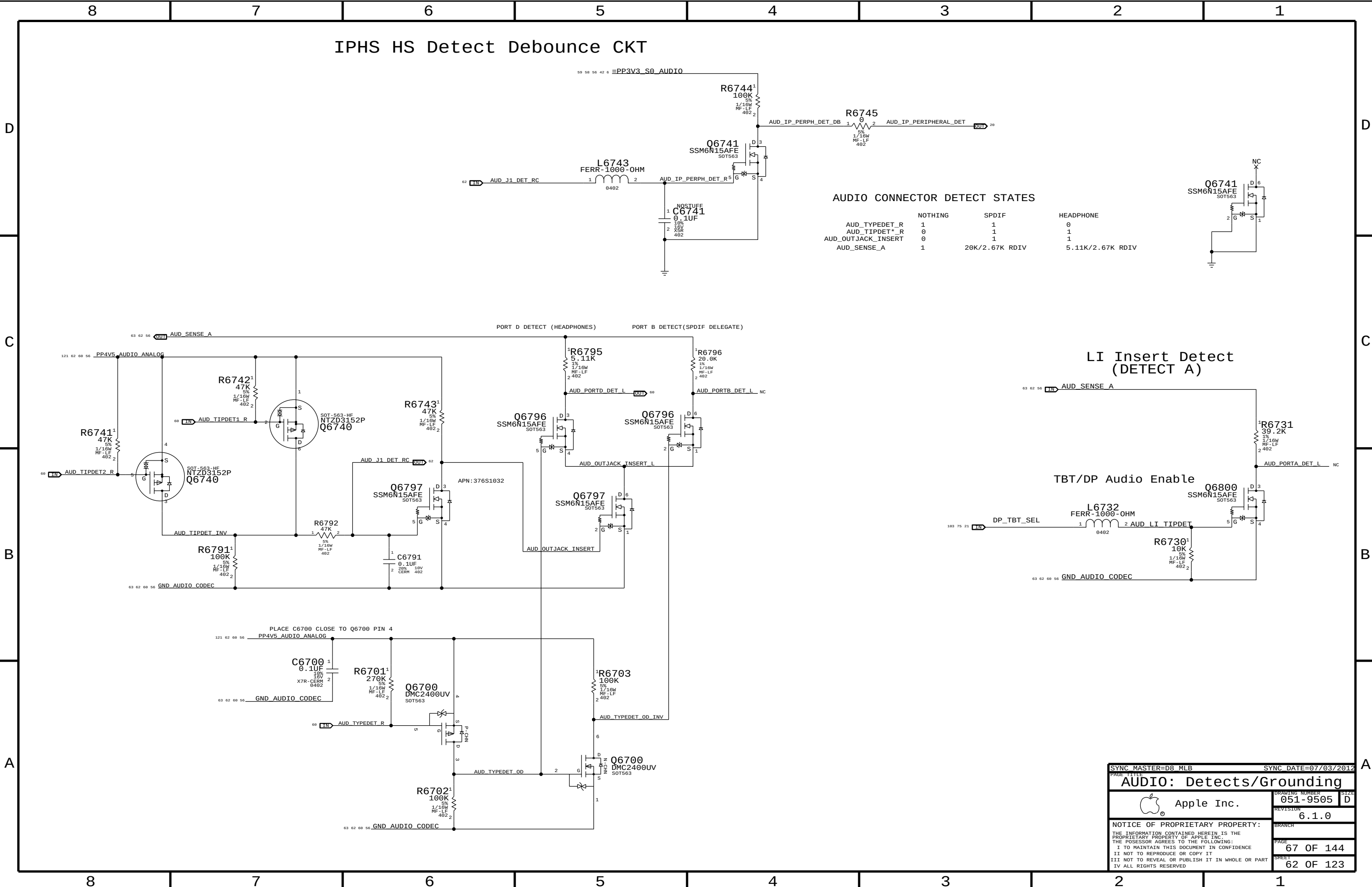
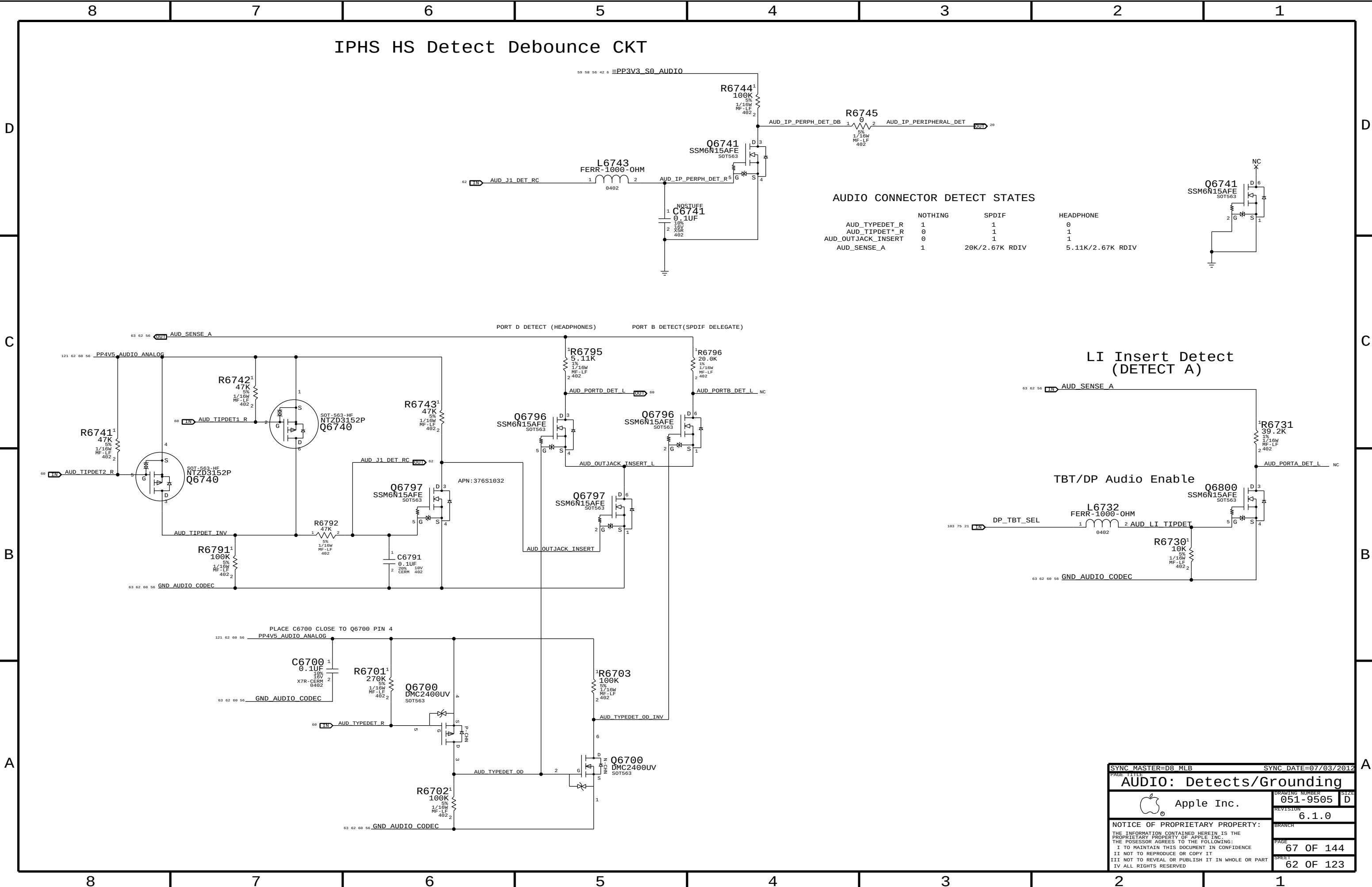
AUDIO CONNECTOR DETECT STATES

	NOTHING	SPDIF	HEADPHONE
AUD_TYPEDET_R	1	1	0
AUD_TIPDET*_R	0	1	1
AUD_OUTJACK_INSERT	0	1	1
AUD_SENSE_A	1	20K/2.67K RDIV	5.11K/2.67K RDIV

LI Insert Detect (DETECT A)

TBT/DP Audio Enable

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IPHS HS Detect Debounce CKT

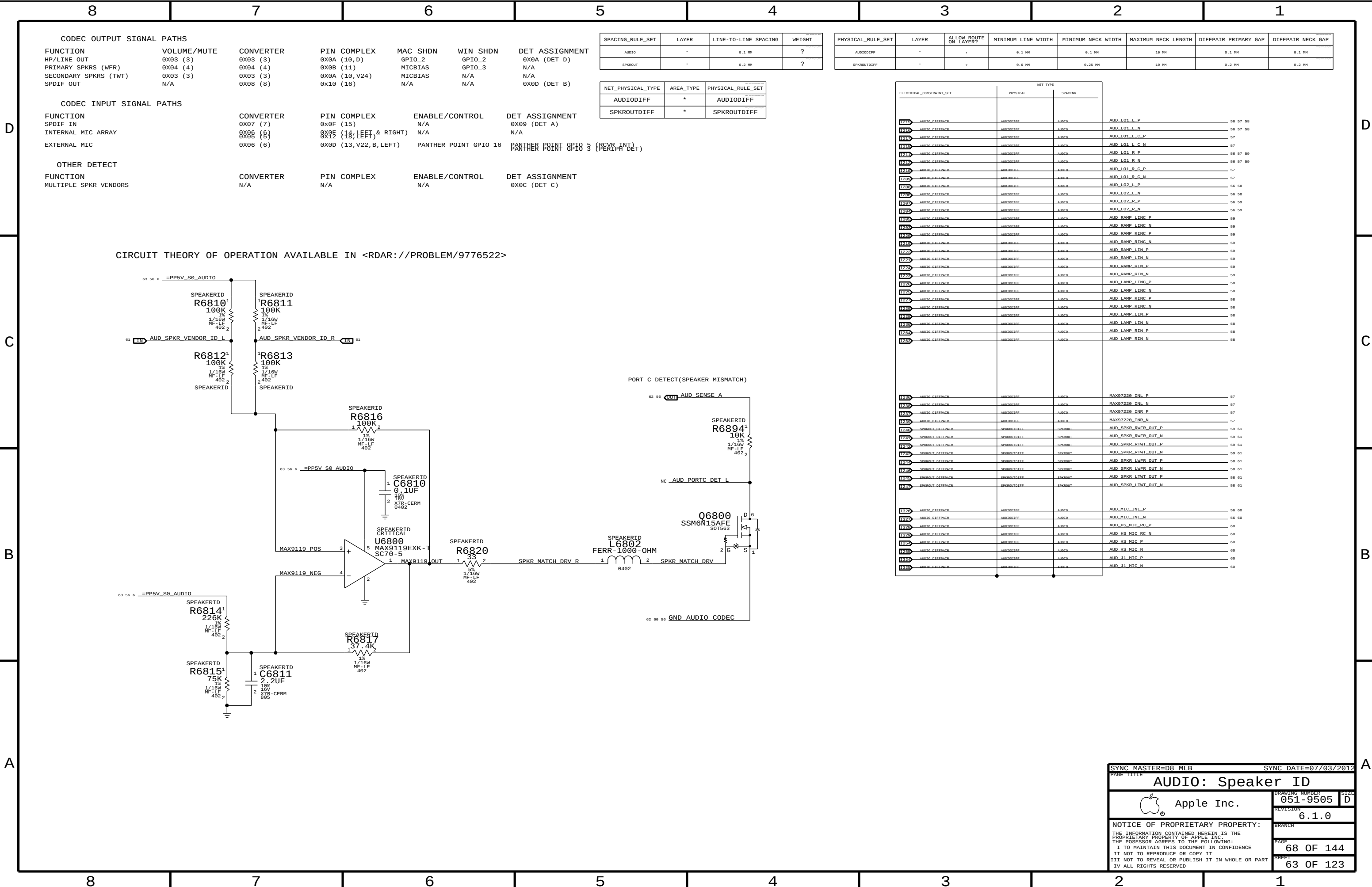
AUDIO CONNECTOR DETECT STATES

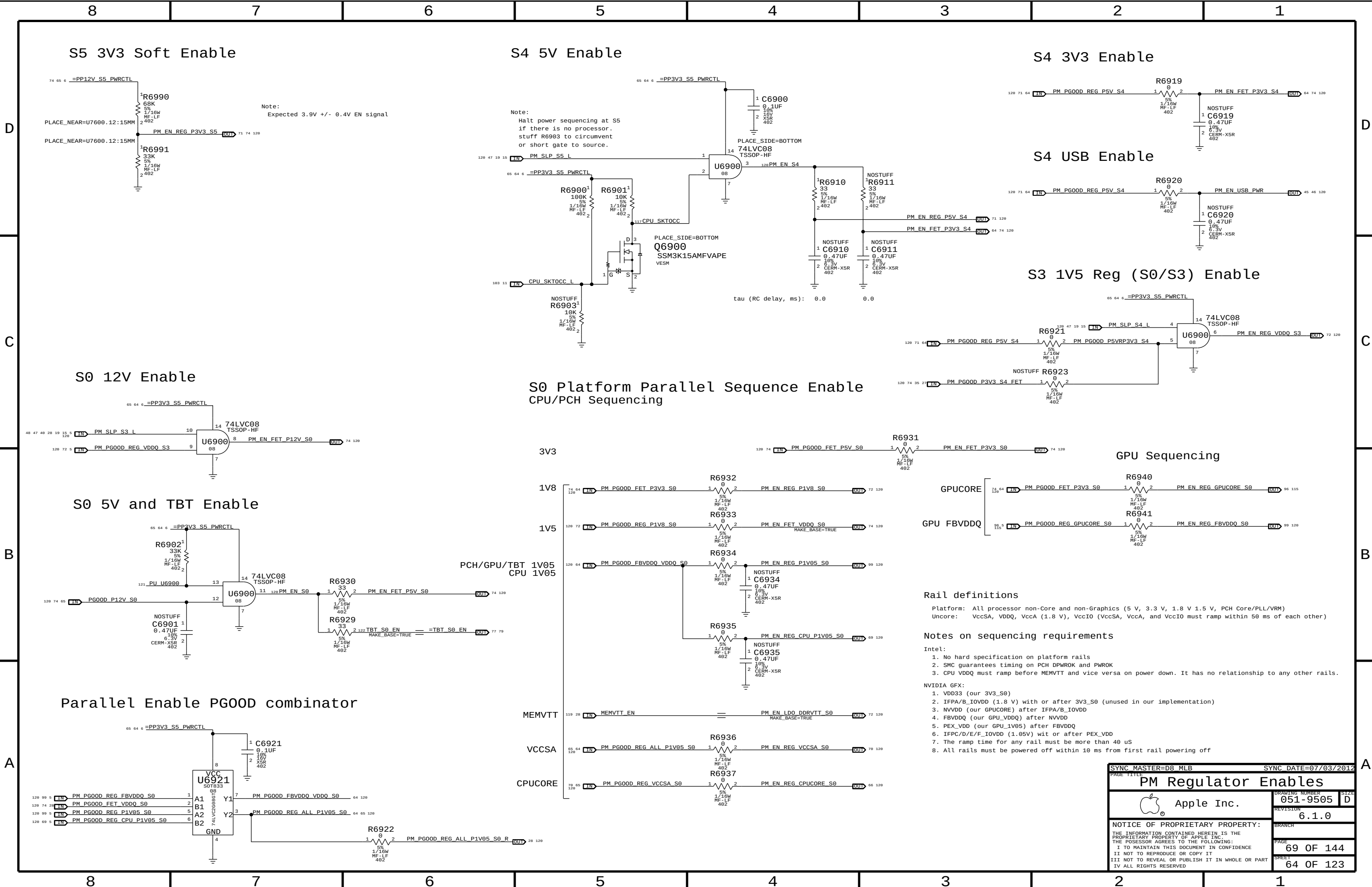
	NOTHING	SPDIF	HEADPHONE
AUD_TYPEDET_R	1	1	0
AUD_TIPDET*_R	0	1	1
AUD_OUTJACK_INSERT	0	1	1
AUD_SENSE_A	1	20K/2.67K RDIV	5.11K/2.67K RDIV

LI Insert Detect (DETECT A)

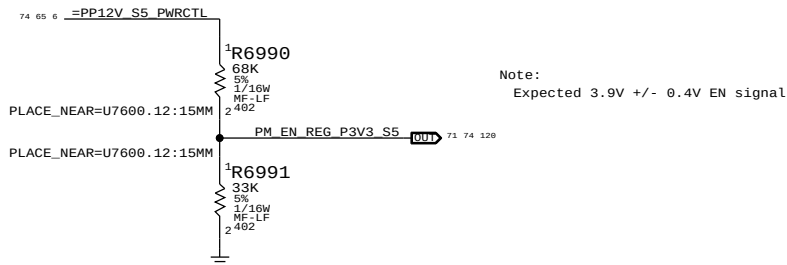
TBT/DP Audio Enable

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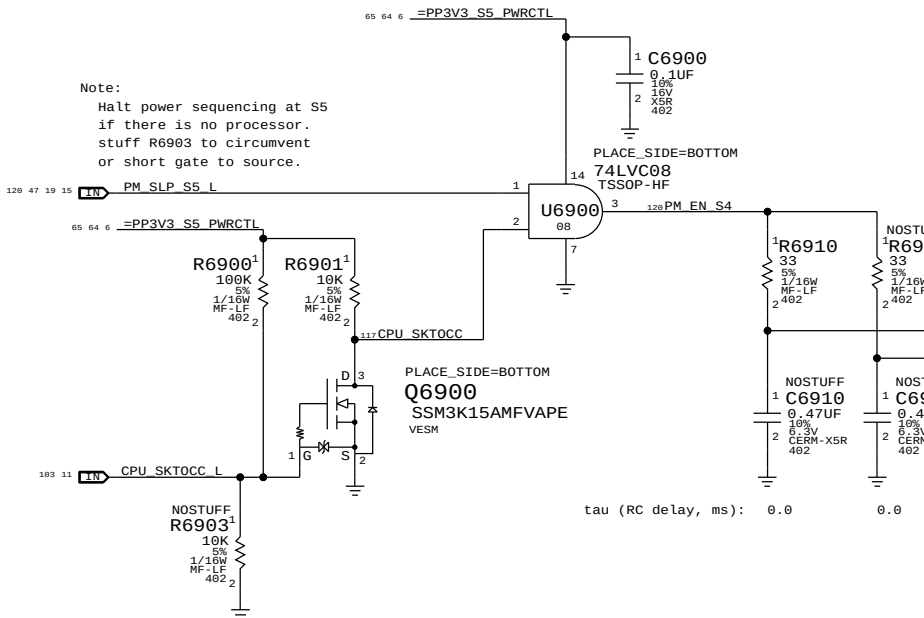




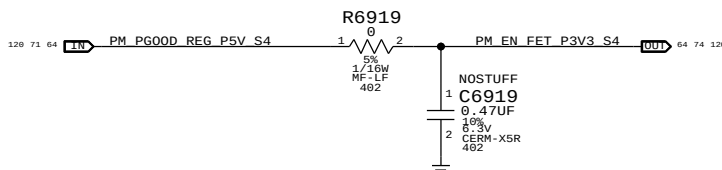
S5 3V3 Soft Enable



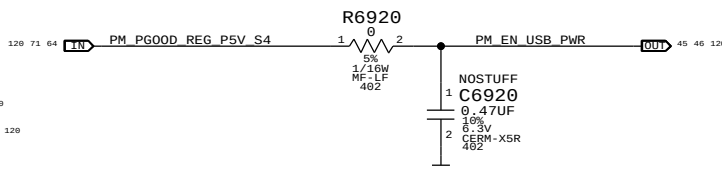
S4 5V Enable



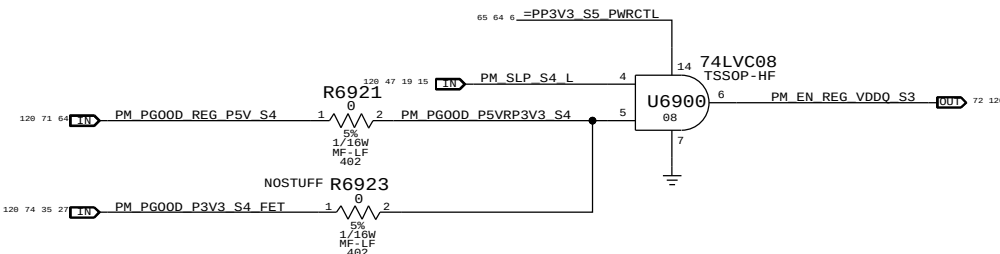
S4 3V3 Enable



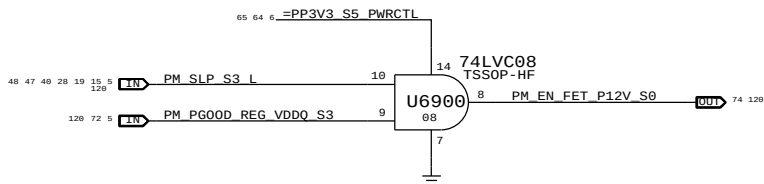
S4 USB Enable



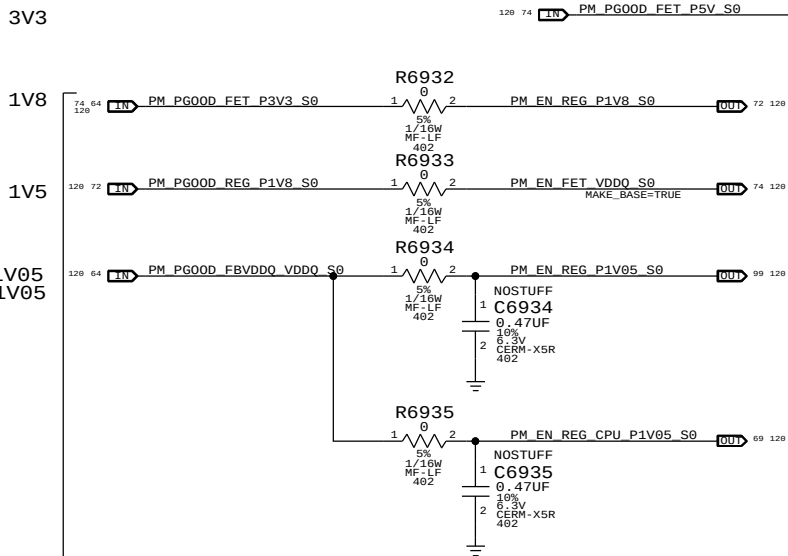
S3 1V5 Reg (S0/S3) Enable



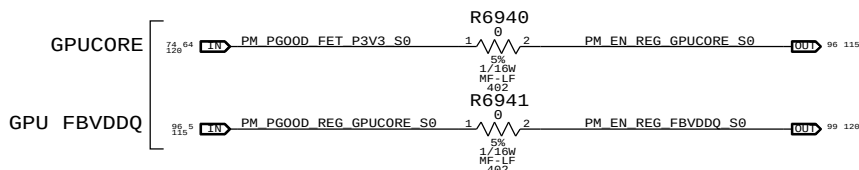
S0 12V Enable



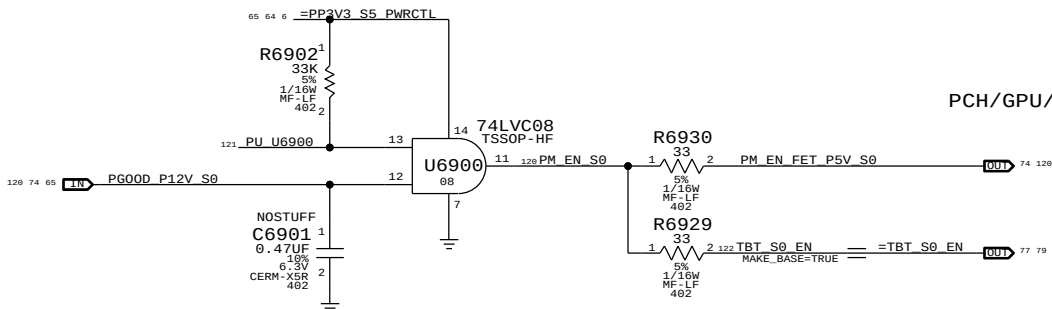
S0 Platform Parallel Sequence Enable CPU/PCH Sequencing



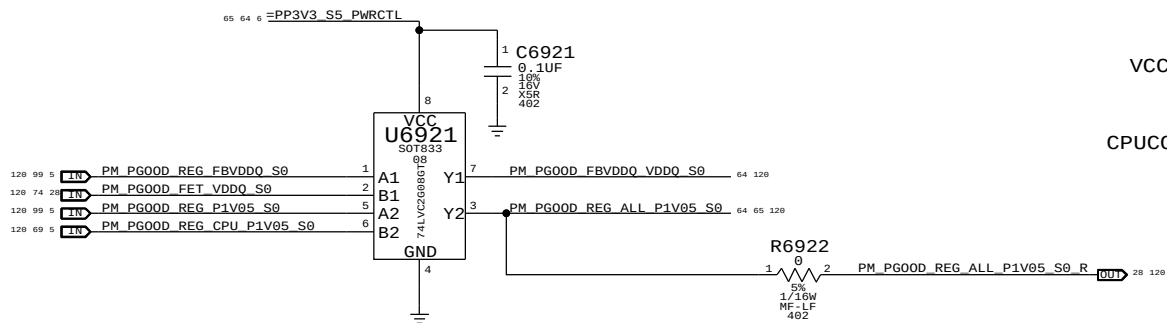
GPU Sequencing



S0 5V and TBT Enable



Parallel Enable PG00D combinator



Rail definitions

Platform: All processor non-Core and non-Graphics (5 V, 3.3 V, 1.8 V 1.5 V, PCH Core/PLL/VRM)
Uncore: VccSA, VDDQ, VccA (1.8 V), VccIO (VccSA, VccA, and VccIO must ramp within 50 ms of each other)

Notes on sequencing requirements

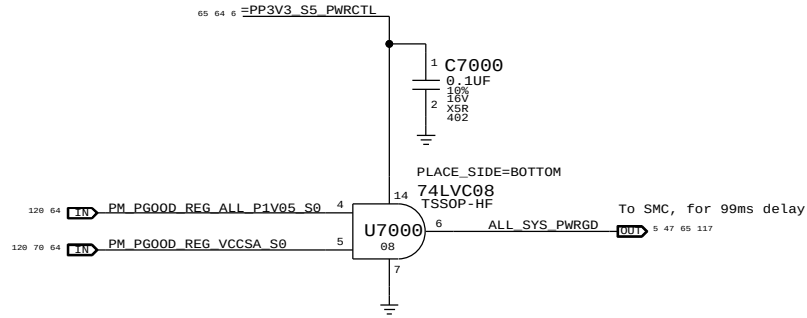
- Intel:
- No hard specification on platform rails
 - SMC guarantees timing on PCH DPWR0K and PWR0K
 - CPU VDDQ must ramp before MEMVTT and vice versa on power down. It has no relationship to any other rails.
- NVIDIA GFX:

- VDD33 (our 3V3_S0)
- IFPA/B_IOVDD (1.8 V) with or after 3V3_S0 (unused in our implementation)
- NVVDQ (our GPU CORE) after IFPA/B_IOVDD
- FBVDDQ (our GPU_VDDQ) after NVVDQ
- PEX_VDD (our GPU_1V05) after FBVDDQ
- IFPC/D/E_F_IOVDD (1.05V) wit or after PEX_VDD
- The ramp time for any rail must be more than 40 uS
- All rails must be powered off within 10 ms from first rail powering off

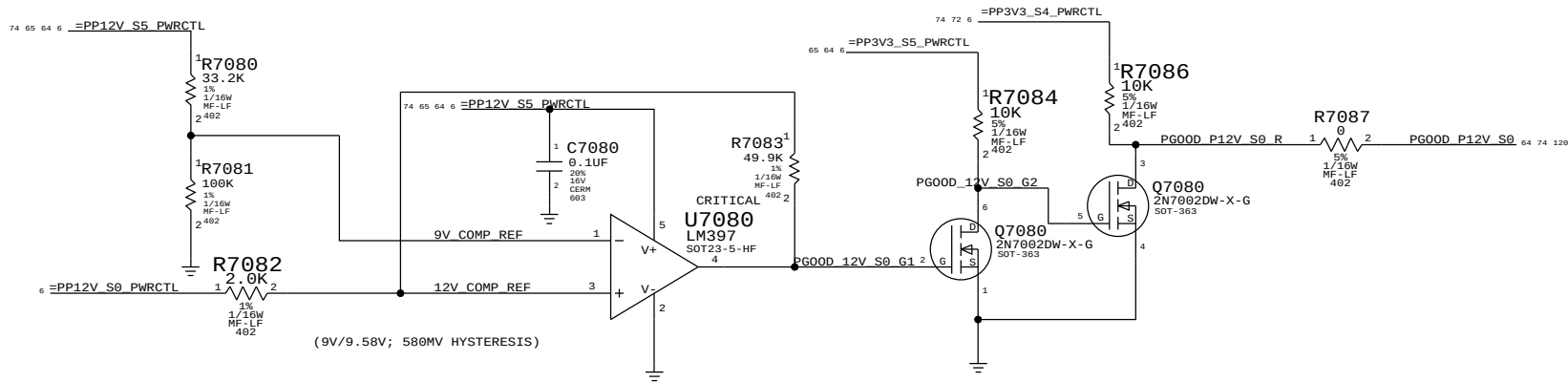
SYNC MASTER=D8_MLB		SYNC DATE=07/03/2012	
PAGE TITLE		PM Regulator Enables	
DRAWING NUMBER		051-9505	SIZE D
REVISION		6.1.0	BRANCH
PAGE		69 OF 144	SHEET
SHEET		64 OF 123	IV ALL RIGHTS RESERVED

Platform Power Good derive SMC ALL_SYS_PWRGD

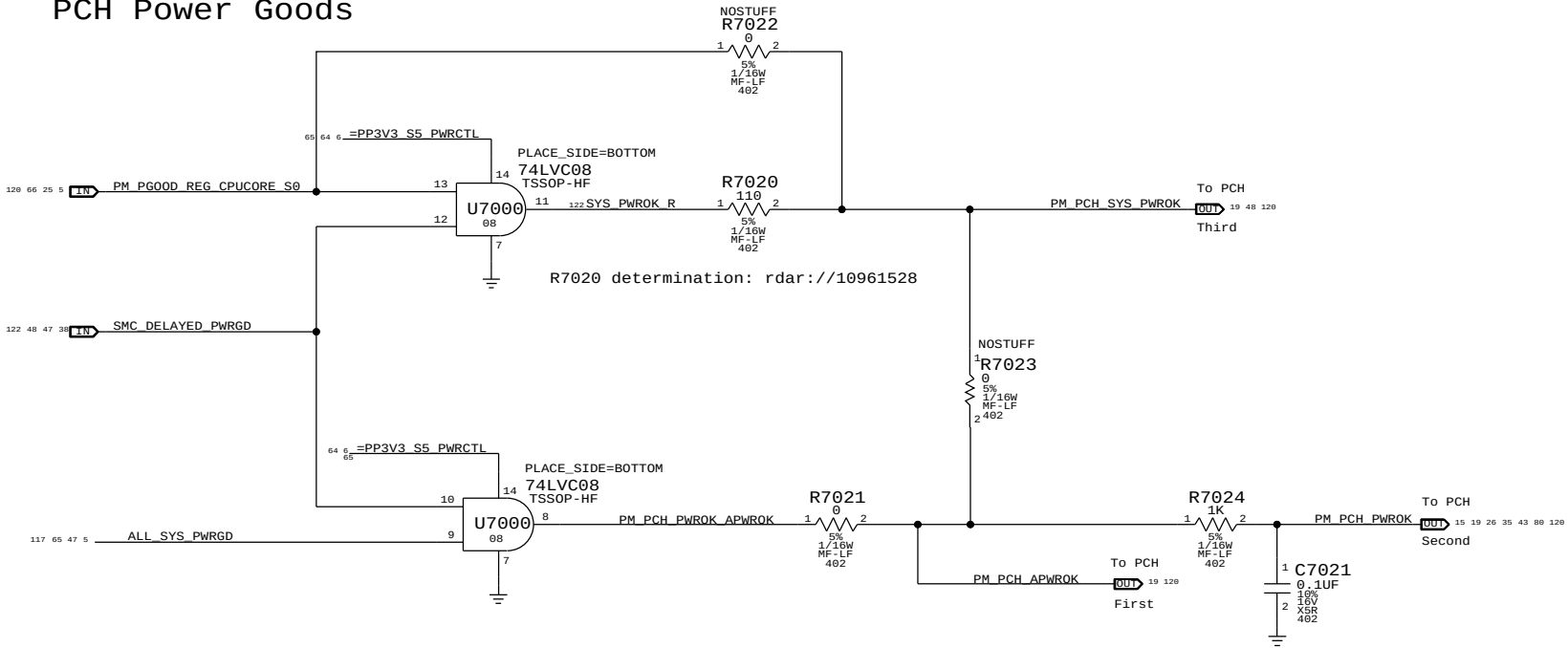
The end of the power sequence for S0 rails except CPU CORE.



PGOOD comparators for PP12V_S0



PCH Power Goods



radar://11043352 Need AND Gate to deassert PM_PCH_PWR0K to PCH when unexpected power loss happens

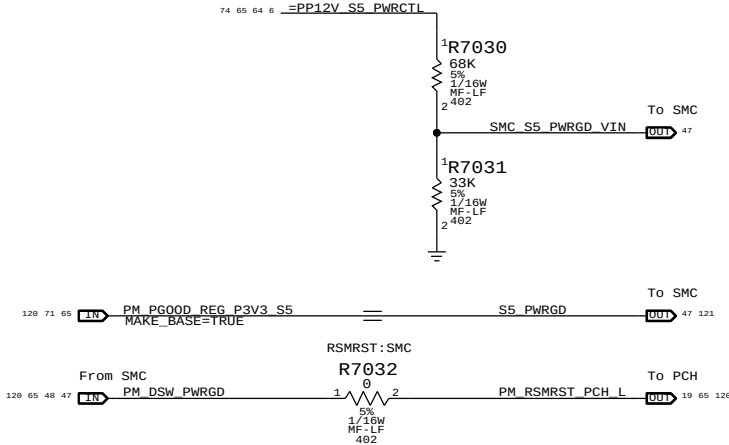
Resume Reset

Intel Doc# 29517 Maho Bay PDG, Section 22.13
Intel Doc# 29562 Panther Point EDS, Section 8.7 and 8.8

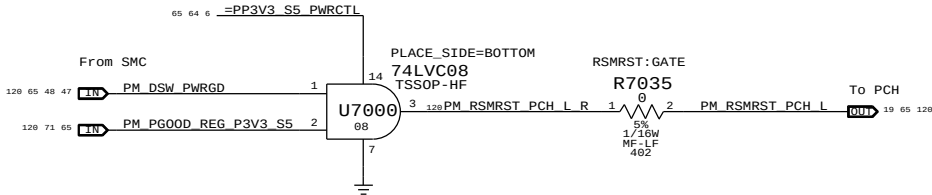
Note:
The iMac K70K72 designs does not support Deep Sx modes so both DPWROK and RSMRST# signals are shorted together

Requirements:
Power on:
Asserted at least 10 ms after all suspend well power is valid
Power off or loss of AC:
Transition to 0.8V or less before VccSUS3_3 drops to 2.90 V
to allow PCH to switch suspend well to battery without excessive loading

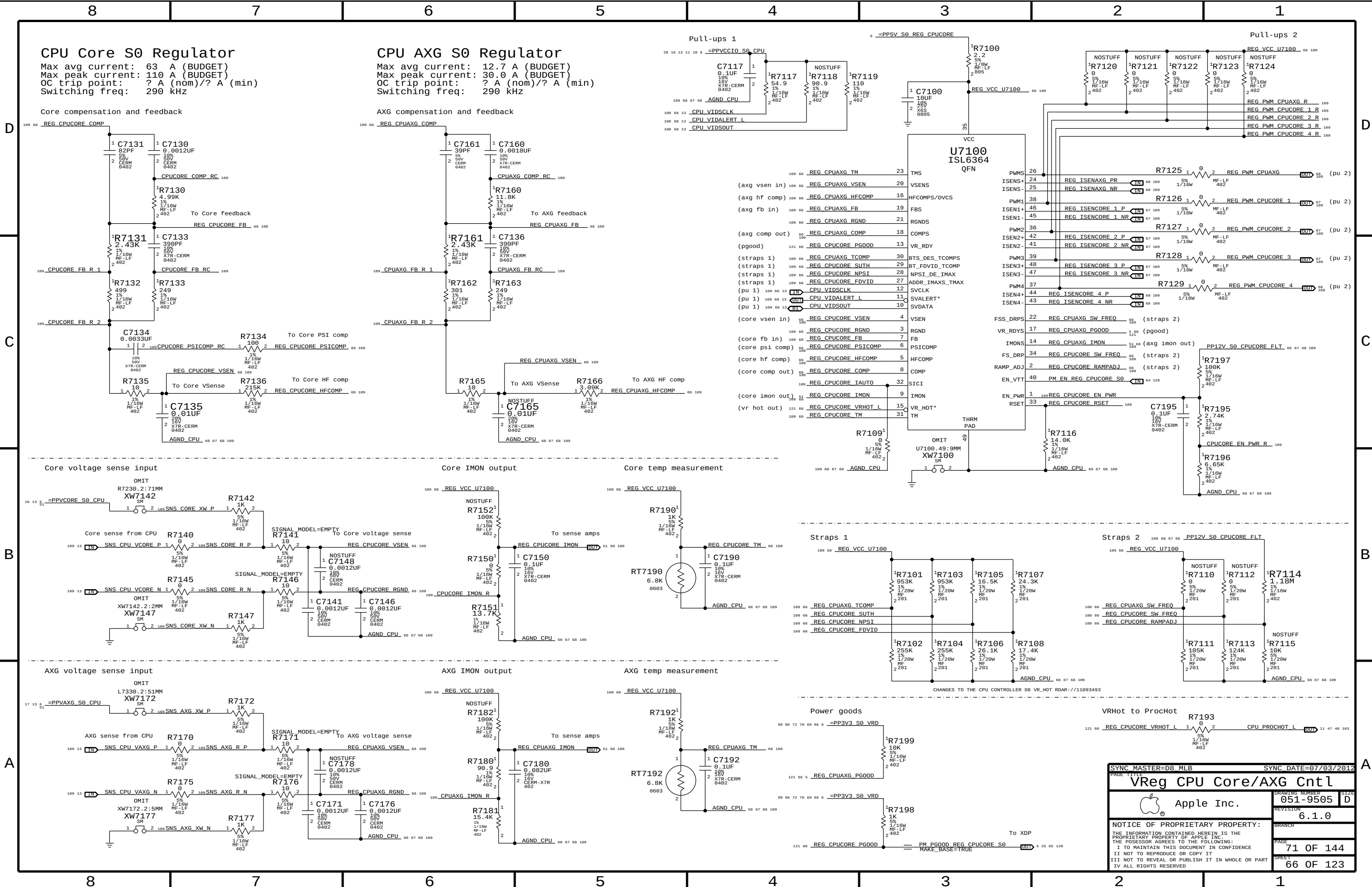
Primary method:
The SMC guarantees proper assertion and de-assertion of RSMRST# for normal operation.
SMC de-asserts RSMRST# (PM_DSW_PWRGD) when S5_PWRGD input is asserted and SMC_S5_PWRGD_VIN input is above comparator input level of 1.5 V.
SMC asserts RSMRST# (PM_DSW_PWRGD) when SMC_S5_PWRGD_VIN input drops from 1.8 V to 1.5 V (as implemented) when 12 V S5 rail drops to 10 V.



Secondary method:
The SMC guarantees proper assertion and de-assertion of RSMRST# for normal operation via PM_DSW_PWRGD.
RSMRST# is asserted when power good from regulator is de-asserted in the event AC is lost. Power good de-assertion should happen quickly enough to meet Intel spec.



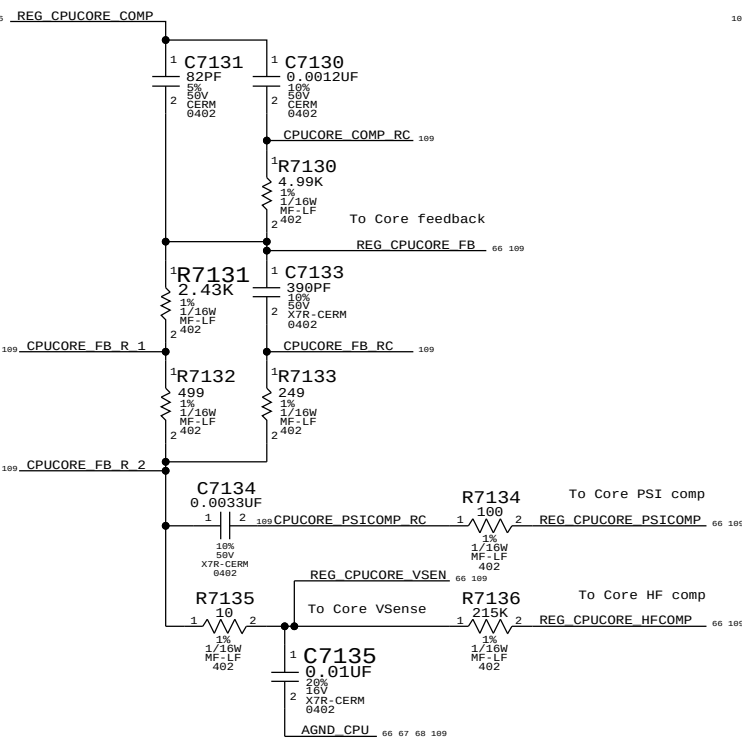
SYNC MASTER=D8_MLB		SYNC DATE=07/03/2012	
PAGE TITLE		PM Power Good	
Apple Inc.		DRAWING NUMBER	051-9505
		REVISION	6.1.0
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CPU Core S0 Regulator

Max avg current: 63 A (BUDGET)
Max peak current: 110 A (BUDGET)
OC trip point: ? A (nom)/? A (min)
Switching freq: 290 kHz

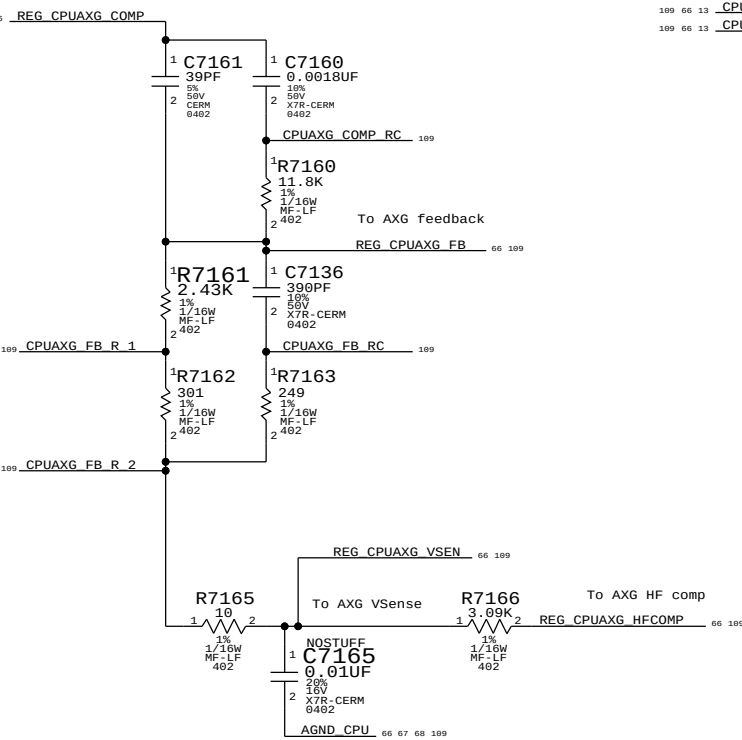
Core compensation and feedback



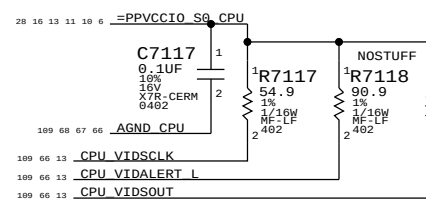
CPU AXG S0 Regulator

Max avg current: 12.7 A (BUDGET)
Max peak current: 30.0 A (BUDGET)
OC trip point: ? A (nom)/? A (min)
Switching freq: 290 kHz

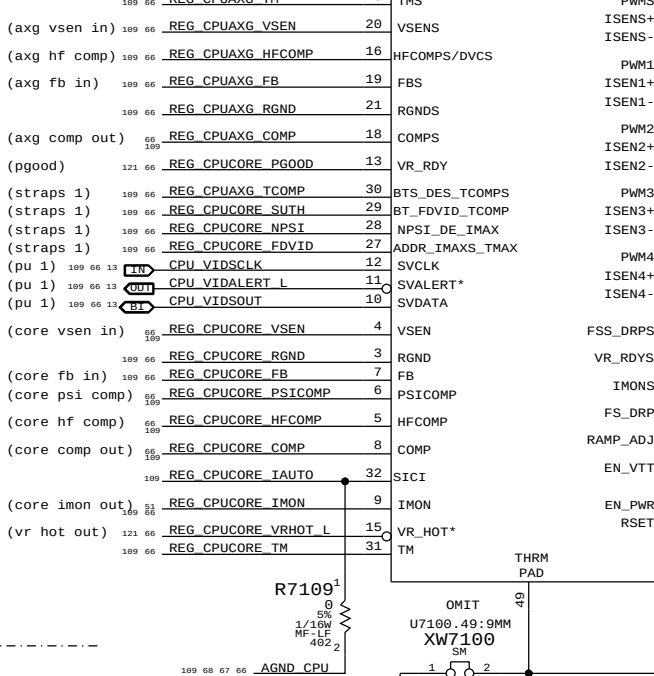
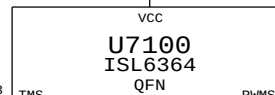
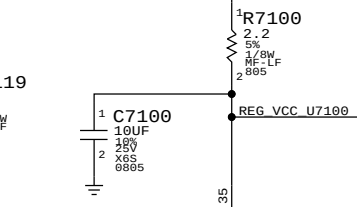
AXG compensation and feedback



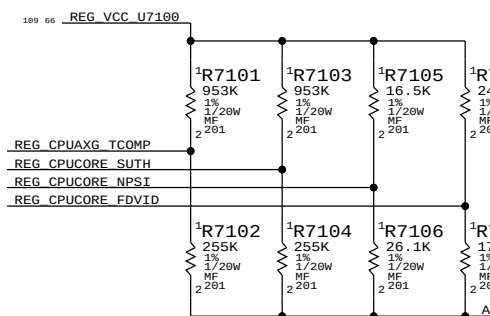
Pull-ups 1



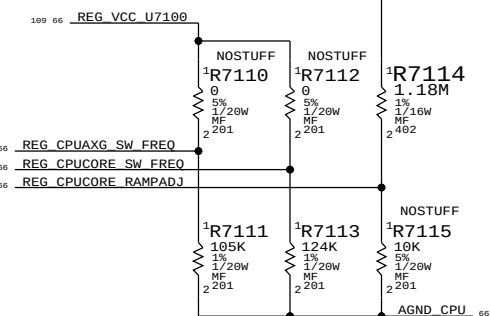
Pull-ups 2



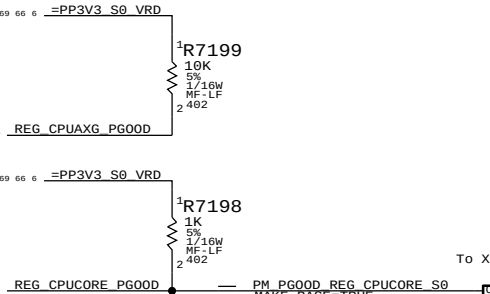
Straps 1



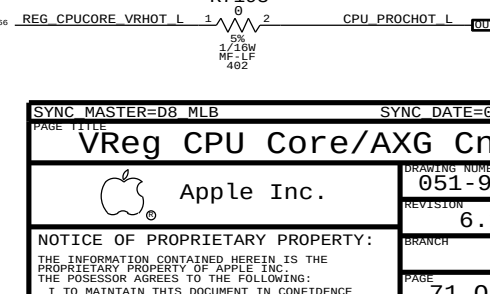
Straps 2



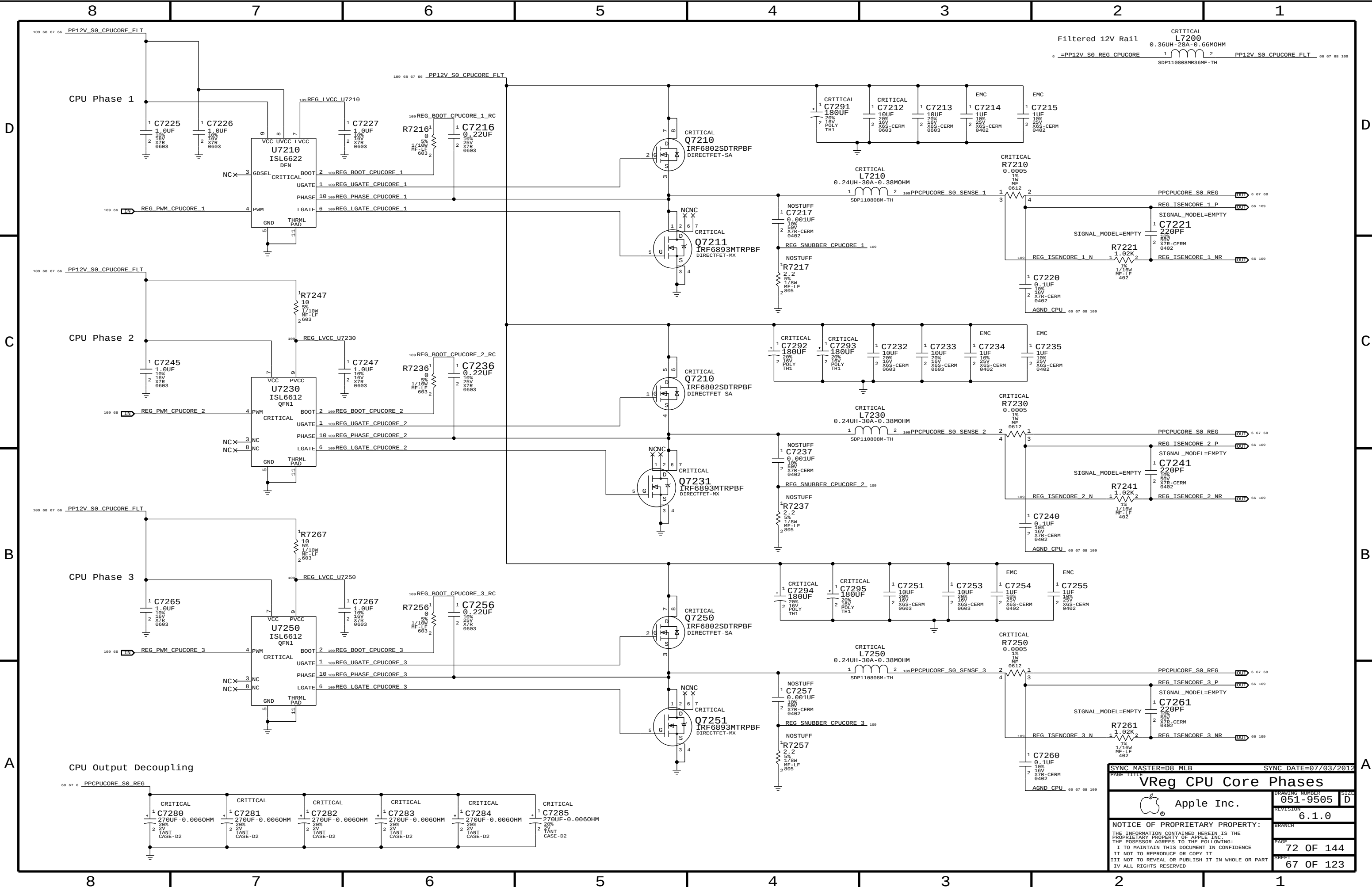
Power goods



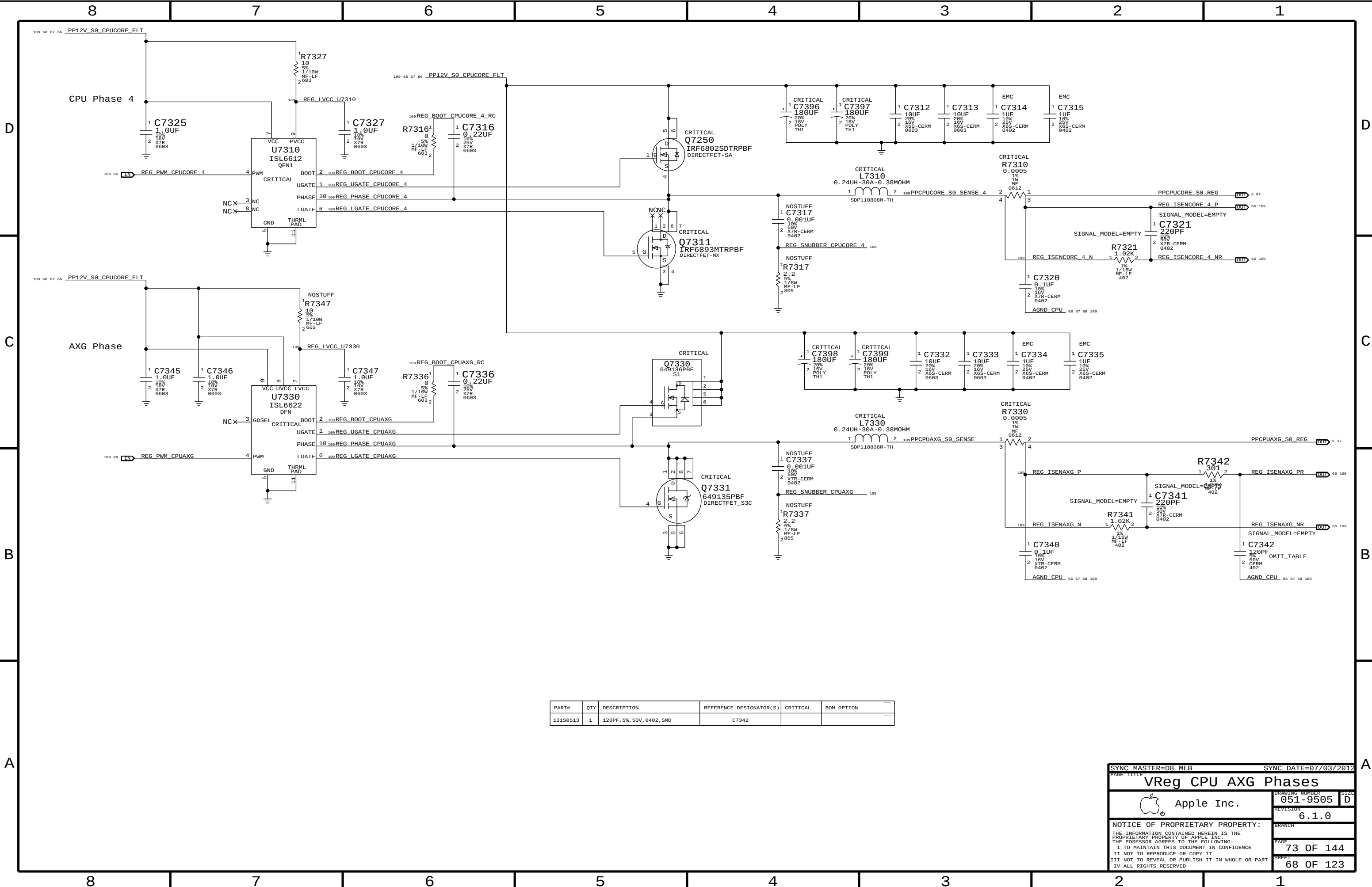
VRHot to Prochot



SYNC MASTER=D8 MLB		SYNC DATE=07/03/2012	
PAGE TITLE		DRAWING NUMBER	
VReg CPU Core/AXG Cntl		051-9505	
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VReg CPU Core Phases			
 Apple Inc.		DRAWING NUMBER	051-9505
		SIZE	D
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PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
131S0513	1	120PF, 5%, 50V, 0402, SMD	C7342		

SYNC MASTER=D8_MLB

SYNC DATE=07/03/2012

VReg CPU AXG Phases

Apple Inc.

051-9505

6.1.0

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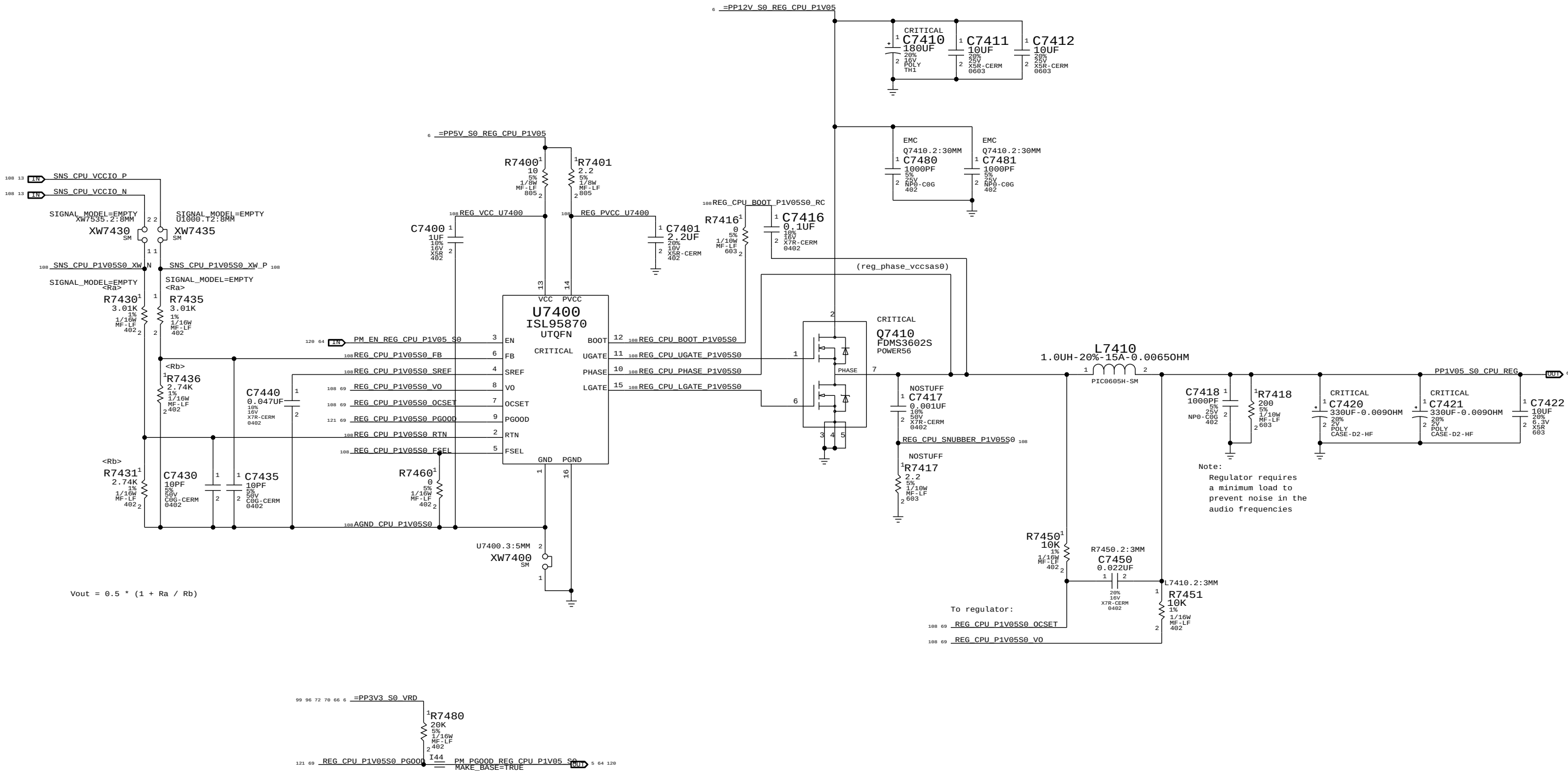
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CPU VccIO (1.05V) S0 Regulator

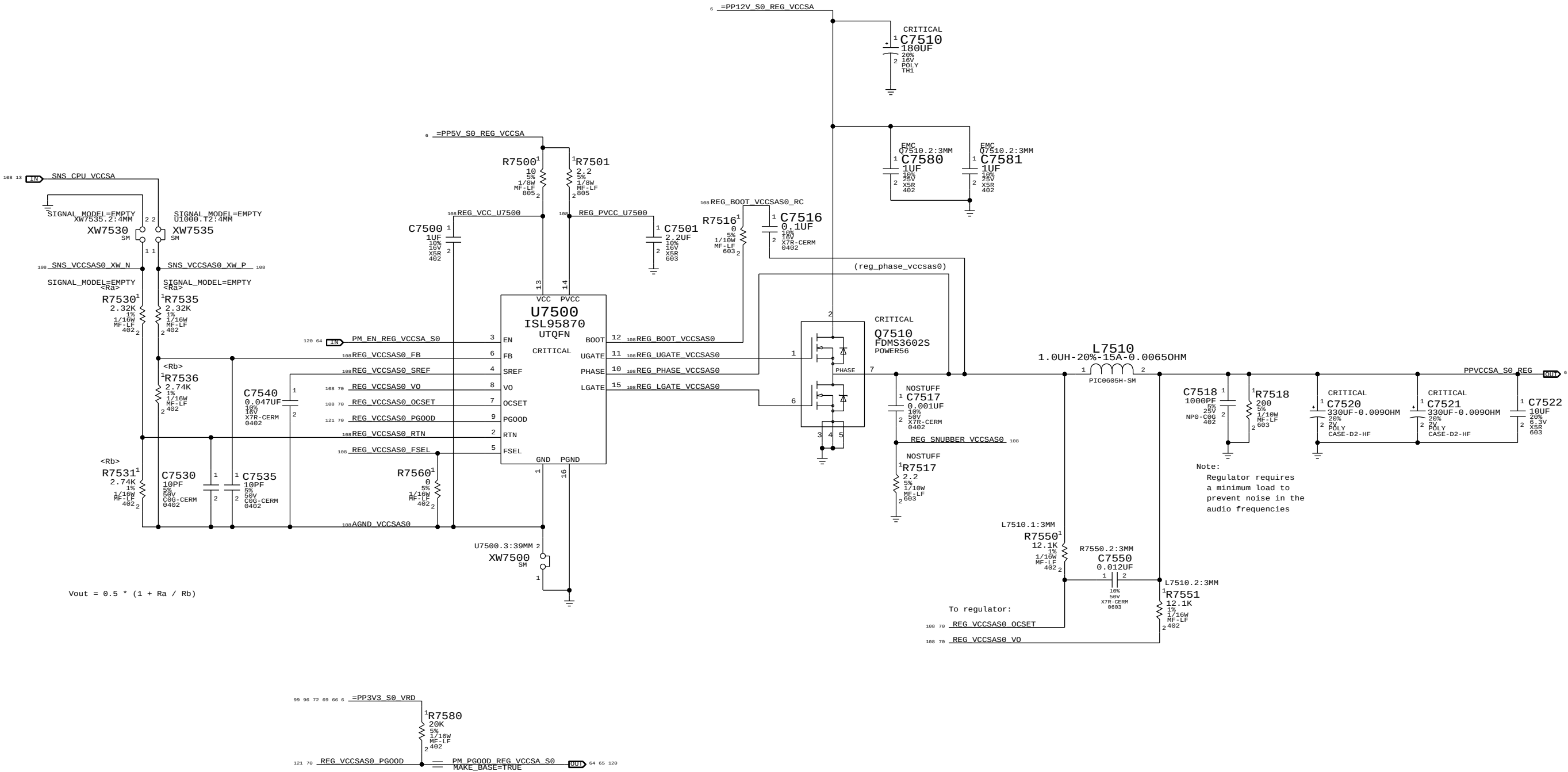
Max avg current: 8.10 A (BUDGET)
Max peak current: 8.50 A (BUDGET)
OC trip point: ? A (min)/? A (max)
Switching freq: 500 kHz



SYNC MASTER=D8_MLB		SYNC DATE=07/03/2012	
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VReg CPU 1.05V S0			
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		REVISION	6.1.0
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		PAGE	74 OF 144
		SHEET	69 OF 123

CPU VccSA (0.925V) S0 Regulator

Max avg current: 12.07 A (BUDGET)
Max peak current: 30 A (BUDGET)
OC trip point: ? A (min)/? A (max)
Switching freq: 500 KHz



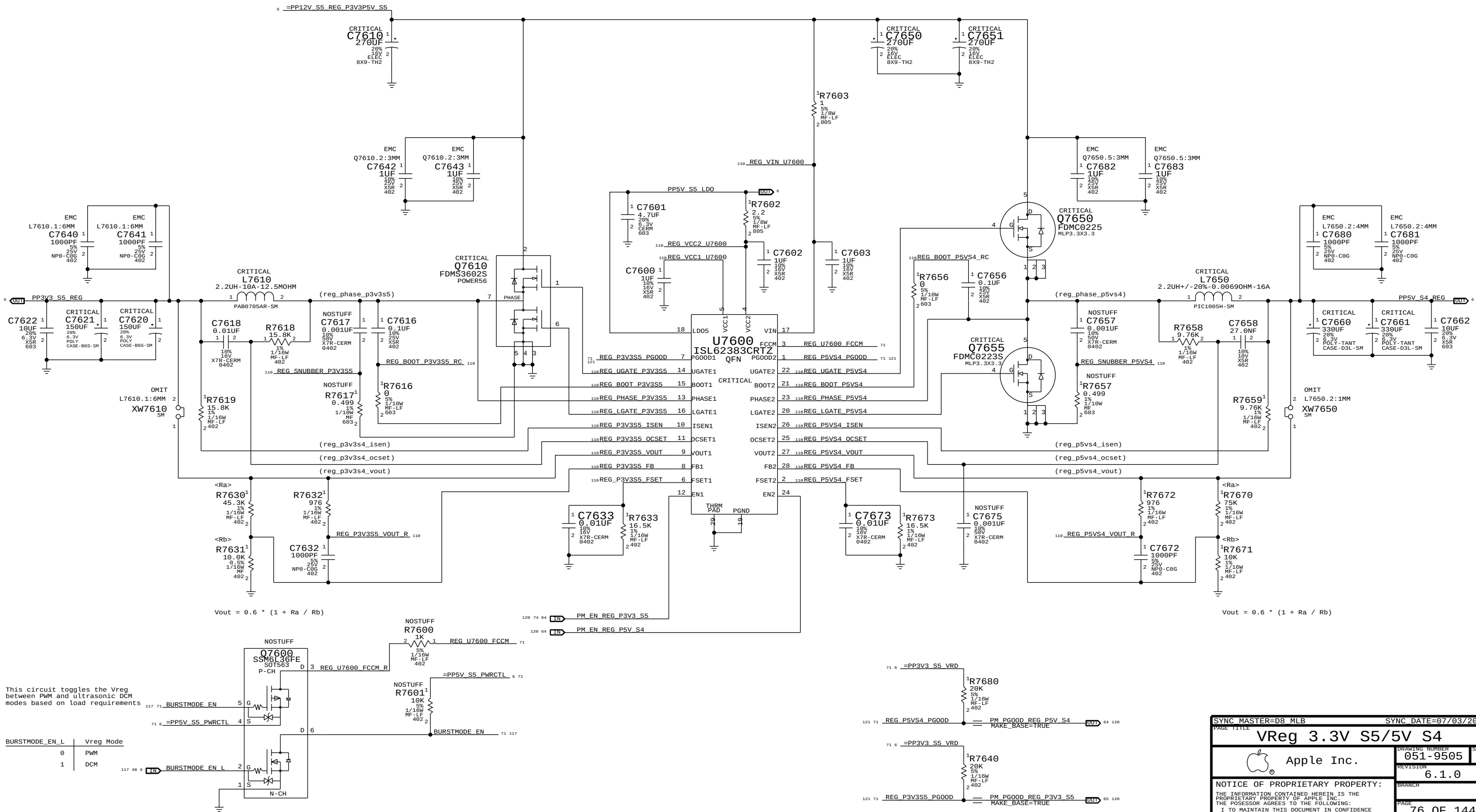
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PAGE TITLE		VReg CPU VccSA S0	
DRAWING NUMBER		051-9505	SIZE D
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3.3V S5 Regulator

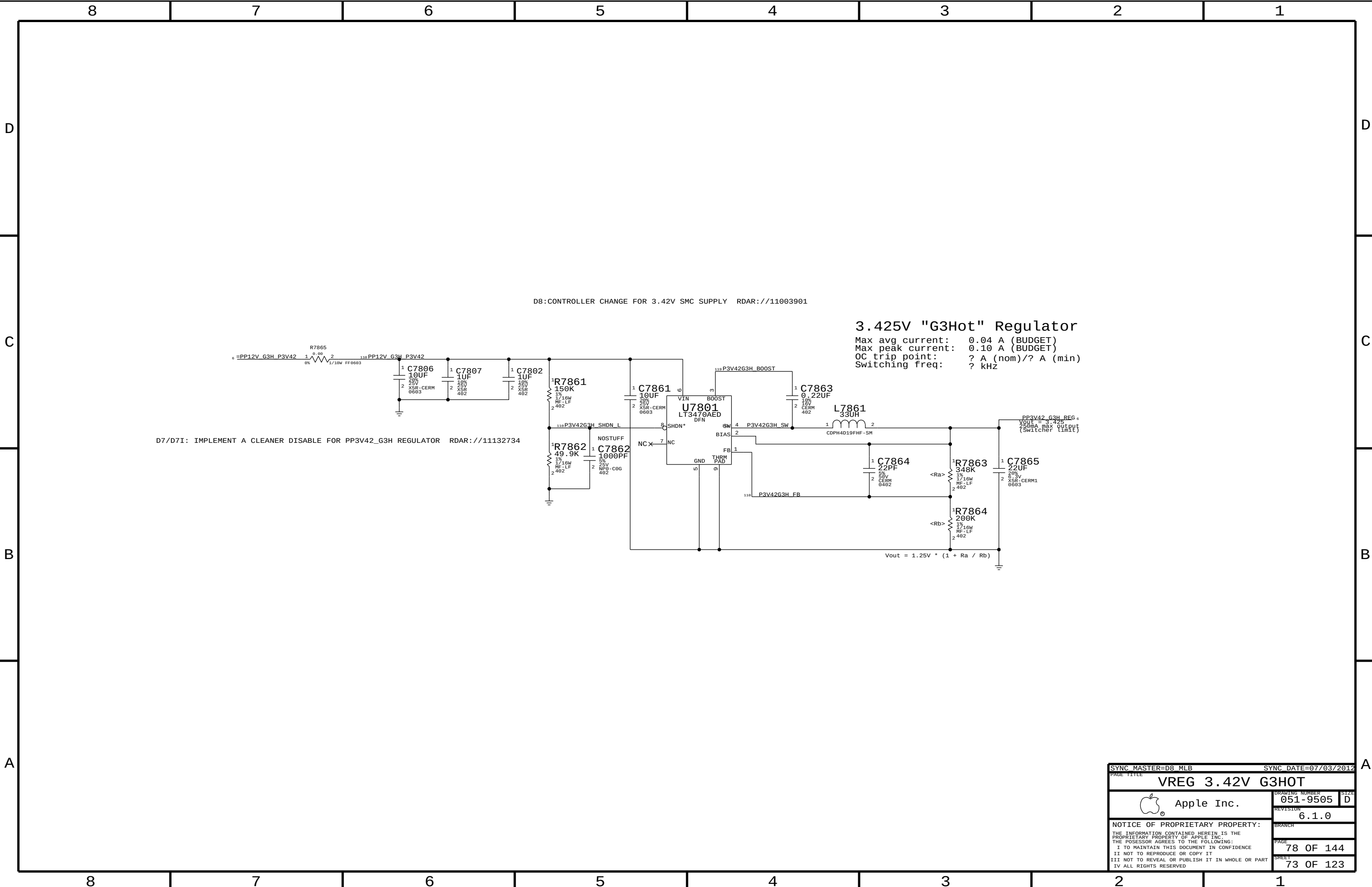
Max avg current: 6 A (design)/ 4.85 A (budget)
Max peak current: ? A (design)/ 6.6 A (budget)
OC trip point: ? A (nom)/? A (min)
Switching freq: 350 kHz

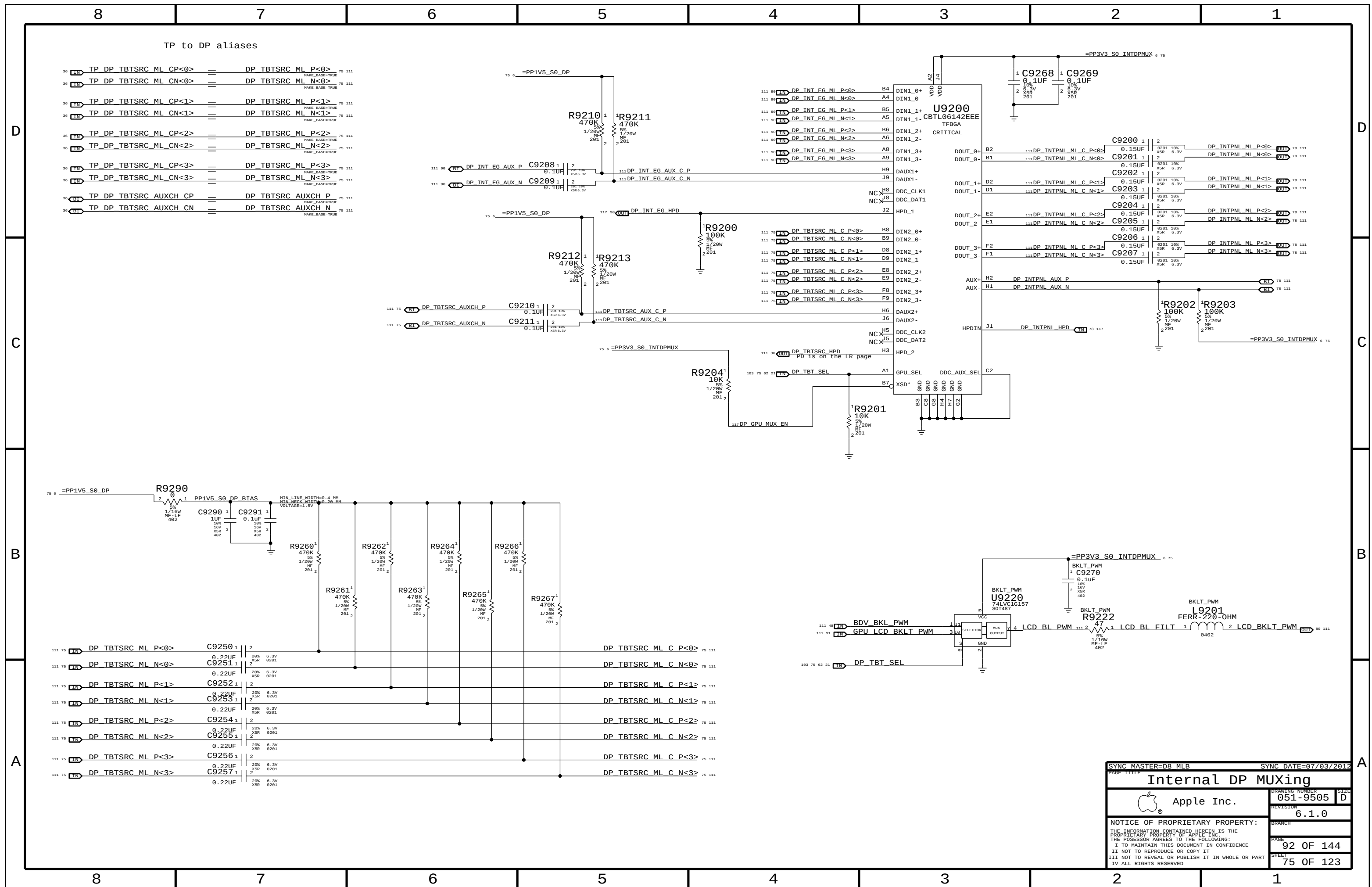
5V S4 Regulator

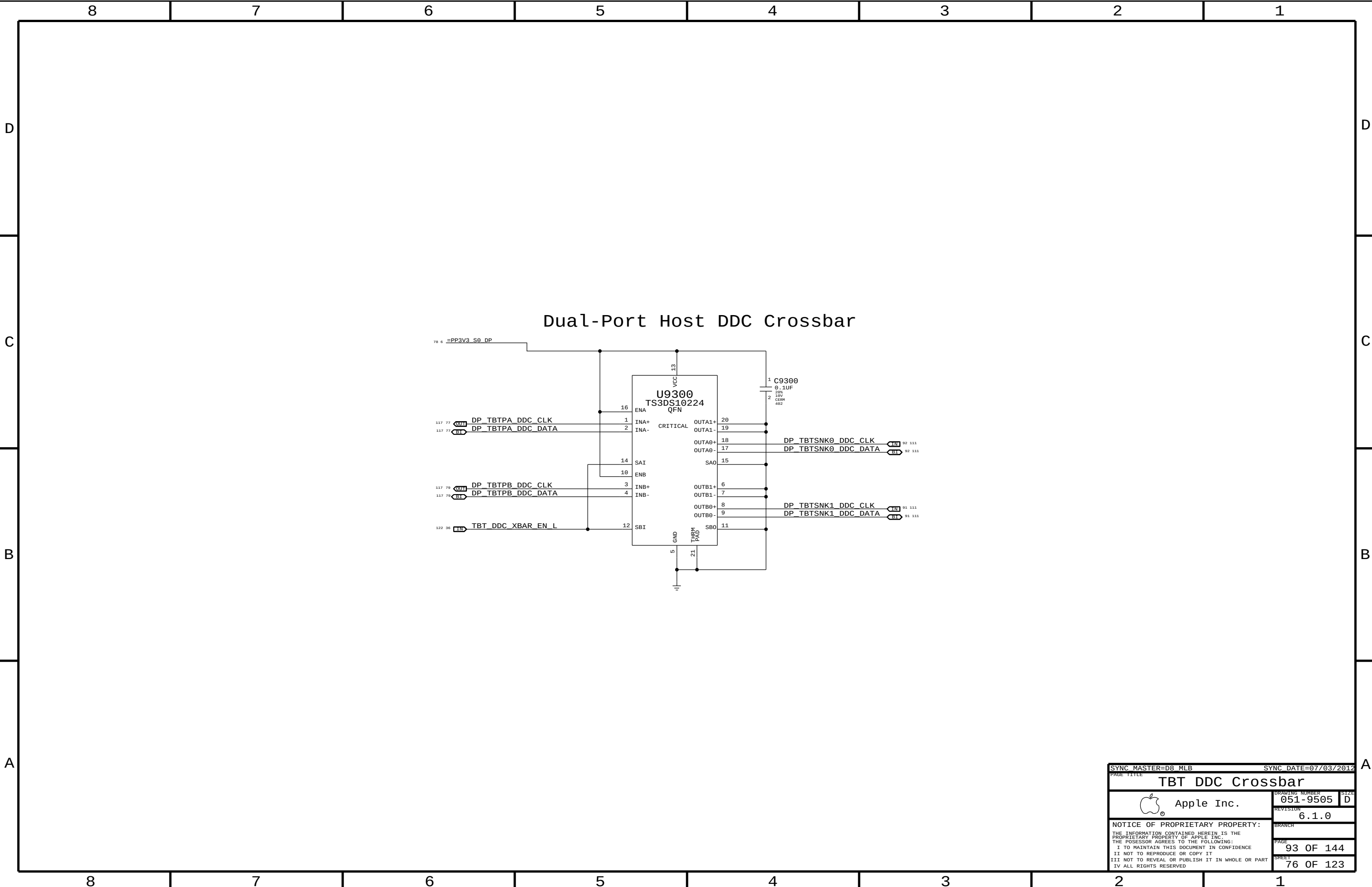
Max avg current: 10 A (design)/ 6.08 A (budget)
Max peak current: ? A (design)/ 6.9 A (budget)
OC trip point: ? A (nom)/? A (min)
Switching freq: 350 kHz



SYNC MASTER=D8 MLB		SYNC DATE=07/03/2012	
PAGE TITLE			
VReg 3.3V S5/5V S4		DRAWING NUMBER	051-9505
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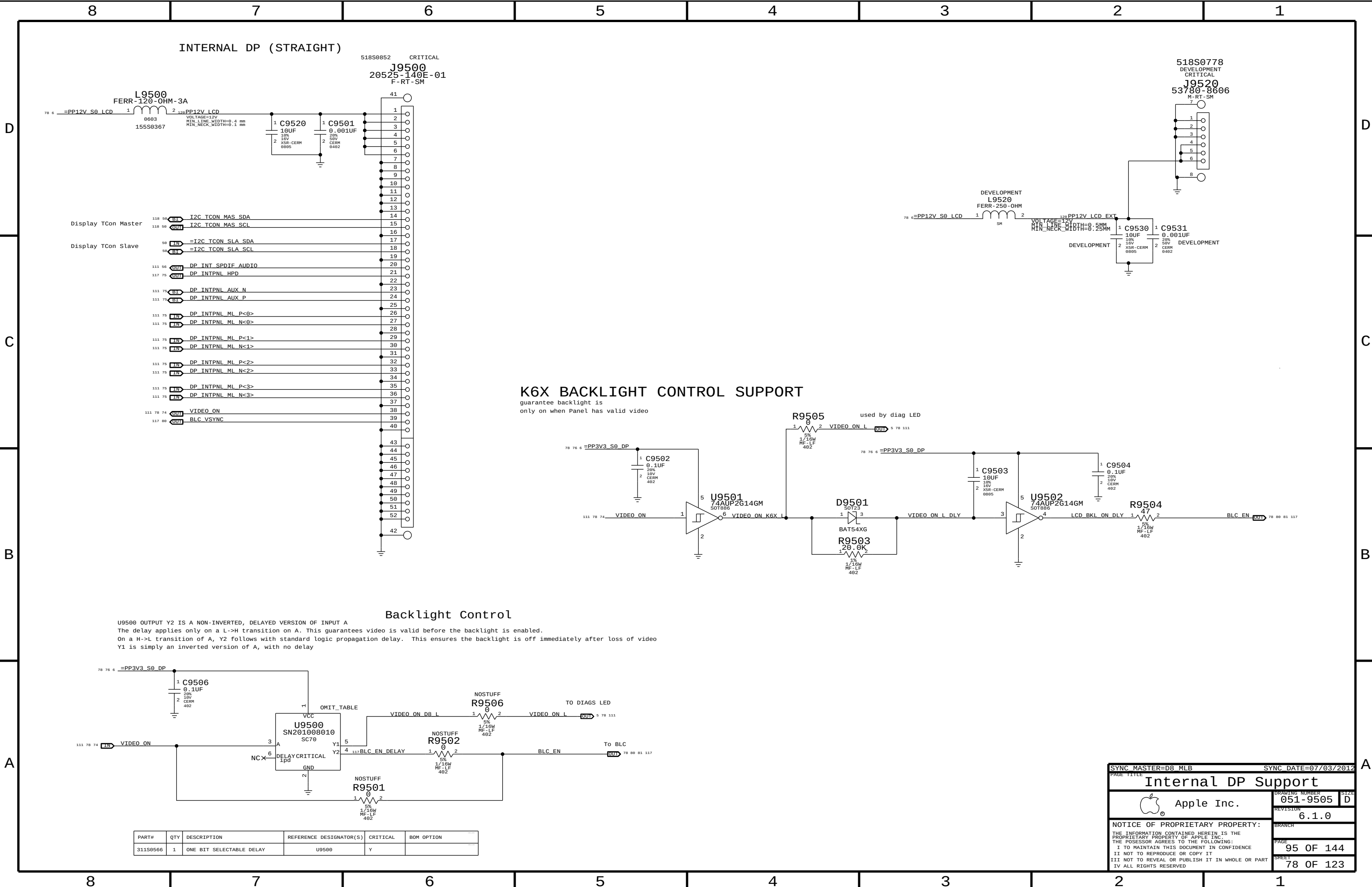




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SYNC MASTER=D8 MLB		SYNC DATE=07/03/2012	
PAGE TITLE			
Thunderbolt Connector A			
 Apple Inc.	DRAWING NUMBER	SIZE	
	051-9505	D	
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INTERNAL DP (STRAIGHT)

518S0852 CRITICAL
J9500
20525-140E-01
F-RT-SM

518S0778 DEVELOPMENT
CRITICAL
J9520
53780-8606
M-RT-SM

K6X BACKLIGHT CONTROL SUPPORT

guarantee backlight is
only on when Panel has valid video

Backlight Control

U9500 OUTPUT Y2 IS A NON-INVERTED, DELAYED VERSION OF INPUT A
The delay applies only on a L->H transition on A. This guarantees video is valid before the backlight is enabled.
On a H->L transition of A, Y2 follows with standard logic propagation delay. This ensures the backlight is off immediately after loss of video
Y1 is simply an inverted version of A, with no delay

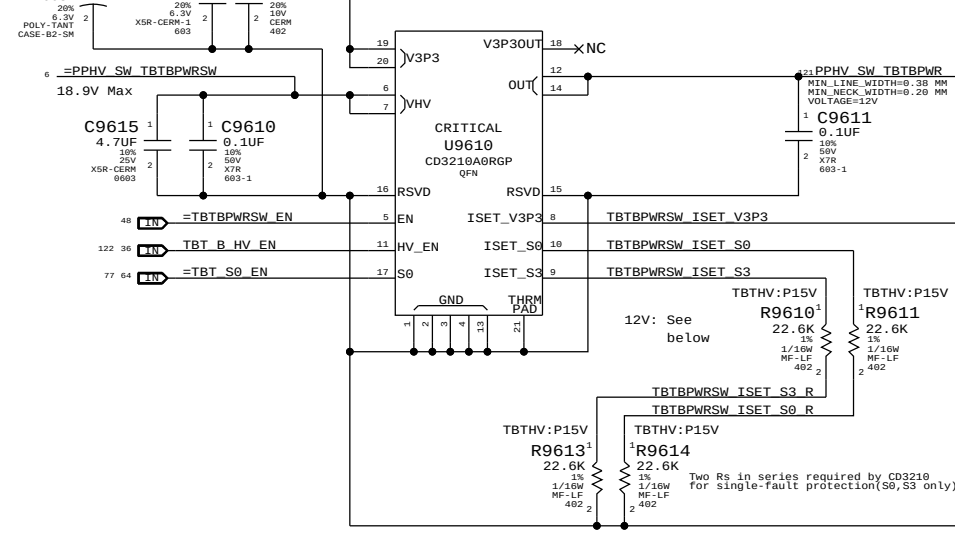
PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
311S0566	1	ONE BIT SELECTABLE DELAY	U9500	Y	

SYNC MASTER=D8 MLB		SYNC DATE=07/03/2012	
PAGE TITLE			
Internal DP Support			
 Apple Inc.		DRAWING NUMBER	051-9505
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3.3V/HV Power MUX

V3P3 must be S4 to support wake from Thunderbolt devices.

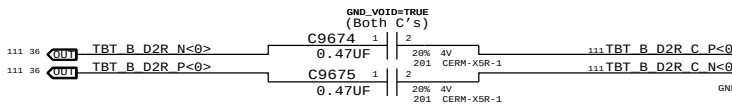
	Nominal	Min	Max
IV3P3	1100mA	1030mA	1200mA
IHV50	890mA	830mA	930mA (assumes 15V, 12W minimum)
IHV53	890mA	830mA	930mA (assumes 3S, 9-12.6V, 7.5-11.7V)



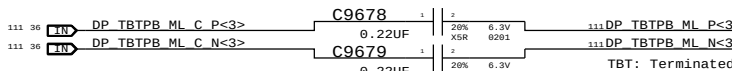
For 12V systems:

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
114S0338	2	RES,MTL FILM,1/16W,17.8K,1,0402,SMD,LF	R9610,R9613		TBTHV:P12V
114S0338	2	RES,MTL FILM,1/16W,17.8K,1,0402,SMD,LF	R9611,R9614		TBTHV:P12V

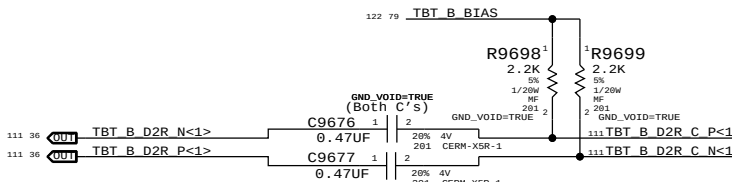
	Nominal	Min	Max
IHV50/S3	1120mA	1090mA	1170mA (12W minimum)



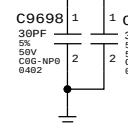
NOTE: Polarity Swapped for Layout!



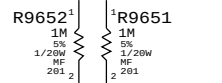
NOTE: Polarity Swapped for Layout!



DP B AUXCH DDC P
DP B AUXCH DDC N



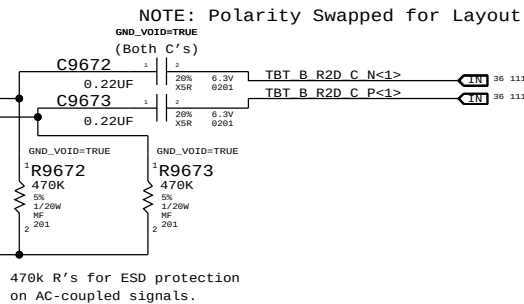
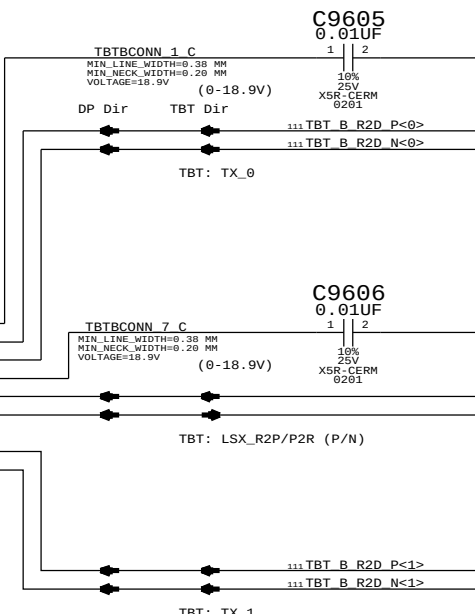
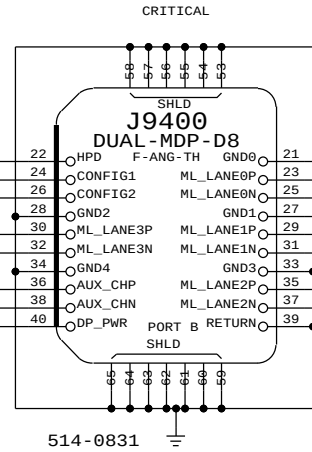
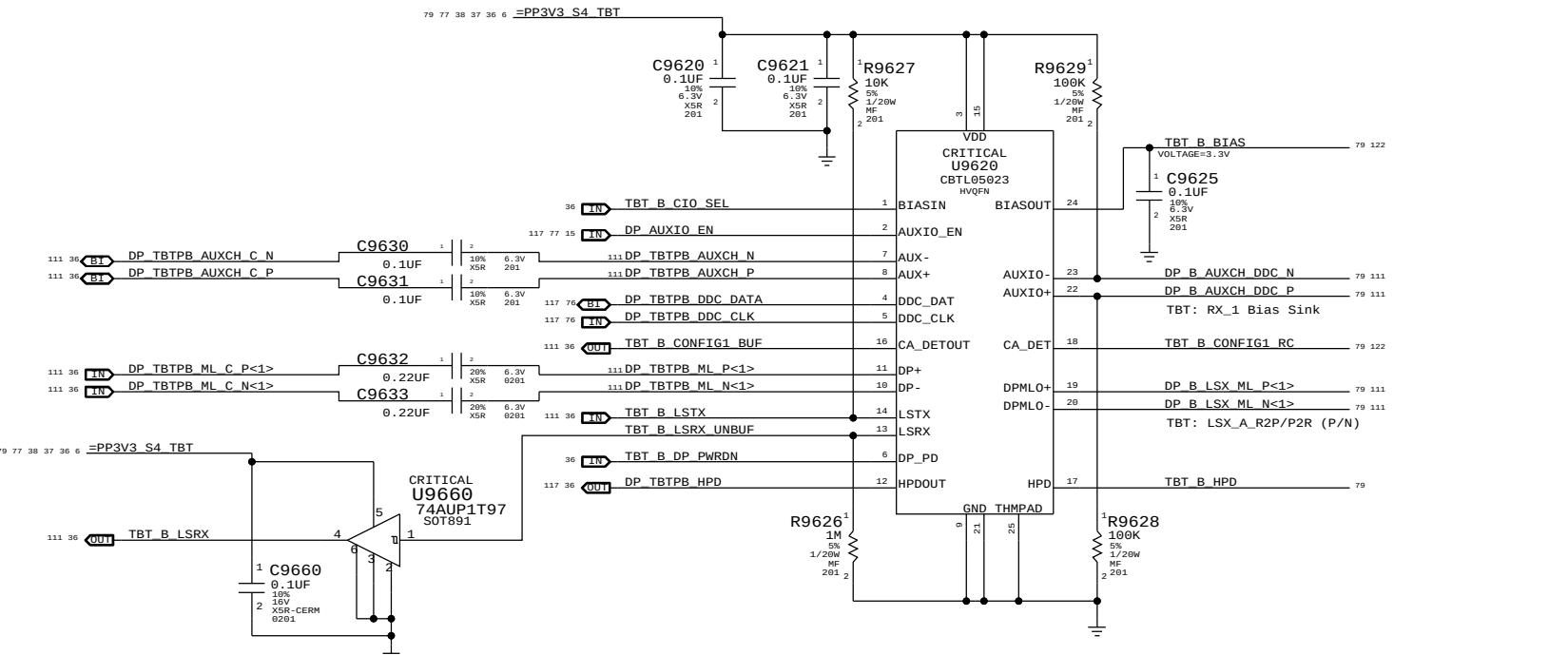
TBT B HPD
TBT B CONFIG1 RC
TBT B CONFIG2 RC



DP Source must pull down HPD input with greater than or equal to 100K (DPv1.1a).

Sink HPD range:
High: 2.0 - 5.0V
Low: 0 - 0.8V

Thunderbolt Connector B



SYNC MASTER=D8 MLB

SYNC DATE=07/03/2012

Thunderbolt Connector B

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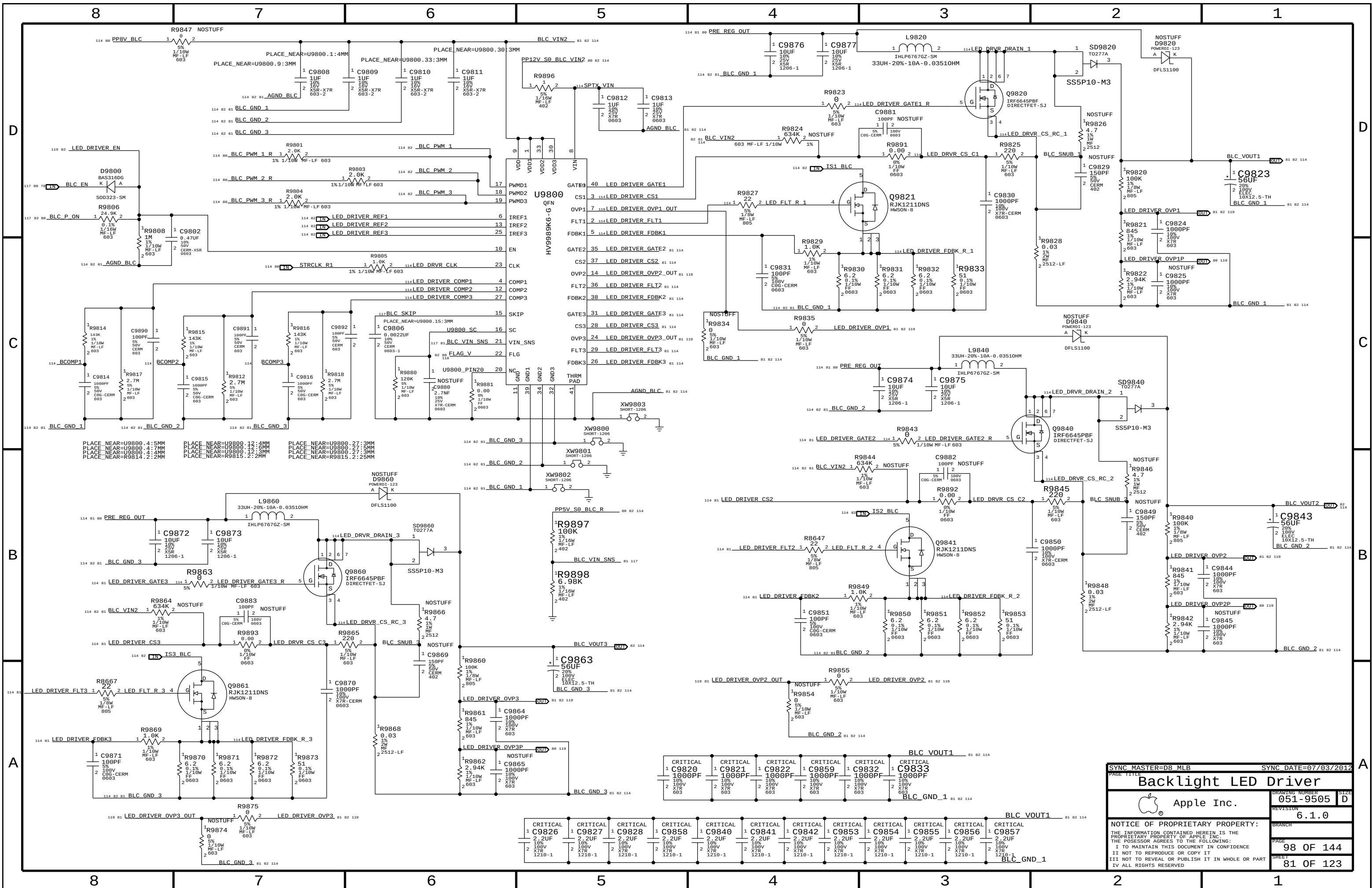
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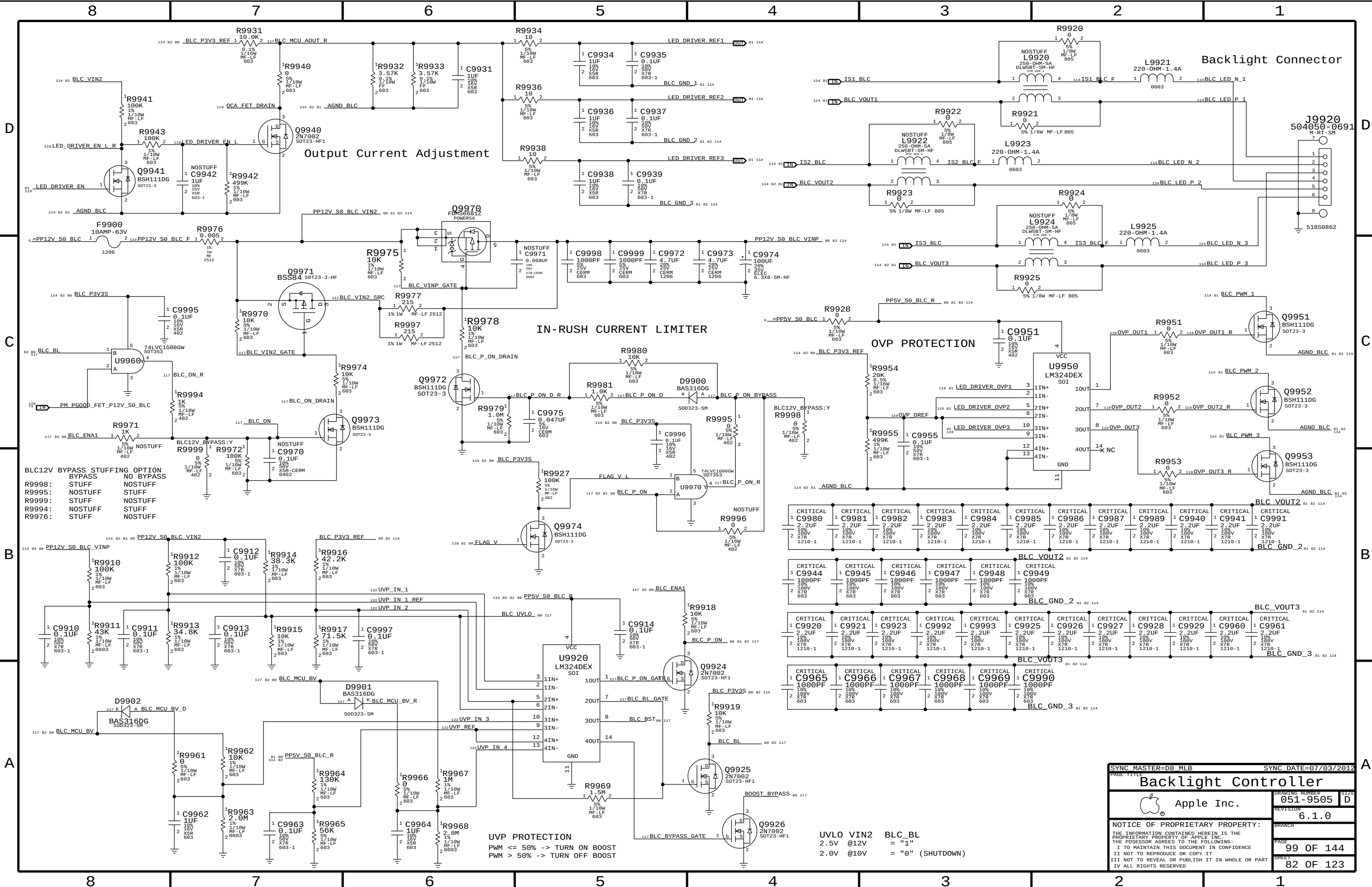
SHEET

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470k R's for ESD protection on AC-coupled signals.



PAGE TITLE		SYNC MASTER=D8_MLB		SYNC DATE=07/03/2012	
Backlight LED Driver		DRAWING NUMBER		SIZE	
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D

C

B

A

D

C

B

A

Output Current Adjustment

IN-RUSH CURRENT LIMITER

OVP PROTECTION

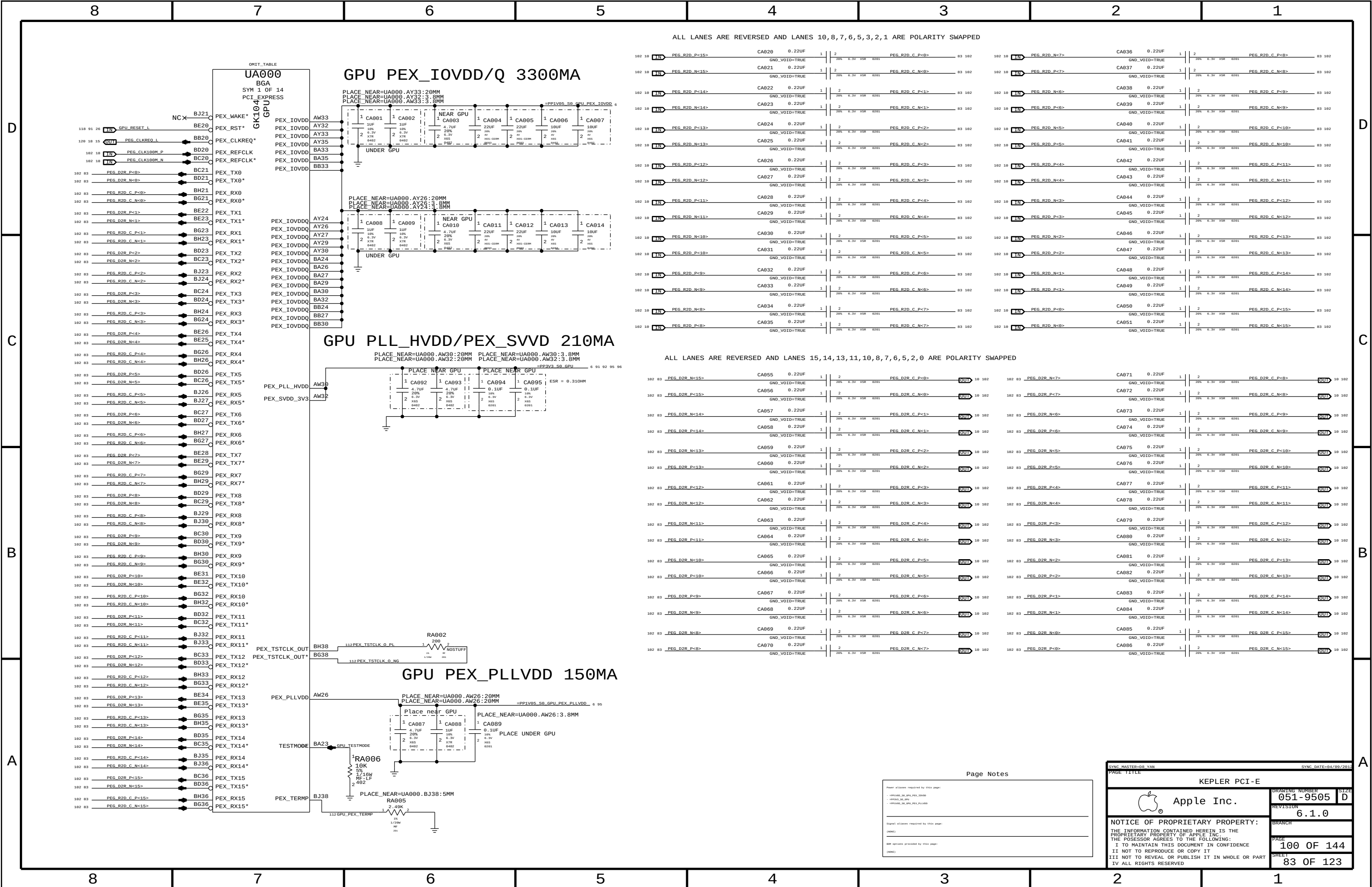
Backlight Connector

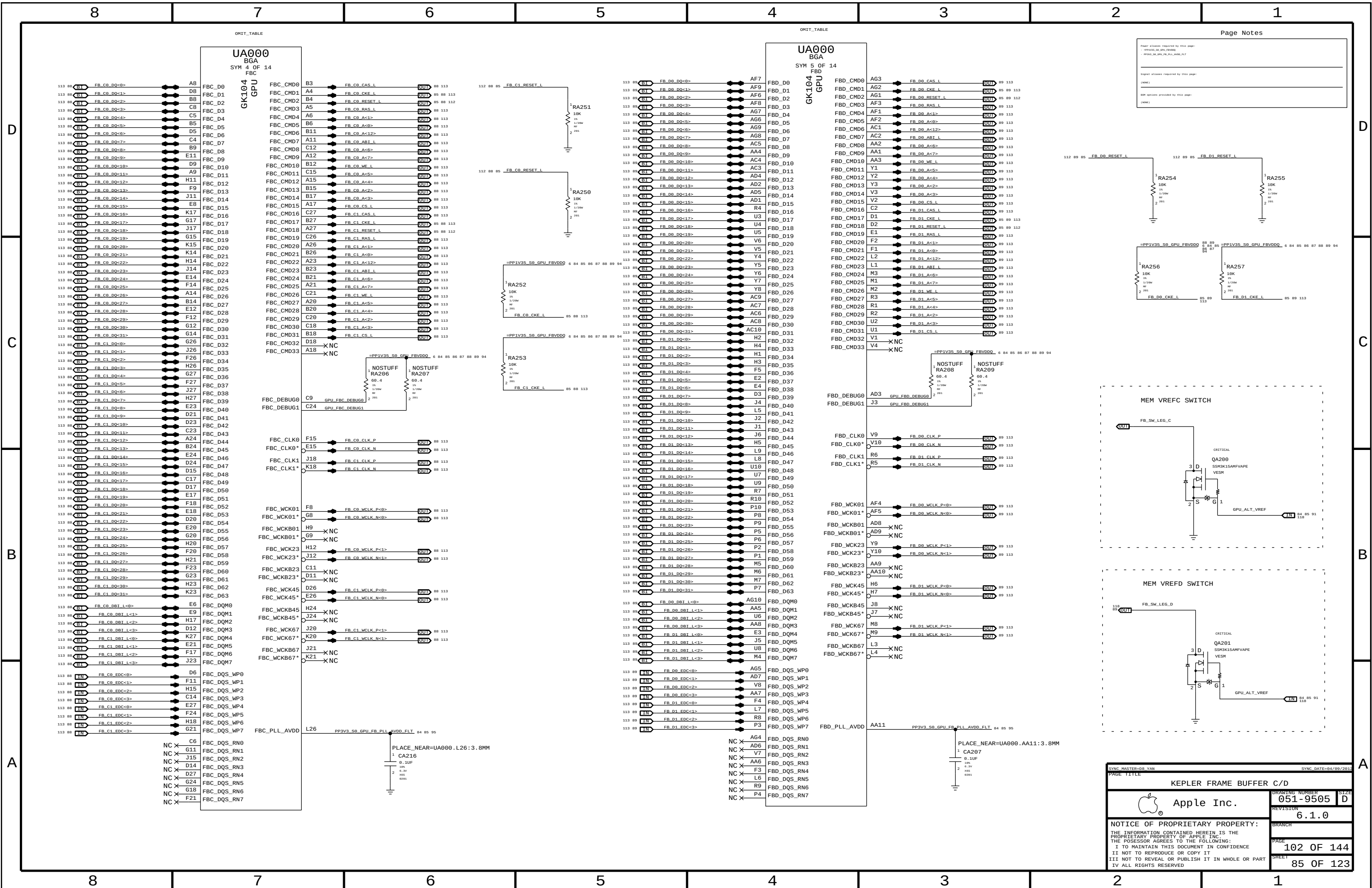
- BLC12V BYPASS STUFFING OPTION
BYPASS NO BYPASS
R9998: STUFF NOSTUFF
R9995: NOSTUFF STUFF
R9999: STUFF NOSTUFF
R9994: NOSTUFF STUFF
R9976: STUFF NOSTUFF

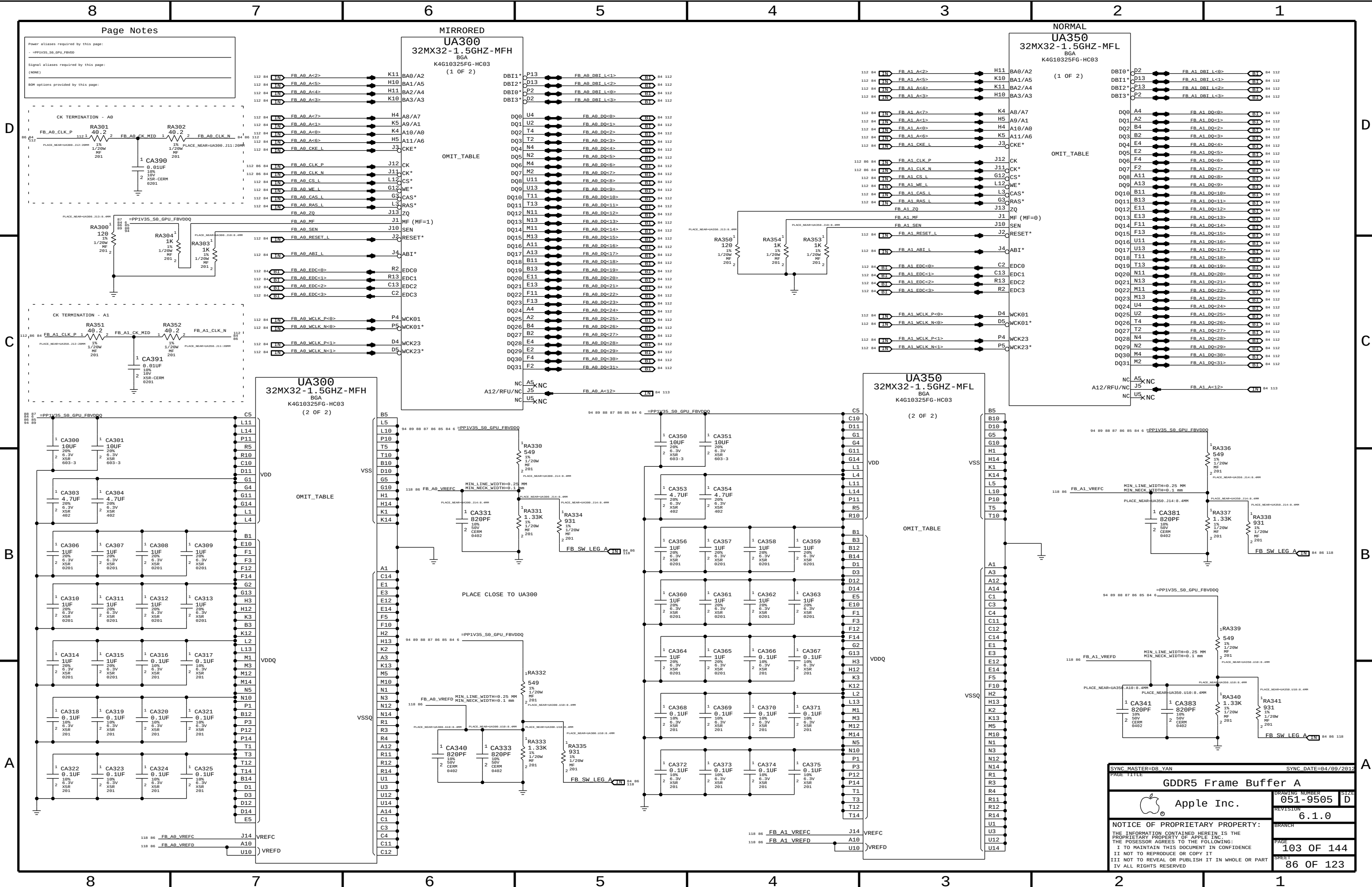
UVP PROTECTION
PWM <= 50% -> TURN ON BOOST
PWM > 50% -> TURN OFF BOOST

UVLO VIN2 BLC_BL
2.5V @12V = "1"
2.0V @10V = "0" (SHUTDOWN)

SYNC MASTER=D8 MLB		SYNC DATE=07/03/2012	
PAGE TITLE			
Backlight Controller		DRAWING NUMBER	051-9505
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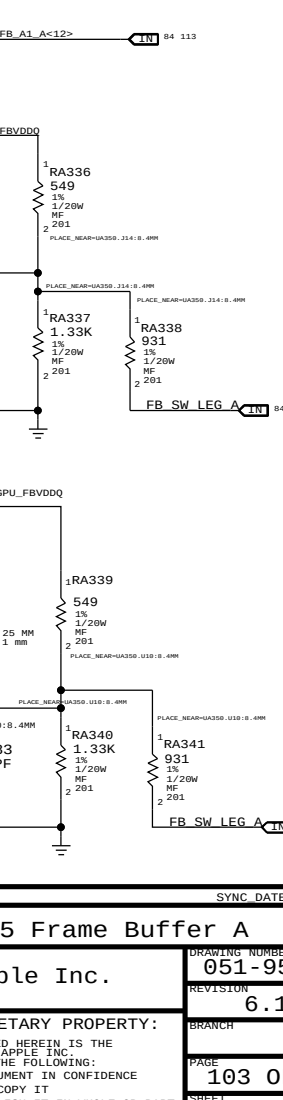
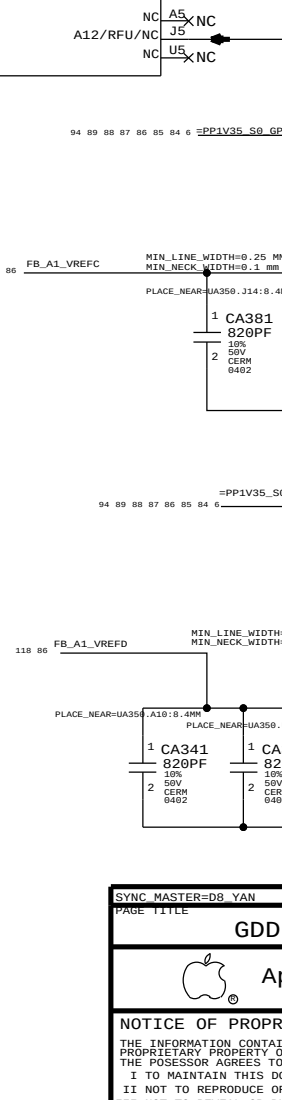
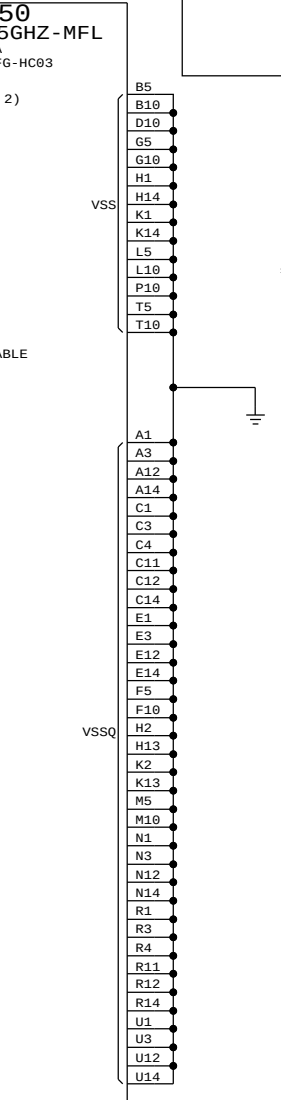
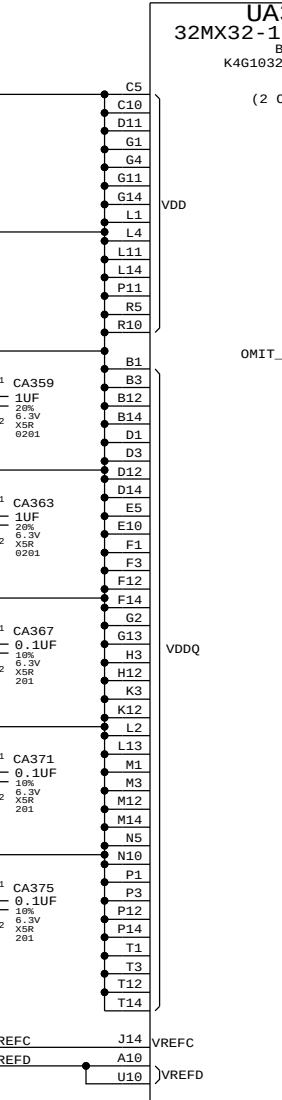
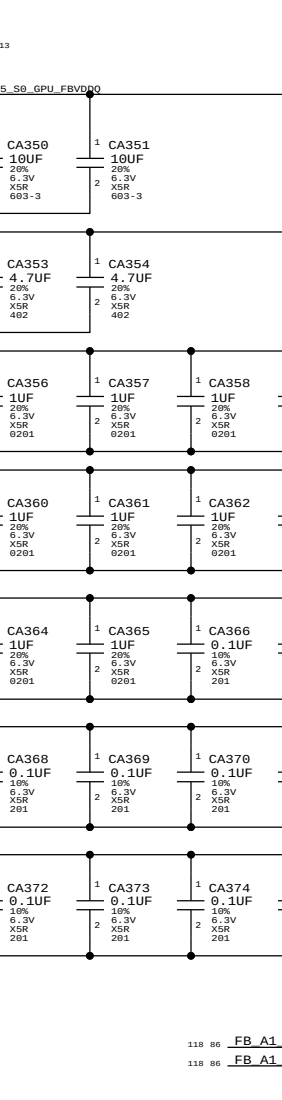
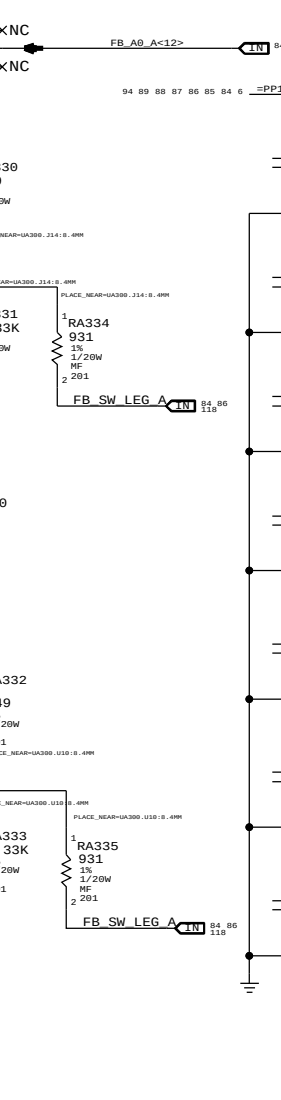
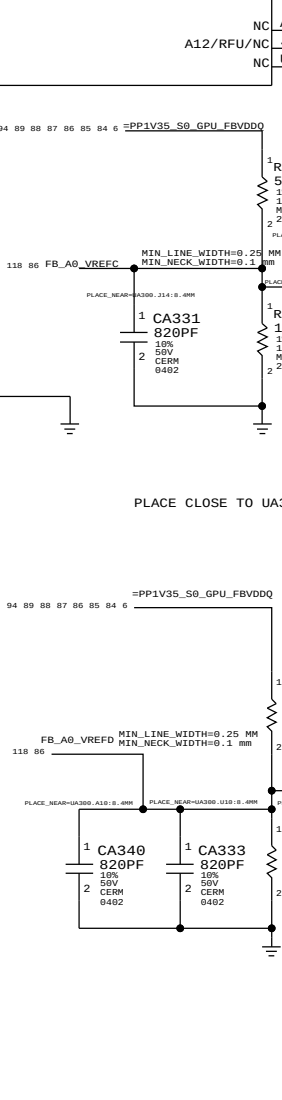
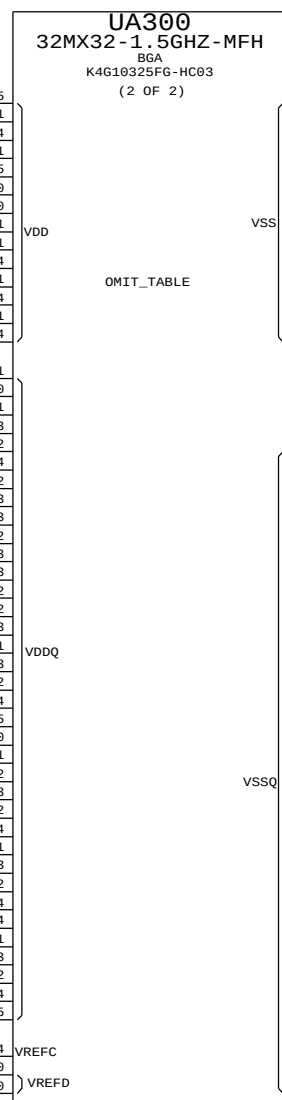
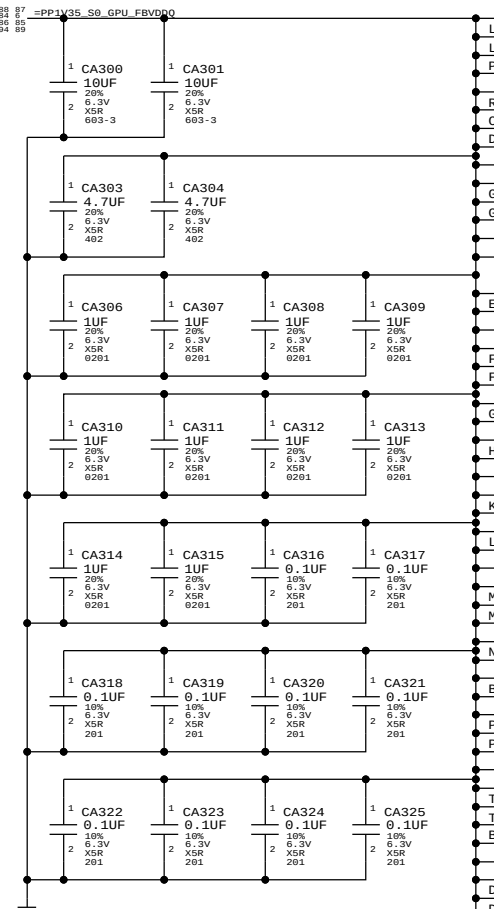
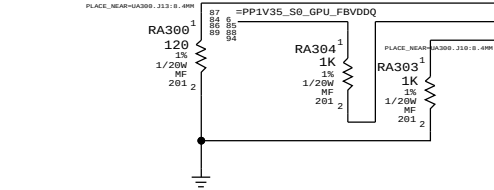
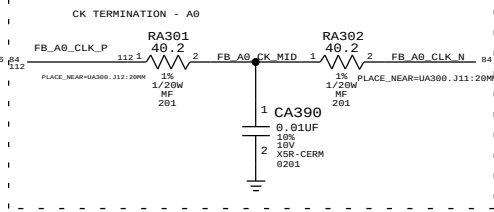




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Signal aliases required by this page:
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GDDR5 Frame Buffer A

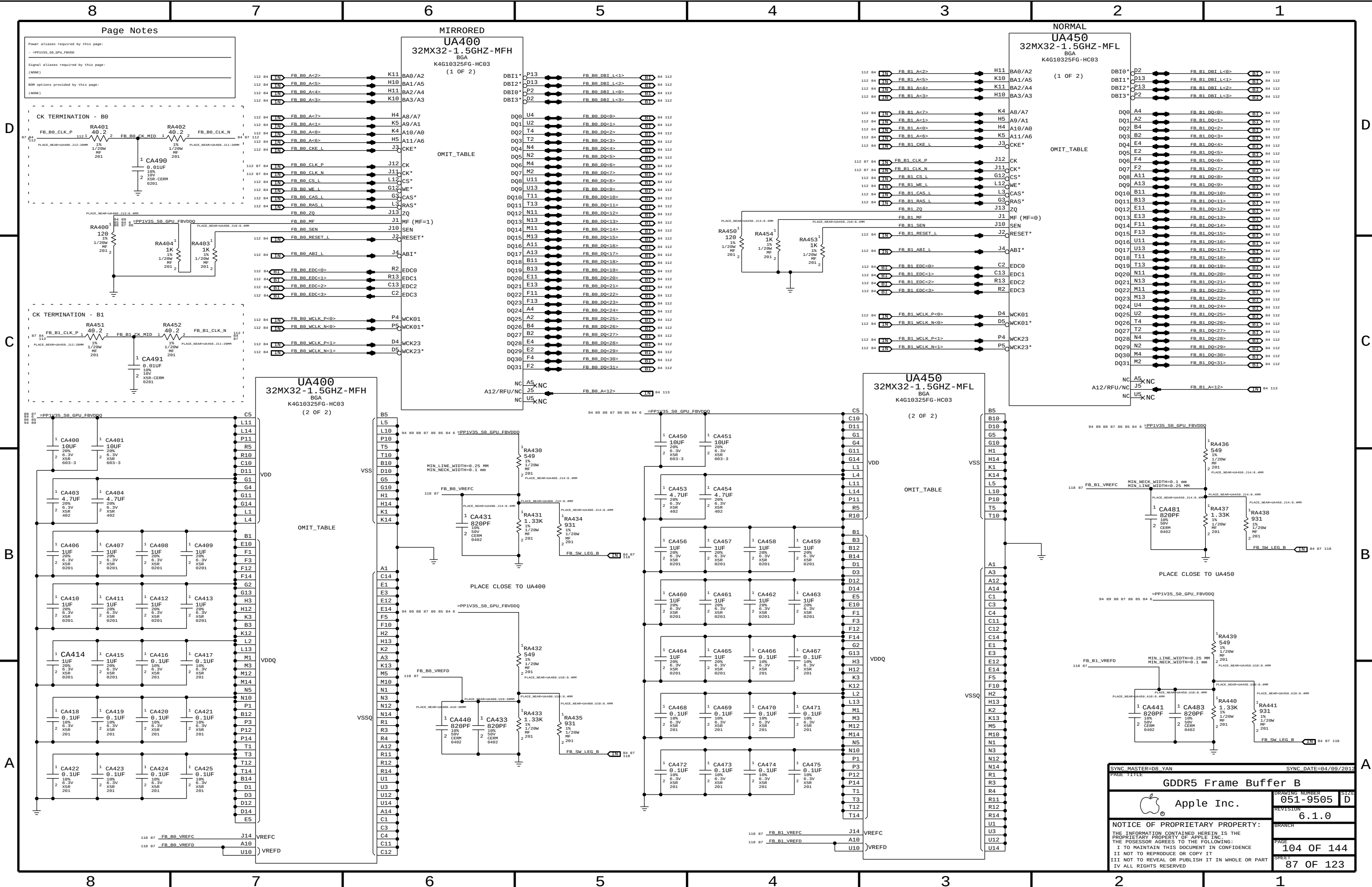
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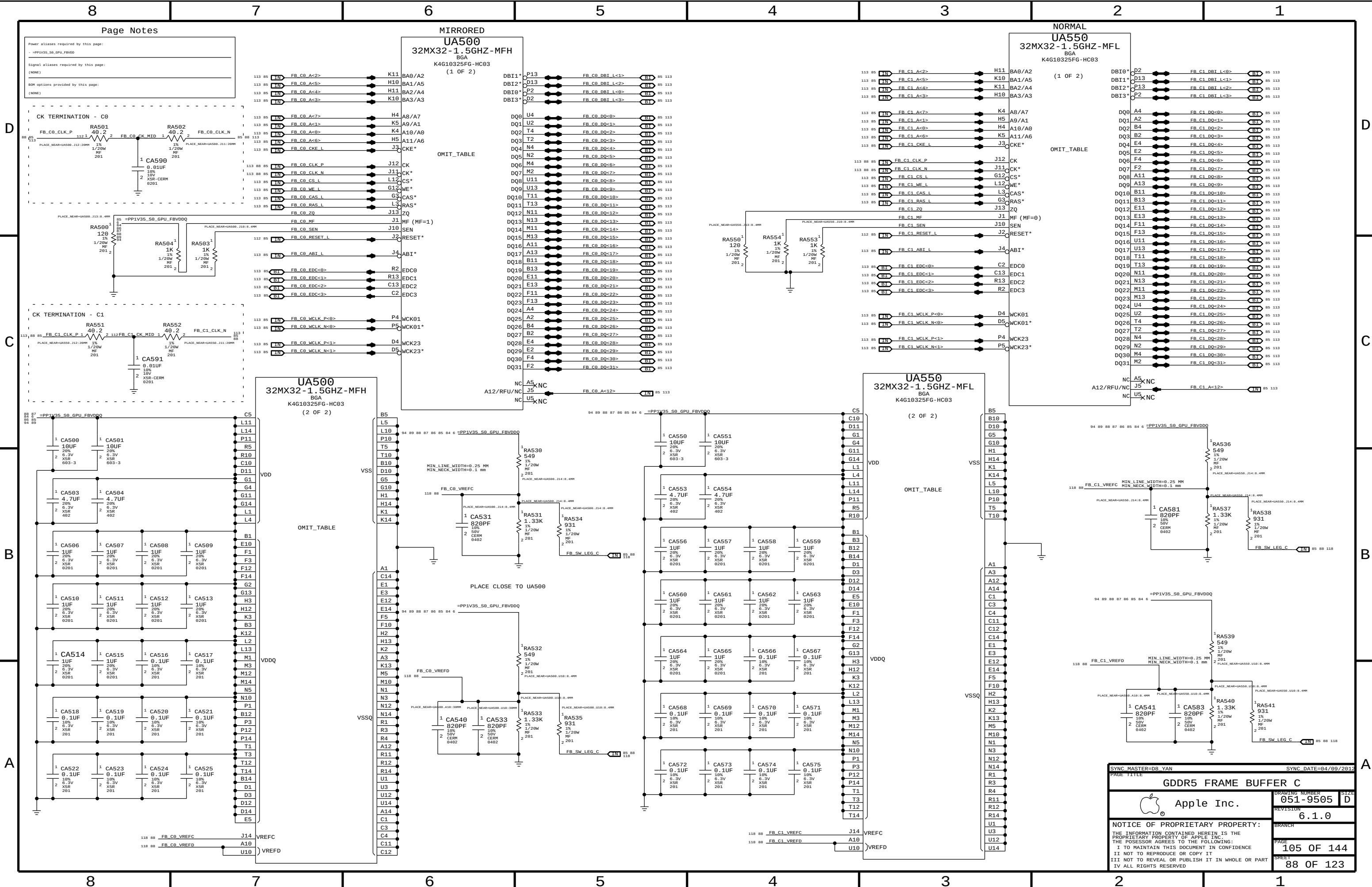
REVISION 6.1.0

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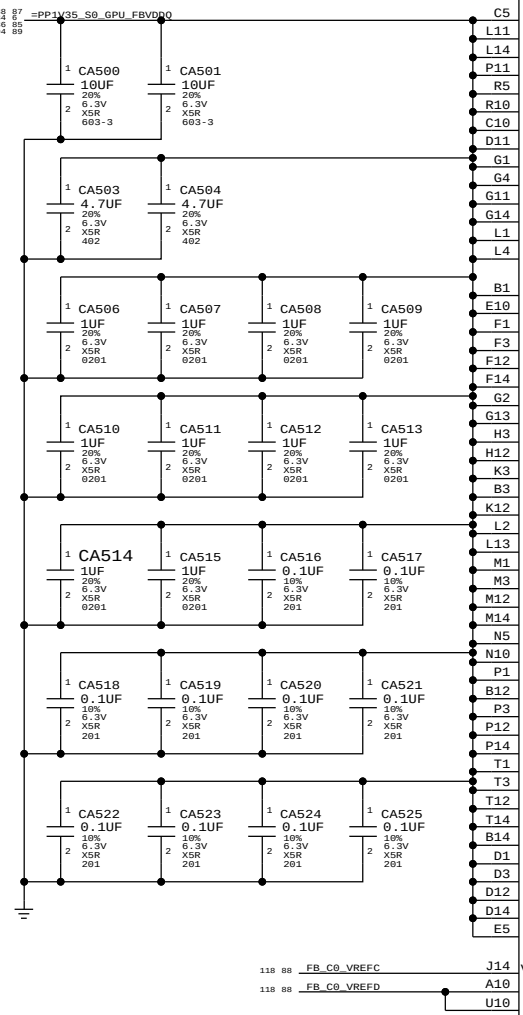
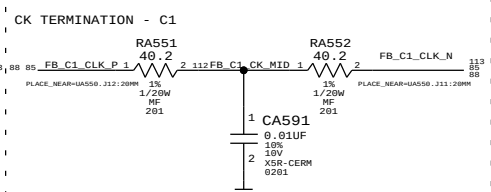
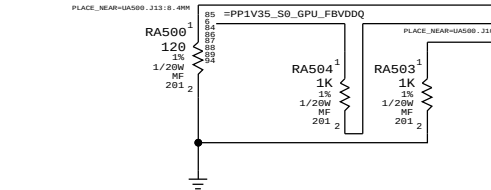
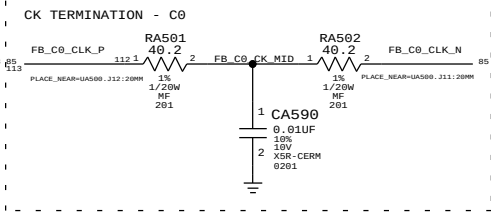




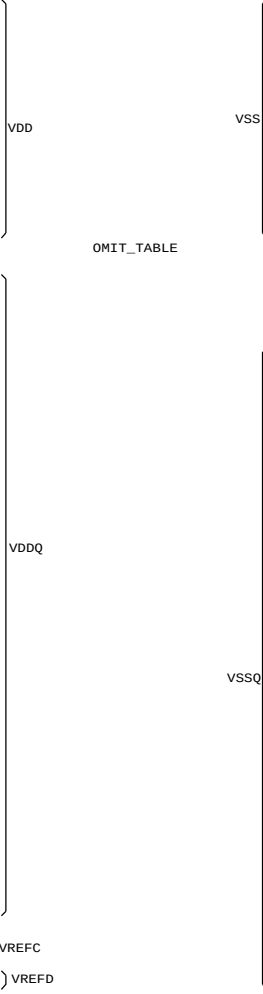
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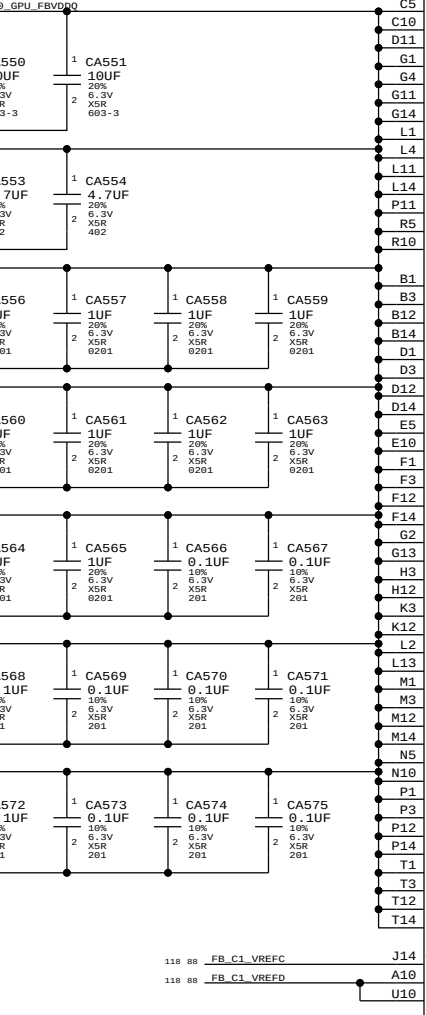
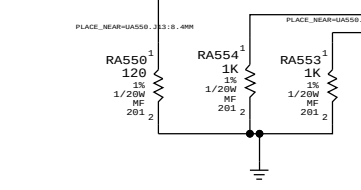
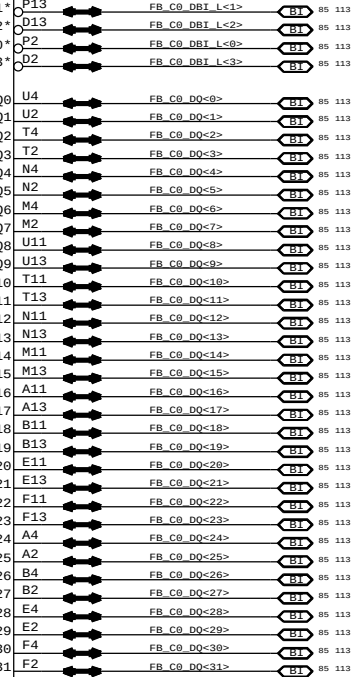
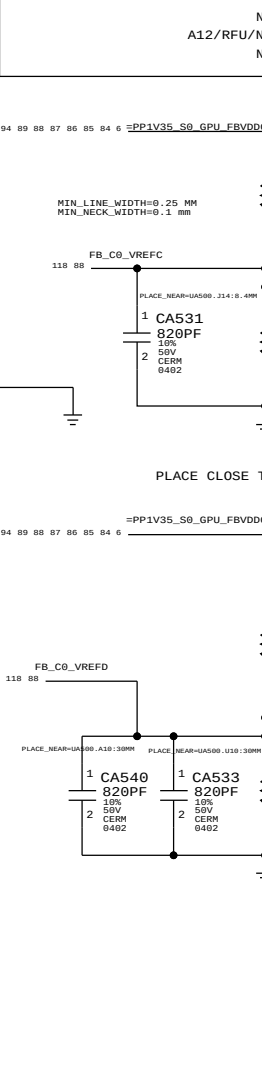
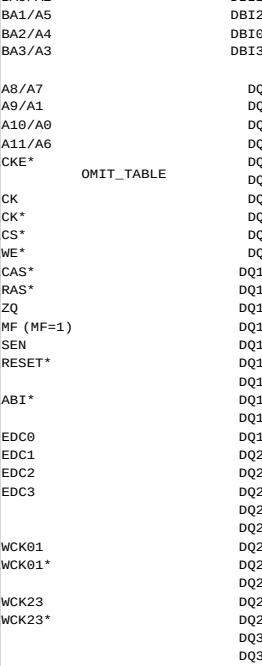
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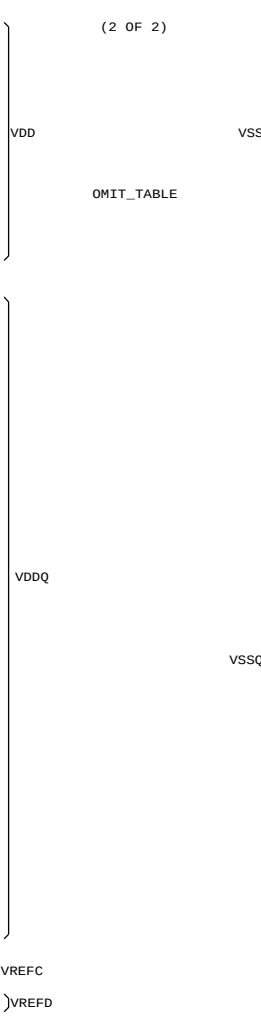
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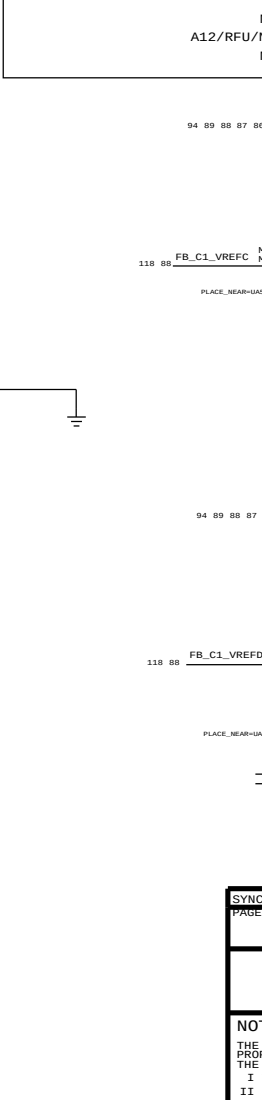
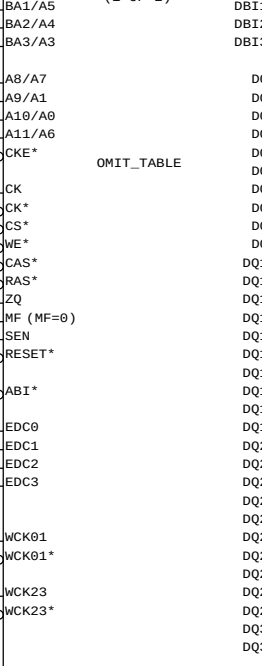
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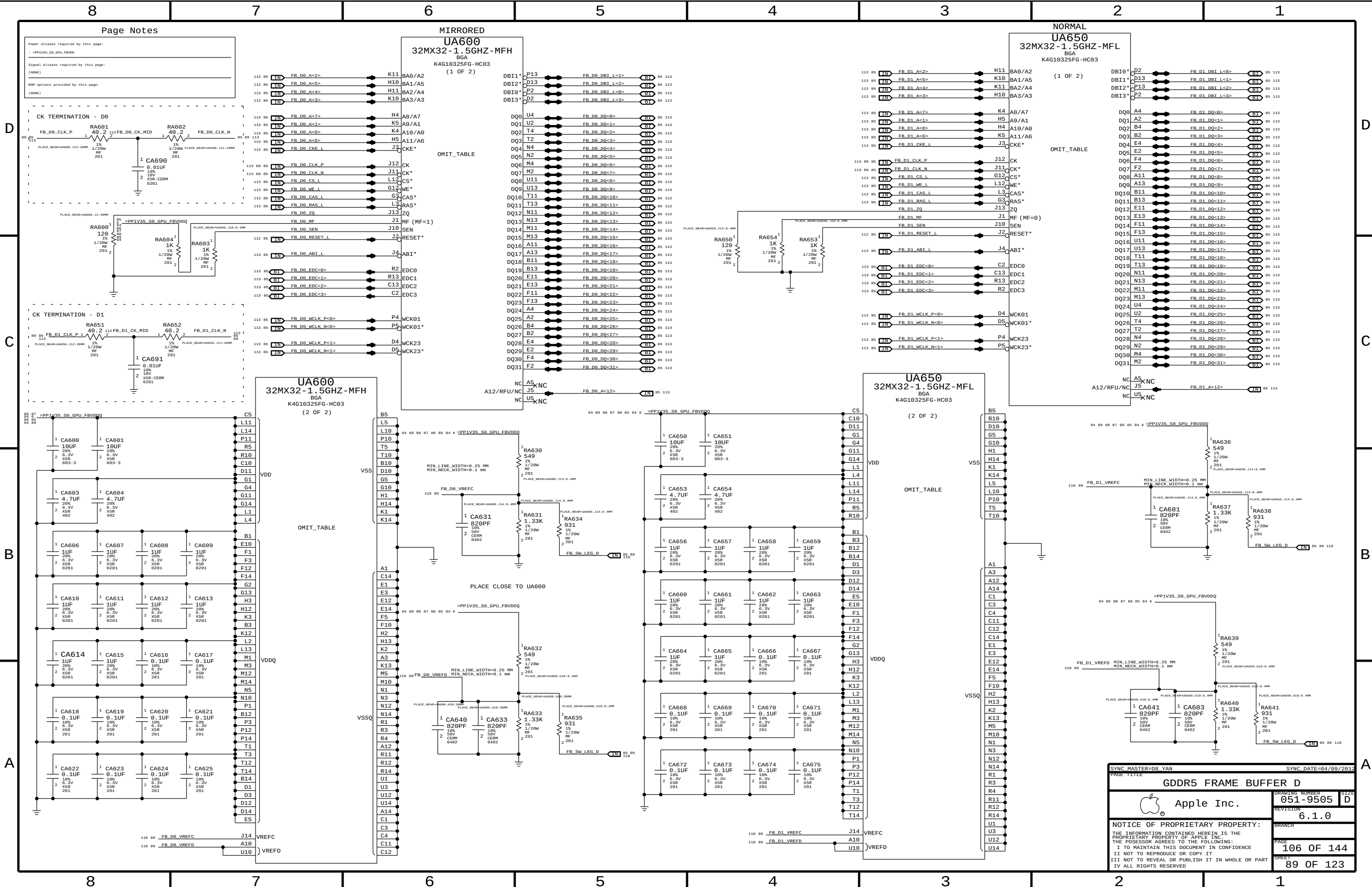


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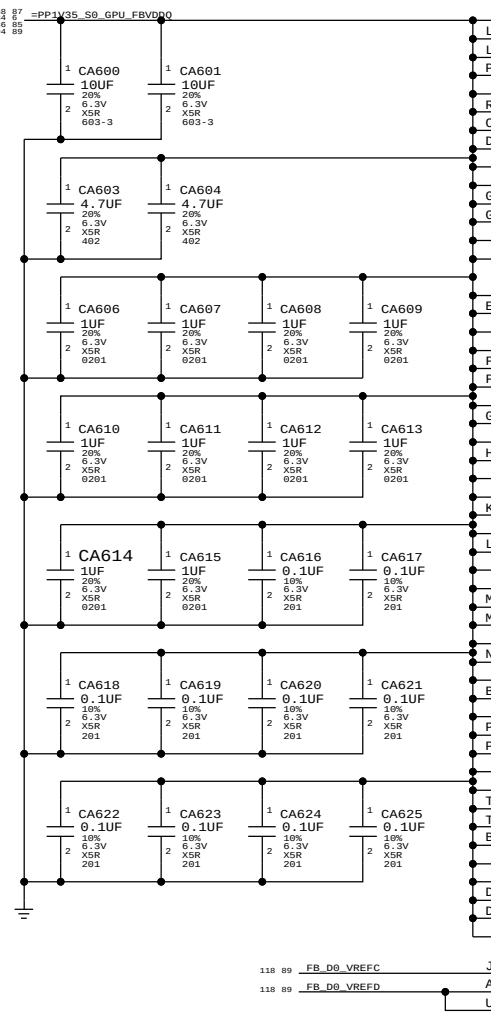
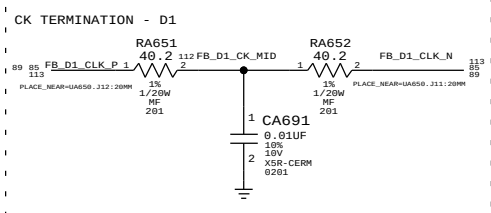
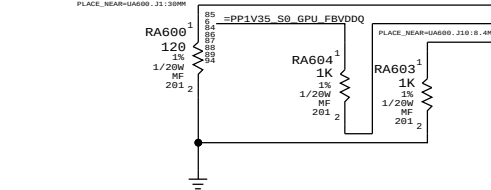
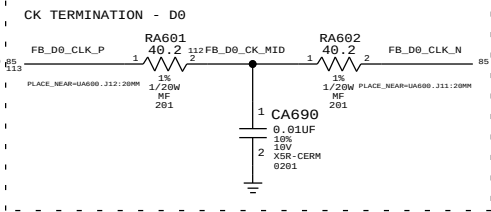




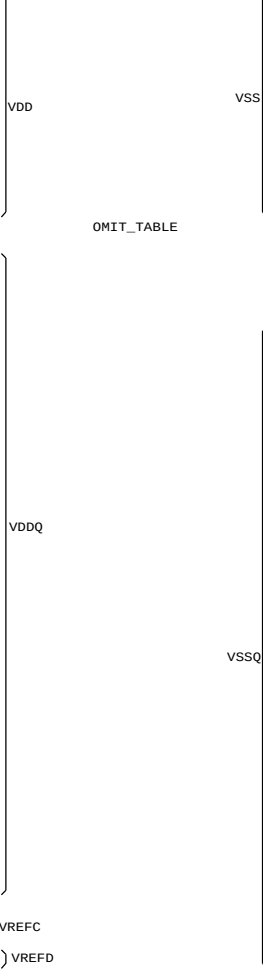
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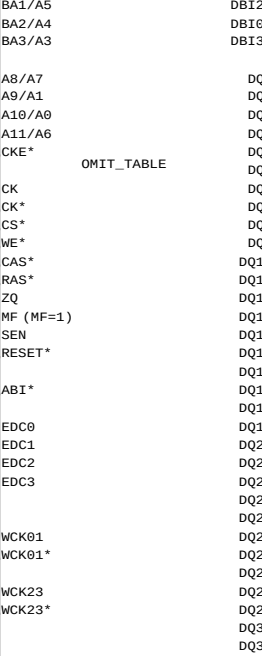
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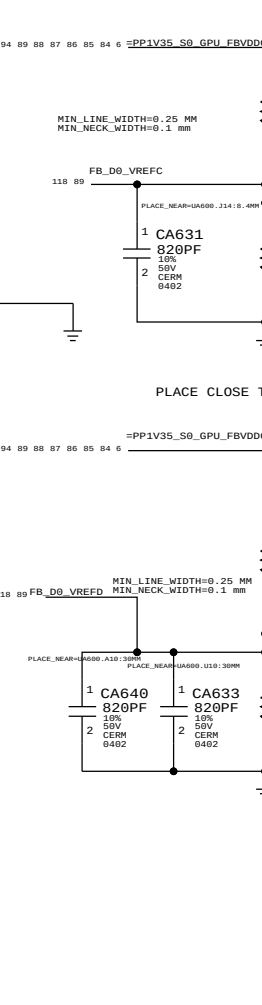
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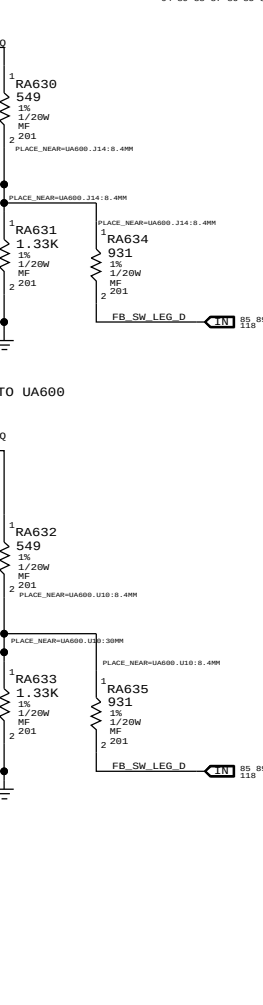
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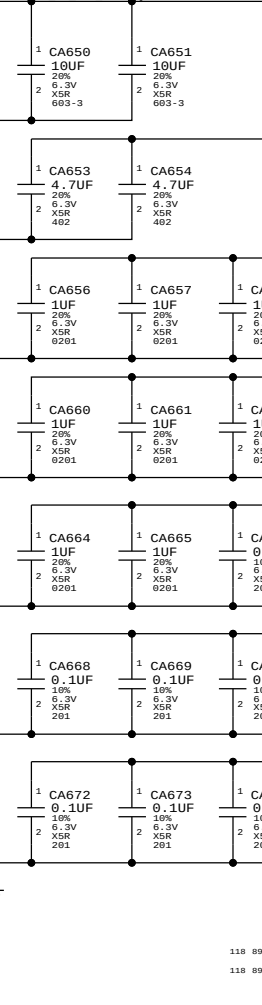
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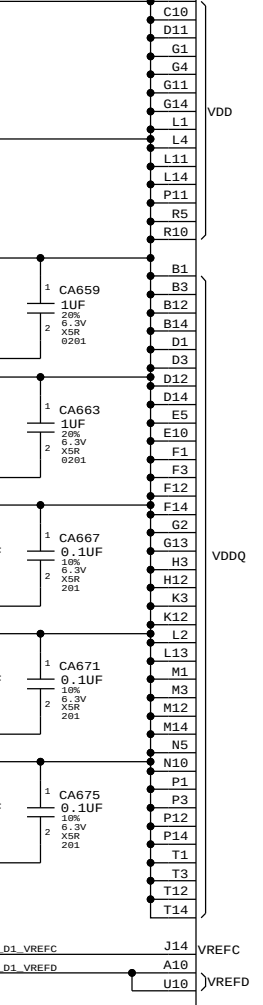
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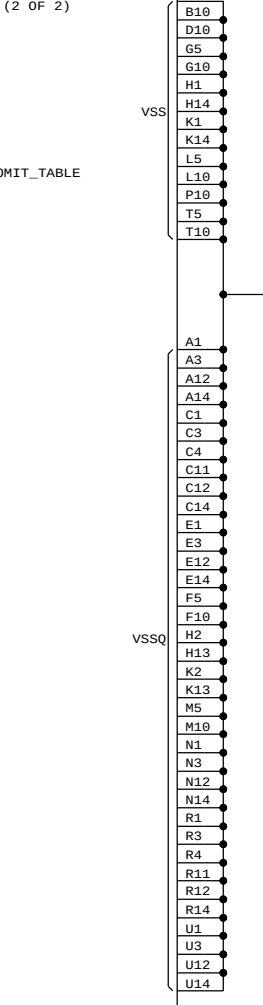
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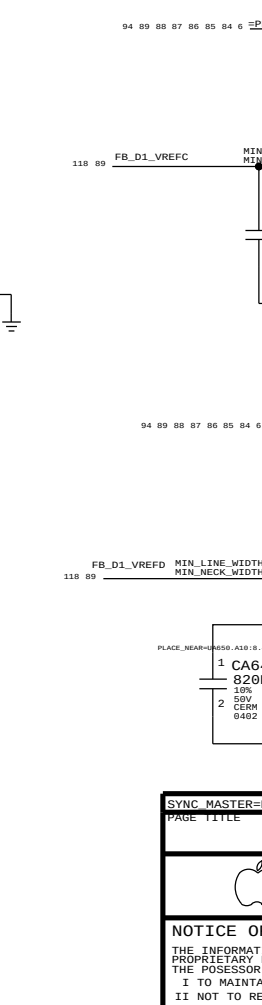
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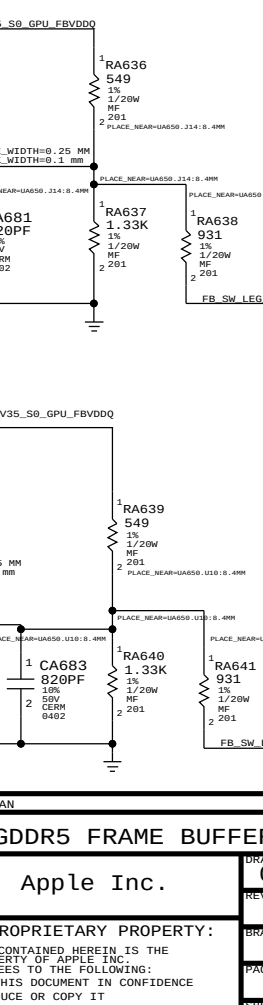
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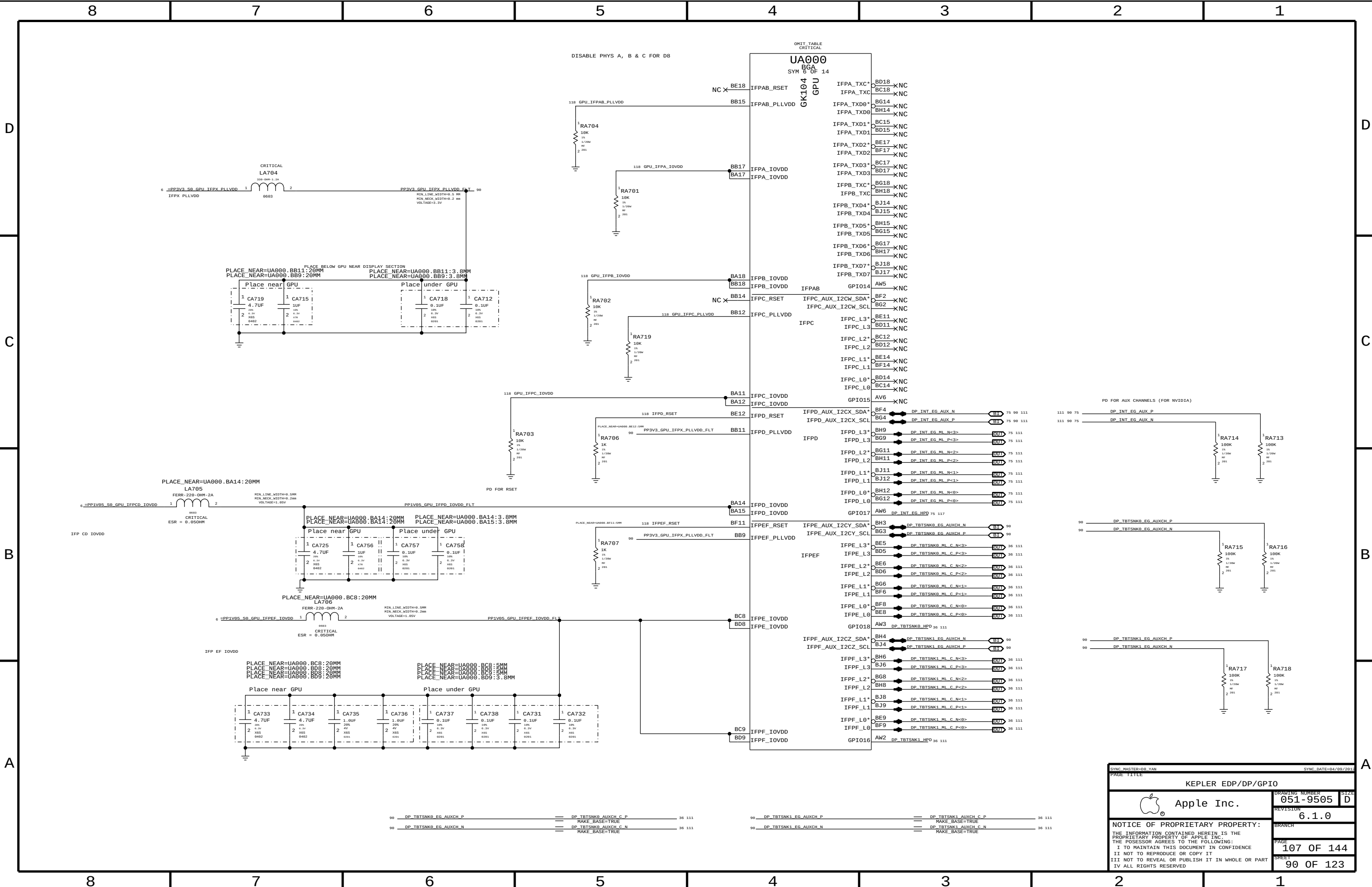


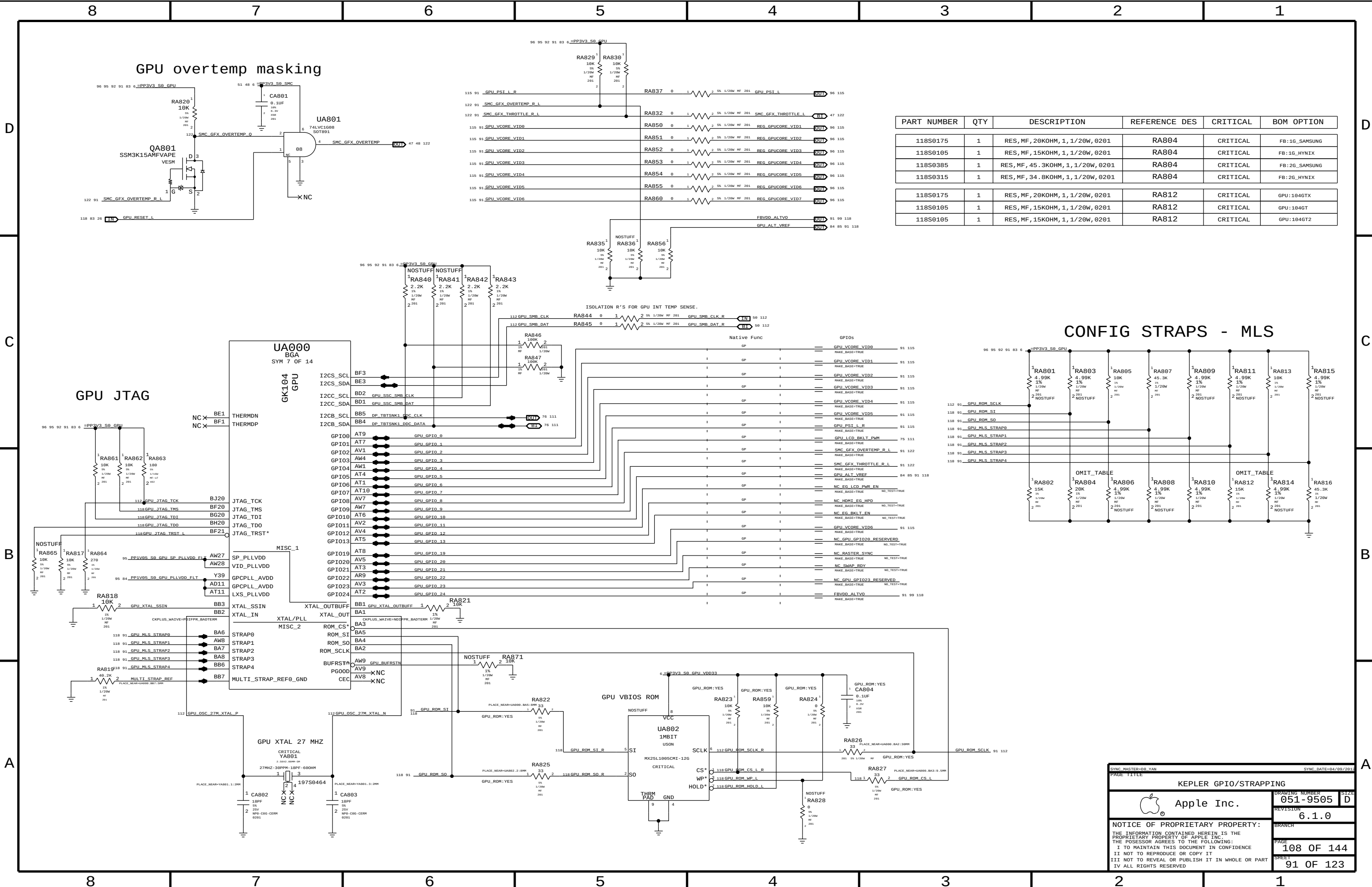
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K4G10325FG-HC03
(2 OF 2)







GPU overtemp masking

CONFIG STRAPS - MLS

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
118S0175	1	RES,MF,20KOHM,1,1/20W,0201	RA804	CRITICAL	FB:1G_SAMSUNG
118S0105	1	RES,MF,15KOHM,1,1/20W,0201	RA804	CRITICAL	FB:1G_HYNIX
118S0385	1	RES,MF,45.3KOHM,1,1/20W,0201	RA804	CRITICAL	FB:2G_SAMSUNG
118S0315	1	RES,MF,34.8KOHM,1,1/20W,0201	RA804	CRITICAL	FB:2G_HYNIX
118S0175	1	RES,MF,20KOHM,1,1/20W,0201	RA812	CRITICAL	GPU:104GTx
118S0105	1	RES,MF,15KOHM,1,1/20W,0201	RA812	CRITICAL	GPU:104GT
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SYNC DATE=04/09/2015

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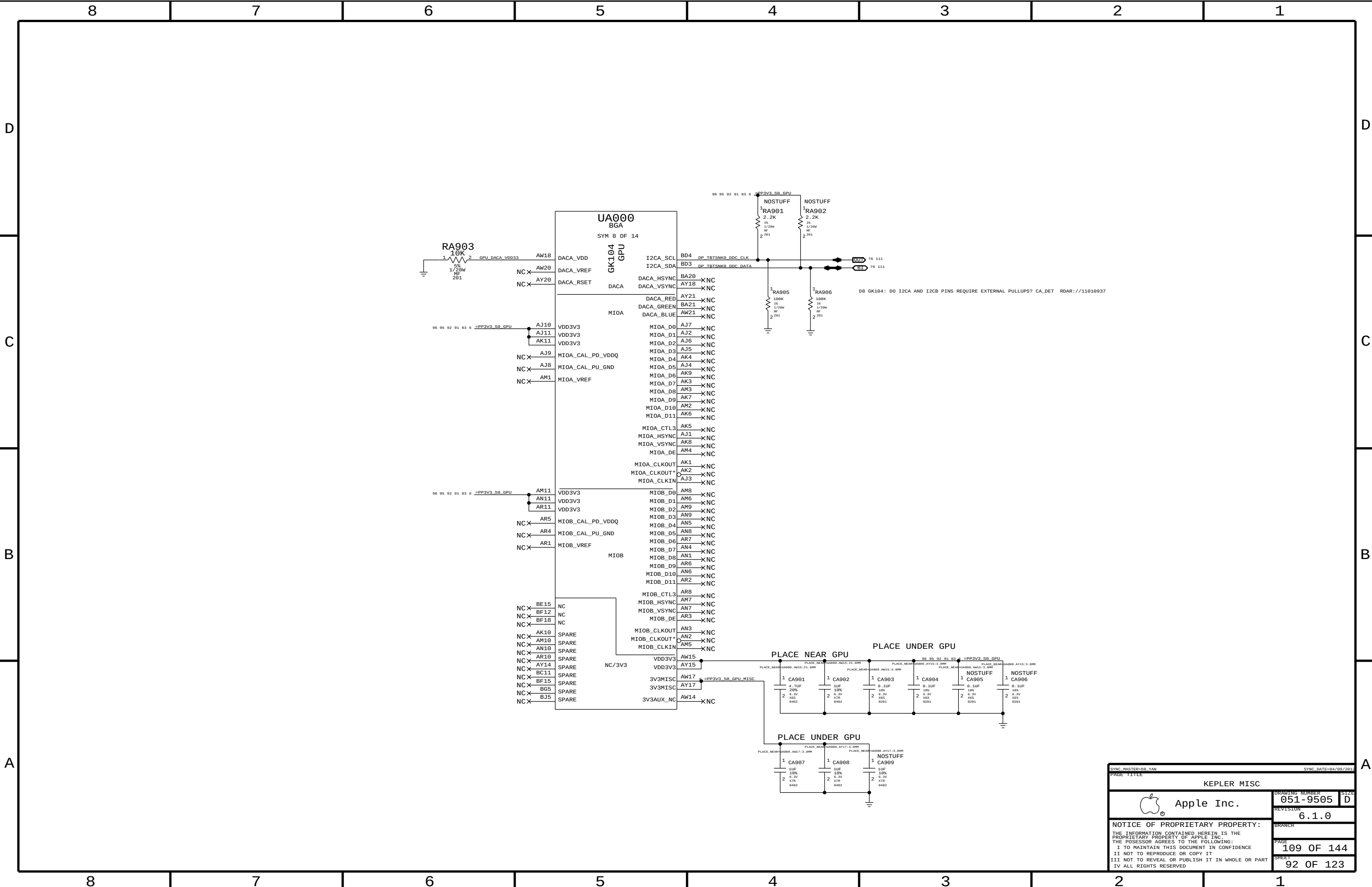
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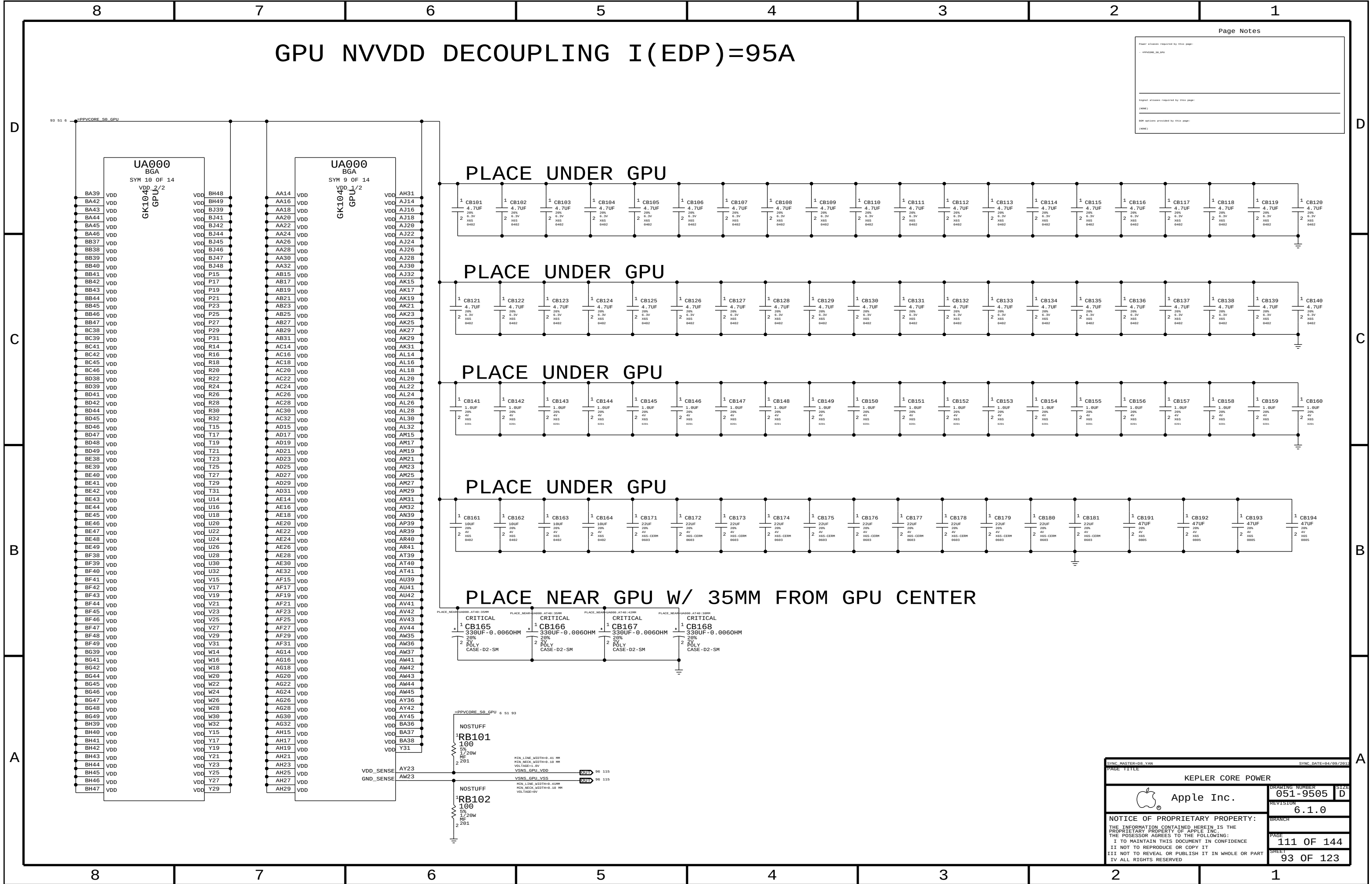
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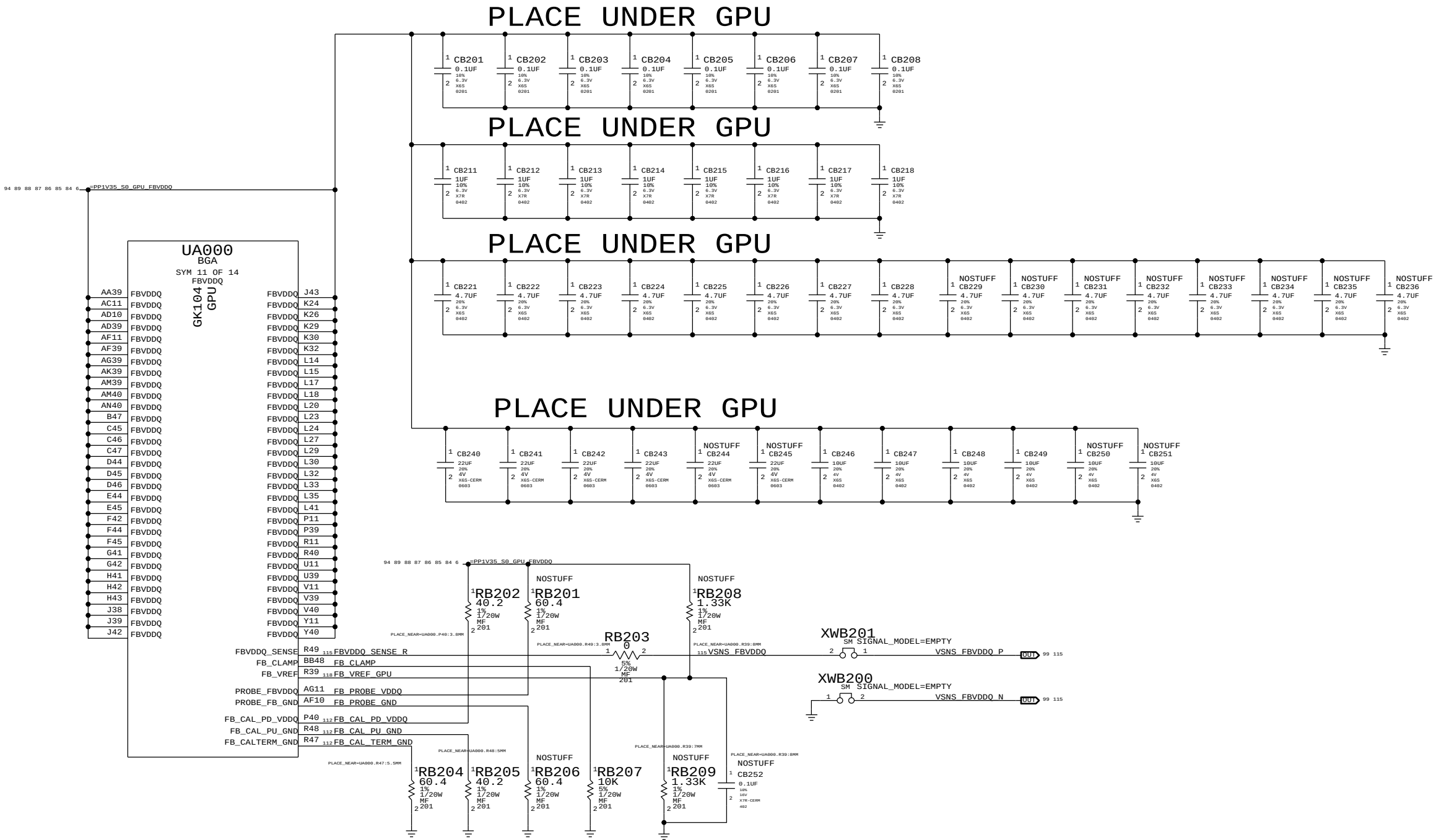
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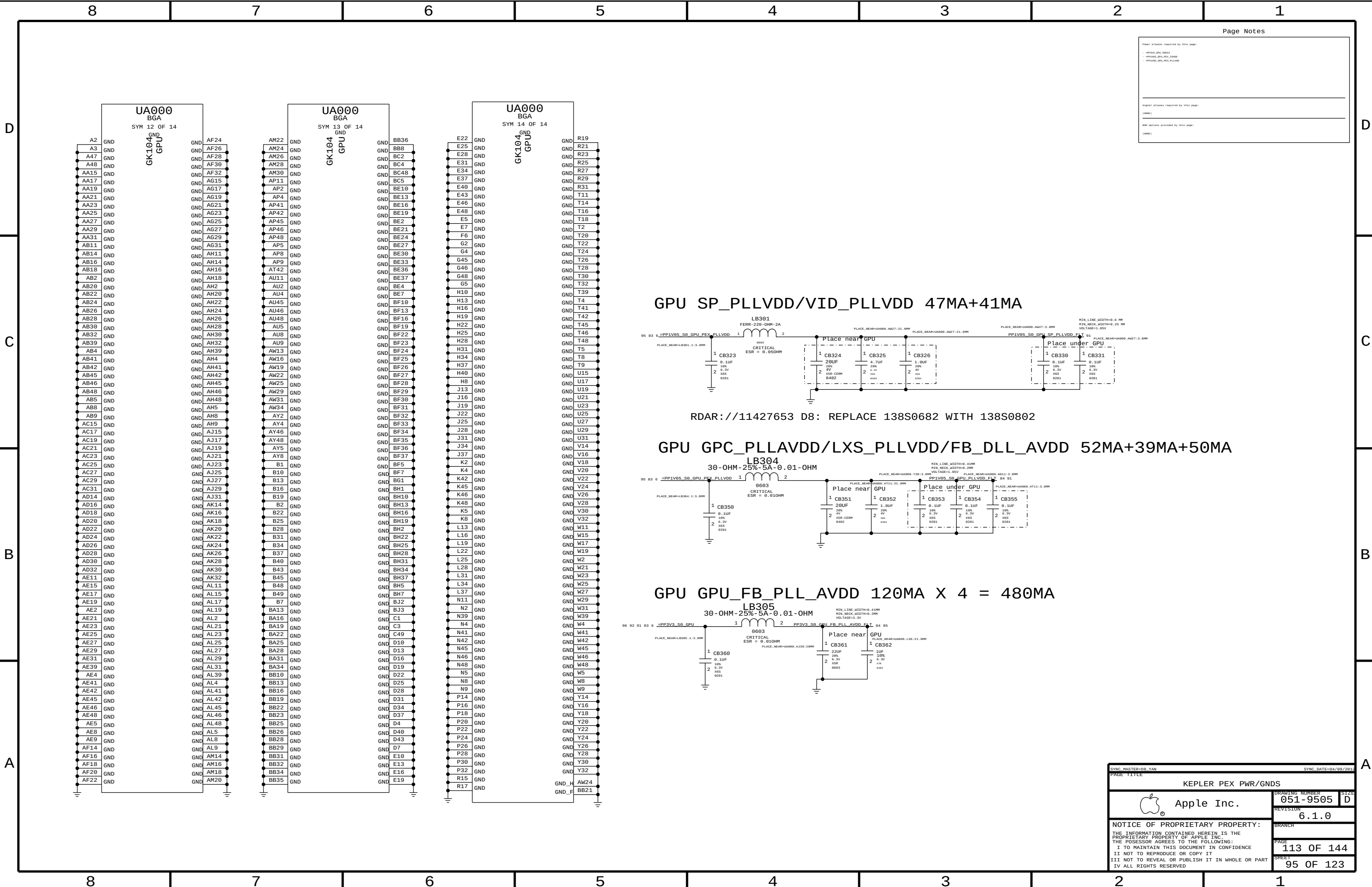


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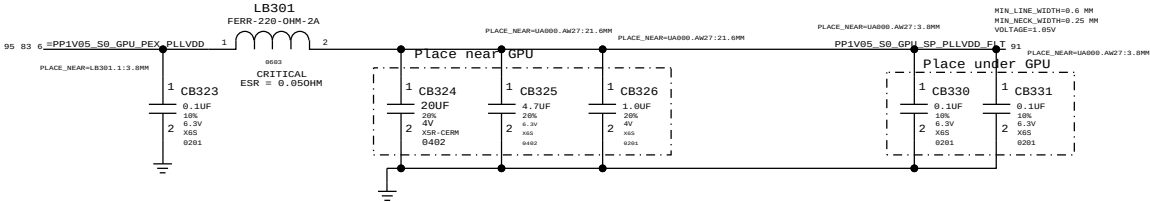
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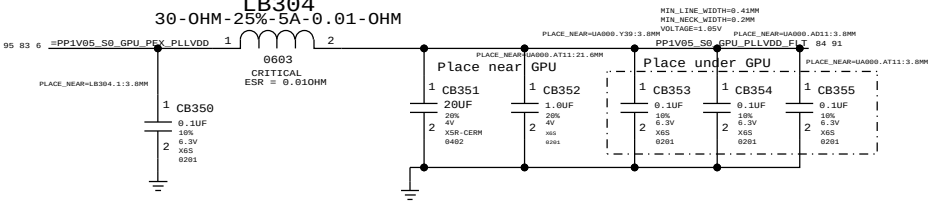
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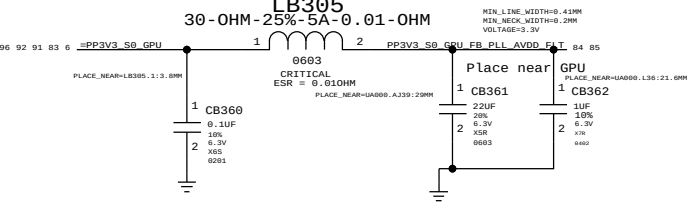


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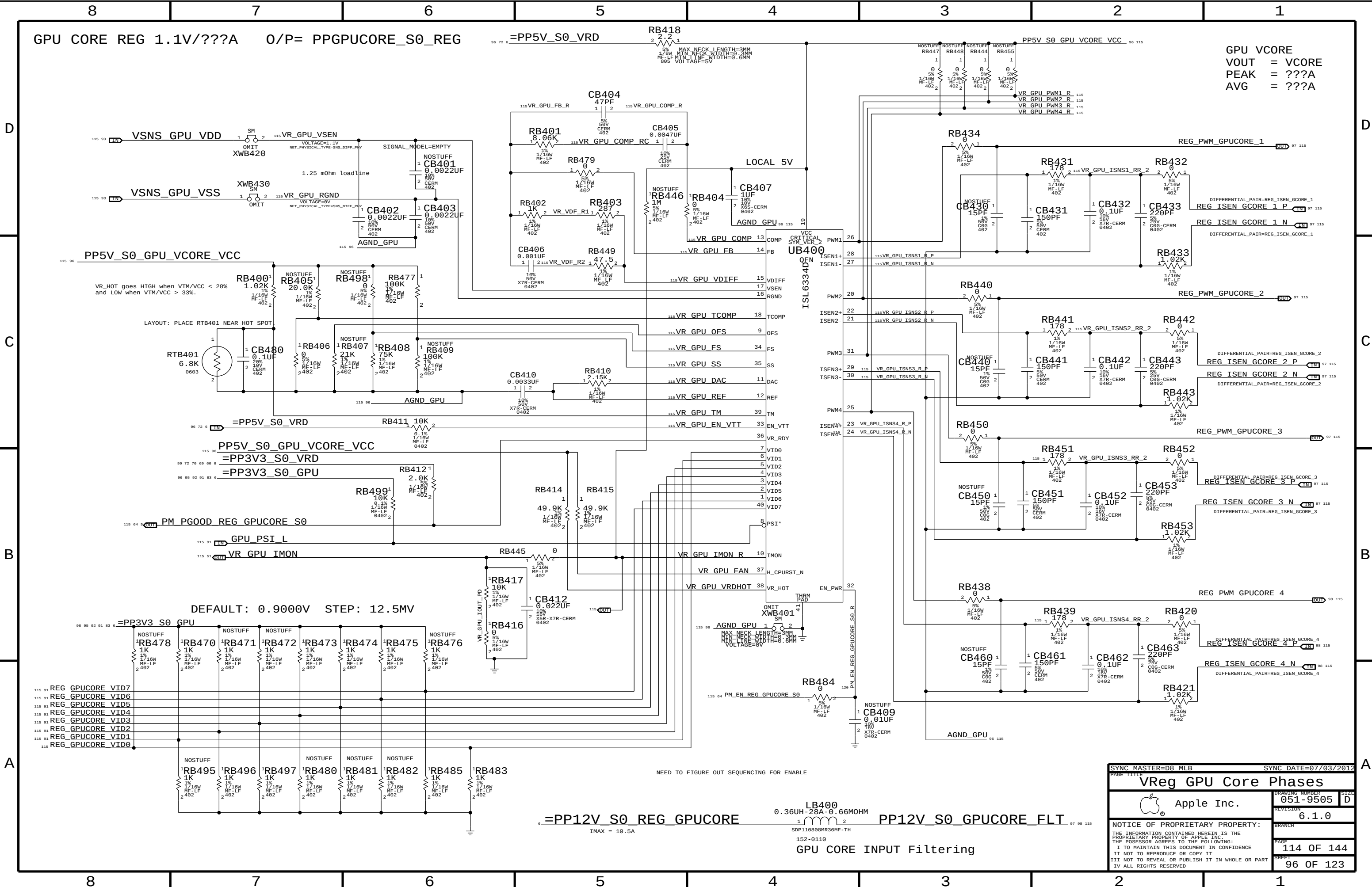
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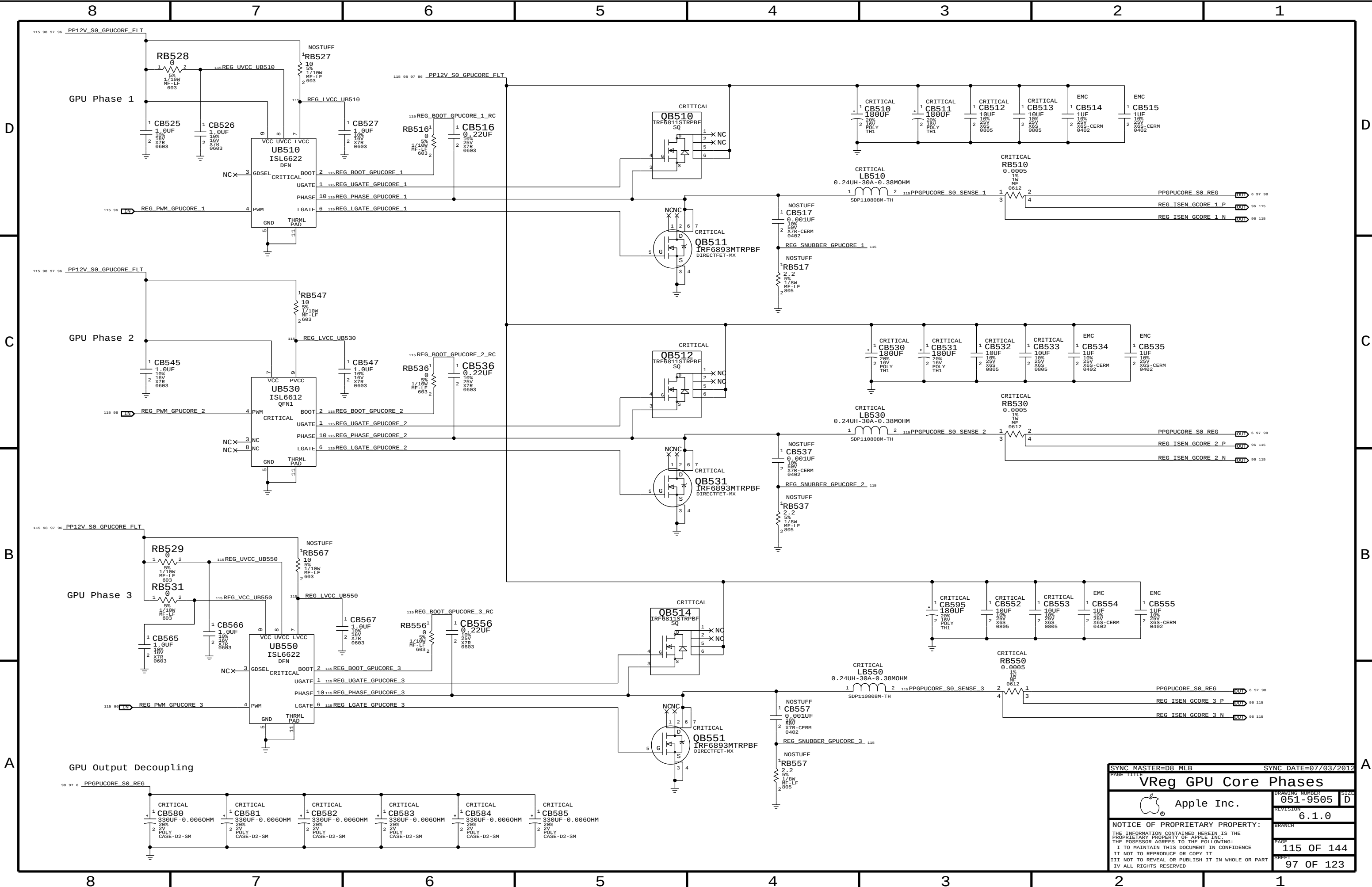
GPU GPU_FB_PLL_AVDD 120MA X 4 = 480MA



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SYNC DATE=07/03/2012

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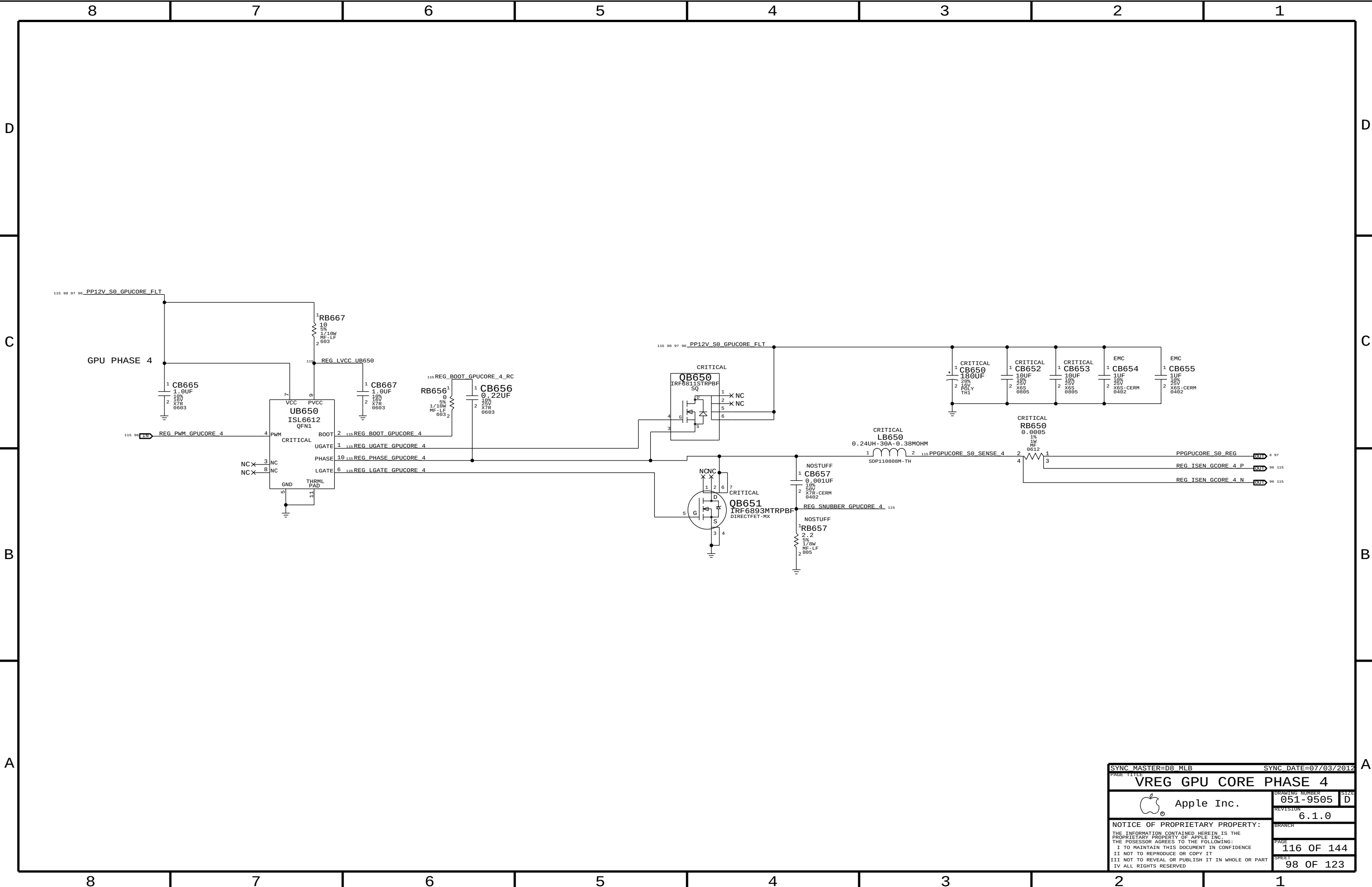
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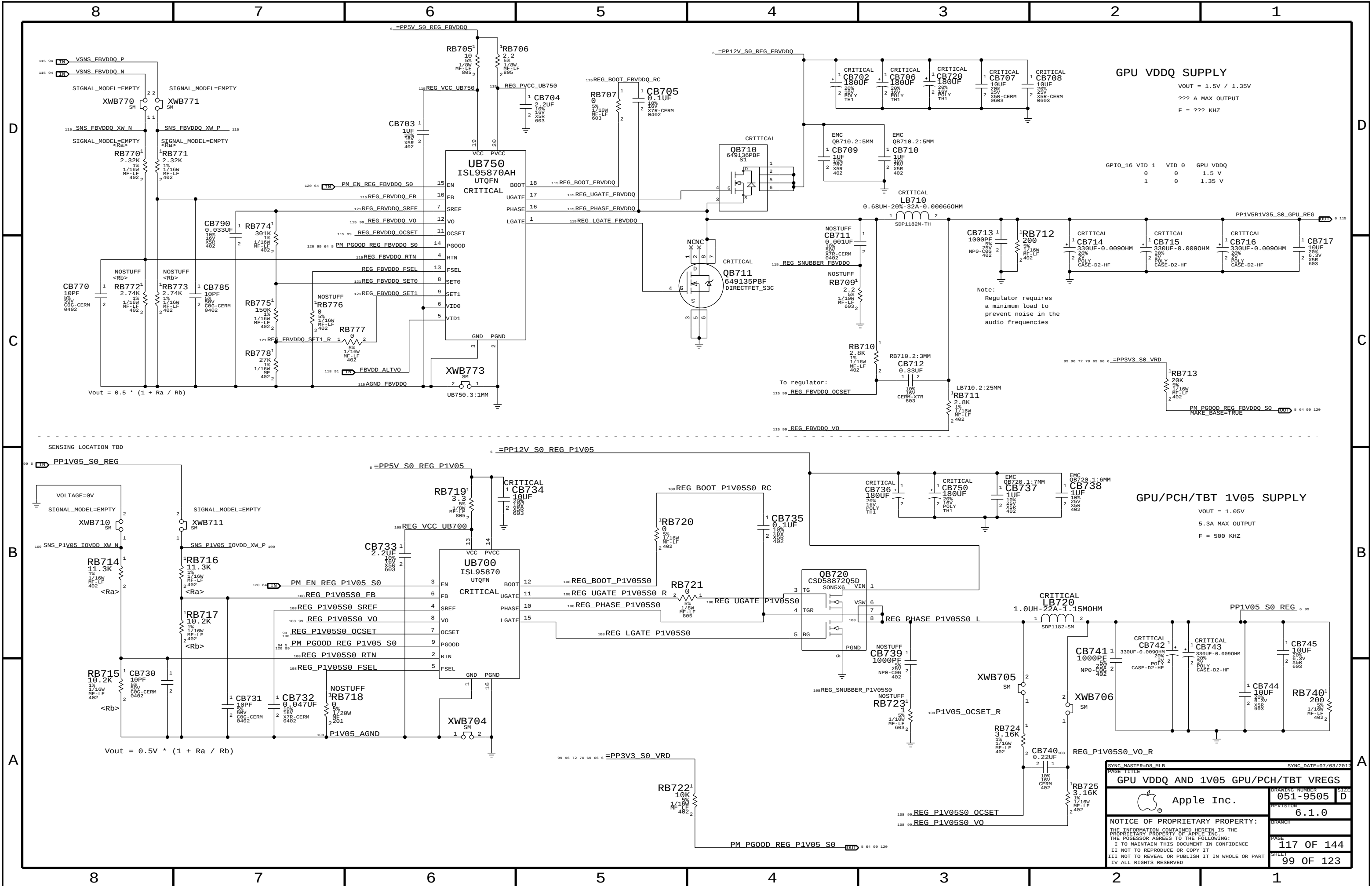
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DDR3

DDR3-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DDR_34S	*	=34_OHM_SE	=34_OHM_SE	=34_OHM_SE	=34_OHM_SE	=STANDARD	=STANDARD
DDR_34S	BOTTOM	=34_OHM_SE	=34_OHM_SE	=34_OHM_SE	2.0 MM	=STANDARD	=STANDARD
DDR_39S	*	=39_OHM_SE	=39_OHM_SE	=39_OHM_SE	=39_OHM_SE	=STANDARD	=STANDARD
DDR_42S	*	=42_OHM_SE	=42_OHM_SE	=42_OHM_SE	=42_OHM_SE	=STANDARD	=STANDARD
DDR_42S	BOTTOM	=42_OHM_SE	=55_OHM_SE	=55_OHM_SE	2.0 MM	=STANDARD	=STANDARD
DDR_42S	ISL5, ISL8	=42_OHM_SE	=55_OHM_SE	=55_OHM_SE	2.0 MM	=STANDARD	=STANDARD
DDR_42S_D	*	=42_OHM_SE	=42_OHM_SE	=42_OHM_SE	=42_OHM_SE	0.1016 MM	0.1016 MM
DDR_42S_D	BOTTOM	=42_OHM_SE	=55_OHM_SE	=55_OHM_SE	2.0 MM	0.1016 MM	0.1016 MM
DDR_42S_D	ISL5, ISL8	=42_OHM_SE	=55_OHM_SE	=55_OHM_SE	2.0 MM	0.1016 MM	0.1016 MM
DDR_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
DDR_68D	*	=68_OHM_DIFF	=68_OHM_DIFF	=68_OHM_DIFF	=68_OHM_DIFF	=68_OHM_DIFF	=68_OHM_DIFF

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
POWER_DDR_P4MM	*	Y	0.400 MM	0.100 MM	3.0 MM	=STANDARD	=STANDARD

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
POWER_DDR_PHY	*	POWER_DDR_P4MM
DDR_CLK_PHY	*	DDR_6BD
DDR_CTRL_PHY	*	DDR_39S
DDR_CMD_PHY	*	DDR_34S
DDR_DQ_PHY	*	DDR_42S
DDR_DQS_PHY	*	DDR_42S_D

DDR3-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DDR_CLK_ISO	TOP, BOTTOM	=5.5X_DIELECTRIC	?
DDR_CLK_ISO	ISL3, ISL10	=4.9X_DIELECTRIC	?
DDR_CLK_ISO	*	=6.5X_DIELECTRIC	?
DDR_CTRL_ISO	TOP, BOTTOM	=4.5X_DIELECTRIC	?
DDR_CTRL_ISO	ISL3, ISL10	=4.0X_DIELECTRIC	?
DDR_CTRL_ISO	*	=5.3X_DIELECTRIC	?
DDR_CTRL2CTRL	TOP, BOTTOM	=3.0X_DIELECTRIC	?
DDR_CTRL2CTRL	ISL3, ISL10	=2.6X_DIELECTRIC	?
DDR_CTRL2CTRL	*	=3.5X_DIELECTRIC	?
DDR_CMD_ISO	TOP, BOTTOM	=4.5X_DIELECTRIC	?
DDR_CMD_ISO	ISL3, ISL10	=4.0X_DIELECTRIC	?
DDR_CMD_ISO	*	=5.3X_DIELECTRIC	?
DDR_CMD2CMD	TOP, BOTTOM	=2.3X_DIELECTRIC	?
DDR_CMD2CMD	ISL3, ISL10	=2.0X_DIELECTRIC	?
DDR_CMD2CMD	*	=2.7X_DIELECTRIC	?
DDR_DATA_ISO	TOP, BOTTOM	=4.5X_DIELECTRIC	?
DDR_DATA_ISO	ISL3, ISL10	=4.0X_DIELECTRIC	?
DDR_DATA_ISO	*	=5.3X_DIELECTRIC	?
DDR_DQ2DQ	TOP, BOTTOM	=3.2X_DIELECTRIC	900
DDR_DQ2DQ	ISL3, ISL10	=2.8X_DIELECTRIC	900
DDR_DQ2DQ	*	=3.8X_DIELECTRIC	900
DDR_DQ2DQS	TOP, BOTTOM	=3.7X_DIELECTRIC	?
DDR_DQ2DQS	ISL3, ISL10	=3.3X_DIELECTRIC	?
DDR_DQ2DQS	*	=4.4X_DIELECTRIC	?
DDR_BL2BL	TOP, BOTTOM	=4.5X_DIELECTRIC	?
DDR_BL2BL	ISL3, ISL10	=4.0X_DIELECTRIC	?
DDR_BL2BL	*	=5.3X_DIELECTRIC	?
DDR_CH2CH	TOP, BOTTOM	=9.1X_DIELECTRIC	?
DDR_CH2CH	ISL3, ISL10	=8.2X_DIELECTRIC	?
DDR_CH2CH	*	=10.9X_DIELECTRIC	?
CMD2DATA_ISO	TOP, BOTTOM	=7X_DIELECTRIC	?
CMD2DATA_ISO	ISL3, ISL10	=5X_DIELECTRIC	?
CMD2DATA_ISO	*	=5X_DIELECTRIC	?

DDR3 Power-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
POWER_DDR_ISO	*	=4.3X_DIELECTRIC	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
POWER_DDR	*	*	POWER_DDR_ISO

Constraints

Clocks: CK[3:0], CK#[3:0]

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
DDR_CLK	*	*	DDR_CLK_ISO

Control: CS#[3:0], CKE[3:0], ODT[3:0]

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
DDR_CTRL	*	*	DDR_CTRL_ISO
DDR_CTRL	DDR_CTRL	*	DDR_CTRL2CTRL

Command: MA[15:0], RAS#, CAS#, WE# BS[2:0]

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
DDR_CMD	*	*	DDR_CMD_ISO
DDR_CMD	DDR_CMD	*	DDR_CMD2CMD
DDR_CMD	DDR_A_DQ_BYTE*	*	CMD2DATA_ISO
DDR_CMD	DDR_A_DQS*	*	CMD2DATA_ISO
DDR_CMD	DDR_B_DQ_BYTE*	*	CMD2DATA_ISO
DDR_CMD	DDR_B_DQS*	*	CMD2DATA_ISO

Data: DQS[7:0], DQS#[7:0], DQ[63:0]

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
DDR_A_DQ_BYTE*	*	*	DDR_DATA_ISO
DDR_A_DQS*	*	*	DDR_DATA_ISO
DDR_B_DQ_BYTE*	*	*	DDR_DATA_ISO
DDR_B_DQS*	*	*	DDR_DATA_ISO
DDR_*_DQ_BYTE*	=SAME	*	DDR_DQ2DQ
DDR_A_DQ_BYTE*	DDR_A_DQS*	*	DDR_DQ2DQS
DDR_A_DQ_BYTE*	DDR_A_DQ_BYTE*	*	DDR_BL2BL
DDR_B_DQ_BYTE*	DDR_B_DQS*	*	DDR_DQ2DQS
DDR_B_DQ_BYTE*	DDR_B_DQ_BYTE*	*	DDR_BL2BL
DDR_A_*	DDR_B_*	*	DDR_CH2CH

Note (1):

Deliberately set DQ to DQS spacing to 3:1 to avoid adding complexity to constraints, even though it can be less. Only one rule per channel is needed by trading off a little space.

Note (2):

Intel suggests 25 mil (0.65 mm) spacing for via to channel, and via to pad to two different channels. D0R3 draws about 20 mA per trace with edge rates in the 100s of ps. The main coupling mechanism is capacitive. A 0.65 mm spacing is used for power nets, which draw far more current (inductive coupling however). These rules are far too conservative. To meet these rules, the spacing must be applied to the net.

DDR3

Electrical Constraint Set	Physical	Spacing		
Channel A				
1E97D DDR_A_CLK0	DDR_CLK_PHY	DDR_CLK	MEM A CLK P<1...0>	12 29
1E97D DDR_A_CLK0	DDR_CLK_PHY	DDR_CLK	MEM A CLK N<1...0>	12 29
1E97D DDR_A_CLK1	DDR_CLK_PHY	DDR_CLK	MEM A CLK P<3...2>	12 30
1E97D DDR_A_CLK1	DDR_CLK_PHY	DDR_CLK	MEM A CLK N<3...2>	12 30
1E97D DDR_A_CTL0	DDR_CTL_PHY	DDR_CTL	MEM A CKE<1...0>	12 29
1E97D DDR_A_CTL0	DDR_CTL_PHY	DDR_CTL	MEM A CS L<1...0>	12 29
1E97D DDR_A_CTL0	DDR_CTL_PHY	DDR_CTL	MEM A ODT<1...0>	12 29
1E97D DDR_A_CTL1	DDR_CTL_PHY	DDR_CTL	MEM A CKE<3...2>	12 30
1E97D DDR_A_CTL1	DDR_CTL_PHY	DDR_CTL	MEM A CS L<3...2>	12 30
1E97D DDR_A_CTL1	DDR_CTL_PHY	DDR_CTL	MEM A ODT<3...2>	12 30
1E97D DDR_A_CMD	DDR_CMD_PHY	DDR_CMD	MEM A A<15...0>	12 29 30
1E97D DDR_A_CMD	DDR_CMD_PHY	DDR_CMD	MEM A BA<2...0>	12 29 30
1E97D DDR_A_CMD	DDR_CMD_PHY	DDR_CMD	MEM A RAS L	12 29 30
1E97D DDR_A_CMD	DDR_CMD_PHY	DDR_CMD	MEM A CAS L	12 29 30
1E97D DDR_A_CMD	DDR_CMD_PHY	DDR_CMD	MEM A WE L	12 29 30
1E97D DDR_A_DQ_BYTE0	DDR_DQ_PHY	DDR_A_DQ_BYTE0	MEM A DQ<7...0>	12 33
1E97D DDR_A_DQ_BYTE1	DDR_DQ_PHY	DDR_A_DQ_BYTE1	MEM A DQ<15...8>	12 33
1E97D DDR_A_DQ_BYTE2	DDR_DQ_PHY	DDR_A_DQ_BYTE2	MEM A DQ<23...16>	12 33
1E97D DDR_A_DQ_BYTE3	DDR_DQ_PHY	DDR_A_DQ_BYTE3	MEM A DQ<31...24>	12 33
1E97D DDR_A_DQ_BYTE4	DDR_DQ_PHY	DDR_A_DQ_BYTE4	MEM A DQ<39...32>	12 33
1E97D DDR_A_DQ_BYTE5	DDR_DQ_PHY	DDR_A_DQ_BYTE5	MEM A DQ<47...40>	12 33
1E97D DDR_A_DQ_BYTE6	DDR_DQ_PHY	DDR_A_DQ_BYTE6	MEM A DQ<55...48>	12 33
1E97D DDR_A_DQ_BYTE7	DDR_DQ_PHY	DDR_A_DQ_BYTE7	MEM A DQ<63...56>	12 33
1E97D DDR_A_DQS0	DDR_DQS_PHY	DDR_A_DQS0	MEM A DQS P<0>	12 33
1E97D DDR_A_DQS0	DDR_DQS_PHY	DDR_A_DQS0	MEM A DQS N<0>	12 33
1E97D DDR_A_DQS1	DDR_DQS_PHY	DDR_A_DQS1	MEM A DQS P<1>	12 33
1E97D DDR_A_DQS1	DDR_DQS_PHY	DDR_A_DQS1	MEM A DQS N<1>	12 33
1E97D DDR_A_DQS2	DDR_DQS_PHY	DDR_A_DQS2	MEM A DQS P<2>	12 33
1E97D DDR_A_DQS2	DDR_DQS_PHY	DDR_A_DQS2	MEM A DQS N<2>	12 33
1E97D DDR_A_DQS3	DDR_DQS_PHY	DDR_A_DQS3	MEM A DQS P<3>	12 33
1E97D DDR_A_DQS3	DDR_DQS_PHY	DDR_A_DQS3	MEM A DQS N<3>	12 33
1E97D DDR_A_DQS4	DDR_DQS_PHY	DDR_A_DQS4	MEM A DQS P<4>	12 33
1E97D DDR_A_DQS4	DDR_DQS_PHY	DDR_A_DQS4	MEM A DQS N<4>	12 33
1E97D DDR_A_DQS5	DDR_DQS_PHY	DDR_A_DQS5	MEM A DQS P<5>	12 33
1E97D DDR_A_DQS5	DDR_DQS_PHY	DDR_A_DQS5	MEM A DQS N<5>	12 33
1E97D DDR_A_DQS6	DDR_DQS_PHY	DDR_A_DQS6	MEM A DQS P<6>	12 33
1E97D DDR_A_DQS6	DDR_DQS_PHY	DDR_A_DQS6	MEM A DQS N<6>	12 33
1E97D DDR_A_DQS7	DDR_DQS_PHY	DDR_A_DQS7	MEM A DQS P<7>	12 33
1E97D DDR_A_DQS7	DDR_DQS_PHY	DDR_A_DQS7	MEM A DQS N<7>	12 33
Channel B				
1E97D DDR_B_CLK0	DDR_CLK_PHY	DDR_CLK	MEM B CLK P<1...0>	12 31
1E97D DDR_B_CLK0	DDR_CLK_PHY	DDR_CLK	MEM B CLK N<1...0>	12 31
1E97D DDR_B_CLK1	DDR_CLK_PHY	DDR_CLK	MEM B CLK P<3...2>	12 32
1E97D DDR_B_CLK1	DDR_CLK_PHY	DDR_CLK	MEM B CLK N<3...2>	12 32
1E97D DDR_B_CTL0	DDR_CTL_PHY	DDR_CTL	MEM B CKE<1...0>	12 31
1E97D DDR_B_CTL0	DDR_CTL_PHY	DDR_CTL	MEM B CS L<1...0>	12 31
1E97D DDR_B_CTL0	DDR_CTL_PHY	DDR_CTL	MEM B ODT<1...0>	12 31
1E97D DDR_B_CTL1	DDR_CTL_PHY	DDR_CTL	MEM B CKE<3...2>	12 32
1E97D DDR_B_CTL1	DDR_CTL_PHY	DDR_CTL	MEM B CS L<3...2>	12 32
1E97D DDR_B_CTL1	DDR_CTL_PHY	DDR_CTL	MEM B ODT<3...2>	12 32
1E97D DDR_B_CMD	DDR_CMD_PHY	DDR_CMD	MEM B A<15...0>	12 31 32
1E97D DDR_B_CMD	DDR_CMD_PHY	DDR_CMD	MEM B BA<2...0>	12 31 32
1E97D DDR_B_CMD	DDR_CMD_PHY	DDR_CMD	MEM B RAS L	12 31 32
1E97D DDR_B_CMD	DDR_CMD_PHY	DDR_CMD	MEM B CAS L	12 31 32
1E97D DDR_B_CMD	DDR_CMD_PHY	DDR_CMD	MEM B WE L	12 31 32
1E97D DDR_B_DQ_BYTE0	DDR_DQ_PHY	DDR_B_DQ_BYTE0	MEM B DQ<7...0>	12 33
1E97D DDR_B_DQ_BYTE1	DDR_DQ_PHY	DDR_B_DQ_BYTE1	MEM B DQ<15...8>	12 33
1E97D DDR_B_DQ_BYTE2	DDR_DQ_PHY	DDR_B_DQ_BYTE2	MEM B DQ<23...16>	12 33
1E97D DDR_B_DQ_BYTE3	DDR_DQ_PHY	DDR_B_DQ_BYTE3	MEM B DQ<31...24>	12 33
1E97D DDR_B_DQ_BYTE4	DDR_DQ_PHY	DDR_B_DQ_BYTE4	MEM B DQ<39...32>	12 33
1E97D DDR_B_DQ_BYTE5	DDR_DQ_PHY	DDR_B_DQ_BYTE5	MEM B DQ<47...40>	12 33
1E97D DDR_B_DQ_BYTE6	DDR_DQ_PHY	DDR_B_DQ_BYTE6	MEM B DQ<55...48>	12 33
1E97D DDR_B_DQ_BYTE7	DDR_DQ_PHY	DDR_B_DQ_BYTE7	MEM B DQ<63...56>	12 33
1E97D DDR_B_DQS0	DDR_DQS_PHY	DDR_B_DQS0	MEM B DQS P<0>	12 33
1E97D DDR_B_DQS0	DDR_DQS_PHY	DDR_B_DQS0	MEM B DQS N<0>	12 33
1E97D DDR_B_DQS1	DDR_DQS_PHY	DDR_B_DQS1	MEM B DQS P<1>	12 33
1E97D DDR_B_DQS1	DDR_DQS_PHY	DDR_B_DQS1	MEM B DQS N<1>	12 33
1E97D DDR_B_DQS2	DDR_DQS_PHY	DDR_B_DQS2	MEM B DQS P<2>	12 33
1E97D DDR_B_DQS2	DDR_DQS_PHY	DDR_B_DQS2	MEM B DQS N<2>	12 33
1E97D DDR_B_DQS3	DDR_DQS_PHY	DDR_B_DQS3	MEM B DQS P<3>	12 33
1E97D DDR_B_DQS3	DDR_DQS_PHY	DDR_B_DQS3	MEM B DQS N<3>	12 33
1E97D DDR_B_DQS4	DDR_DQS_PHY	DDR_B_DQS4	MEM B DQS P<4>	12 33
1E97D DDR_B_DQS4	DDR_DQS_PHY	DDR_B_DQS4	MEM B DQS N<4>	12 33
1E97D DDR_B_DQS5	DDR_DQS_PHY	DDR_B_DQS5	MEM B DQS P<5>	12 33
1E97D DDR_B_DQS5	DDR_DQS_PHY	DDR_B_DQS5	MEM B DQS N<5>	12 33
1E97D DDR_B_DQS6	DDR_DQS_PHY	DDR_B_DQS6	MEM B DQS P<6>	12 33
1E97D DDR_B_DQS6	DDR_DQS_PHY	DDR_B_DQS6	MEM B DQS N<6>	12 33
1E97D DDR_B_DQS7	DDR_DQS_PHY	DDR_B_DQS7	MEM B DQS P<7>	12 33
1E97D DDR_B_DQS7	DDR_DQS_PHY	DDR_B_DQS7	MEM B DQS N<7>	12 33
Reset				
1E97D	DDR_S0S	CPU	MEM RESET L	28 29 30 31 32

Note (3):

In order for the constraints $DDR_*_DQ_BYTE*$ to =SAME to win out over $DDR_ \{A,B\}_DQ_BYTE*$ to $DDR_ \{A,B\}_DQ_BYTE*$ so that the small intra-bytelane spacing is used, the spacing rule DDR_DQ2DQ must have a weight greater than DDR_BL2BL .

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DDR3 Constraints			
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SATA

SATA-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SATA_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
SATA_90D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
SATA_PHY	*	SATA_90D
COMP_SATA_PHY	*	SATA_50S

SATA-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SATA_ISO	*	=7.2X_DIELECTRIC	?
COMP_SATA_ISO	*	=5.4X_DIELECTRIC	?

Constraints

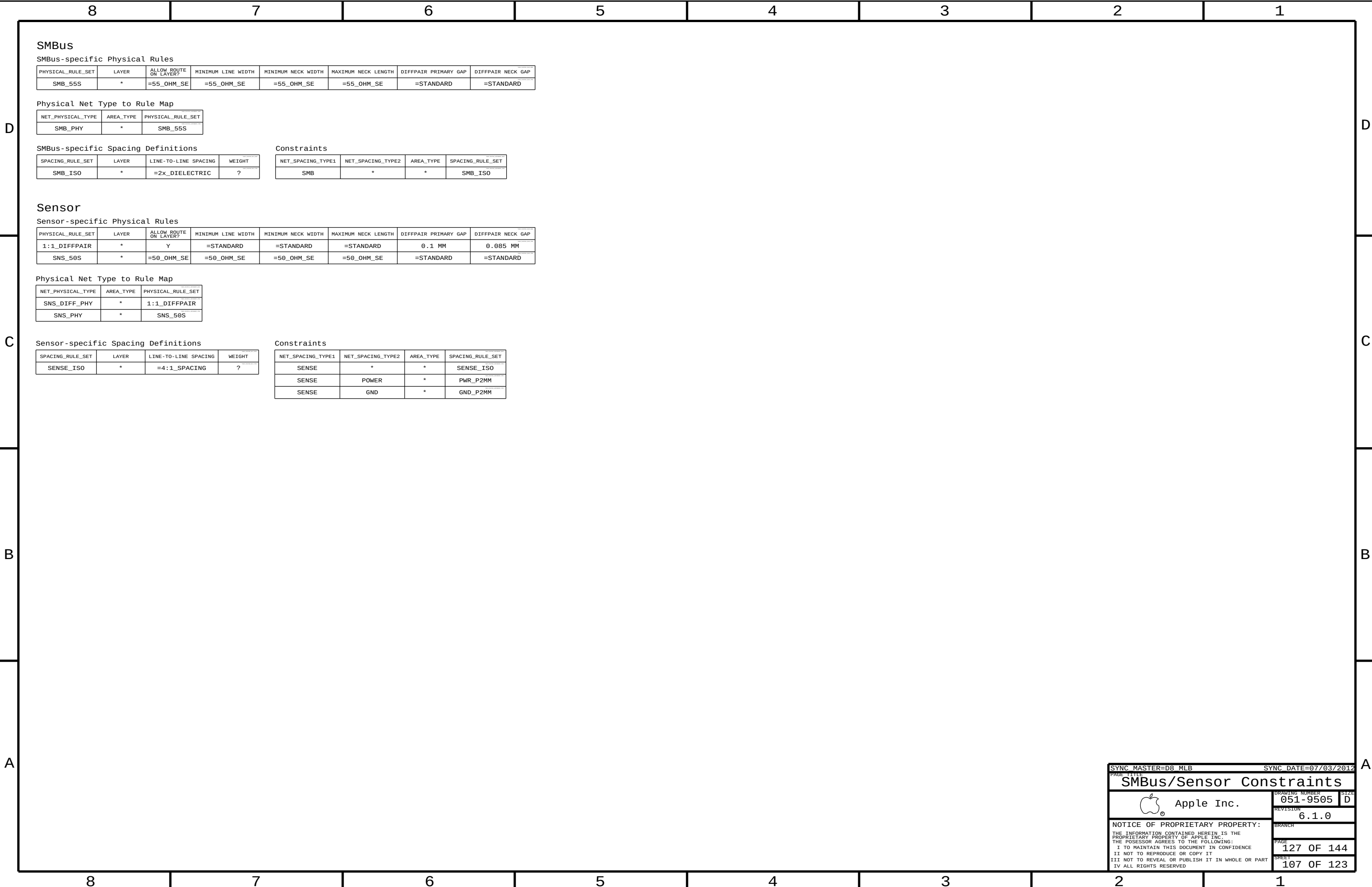
NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
SATA	*	*	SATA_ISO
COMP_SATA	*	*	COMP_SATA_ISO

SATA

Electrical Constraint Set	Physical	Spacing	
PCH SATA Port 0 (HDD)			
EE01 SATA_R2D	SATA_PHY	SATA	SATA HDD R2D P 44
EE02 SATA_R2D	SATA_PHY	SATA	SATA HDD R2D N 44
EE03 SATA_R2D	SATA_PHY	SATA	SATA HDD R2D C P 19 44
EE04 SATA_R2D	SATA_PHY	SATA	SATA HDD R2D C N 19 44
EE05 SATA_D2R	SATA_PHY	SATA	SATA HDD D2R P 19 44
EE06 SATA_D2R	SATA_PHY	SATA	SATA HDD D2R N 19 44
EE07 SATA_D2R	SATA_PHY	SATA	SATA HDD D2R C P 44
EE08 SATA_D2R	SATA_PHY	SATA	SATA HDD D2R C N 44
PCH SATA Port 1 (SSD)			
EE09 SATA_R2D_MUX_SSD	SATA_PHY	SATA	SATA SSD R2D P 19 44
EE10 SATA_R2D_MUX_SSD	SATA_PHY	SATA	SATA SSD R2D N 19 44
EE11 SATA_D2R_MUX_SSD	SATA_PHY	SATA	SATA SSD D2R P 19 44
EE12 SATA_D2R_MUX_SSD	SATA_PHY	SATA	SATA SSD D2R N 19 44
PCH SATA Compensation			
EE13	COMP_SATA_PHY	COMP_SATA	PCH SATAICOMP 19
EE14	COMP_SATA_PHY	COMP_SATA	PCH SATA3COMP 19
EE15	COMP_SATA_PHY	COMP_SATA	PCH SATA3RBIAS 19

Unused

Electrical Constraint Set	Physical	Spacing



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SMBus/Sensor Constraints

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DC - DC

Power-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
GND_P3MM	*	Y	0.300 MM	0.200 MM	3.0 MM	=STANDARD	=STANDARD
GND_P5MM	*	Y	0.500 MM	0.200 MM	3.0 MM	=STANDARD	=STANDARD
POWER_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
POWER_P3MM	*	Y	0.300 MM	0.200 MM	3.0 MM	=STANDARD	=STANDARD
POWER_P6MM	*	Y	0.600 MM	0.200 MM	3.0 MM	=STANDARD	=STANDARD
POWER_P5MM	*	Y	0.500 MM	0.200 MM	3.0 MM	=STANDARD	=STANDARD

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
VR_CTL_PHY	*	POWER_P5MM
VR_CTL_PHY	BGA	STANDARD
VR_VID_PHY	*	POWER_50S
POWER_PHY	*	POWER_P6MM
GND_PHY	*	GND_P5MM

Constraints

Power and Common

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
POWER	*	*	POWER_ISO
GND	*	*	GND_ISO

Power-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
POWER_ISO	*	=STANDARD	?

DC-DC Baddies

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SWNODE_ISO	*	=8:1_SPACING	1000
SWNODE_SW2SW	*	=1:1_SPACING	?
SWNODE_SW2PWR	*	=2:1_SPACING	?
SWNODE_SW2GND	*	=2:1_SPACING	?
SWNODE_LG2SW	*	=3:1_SPACING	?

DC-DC Control

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
VR_CTL_ISO	*	=3:1_SPACING	?
VR_VID_ISO	*	=4X_DIELECTRIC	?

CPU VccSA

Electrical Constraint Set	Physical	Spacing	Voltage	IDIT	NO_TEST	
Input Bus						
REG	POWER_PHY	POWER	5V			REG VCC U7500
REG	POWER_PHY	POWER	5V			REG PVCC U7500
Local Ground						
AGND	GND_PHY	GND	0V			AGND VCCSA0
VCCIO						
REG	POWER_PHY	VR_SWITCH	12V	TRUE		REG PHASE VCCSA0
REG	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG BOOT VCCSA0
REG	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG BOOT VCCSA0_RC
REG	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG UGATE VCCSA0
REG	VR_CTL_PHY	VR_LGATE	12V	TRUE		REG LGATE VCCSA0
REG	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG SNUBBER VCCSA0
REG	VR_CTL_PHY	VR_CTL				REG VCCSA0_OCSET
REG	VR_CTL_PHY	VR_CTL				REG VCCSA0_VO
SNS		SENSE				SNS_CPU_VCCSA
REG	VSNM_CPU_VCCSA	SNS_DIFF_PHY	SENSE			SNS VCCSA0_XW_P
REG	VSNM_CPU_VCCSA	SNS_DIFF_PHY	SENSE			SNS VCCSA0_XW_N
REG		SENSE				REG VCCSA0_FB
REG		SENSE				REG VCCSA0_RTN
REG	VR_CTL_PHY	VR_CTL				REG VCCSA0_SREF
REG	VR_CTL_PHY	VR_CTL				REG VCCSA0_FSEL
Output Bus						
REG	POWER_PHY	POWER	0.925V			PPVCCSA_S0

CPU VccIO PCH 1.05V S0

Electrical Constraint Set	Physical	Spacing	Voltage	IDIT	NO_TEST	
Input Bus						
H1	POWER_PHY	POWER	5V			REG VCC U7400
H2	POWER_PHY	POWER	5V			REG PVCC U7400
Local Ground						
H3	GND_PHY	GND	0V			AGND CPU P1V05S0
1.05V S0						
H5	POWER_PHY	VR_SWITCH	12V	TRUE		REG CPU PHASE P1V05S0
H7	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG CPU BOOT P1V05S0
H8	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG CPU BOOT P1V05S0_RC
H9	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG CPU UGATE P1V05S0
H10	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG CPU UGATE P1V05S0_R
H11	VR_CTL_PHY	VR_LGATE	12V	TRUE		REG CPU LGATE P1V05S0
H12	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG CPU SNUBBER P1V05S0
H13	VR_CTL_PHY	VR_CTL				REG CPU P1V05S0_OCSET
H14	VR_CTL_PHY	VR_CTL				REG CPU P1V05S0_V0
H15 VSNS_CPU_VCC10	SNS_DIFF_PHY	SENSE				SNS_CPU_VCC10_P
H16 VSNS_CPU_VCC10	SNS_DIFF_PHY	SENSE				SNS_CPU_VCC10_N
H17	SNS_DIFF_PHY	SENSE				SNS_CPU P1V05S0_XW_P
H18		SENSE				SNS_CPU P1V05S0_XW_N
H19		SENSE				REG CPU P1V05S0_FB
H20		SENSE				REG CPU P1V05S0_RTN
H21	VR_CTL_PHY	VR_CTL				REG CPU P1V05S0_SREF
H22	VR_CTL_PHY	VR_CTL				REG CPU P1V05S0_FSEL
Output Bus						
H23	POWER_PHY	POWER	1.05V			PP1V05_S0_CPU

PCH/GPU/TBT 1.05V S0

Electrical Constraint Set	Physical	Spacing	Voltage	DIDT	NO_TEST	
Input Bus						
E44	POWER_PHY	POWER	5V			REG_VCC_UB700 99
Local Ground						
E46	GND_PHY	GND	0V			P1V05_AGND 99
1.05V S0						
E49	POWER_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_P1V05S0 99
E49	POWER_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_P1V05S0_L 99
E49	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_B00T_P1V05S0 99
E50	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_B00T_P1V05S0_RC 99
E50	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_P1V05S0 99
E50	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_P1V05S0_R 99
E50	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_LGATE_P1V05S0 99
E50	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_SNUBBER_P1V05S0 99
E50	VR_CTL_PHY	VR_CTL				REG_P1V05S0_OCSET 99
E50	VR_CTL_PHY	VR_CTL				REG_P1V05S0_V0 99
E72	POWER_PHY	VR_SWITCH	12V			REG_P1V05S0_V0_R 99
E78	POWER_PHY	VR_SWITCH	12V			P1V05_OCSET_R 99
E61		SENSE				REG_P1V05S0_FB 99
E62		SENSE				REG_P1V05S0_RTIN 99
E63	VR_CTL_PHY	VR_CTL				REG_P1V05S0_SREF 99
E64	VR_CTL_PHY	VR_CTL				REG_P1V05S0_FSEL 99
Output Bus						
E68	POWER_PHY	POWER	1.05V			PP1V05_S0_PCH 6
E68	POWER_PHY	POWER	1.05V			PP1V05_S0 6
FET Switched						
E69	POWER_PHY	POWER	1.05V			PP1V05_TBTLCL 6 38
E69	POWER_PHY	POWER	1.05V			PP1V05_TBTCIO 6

SYNCH MASTER-D8 MLB		SYNCH DATE=07/03/2012	
PAGE TITLE			
VReg Constraints			
 Apple Inc.		DRAWING NUMBER	051-9505
		SIZE	D
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<table><tr><th>Electrical Constraint Set</th><th>Physical</th><th>Spacing</th><th>Voltage</th><th>DIDT</th><th>NO_TEST</th><th></th></tr><tr><td colspan="7">Input Bus</td></tr><tr><td>1E88P</td><td>POWER_PHY</td><td>POWER</td><td>12V</td><td></td><td></td><td>PP12V_S0_CPUCORE_FLT</td></tr><tr><td>1E89P</td><td>POWER_PHY</td><td>POWER</td><td>5V</td><td></td><td></td><td>REG_VCC_U7100</td></tr><tr><td colspan="7">Local Ground</td></tr><tr><td>1E90P</td><td>GND_PHY</td><td>GND</td><td>0V</td><td></td><td></td><td>AGND_CPU</td></tr><tr><td colspan="7">Phase 1</td></tr><tr><td>1E88P</td><td>POWER_PHY</td><td>POWER</td><td>12V</td><td></td><td></td><td>REG_LVCC_U7210</td></tr><tr><td>1E89P</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_PWM_CPUCORE_1</td></tr><tr><td>1E89P</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_PWM_CPUCORE_1_R</td></tr><tr><td>1E89P</td><td>POWER_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td><td>REG_PHASE_CPUCORE_1</td></tr><tr><td>1E89P</td><td>VR_CTL_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td><td>REG_BOOT_CPUCORE_1</td></tr><tr><td>1E89P</td><td>VR_CTL_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td>TRUE</td><td>REG_BOOT_CPUCORE_1_RC</td></tr><tr><td>1E89P</td><td>VR_CTL_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td><td>REG_UGATE_CPUCORE_1</td></tr><tr><td>1E89P</td><td>VR_CTL_PHY</td><td>VR_LGATE</td><td>12V</td><td>TRUE</td><td></td><td>REG_LGATE_CPUCORE_1</td></tr><tr><td>1E89P</td><td>VR_CTL_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td>TRUE</td><td>REG_SNUBBER_CPUCORE_1</td></tr><tr><td>1E89P</td><td>POWER_PHY</td><td>POWER</td><td>1.1V</td><td></td><td></td><td>PPCPUCORE_S0_SENSE_1</td></tr><tr><td>1E89P</td><td>TSNS_CPU_CORE</td><td>SNS_DIFF_PHY</td><td>SENSE</td><td></td><td></td><td>REG_ISENCORE_1_P</td></tr><tr><td>1E89P</td><td>TSNS_CPU_CORE</td><td>SNS_DIFF_PHY</td><td>SENSE</td><td></td><td></td><td>REG_ISENCORE_1_N</td></tr><tr><td>1E89P</td><td></td><td>SENSE</td><td></td><td></td><td></td><td>REG_ISENCORE_1_NR</td></tr><tr><td colspan="7">Phase 2</td></tr><tr><td>1E88P</td><td>POWER_PHY</td><td>POWER</td><td>12V</td><td></td><td></td><td>REG_LVCC_U7230</td></tr><tr><td>1E89P</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_PWM_CPUCORE_2</td></tr><tr><td>1E89P</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_PWM_CPUCORE_2_R</td></tr><tr><td>1E89P</td><td>POWER_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td><td>REG_PHASE_CPUCORE_2</td></tr><tr><td>1E89P</td><td>VR_CTL_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td><td>REG_BOOT_CPUCORE_2</td></tr><tr><td>1E89P</td><td>VR_CTL_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td>TRUE</td><td>REG_BOOT_CPUCORE_2_RC</td></tr><tr><td>1E89P</td><td>VR_CTL_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td><td>REG_UGATE_CPUCORE_2</td></tr><tr><td>1E89P</td><td>VR_CTL_PHY</td><td>VR_LGATE</td><td>12V</td><td>TRUE</td><td></td><td>REG_LGATE_CPUCORE_2</td></tr><tr><td>1E89P</td><td>VR_CTL_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td>TRUE</td><td>REG_SNUBBER_CPUCORE_2</td></tr><tr><td>1E89P</td><td>POWER_PHY</td><td>POWER</td><td>1.1V</td><td></td><td></td><td>PPCPUCORE_S0_SENSE_2</td></tr><tr><td>1E89P</td><td>TSNS_CPU_CORE</td><td>SNS_DIFF_PHY</td><td>SENSE</td><td></td><td></td><td>REG_ISENCORE_2_P</td></tr><tr><td>1E89P</td><td>TSNS_CPU_CORE</td><td>SNS_DIFF_PHY</td><td>SENSE</td><td></td><td></td><td>REG_ISENCORE_2_N</td></tr><tr><td>1E89P</td><td></td><td>SENSE</td><td></td><td></td><td></td><td>REG_ISENCORE_2_NR</td></tr><tr><td colspan="7">Phase 3</td></tr><tr><td>1E88P</td><td>POWER_PHY</td><td>POWER</td><td>12V</td><td></td><td></td><td>REG_LVCC_U7250</td></tr><tr><td>1E89P</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_PWM_CPUCORE_3</td></tr><tr><td>1E89P</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_PWM_CPUCORE_3_R</td></tr><tr><td>1E89P</td><td>POWER_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td><td>REG_PHASE_CPUCORE_3</td></tr><tr><td>1E89P</td><td>VR_CTL_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td><td>REG_BOOT_CPUCORE_3</td></tr><tr><td>1E89P</td><td>VR_CTL_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td>TRUE</td><td>REG_BOOT_CPUCORE_3_RC</td></tr><tr><td>1E89P</td><td>VR_CTL_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td><td>REG_UGATE_CPUCORE_3</td></tr><tr><td>1E89P</td><td>VR_CTL_PHY</td><td>VR_LGATE</td><td>12V</td><td>TRUE</td><td></td><td>REG_LGATE_CPUCORE_3</td></tr><tr><td>1E89P</td><td>VR_CTL_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td>TRUE</td><td>REG_SNUBBER_CPUCORE_3</td></tr><tr><td>1E89P</td><td>POWER_PHY</td><td>POWER</td><td>1.1V</td><td></td><td></td><td>PPCPUCORE_S0_SENSE_3</td></tr><tr><td>1E89P</td><td>TSNS_CPU_CORE</td><td>SNS_DIFF_PHY</td><td>SENSE</td><td></td><td></td><td>REG_ISENCORE_3_P</td></tr><tr><td>1E89P</td><td>TSNS_CPU_CORE</td><td>SNS_DIFF_PHY</td><td>SENSE</td><td></td><td></td><td>REG_ISENCORE_3_N</td></tr><tr><td>1E89P</td><td></td><td>SENSE</td><td></td><td></td><td></td><td>REG_ISENCORE_3_NR</td></tr><tr><td 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4</td></tr><tr><td>1E88P</td><td>POWER_PHY</td><td>POWER</td><td>12V</td><td></td><td></td><td>REG_LVCC_U7310</td></tr><tr><td>1E89P</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_PWM_CPUCORE_4</td></tr><tr><td>1E89P</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_PWM_CPUCORE_4_R</td></tr><tr><td>1E89P</td><td>POWER_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td><td>REG_PHASE_CPUCORE_4</td></tr><tr><td>1E89P</td><td>VR_CTL_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td><td>REG_BOOT_CPUCORE_4</td></tr><tr><td>1E89P</td><td>VR_CTL_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td>TRUE</td><td>REG_BOOT_CPUCORE_4_RC</td></tr><tr><td>1E89P</td><td>VR_CTL_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td><td>REG_UGATE_CPUCORE_4</td></tr><tr><td>1E89P</td><td>VR_CTL_PHY</td><td>VR_LGATE</td><td>12V</td><td>TRUE</td><td></td><td>REG_LGATE_CPUCORE_4</td></tr><tr><td>1E89P</td><td>VR_CTL_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td>TRUE</td><td>REG_SNUBBER_CPUCORE_4</td></tr><tr><td>1E89P</td><td>POWER_PHY</td><td>POWER</td><td>1.1V</td><td></td><td></td><td>PPCPUCORE_S0_SENSE_4</td></tr><tr><td>1E89P</td><td>TSNS_CPU_CORE</td><td>SNS_DIFF_PHY</td><td>SENSE</td><td></td><td></td><td>REG_ISENCORE_4_P</td></tr><tr><td>1E89P</td><td>TSNS_CPU_CORE</td><td>SNS_DIFF_PHY</td><td>SENSE</td><td></td><td></td><td>REG_ISENCORE_4_N</td></tr><tr><td>1E89P</td><td></td><td>SENSE</td><td></td><td></td><td></td><td>REG_ISENCORE_4_NR</td></tr><tr><td>1E89P</td><td>TSNS_CPU_CORE</td><td>SNS_DIFF_PHY</td><td>SENSE</td><td></td><td></td><td>SNS_CORE_XW_P</td></tr><tr><td>1E89P</td><td>TSNS_CPU_CORE</td><td>SNS_DIFF_PHY</td><td>SENSE</td><td></td><td></td><td>SNS_CORE_XW_N</td></tr><tr><td>1E89P</td><td>TSNS_CPU_CORE</td><td>SNS_DIFF_PHY</td><td>SENSE</td><td></td><td></td><td>SNS_CORE_R_P</td></tr><tr><td>1E89P</td><td>TSNS_CPU_CORE</td><td>SNS_DIFF_PHY</td><td>SENSE</td><td></td><td></td><td>SNS_CORE_R_N</td></tr><tr><td>1E89P</td><td>TSNS_CPU_CORE</td><td>SNS_DIFF_PHY</td><td>SENSE</td><td></td><td></td><td>SNS_CPU_VAXG_P</td></tr><tr><td>1E89P</td><td>TSNS_CPU_CORE</td><td>SNS_DIFF_PHY</td><td>SENSE</td><td></td><td></td><td>SNS_CPU_VAXG_N</td></tr><tr><td>1E89P</td><td>TSNS_CPU_CORE</td><td>SNS_DIFF_PHY</td><td>SENSE</td><td></td><td></td><td>SNS_CPU_VCORE_P</td></tr><tr><td>1E89P</td><td>TSNS_CPU_CORE</td><td>SNS_DIFF_PHY</td><td>SENSE</td><td></td><td></td><td>SNS_CPU_VCORE_N</td></tr><tr><td>1E89P</td><td>TSNS_CPU_CORE</td><td>SNS_DIFF_PHY</td><td>SENSE</td><td></td><td></td><td>SNS_P1V05_IOVDD_XW_P</td></tr><tr><td>1E89P</td><td>TSNS_CPU_CORE</td><td>SNS_DIFF_PHY</td><td>SENSE</td><td></td><td></td><td>SNS_P1V05_IOVDD_XW_N</td></tr></table>							Electrical Constraint Set	Physical	Spacing	Voltage	DIDT	NO_TEST		Input Bus							1E88P	POWER_PHY	POWER	12V			PP12V_S0_CPUCORE_FLT	1E89P	POWER_PHY	POWER	5V			REG_VCC_U7100	Local Ground							1E90P	GND_PHY	GND	0V			AGND_CPU	Phase 1							1E88P	POWER_PHY	POWER	12V			REG_LVCC_U7210	1E89P	VR_CTL_PHY	VR_CTL				REG_PWM_CPUCORE_1	1E89P	VR_CTL_PHY	VR_CTL				REG_PWM_CPUCORE_1_R	1E89P	POWER_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_CPUCORE_1	1E89P	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_CPUCORE_1	1E89P	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT_CPUCORE_1_RC	1E89P	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_CPUCORE_1	1E89P	VR_CTL_PHY	VR_LGATE	12V	TRUE		REG_LGATE_CPUCORE_1	1E89P	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_SNUBBER_CPUCORE_1	1E89P	POWER_PHY	POWER	1.1V			PPCPUCORE_S0_SENSE_1	1E89P	TSNS_CPU_CORE	SNS_DIFF_PHY	SENSE			REG_ISENCORE_1_P	1E89P	TSNS_CPU_CORE	SNS_DIFF_PHY	SENSE			REG_ISENCORE_1_N	1E89P		SENSE				REG_ISENCORE_1_NR	Phase 2							1E88P	POWER_PHY	POWER	12V			REG_LVCC_U7230	1E89P	VR_CTL_PHY	VR_CTL				REG_PWM_CPUCORE_2	1E89P	VR_CTL_PHY	VR_CTL				REG_PWM_CPUCORE_2_R	1E89P	POWER_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_CPUCORE_2	1E89P	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_CPUCORE_2	1E89P	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT_CPUCORE_2_RC	1E89P	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_CPUCORE_2	1E89P	VR_CTL_PHY	VR_LGATE	12V	TRUE		REG_LGATE_CPUCORE_2	1E89P	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_SNUBBER_CPUCORE_2	1E89P	POWER_PHY	POWER	1.1V			PPCPUCORE_S0_SENSE_2	1E89P	TSNS_CPU_CORE	SNS_DIFF_PHY	SENSE			REG_ISENCORE_2_P	1E89P	TSNS_CPU_CORE	SNS_DIFF_PHY	SENSE			REG_ISENCORE_2_N	1E89P		SENSE				REG_ISENCORE_2_NR	Phase 3							1E88P	POWER_PHY	POWER	12V			REG_LVCC_U7250	1E89P	VR_CTL_PHY	VR_CTL				REG_PWM_CPUCORE_3	1E89P	VR_CTL_PHY	VR_CTL				REG_PWM_CPUCORE_3_R	1E89P	POWER_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_CPUCORE_3	1E89P	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_CPUCORE_3	1E89P	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT_CPUCORE_3_RC	1E89P	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_CPUCORE_3	1E89P	VR_CTL_PHY	VR_LGATE	12V	TRUE		REG_LGATE_CPUCORE_3	1E89P	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_SNUBBER_CPUCORE_3	1E89P	POWER_PHY	POWER	1.1V			PPCPUCORE_S0_SENSE_3	1E89P	TSNS_CPU_CORE	SNS_DIFF_PHY	SENSE			REG_ISENCORE_3_P	1E89P	TSNS_CPU_CORE	SNS_DIFF_PHY	SENSE			REG_ISENCORE_3_N	1E89P		SENSE				REG_ISENCORE_3_NR	Phase 4							1E88P	POWER_PHY	POWER	12V			REG_LVCC_U7310	1E89P	VR_CTL_PHY	VR_CTL				REG_PWM_CPUCORE_4	1E89P	VR_CTL_PHY	VR_CTL				REG_PWM_CPUCORE_4_R	1E89P	POWER_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_CPUCORE_4	1E89P	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_CPUCORE_4	1E89P	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT_CPUCORE_4_RC	1E89P	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_CPUCORE_4	1E89P	VR_CTL_PHY	VR_LGATE	12V	TRUE		REG_LGATE_CPUCORE_4	1E89P	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_SNUBBER_CPUCORE_4	1E89P	POWER_PHY	POWER	1.1V			PPCPUCORE_S0_SENSE_4	1E89P	TSNS_CPU_CORE	SNS_DIFF_PHY	SENSE			REG_ISENCORE_4_P	1E89P	TSNS_CPU_CORE	SNS_DIFF_PHY	SENSE			REG_ISENCORE_4_N	1E89P		SENSE				REG_ISENCORE_4_NR	1E89P	TSNS_CPU_CORE	SNS_DIFF_PHY	SENSE			SNS_CORE_XW_P	1E89P	TSNS_CPU_CORE	SNS_DIFF_PHY	SENSE			SNS_CORE_XW_N	1E89P	TSNS_CPU_CORE	SNS_DIFF_PHY	SENSE			SNS_CORE_R_P	1E89P	TSNS_CPU_CORE	SNS_DIFF_PHY	SENSE			SNS_CORE_R_N	1E89P	TSNS_CPU_CORE	SNS_DIFF_PHY	SENSE			SNS_CPU_VAXG_P	1E89P	TSNS_CPU_CORE	SNS_DIFF_PHY	SENSE			SNS_CPU_VAXG_N	1E89P	TSNS_CPU_CORE	SNS_DIFF_PHY	SENSE			SNS_CPU_VCORE_P	1E89P	TSNS_CPU_CORE	SNS_DIFF_PHY	SENSE			SNS_CPU_VCORE_N	1E89P	TSNS_CPU_CORE	SNS_DIFF_PHY	SENSE			SNS_P1V05_IOVDD_XW_P	1E89P	TSNS_CPU_CORE	SNS_DIFF_PHY	SENSE			SNS_P1V05_IOVDD_XW_N	<table><tr><th>Electrical Constraint Set</th><th>Physical</th><th>Spacing</th><th>Voltage</th><th>DIDT</th><th>NO_TEST</th><th></th></tr><tr><td colspan="7">AXG</td></tr><tr><td>1E88P</td><td>POWER_PHY</td><td>POWER</td><td>12V</td><td></td><td></td><td>REG_LVCC_U7330</td></tr><tr><td>1E89P</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_PWM_CPUAXG</td></tr><tr><td>1E89P</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_PWM_CPUAXG_R</td></tr><tr><td>1E89P</td><td>POWER_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td><td>REG_PHASE_CPUAXG</td></tr><tr><td>1E89P</td><td>VR_CTL_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td><td>REG_BOOT_CPUAXG</td></tr><tr><td>1E89P</td><td>VR_CTL_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td>TRUE</td><td>REG_BOOT_CPUAXG_RC</td></tr><tr><td>1E89P</td><td>VR_CTL_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td></td><td>REG_UGATE_CPUAXG</td></tr><tr><td>1E89P</td><td>VR_CTL_PHY</td><td>VR_LGATE</td><td>12V</td><td>TRUE</td><td></td><td>REG_LGATE_CPUAXG</td></tr><tr><td>1E89P</td><td>VR_CTL_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td>TRUE</td><td>REG_SNUBBER_CPUAXG</td></tr><tr><td>1E89P</td><td>POWER_PHY</td><td>POWER</td><td>1.1V</td><td></td><td></td><td>PPCPUAXG_S0_SENSE</td></tr><tr><td>1E89P</td><td>TSNS_CPU_AXG</td><td>SNS_DIFF_PHY</td><td>SENSE</td><td></td><td></td><td>REG_ISENAXG_P</td></tr><tr><td>1E89P</td><td>TSNS_CPU_AXG</td><td>SNS_DIFF_PHY</td><td>SENSE</td><td></td><td></td><td>REG_ISENAXG_N</td></tr><tr><td>1E89P</td><td></td><td>SENSE</td><td></td><td></td><td></td><td>REG_ISENAXG_PR</td></tr><tr><td>1E89P</td><td></td><td>SENSE</td><td></td><td></td><td></td><td>REG_ISENAXG_NR</td></tr><tr><td>1E89P</td><td>TSNS_CPU_AXG</td><td>SNS_DIFF_PHY</td><td>SENSE</td><td></td><td></td><td>SNS_AXG_R_P</td></tr><tr><td>1E89P</td><td>TSNS_CPU_AXG</td><td>SNS_DIFF_PHY</td><td>SENSE</td><td></td><td></td><td>SNS_AXG_R_N</td></tr><tr><td>1E89P</td><td>TSNS_CPU_AXG</td><td>SNS_DIFF_PHY</td><td>SENSE</td><td></td><td></td><td>SNS_AXG_XW_P</td></tr><tr><td>1E89P</td><td>TSNS_CPU_AXG</td><td>SNS_DIFF_PHY</td><td>SENSE</td><td></td><td></td><td>SNS_AXG_XW_N</td></tr><tr><td colspan="7">ISL6364</td></tr><tr><td>1E89P</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_CPUCORE_COMP</td></tr><tr><td>1E89P</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>CPUCORE_COMP_RC</td></tr><tr><td>1E89P</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_CPUCORE_FB</td></tr><tr><td>1E89P</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>CPUCORE_FB_RC</td></tr><tr><td>1E89P</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>CPUCORE_FB_R_1</td></tr><tr><td>1E89P</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>CPUCORE_FB_R_2</td></tr><tr><td>1E89P</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>CPUCORE_PSICOMP_RC</td></tr><tr><td>1E89P</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_CPUCORE_PSICOMP</td></tr><tr><td>1E89P</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_CPUCORE_HFCOMP</td></tr><tr><td>1E89P</td><td></td><td>SENSE</td><td></td><td></td><td></td><td>REG_CPUCORE_VSEN</td></tr><tr><td>1E89P</td><td></td><td>SENSE</td><td></td><td></td><td></td><td>REG_CPUCORE_RGND</td></tr><tr><td>1E89P</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_CPUCORE_IMON</td></tr><tr><td>1E89P</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>CPUCORE_IMON_R</td></tr><tr><td>1E89P</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_CPUCORE_TM</td></tr><tr><td>1E89P</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_CPUCORE_SUTH</td></tr><tr><td>1E89P</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_CPUCORE_NPSI</td></tr><tr><td>1E89P</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_CPUCORE_FVID</td></tr><tr><td>1E89P</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_CPUCORE_IAUTO</td></tr><tr><td>1E89P</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_CPUCORE_SW_FREQ</td></tr><tr><td>1E89P</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_CPUCORE_RAMPADJ</td></tr><tr><td>1E89P</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_CPUCORE_EN_PWR</td></tr><tr><td>1E89P</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>CPUCORE_EN_PWR_R</td></tr><tr><td>1E89P</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_CPUCORE_RSET</td></tr><tr><td>1E89P</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_CPUAXG_COMP</td></tr><tr><td>1E89P</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>CPUAXG_COMP_RC</td></tr><tr><td>1E89P</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_CPUAXG_FB</td></tr><tr><td>1E89P</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>CPUAXG_FB_RC</td></tr><tr><td>1E89P</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>CPUAXG_FB_R_1</td></tr><tr><td>1E89P</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>CPUAXG_FB_R_2</td></tr><tr><td>1E89P</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_CPUAXG_HFCOMP</td></tr><tr><td>1E89P</td><td></td><td>SENSE</td><td></td><td></td><td></td><td>REG_CPUAXG_VSEN</td></tr><tr><td>1E89P</td><td></td><td>SENSE</td><td></td><td></td><td></td><td>REG_CPUAXG_RGND</td></tr><tr><td>1E89P</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_CPUAXG_IMON</td></tr><tr><td>1E89P</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>CPUAXG_IMON_R</td></tr><tr><td>1E89P</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_CPUAXG_TM</td></tr><tr><td>1E89P</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_CPUAXG_TCOMP</td></tr><tr><td>1E89P</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td></td><td>REG_CPUAXG_SW_FREQ</td></tr><tr><td>1E89P</td><td>VR_VTD_PHY</td><td>VR_VTD</td><td></td><td></td><td></td><td>CPU_VIDSCLK</td></tr><tr><td>1E89P</td><td>VR_VTD_PHY</td><td>VR_VTD</td><td></td><td></td><td></td><td>CPU_VIDSCLK_R</td></tr><tr><td>1E89P</td><td>VR_VTD_PHY</td><td>VR_VTD</td><td></td><td></td><td></td><td>CPU_VIDALERT_L</td></tr><tr><td>1E89P</td><td>VR_VTD_PHY</td><td>VR_VTD</td><td></td><td></td><td></td><td>CPU_VIDALERT_R_L</td></tr><tr><td>1E89P</td><td>VR_VTD_PHY</td><td>VR_VTD</td><td></td><td></td><td></td><td>CPU_VIDSOUT</td></tr><tr><td>1E89P</td><td>VR_VTD_PHY</td><td>VR_VTD</td><td></td><td></td><td></td><td>CPU_VIDSOUT_R</td></tr><tr><td colspan="7">Output Bus</td></tr><tr><td>1E89P</td><td>POWER_PHY</td><td>POWER</td><td>1.1V</td><td></td><td></td><td>PPVCORE_S0_CPU</td></tr><tr><td>1E89P</td><td>POWER_PHY</td><td>POWER</td><td>1.1V</td><td></td><td></td><td>PPVAXG_S0</td></tr></table>	Electrical Constraint Set	Physical	Spacing	Voltage	DIDT	NO_TEST		AXG							1E88P	POWER_PHY	POWER	12V			REG_LVCC_U7330	1E89P	VR_CTL_PHY	VR_CTL				REG_PWM_CPUAXG	1E89P	VR_CTL_PHY	VR_CTL				REG_PWM_CPUAXG_R	1E89P	POWER_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_CPUAXG	1E89P	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_CPUAXG	1E89P	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT_CPUAXG_RC	1E89P	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_CPUAXG	1E89P	VR_CTL_PHY	VR_LGATE	12V	TRUE		REG_LGATE_CPUAXG	1E89P	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_SNUBBER_CPUAXG	1E89P	POWER_PHY	POWER	1.1V			PPCPUAXG_S0_SENSE	1E89P	TSNS_CPU_AXG	SNS_DIFF_PHY	SENSE			REG_ISENAXG_P	1E89P	TSNS_CPU_AXG	SNS_DIFF_PHY	SENSE			REG_ISENAXG_N	1E89P		SENSE				REG_ISENAXG_PR	1E89P		SENSE				REG_ISENAXG_NR	1E89P	TSNS_CPU_AXG	SNS_DIFF_PHY	SENSE			SNS_AXG_R_P	1E89P	TSNS_CPU_AXG	SNS_DIFF_PHY	SENSE			SNS_AXG_R_N	1E89P	TSNS_CPU_AXG	SNS_DIFF_PHY	SENSE			SNS_AXG_XW_P	1E89P	TSNS_CPU_AXG	SNS_DIFF_PHY	SENSE			SNS_AXG_XW_N	ISL6364							1E89P	VR_CTL_PHY	VR_CTL				REG_CPUCORE_COMP	1E89P	VR_CTL_PHY	VR_CTL				CPUCORE_COMP_RC	1E89P	VR_CTL_PHY	VR_CTL				REG_CPUCORE_FB	1E89P	VR_CTL_PHY	VR_CTL				CPUCORE_FB_RC	1E89P	VR_CTL_PHY	VR_CTL				CPUCORE_FB_R_1	1E89P	VR_CTL_PHY	VR_CTL				CPUCORE_FB_R_2	1E89P	VR_CTL_PHY	VR_CTL				CPUCORE_PSICOMP_RC	1E89P	VR_CTL_PHY	VR_CTL				REG_CPUCORE_PSICOMP	1E89P	VR_CTL_PHY	VR_CTL				REG_CPUCORE_HFCOMP	1E89P		SENSE				REG_CPUCORE_VSEN	1E89P		SENSE				REG_CPUCORE_RGND	1E89P	VR_CTL_PHY	VR_CTL				REG_CPUCORE_IMON	1E89P	VR_CTL_PHY	VR_CTL				CPUCORE_IMON_R	1E89P	VR_CTL_PHY	VR_CTL				REG_CPUCORE_TM	1E89P	VR_CTL_PHY	VR_CTL				REG_CPUCORE_SUTH	1E89P	VR_CTL_PHY	VR_CTL				REG_CPUCORE_NPSI	1E89P	VR_CTL_PHY	VR_CTL				REG_CPUCORE_FVID	1E89P	VR_CTL_PHY	VR_CTL				REG_CPUCORE_IAUTO	1E89P	VR_CTL_PHY	VR_CTL				REG_CPUCORE_SW_FREQ	1E89P	VR_CTL_PHY	VR_CTL				REG_CPUCORE_RAMPADJ	1E89P	VR_CTL_PHY	VR_CTL				REG_CPUCORE_EN_PWR	1E89P	VR_CTL_PHY	VR_CTL				CPUCORE_EN_PWR_R	1E89P	VR_CTL_PHY	VR_CTL				REG_CPUCORE_RSET	1E89P	VR_CTL_PHY	VR_CTL				REG_CPUAXG_COMP	1E89P	VR_CTL_PHY	VR_CTL				CPUAXG_COMP_RC	1E89P	VR_CTL_PHY	VR_CTL				REG_CPUAXG_FB	1E89P	VR_CTL_PHY	VR_CTL				CPUAXG_FB_RC	1E89P	VR_CTL_PHY	VR_CTL				CPUAXG_FB_R_1	1E89P	VR_CTL_PHY	VR_CTL				CPUAXG_FB_R_2	1E89P	VR_CTL_PHY	VR_CTL				REG_CPUAXG_HFCOMP	1E89P		SENSE				REG_CPUAXG_VSEN	1E89P		SENSE				REG_CPUAXG_RGND	1E89P	VR_CTL_PHY	VR_CTL				REG_CPUAXG_IMON	1E89P	VR_CTL_PHY	VR_CTL				CPUAXG_IMON_R	1E89P	VR_CTL_PHY	VR_CTL				REG_CPUAXG_TM	1E89P	VR_CTL_PHY	VR_CTL				REG_CPUAXG_TCOMP	1E89P	VR_CTL_PHY	VR_CTL				REG_CPUAXG_SW_FREQ	1E89P	VR_VTD_PHY	VR_VTD				CPU_VIDSCLK	1E89P	VR_VTD_PHY	VR_VTD				CPU_VIDSCLK_R	1E89P	VR_VTD_PHY	VR_VTD				CPU_VIDALERT_L	1E89P	VR_VTD_PHY	VR_VTD				CPU_VIDALERT_R_L	1E89P	VR_VTD_PHY	VR_VTD				CPU_VIDSOUT	1E89P	VR_VTD_PHY	VR_VTD				CPU_VIDSOUT_R	Output Bus							1E89P	POWER_PHY	POWER	1.1V			PPVCORE_S0_CPU	1E89P	POWER_PHY	POWER	1.1V			PPVAXG_S0
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1E89P	VR_CTL_PHY	VR_CTL				REG_CPUAXG_COMP																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E89P	VR_CTL_PHY	VR_CTL				CPUAXG_COMP_RC																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
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1E89P	VR_CTL_PHY	VR_CTL				CPUAXG_FB_R_1																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E89P	VR_CTL_PHY	VR_CTL				CPUAXG_FB_R_2																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E89P	VR_CTL_PHY	VR_CTL				REG_CPUAXG_HFCOMP																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E89P		SENSE				REG_CPUAXG_VSEN																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E89P		SENSE				REG_CPUAXG_RGND																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
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1E89P	VR_CTL_PHY	VR_CTL				REG_CPUAXG_SW_FREQ																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E89P	VR_VTD_PHY	VR_VTD				CPU_VIDSCLK																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
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1E89P	VR_VTD_PHY	VR_VTD				CPU_VIDALERT_L																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
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1E89P	VR_VTD_PHY	VR_VTD				CPU_VIDSOUT																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
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1E89P	POWER_PHY	POWER	1.1V			PPVCORE_S0_CPU																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
1E89P	POWER_PHY	POWER	1.1V			PPVAXG_S0																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														

SYNC_MASTER=D8_MLB

SYNC_DATE=07/03/2012

PAGE TITLE

CPU VReg Constraints

 Apple Inc.

DRAWING NUMBER
051-9505

REVISION
6.1.0

BRANCH

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3.3V S5/5V S4

	Physical	Spacing	Voltage	D1D2	NO_TEST	
Input Bus						
H065	POWER_PHY	POWER	12V			REG_VIN_U7600
H125	POWER_PHY	POWER	5V			REG_VCC1_U7600
H135	POWER_PHY	POWER	5V			REG_VCC2_U7600
3.3V S5						
H035	POWER_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_P3V3S5
H075	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_B00T_P3V3S5
H085	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_B00T_P3V3S5_RC
H045	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_P3V3S5
H055	VR_CTL_PHY	VR_IGATE	12V	TRUE		REG_LGATE_P3V3S5
H095	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_SNUBBER_P3V3S5
H107	VR_CTL_PHY	VR_CTL				REG_P3V3S5_ISEN
H107	VR_CTL_PHY	VR_CTL				REG_P3V3S5_OCSET
H098	VR_CTL_PHY	VR_CTL				REG_P3V3S5_FSET
H098	VR_CTL_PHY	VR_CTL				REG_P3V3S5_VOUT
H109	VR_CTL_PHY	VR_CTL				REG_P3V3S5_VOUT_R
H109	VR_CTL_PHY	VR_CTL				REG_P3V3S5_FB
5V S3						
H065	POWER_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_P5VS4
H125	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_B00T_P5VS4
H135	VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_B00T_P5VS4_RC
H045	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_P5VS4
H055	VR_CTL_PHY	VR_IGATE	12V	TRUE		REG_LGATE_P5VS4
H095	VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_SNUBBER_P5VS4
H107	VR_CTL_PHY	VR_CTL				REG_P5VS4_ISEN
H107	VR_CTL_PHY	VR_CTL				REG_P5VS4_OCSET
H098	VR_CTL_PHY	VR_CTL				REG_P5VS4_FSET
H098	VR_CTL_PHY	VR_CTL				REG_P5VS4_VOUT
H109	VR_CTL_PHY	VR_CTL				REG_P5VS4_VOUT_R
H109	VR_CTL_PHY	VR_CTL				REG_P5VS4_FB
Output Bus						
H035	POWER_PHY	POWER	5V			PP5V_S5
H125	POWER_PHY	POWER	5V			PP5V_S4
H135	POWER_PHY	POWER	3.3V			PP3V3_S5
FET Switched						
H122	POWER_PHY	POWER	5V			PP5V_S0
H129	POWER_PHY	POWER	3.3V			PP3V3_S4
H132	POWER_PHY	POWER	3.3V			PP3V3_S0
H132	POWER_PHY	POWER	3.3V			PP3V3_S4_ENET
H135	POWER_PHY	POWER	3.3V			PP3V3_TBTLIC
H095	POWER_PHY	POWER	3.3V			PP3V3_S4_AP
H107	POWER_PHY	POWER	3.3V			PP3V3_S0_SSD

VDDQ S3 (1.5V)/VTT S0

Physical	Spacing	Voltage	DIDT	NO_TEST	
Input Bus					
H139 POWER_PHY	POWER	5V			REG_V5IN_U7700 72
Local Ground					
H139 GND_PHY	GND	0V			AGND_VDDQ53 73
VDDQ S3					
H139 POWER_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_VDDQ53 72
H139 POWER_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_VDDQ53_L 72
H139 VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_B00T_VDDQ53 72
H139 VR_CTL_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_B00T_VDDQ53_RC 72
H139 VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_VDDQ53 72
H139 VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_VDDQ53_R 72
H139 VR_CTL_PHY	VR_LGATE	12V	TRUE		REG_LGATE_VDDQ53 72
H139 VR_CTL_PHY	VR_SWITCH	12V	TRUE		REG_SNUBBER_VDDQ53 72
H139 POWER_PHY	POWER	1.5V			PPVDDQ_S3_SENSE 72
H134 VR_CTL_PHY	VR_CTL				REG_VDDQ53_VDDQ5NS 72
H139 VR_CTL_PHY	VR_CTL				REG_VDDQ53_VREF 72
H139 VR_CTL_PHY	VR_CTL				REG_VDDQ53_REFIN 72
H139 VR_CTL_PHY	VR_CTL				REG_VDDQ53_MODE 72
H139 VR_CTL_PHY	VR_CTL				REG_VDDQ53_TRIP 72
H139 VR_CTL_PHY	VR_CTL				LDO_DDRVITS0_SNS 72
Output Bus					
H139 POWER_PHY	POWER	1.5V			PPVDDQ_S3 6
H139 POWER_DDR_PHY	POWER_DDR	0.75V			PPDDRVTT_S0 6
FET Switched					
H139 POWER_PHY	POWER	1.5V			PP1V5_S0 6
Sensed					
H139 POWER_PHY	POWER	1.5V			PPVDDQ_S3_DDR 6
H139 POWER_PHY	POWER	1.5V			PP1V5_S0_CPU_MEM 6
H139 POWER_PHY	POWER	1.5V			PPFBVDDQ_S0_GPU 6

DDR3 Vref

	Physical	Spacing	Voltage	DIDT	NO_TEST	
Memory Vref						
PP3V3	POWER_PHY	POWER	3.3V			PP3V3 S4 VREFMRGN DAC
PP3V3	POWER_PHY	POWER	3.3V			PP3V3 S4 VREFMRGN CTRL
PPDDR	POWER_DDR_PHY	POWER_DDR	0.75V			PPDDR VREF DQ MEM A S3
PPDDR	POWER_DDR_PHY	POWER_DDR	0.75V			PPDDR VREF DQ MEM B S3
CPU	POWER_DDR_PHY	POWER_DDR	0.75V			CPU DIMM VREF DAC A
CPU	POWER_DDR_PHY	POWER_DDR	0.75V			CPU DIMM VREF DAC B
PPDDR	POWER_DDR_PHY	POWER_DDR	0.75V			PPDDR VREF CA MEM A S3
PPDDR	POWER_DDR_PHY	POWER_DDR	0.75V			PPDDR VREF CA MEM B S3
CPU	POWER_DDR_PHY	POWER_DDR	0.75V			CPU DDR VREF
PPDDR	POWER_DDR_PHY	POWER_DDR	0.75V			PPDDR VTT S3

1.8V S0

Physical		Spacing	Voltage	DIDT	NO_TEST	
1.8V S0						
 POWER_PHY	VR_SWITCH	5V	TRUE		REG_PHASE_P1V8S0	7
 VR_CTL_PHY	VR_CTL				REG_P1V8S0_VFB	7
 VR_CTL_PHY	VR_CTL				REG_P1V8S0_SYNC	7
Output Bus						
 POWER_PHY	POWER	1.8V			PP1V8_S0	6

HDD S0

Physical	Spacing	Voltage	IDIT	NO_TEST	
HDD S0 POWER_PHY	POWER	5V			PP5V_S0_HDD 0

12V S5

Physical	Spacing	Voltage	DIOT	NO_TEST
Input Bus				
 POWER_PHY	POWER	12V		PP12V ACDC
FET Switched				
 POWER_PHY	POWER	12V		PP12V S0
Sensed				
 POWER_PHY	POWER	12V		PP12V S5
 POWER_PHY	POWER	12V		PP12V G3H
 POWER_PHY	POWER	12V		PP12V G3H P3V42
 POWER_PHY	POWER	12V		PP12V S0 FBVDDQ
 POWER_PHY	POWER	12V		PP12V S0 CPU P1V05
 POWER_PHY	POWER	12V		PP12V S0 VCCSA
 POWER_PHY	POWER	12V		PP12V S0 P1V05
 POWER_PHY	POWER	12V		PP12V S0 HDD
 POWER_PHY	POWER	12V		PP12V S0 BLC

Ground/Common

Physical	Spacing	Voltage	IDT	NO_TEST	
Common					
1820 GND_PHY	GND	6V			GND

3V42 S0

Physical	Spacing	Voltage	IDT	NO_TEST	
3V42 S0					
TEST POWER_PHY	VR_SWITCH	5V	TRUE		P3V42G3H_SW 7
TEST VR_CTL_PHY	VR_CTL				P3V42G3H_FB 7
TEST VR_CTL_PHY	VR_CTL				P3V42G3H_SHDN_L 7
TEST POWER_PHY	POWER	3.3V			PP3V3_G3 6
TEST POWER_PHY	POWER	3.3V			PP3V42_G3H 6

GDDR5

GDDR5-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
GDDR_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD
GDDR_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	12 MM	=STANDARD	=STANDARD
GDDR_80D	*	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
GDDR_MA_PHY	*	GDDR_45S
GDDR_ADBI_PHY	*	GDDR_45S
GDDR_CTRL_PHY	*	GDDR_45S
GDDR_CLK_PHY	*	GDDR_80D
GDDR_DQ_PHY	*	GDDR_45S
GDDR_EDC_PHY	*	GDDR_45S
GDDR_DBI_PHY	*	GDDR_45S
GDDR_WCK_PHY	*	GDDR_80D

Main Segment Min Spacing Rules for 4.5 Gbps or Less (AMD Doc# 49919)

[illegible]

GDDR5-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GDDR_ISO	*	=3X_DIELECTRIC	?
GDDR_ISO	TOP, BOTTOM	=5x_DIELECTRIC	?
GDDR_MA2MA	*	=2x_DIELECTRIC	?
GDDR_MA2MA	TOP, BOTTOM	=3X_DIELECTRIC	?
GDDR_ADBI2ADBI	*	=2x_DIELECTRIC	?
GDDR_ADBI2ADBI	TOP, BOTTOM	=2x_DIELECTRIC	?
GDDR_CTRL2CTRL	*	=2x_DIELECTRIC	?
GDDR_CTRL2CTRL	TOP, BOTTOM	=2x_DIELECTRIC	?
GDDR_CLK2CLK	*	=3X_DIELECTRIC	?
GDDR_CLK2CLK	TOP, BOTTOM	=5x_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GDDR_DQ2DQ	*	=3x_DIELECTRIC	?
GDDR_DQ2DQ	TOP, BOTTOM	=3x_DIELECTRIC	?
GDDR_EDC_ISO	*	=3X_DIELECTRIC	?
GDDR_EDC_ISO	TOP, BOTTOM	=5X_DIELECTRIC	?
GDDR_EDC2EDC	*	=3X_DIELECTRIC	?
GDDR_EDC2EDC	TOP, BOTTOM	=5X_DIELECTRIC	?
GDDR_DBI2DBI	*	=3x_DIELECTRIC	?
GDDR_DBI2DBI	TOP, BOTTOM	=3x_DIELECTRIC	?
GDDR_WCK2WCK	*	=3X_DIELECTRIC	?
GDDR_WCK2WCK	TOP, BOTTOM	=5x_DIELECTRIC	?

Constraints ($x \in \{A, B\}, y \in \{0, 1\}$)

Memory Address: M_{Axy}[8:0]

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GDDR_*_*_MA	*	*	GDDR_ISO
GDDR_*_*_MA	=SAME	*	GDDR_MA2MA

Address Dynamic Bus Inversion: ADBIx

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GDDR_*_*_ADBI	*	*	GDDR_ISO
GDDR_*_*_ADBI	=SAME	*	GDDR_ADBI2ADBI

Control: Reset, CKExy, CSxy, WExy, RASxy, CASxy

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GDDR_CTRL	*	*	GDDR_ISO
GDDR_*_*_CTRL	*	*	GDDR_ISO
GDDR_*_*_CTRL	=SAME	*	GDDR_CTRL2CTRL

Clock: CKxy

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GDDR_*_*_CLK	*	*	GDDR_ISO
GDDR_*_*_CLK	=SAME	*	GDDR_CLK2CLK

GPU

GPU-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CLK_GPU_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

GPU-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_GPU_ISO	*	=4:1_SPACING	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CLK_GPU	*	*	CLK_GPU_ISO

GDDR5 Frame Buffer A

Electrical Constraint Set	Physical	Spacing		
Memory Address				
MEM001 GDDR A0 MA	GDDR_MA_PHY	GDDR_A_0_MA	FB A0 A<8..0>	NO TEST=TRUE 04 06
MEM002 GDDR A1 MA	GDDR_MA_PHY	GDDR_A_1_MA	FB A1 A<8..0>	NO TEST=TRUE 04 06
Address Dynamic Bus Inv				
FE001 GDDR A0 ADBT	GDDR_ADBT_PHY	GDDR_A_0_ADBT	FB A0 ABT L	NO TEST=TRUE 04 06
FE002 GDDR A1 ADBT	GDDR_ADBT_PHY	GDDR_A_1_ADBT	FB A1 ABT L	NO TEST=TRUE 04 06
Control				
FE003 GDDR A0 CKE	GDDR_CTRL_PHY	GDDR_A_0_CTRL	FB A0 CKE L	NO TEST=TRUE 04 06
FE004 GDDR A0 CTBI	GDDR_CTRL_PHY	GDDR_A_0_CTBI	FB A0 CS L	NO TEST=TRUE 04 06
FE005 GDDR A0 CTBI	GDDR_CTRL_PHY	GDDR_A_0_CTBI	FB A0 WE L	NO TEST=TRUE 04 06
FE006 GDDR A0 CTBI	GDDR_CTRL_PHY	GDDR_A_0_CTBI	FB A0 CAS L	NO TEST=TRUE 04 06
FE007 GDDR A0 CTBI	GDDR_CTRL_PHY	GDDR_A_0_CTBI	FB A0 RAS L	NO TEST=TRUE 04 06
FE008 GDDR A1 CKE	GDDR_CTRL_PHY	GDDR_A_1_CTRL	FB A1 CKE L	NO TEST=TRUE 04 06
FE009 GDDR A1 CTBI	GDDR_CTRL_PHY	GDDR_A_1_CTBI	FB A1 CS L	NO TEST=TRUE 04 06
FE010 GDDR A1 CTBI	GDDR_CTRL_PHY	GDDR_A_1_CTBI	FB A1 WE L	NO TEST=TRUE 04 06
FE011 GDDR A1 CTBI	GDDR_CTRL_PHY	GDDR_A_1_CTBI	FB A1 CAS L	NO TEST=TRUE 04 06
FE012 GDDR A1 CTBI	GDDR_CTRL_PHY	GDDR_A_1_CTBI	FB A1 RAS L	NO TEST=TRUE 04 06
Clock				
FE013 GDDR A0 CLK	GDDR_CLK_PHY	GDDR_A_0_CLK	FB A0 CLK P	NO TEST=TRUE 04 06
FE014 GDDR A0 CLK	GDDR_CLK_PHY	GDDR_A_0_CLK	FB A0 CLK N	NO TEST=TRUE 04 06
FE015 GDDR A1 CLK	GDDR_CLK_PHY	GDDR_A_1_CLK	FB A1 CLK P	NO TEST=TRUE 04 06
FE016 GDDR A1 CLK	GDDR_CLK_PHY	GDDR_A_1_CLK	FB A1 CLK N	NO TEST=TRUE 04 06
Data				
FE017 GDDR A0 DQ BYTE0	GDDR_DQ_PHY	GDDR_A_0_DQ	FB A0 DQ<7..0>	NO TEST=TRUE 04 06
FE018 GDDR A0 DQ BYTE1	GDDR_DQ_PHY	GDDR_A_0_DQ	FB A0 DQ<15..8>	NO TEST=TRUE 04 06
FE019 GDDR A0 DQ BYTE2	GDDR_DQ_PHY	GDDR_A_0_DQ	FB A0 DQ<23..16>	NO TEST=TRUE 04 06
FE020 GDDR A0 DQ BYTE3	GDDR_DQ_PHY	GDDR_A_0_DQ	FB A0 DQ<31..24>	NO TEST=TRUE 04 06
FE021 GDDR A1 DQ BYTE0	GDDR_DQ_PHY	GDDR_A_1_DQ	FB A1 DQ<7..0>	NO TEST=TRUE 04 06
FE022 GDDR A1 DQ BYTE1	GDDR_DQ_PHY	GDDR_A_1_DQ	FB A1 DQ<15..8>	NO TEST=TRUE 04 06
FE023 GDDR A1 DQ BYTE2	GDDR_DQ_PHY	GDDR_A_1_DQ	FB A1 DQ<23..16>	NO TEST=TRUE 04 06
FE024 GDDR A1 DQ BYTE3	GDDR_DQ_PHY	GDDR_A_1_DQ	FB A1 DQ<31..24>	NO TEST=TRUE 04 06
Error Detection				
FE025 GDDR A0 EDC0	GDDR_EDC_PHY	GDDR_A_0_EDC	FB A0 EDC<0>	NO TEST=TRUE 04 06
FE026 GDDR A0 EDC1	GDDR_EDC_PHY	GDDR_A_0_EDC	FB A0 EDC<1>	NO TEST=TRUE 04 06
FE027 GDDR A0 EDC2	GDDR_EDC_PHY	GDDR_A_0_EDC	FB A0 EDC<2>	NO TEST=TRUE 04 06
FE028 GDDR A0 EDC3	GDDR_EDC_PHY	GDDR_A_0_EDC	FB A0 EDC<3>	NO TEST=TRUE 04 06
FE029 GDDR A1 EDC0	GDDR_EDC_PHY	GDDR_A_1_EDC	FB A1 EDC<0>	NO TEST=TRUE 04 06
FE030 GDDR A1 EDC1	GDDR_EDC_PHY	GDDR_A_1_EDC	FB A1 EDC<1>	NO TEST=TRUE 04 06
FE031 GDDR A1 EDC2	GDDR_EDC_PHY	GDDR_A_1_EDC	FB A1 EDC<2>	NO TEST=TRUE 04 06
FE032 GDDR A1 EDC3	GDDR_EDC_PHY	GDDR_A_1_EDC	FB A1 EDC<3>	NO TEST=TRUE 04 06
Data Dynamic Bus Inv				
FE033 GDDR A0 DBT0	GDDR_DBT_PHY	GDDR_A_0_DBT	FB A0 DBT L<0>	NO TEST=TRUE 04 06
FE034 GDDR A0 DBT1	GDDR_DBT_PHY	GDDR_A_0_DBT	FB A0 DBT L<1>	NO TEST=TRUE 04 06
FE035 GDDR A0 DBT2	GDDR_DBT_PHY	GDDR_A_0_DBT	FB A0 DBT L<2>	NO TEST=TRUE 04 06
FE036 GDDR A0 DBT3	GDDR_DBT_PHY	GDDR_A_0_DBT	FB A0 DBT L<3>	NO TEST=TRUE 04 06
FE037 GDDR A1 DBT0	GDDR_DBT_PHY	GDDR_A_1_DBT	FB A1 DBT L<0>	NO TEST=TRUE 04 06
FE038 GDDR A1 DBT1	GDDR_DBT_PHY	GDDR_A_1_DBT	FB A1 DBT L<1>	NO TEST=TRUE 04 06
FE039 GDDR A1 DBT2	GDDR_DBT_PHY	GDDR_A_1_DBT	FB A1 DBT L<2>	NO TEST=TRUE 04 06
FE040 GDDR A1 DBT3	GDDR_DBT_PHY	GDDR_A_1_DBT	FB A1 DBT L<3>	NO TEST=TRUE 04 06
Forwarded Clock				
FE041 GDDR A0 WCK0	GDDR_WCK_PHY	GDDR_A_0_WCK	FB A0 WCLK P<0>	NO TEST=TRUE 04 06
FE042 GDDR A0 WCK0	GDDR_WCK_PHY	GDDR_A_0_WCK	FB A0 WCLK N<0>	NO TEST=TRUE 04 06
FE043 GDDR A0 WCK1	GDDR_WCK_PHY	GDDR_A_0_WCK	FB A0 WCLK P<1>	NO TEST=TRUE 04 06
FE044 GDDR A0 WCK1	GDDR_WCK_PHY	GDDR_A_0_WCK	FB A0 WCLK N<1>	NO TEST=TRUE 04 06
FE045 GDDR A1 WCK0	GDDR_WCK_PHY	GDDR_A_1_WCK	FB A1 WCLK P<0>	NO TEST=TRUE 04 06
FE046 GDDR A1 WCK0	GDDR_WCK_PHY	GDDR_A_1_WCK	FB A1 WCLK N<0>	NO TEST=TRUE 04 06
FE047 GDDR A1 WCK1	GDDR_WCK_PHY	GDDR_A_1_WCK	FB A1 WCLK P<1>	NO TEST=TRUE 04 06
FE048 GDDR A1 WCK1	GDDR_WCK_PHY	GDDR_A_1_WCK	FB A1 WCLK N<1>	NO TEST=TRUE 04 06

GPU

Electrical Constraint Set	Physical	Spacing	
Clocks			
PEX0	CLK_GPU_55S	CLK_GPU	PEX_TSTCLK_0_PL 83
PEX0	CLK_GPU_55S	CLK_GPU	PEX_TSTCLK_0_NG 83
PEX0	CLK_PCTE_PHY	CLK_PCTE	GPU_TESTMODE 83
PEX0	CLK_PCTE_PHY	CLK_PCTE	GPU_PEX_TERM 83
PEX0	CLK_GPU_55S	CLK_GPU	FB_A0_CK_MID 86
PEX0	CLK_GPU_55S	CLK_GPU	FB_A1_CK_MID NO_TEST=TRUE 86
PEX0	CLK_GPU_55S	CLK_GPU	FB_B0_CK_MID NO_TEST=TRUE 87
PEX0	CLK_GPU_55S	CLK_GPU	FB_B1_CK_MID NO_TEST=TRUE 87
PEX0	CLK_GPU_55S	CLK_GPU	FB_C0_CK_MID NO_TEST=TRUE 88
PEX0	CLK_GPU_55S	CLK_GPU	FB_C1_CK_MID NO_TEST=TRUE 88
PEX0	CLK_GPU_55S	CLK_GPU	FB_D0_CK_MID NO_TEST=TRUE 89
PEX0	CLK_GPU_55S	CLK_GPU	FB_D1_CK_MID NO_TEST=TRUE 89
PEX0	CLK_GPU_55S	CLK_GPU	GPU_JTAG_TCK 91
PEX0	CLK_GPU_55S	CLK_GPU	GPU_ROM_SCLK 91
PEX0	CLK_GPU_55S	CLK_GPU	GPU_ROM_SCLK_R 91
PEX0	CLK_GPU_55S	CLK_GPU	GPU_OSC_27M_XTAL_P 91
PEX0	CLK_GPU_55S	CLK_GPU	GPU_OSC_27M_XTAL_N 91
SMB			
ES00	SMB_PHY	SMB	GPU_SMB_CLK 91
ES00	SMB_PHY	SMB	GPU_SMB_DAT 91
ES00	SMB_PHY	SMB	GPU_SMB_CLK_R 508 91
ES00	SMB_PHY	SMB	GPU_SMB_DAT_R 508 91
PCIe Compensation			
ES00	PCIE_S0S	COMP_PCIE	FB_CAL_PD_VDDQ 94
ES00	PCIE_S0S	COMP_PCIE	FB_CAL_PU_GND 94
ES00	PCIE_S0S	COMP_PCIE	FB_CAL_TERM_GND 94

GDDR5 Frame Buffer B

Electrical Constraint Set	Physical	Spacing		
Memory Address				
H509 GDDR B0 MA	GDDR MA PHY	GDDR B 0 MA	FB B0 A<8..0>	NO TEST-TRUE
H599 GDDR B1 MA	GDDR MA PHY	GDDR B 1 MA	FB B1 A<8..0>	NO TEST-TRUE
Address Dynamic Bus Inv				
H509 GDDR B0 ADBI	GDDR ADBI PHY	GDDR B 0 ADBI	FB B0 ABI L	NO TEST-TRUE
H599 GDDR B1 ADBI	GDDR ADBI PHY	GDDR B 1 ADBI	FB B1 ABI L	NO TEST-TRUE
Control				
H509 GDDR B0_CKE	GDDR CTRL PHY	GDDR B 0 CTRL	FB B0 CKE L	NO TEST-TRUE
H509 GDDR B0_CTRL	GDDR_CTRL PHY	GDDR B 0_CTRL	FB B0 CS L	NO TEST-TRUE
H509 GDDR B0_CTRL	GDDR_CTRL PHY	GDDR B 0_CTRL	FB B0 WE L	NO TEST-TRUE
H509 GDDR B0_CTRL	GDDR_CTRL PHY	GDDR B 0_CTRL	FB B0 CAS L	NO TEST-TRUE
H509 GDDR B0_CTRL	GDDR_CTRL PHY	GDDR B 0_CTRL	FB B0 RAS L	NO TEST-TRUE
H599 GDDR B1_CKE	GDDR_CTRL PHY	GDDR B 1_CTRL	FB B1 CKE L	NO TEST-TRUE
H599 GDDR B1_CTRL	GDDR_CTRL PHY	GDDR B 1_CTRL	FB B1 CS L	NO TEST-TRUE
H599 GDDR B1_CTRL	GDDR_CTRL PHY	GDDR B 1_CTRL	FB B1 WE L	NO TEST-TRUE
H599 GDDR B1_CTRL	GDDR_CTRL PHY	GDDR B 1_CTRL	FB B1 CAS L	NO TEST-TRUE
H599 GDDR B1_CTRL	GDDR_CTRL PHY	GDDR B 1_CTRL	FB B1 RAS L	NO TEST-TRUE
Clock				
H509 GDDR B0_CLK	GDDR_CLK PHY	GDDR B 0_CLK	FB B0_CLK P	NO TEST-TRUE
H599 GDDR B0_CLK	GDDR_CLK PHY	GDDR B 0_CLK	FB B0_CLK N	NO TEST-TRUE
H509 GDDR B1_CLK	GDDR_CLK PHY	GDDR B 1_CLK	FB B1_CLK P	NO TEST-TRUE
H599 GDDR B1_CLK	GDDR_CLK PHY	GDDR B 1_CLK	FB B1_CLK N	NO TEST-TRUE
Data				
H509 GDDR B0_DQ_BYTE0	GDDR_DQ PHY	GDDR B 0_DQ	FB B0_DQ<7..0>	NO TEST-TRUE
H509 GDDR B0_DQ_BYTE1	GDDR_DQ PHY	GDDR B 0_DQ	FB B0_DQ<15..8>	NO TEST-TRUE
H509 GDDR B0_DQ_BYTE2	GDDR_DQ PHY	GDDR B 0_DQ	FB B0_DQ<23..16>	NO TEST-TRUE
H509 GDDR B0_DQ_BYTE3	GDDR_DQ PHY	GDDR B 0_DQ	FB B0_DQ<31..24>	NO TEST-TRUE
H599 GDDR B1_DQ_BYTE0	GDDR_DQ PHY	GDDR B 1_DQ	FB B1_DQ<7..0>	NO TEST-TRUE
H599 GDDR B1_DQ_BYTE1	GDDR_DQ PHY	GDDR B 1_DQ	FB B1_DQ<15..8>	NO TEST-TRUE
H599 GDDR B1_DQ_BYTE2	GDDR_DQ PHY	GDDR B 1_DQ	FB B1_DQ<23..16>	NO TEST-TRUE
H599 GDDR B1_DQ_BYTE3	GDDR_DQ PHY	GDDR B 1_DQ	FB B1_DQ<31..24>	NO TEST-TRUE
Error Detection				
H520 GDDR B0_EDC0	GDDR_EDC PHY	GDDR B 0_EDC	FB B0_EDC<0>	NO TEST-TRUE
H520 GDDR B0_EDC1	GDDR_EDC PHY	GDDR B 0_EDC	FB B0_EDC<1>	NO TEST-TRUE
H520 GDDR B0_EDC2	GDDR_EDC PHY	GDDR B 0_EDC	FB B0_EDC<2>	NO TEST-TRUE
H520 GDDR B0_EDC3	GDDR_EDC PHY	GDDR B 0_EDC	FB B0_EDC<3>	NO TEST-TRUE
H520 GDDR B1_EDC0	GDDR_EDC PHY	GDDR B 1_EDC	FB B1_EDC<0>	NO TEST-TRUE
H520 GDDR B1_EDC1	GDDR_EDC PHY	GDDR B 1_EDC	FB B1_EDC<1>	NO TEST-TRUE
H520 GDDR B1_EDC2	GDDR_EDC PHY	GDDR B 1_EDC	FB B1_EDC<2>	NO TEST-TRUE
H520 GDDR B1_EDC3	GDDR_EDC PHY	GDDR B 1_EDC	FB B1_EDC<3>	NO TEST-TRUE
Data Dynamic Bus Inv				
H520 GDDR B0_DBI0	GDDR_DBI PHY	GDDR B 0_DBI	FB B0_DBI L<0>	NO TEST-TRUE
H520 GDDR B0_DBI1	GDDR_DBI PHY	GDDR B 0_DBI	FB B0_DBI L<1>	NO TEST-TRUE
H520 GDDR B0_DBI2	GDDR_DBI PHY	GDDR B 0_DBI	FB B0_DBI L<2>	NO TEST-TRUE
H520 GDDR B0_DBI3	GDDR_DBI PHY	GDDR B 0_DBI	FB B0_DBI L<3>	NO TEST-TRUE
H520 GDDR B1_DBI0	GDDR_DBI PHY	GDDR B 1_DBI	FB B1_DBI L<0>	NO TEST-TRUE
H520 GDDR B1_DBI1	GDDR_DBI PHY	GDDR B 1_DBI	FB B1_DBI L<1>	NO TEST-TRUE
H520 GDDR B1_DBI2	GDDR_DBI PHY	GDDR B 1_DBI	FB B1_DBI L<2>	NO TEST-TRUE
H520 GDDR B1_DBI3	GDDR_DBI PHY	GDDR B 1_DBI	FB B1_DBI L<3>	NO TEST-TRUE
Forwarded Clock				
H509 GDDR B0_WCK0	GDDR_WCK PHY	GDDR B 0_WCK	FB B0_WCK P<0>	NO TEST-TRUE
H509 GDDR B0_WCK0	GDDR_WCK PHY	GDDR B 0_WCK	FB B0_WCK N<0>	NO TEST-TRUE
H509 GDDR B0_WCK1	GDDR_WCK PHY	GDDR B 0_WCK	FB B0_WCK P<1>	NO TEST-TRUE
H509 GDDR B0_WCK1	GDDR_WCK PHY	GDDR B 0_WCK	FB B0_WCK N<1>	NO TEST-TRUE
H599 GDDR B1_WCK0	GDDR_WCK PHY	GDDR B 1_WCK	FB B1_WCK P<0>	NO TEST-TRUE
H599 GDDR B1_WCK0	GDDR_WCK PHY	GDDR B 1_WCK	FB B1_WCK N<0>	NO TEST-TRUE
H599 GDDR B1_WCK1	GDDR_WCK PHY	GDDR B 1_WCK	FB B1_WCK P<1>	NO TEST-TRUE
H599 GDDR B1_WCK1	GDDR_WCK PHY	GDDR B 1_WCK	FB B1_WCK N<1>	NO TEST-TRUE

Frame Buffer Reset

Electrical Constraint Set	Physical	Spacing	
Reset			
H045 GDDR_A0_RESET	GDDR_S0S	GDDR_CTLR1	FB_A0 RESET L
H046 GDDR_A1_RESET	GDDR_S0S	GDDR_CTLR1	FB_A1 RESET L
H047 GDDR_B0_RESET	GDDR_S0S	GDDR_CTLR1	FB_B0 RESET L
H048 GDDR_B1_RESET	GDDR_S0S	GDDR_CTLR1	FB_B1 RESET L
H049 GDDR_C0_RESET	GDDR_S0S	GDDR_CTLR1	FB_C0 RESET L
H050 GDDR_C1_RESET	GDDR_S0S	GDDR_CTLR1	FB_C1 RESET L
H051 GDDR_D0_RESET	GDDR_S0S	GDDR_CTLR1	FB_D0 RESET L
H052 GDDR_D1_RESET	GDDR_S0S	GDDR_CTLR1	FB_D1 RESET L

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C

B

A

Constraints

BLC High Voltage Output

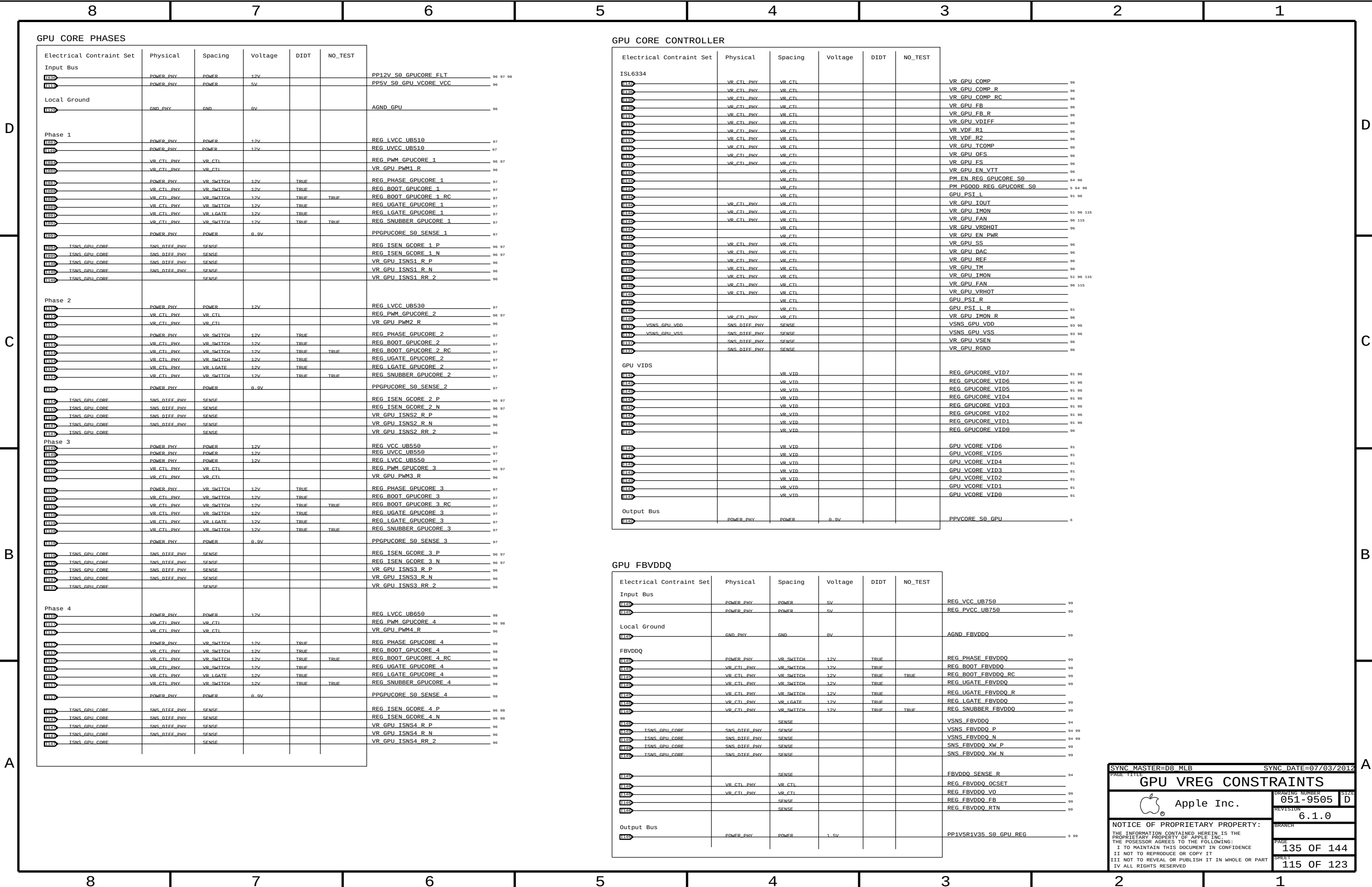
NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
BLC_HV	BLC_HV	*	BLC_CTL_ISO
BLC_HV	*	*	BLC_HV_ISO

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
BLC_PHASE	*	*	PHASE_ISO
BLC_PHASE	BLC_PHASE	*	PHASE_SW2SW
BLC_PHASE	POWER	*	PHASE_SW2PWR
BLC_PHASE	GND	*	PHASE_SW2GND

BLC Control			
NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
BLC_CTL	*	*	BLC_CTL_ISO

Physical		Spacing	Voltage	DI0T	NO_TEST		
Input	Bus						
H760	POWER_PHY	POWER	12V			PP12V S0 BLC VIN2	00 01 02
H770	POWER_PHY	POWER	12V			PP12V S0 BLC VINP	00 02
H800	POWER_PHY	POWER	14V			PRE REG_OUT	00 01
H800	POWER_PHY	POWER	3.3V			BLC P3V3S	00 02
H800	POWER_PHY	POWER	3.3V			BLC P3V3 REF	00 02
H800	POWER_PHY	POWER	3.3V			BLC P3V3	00 02
H860	POWER_PHY	POWER	3.3V			PP3V3 S0 BLC R	00
H840	POWER_PHY	POWER	0V			SPTX VIN	01
H810	POWER_PHY	POWER	0V			PP0V BLC	00 01
H810	POWER_PHY	POWER	0V			BLC VIN2	01 02
H800	POWER_PHY	POWER	14V			B00ST_FET_DRAIN	00
H800	POWER_PHY	POWER	12V			B00ST_VDD	00
H800	POWER_PHY	POWER	0V			PP5V S0 BLC R	00 01 02
H800	POWER_PHY	POWER	12V			PP12V S0 BLC F	02
Local Ground							
H790	BLC_CTL_PHY	BLC_PHASE	0V			BLC_GND_1	01 02
H800	BLC_CTL_PHY	BLC_PHASE	0V			BLC_GND_2	01 02
H800	BLC_CTL_PHY	BLC_PHASE	0V			BLC_GND_3	01 02
H810	GND_PHY	GND	0V			AGND_BLC	01 02
Backlight							
H750	BLC_CTL_PHY	BLC_PHASE				LED_DRIVER_GATE1	01
H760	BLC_CTL_PHY	BLC_PHASE				LED_DRIVER_GATE1_R	01
H750	BLC_CTL_PHY	BLC_PHASE				LED_DRIVER_GATE2	01
H760	BLC_CTL_PHY	BLC_PHASE				LED_DRIVER_GATE2_R	01
H800	BLC_CTL_PHY	BLC_PHASE				LED_DRIVER_GATE3	01
H760	BLC_CTL_PHY	BLC_PHASE				LED_DRIVER_GATE3_R	01
H750	BLC_CTL_PHY	BLC_CTL				LED_DRVRS_CS_RC_1	01
H750	BLC_CTL_PHY	BLC_CTL				LED_DRVRS_CS_RC_2	01
H750	BLC_CTL_PHY	BLC_CTL				LED_DRVRS_CS_RC_3	01
H800	BLC_CTL_PHY	BLC_CTL				LED_DRIVER_CS1	01
H800	BLC_CTL_PHY	BLC_CTL				LED_DRIVER_CS2	01
H800	BLC_CTL_PHY	BLC_CTL				LED_DRIVER_CS3	01
H820	BLC_CTL_PHY	BLC_CTL				LED_DRVRS_CS_C1	01
H820	BLC_CTL_PHY	BLC_CTL				LED_DRVRS_CS_C2	01
H820	BLC_CTL_PHY	BLC_CTL				LED_DRVRS_CS_C3	01
H830	BLC_CTL_PHY	BLC_HV	00V			LED_DRIVER_FDBK_R_1	01
H830	BLC_CTL_PHY	BLC_HV	00V			LED_DRIVER_FDBK_R_2	01
H830	BLC_CTL_PHY	BLC_HV	00V			LED_DRIVER_FDBK_R_3	01
H830	BLC_CTL_PHY	BLC_CTL	00V			LED_DRIVER_FDBK1	01
H830	BLC_CTL_PHY	BLC_CTL	00V			LED_DRIVER_FDBK2	01
H820	BLC_CTL_PHY	BLC_CTL	00V			LED_DRIVER_FDBK3	01
H830	BLC_CTL_PHY	BLC_CTL				BLC_PWM_1_R	00 01
H830	BLC_CTL_PHY	BLC_CTL				BLC_PWM_2_R	00 01
H830	BLC_CTL_PHY	BLC_CTL				BLC_PWM_3_R	00 01
H830	BLC_CTL_PHY	BLC_CTL				BLC_PWM_1	01 02
H830	BLC_CTL_PHY	BLC_CTL				BLC_PWM_2	01 02
H830	BLC_CTL_PHY	BLC_CTL				BLC_PWM_3	01 02
H810	BLC_CTL_PHY	BLC_CTL				LED_DRIVER_REF1	01 02
H810	BLC_CTL_PHY	BLC_CTL				LED_DRIVER_REF2	01 02
H810	BLC_CTL_PHY	BLC_CTL				LED_DRIVER_REF3	01 02
H760	BLC_CTL_PHY	BLC_CTL				LED_DRIVER_COMP1	01
H760	BLC_CTL_PHY	BLC_CTL				LED_DRIVER_COMP2	01
H770	BLC_CTL_PHY	BLC_CTL				LED_DRIVER_COMP3	01
H820	BLC_CTL_PHY	BLC_CTL				BCOMP1	01
H820	BLC_CTL_PHY	BLC_CTL				BCOMP2	01
H820	BLC_CTL_PHY	BLC_CTL				BCOMP3	01
H830	BLC_CTL_PHY	BLC_CTL				LED_DRIVER_FLT1	01
H830	BLC_CTL_PHY	BLC_CTL				LED_DRIVER_FLT2	01
H830	BLC_CTL_PHY	BLC_CTL				LED_DRIVER_FLT3	01
H830	BLC_CTL_PHY	BLC_CTL				LED_FLT_R_1	01
H830	BLC_CTL_PHY	BLC_CTL				LED_FLT_R_2	01
H830	BLC_CTL_PHY	BLC_CTL				LED_FLT_R_3	01
H850	BLC_CTL_PHY	BLC_CTL				PRE_REG_OUT_R	00
H850	BLC_CTL_PHY	BLC_CTL				B00ST_FB	00
H850	BLC_CTL_PHY	BLC_CTL				B00ST_COMP	00
H850	BLC_CTL_PHY	BLC_CTL				B00ST_COMP_C	00
H850	BLC_CTL_PHY	BLC_CTL			TRUE	B00ST_GDRV	00
H850	BLC_CTL_PHY	BLC_CTL			TRUE	B00ST_GDRV_R	00
H850	BLC_CTL_PHY	BLC_CTL				B00ST_ISNS	00
H850	BLC_CTL_PHY	BLC_CTL				B00ST_ISNS_R	

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GPU VREG CONSTRAINTS

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GPU VREG CONSTRAINTS

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Caesar IV (Ethernet/SD)

CIV-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
ENET_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
ENET_100D	*	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF
SD_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
ENET_COMP_PHY	*	ENET_50S
ENET_DIFF_PHY	*	ENET_100D
SD_PHY	*	SD_50S

CIV-specific Spacing Definitions

Ethernet

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
ENET_DIFF_ISO	*	=6:1_SPACING	?
ENET_DIFF2DIFF	*	=3:1_SPACING	?
ENET_TRANS_ISO	*	1.27 MM	?
COMP_ENET_ISO	*	=4:1_SPACING	?

SD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SD_ISO	*	=3:1_SPACING	?

Constraints Ethernet

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
ENET_DIFF	*	*	ENET_DIFF_ISO
ENET_DIFF	ENET_DIFF	*	ENET_DIFF2DIFF
ENET_TRANS	*	*	ENET_TRANS_ISO
COMP_ENET	*	*	COMP_ENET_ISO
ENET_TRANS	ENET_TRANS	*	ENET_DIFF2DIFF

2 kV isolation

SD

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
SD	*	*	SD_ISO

VENI, VIDI, VICI!

Electrical Constraint Set Ethernet		Physical	Spacing		
ENET_MDI_NORM	ENET_MDI_NORM	ENET_DIFF_PHY	ENET_DIFF	ENETCONN_MDI_P<3..0>	39 40
ENET_MDI_NORM	ENET_MDI_NORM	ENET_DIFF_PHY	ENET_DIFF	ENETCONN_MDI_N<3..0>	39 40
ENET_MDI_NORM	ENET_MDI_NORM	ENET_DIFF_PHY	ENET_TRANS	ENETCONN_MDI_T_P<3..0>	39 40
ENET_MDI_NORM	ENET_MDI_NORM	ENET_DIFF_PHY	ENET_TRANS	ENETCONN_MDI_T_N<3..0>	40
			ENET_TRANS	ENETCONN_MCT3	40
			ENET_TRANS	ENETCONN_MCT2	40
			ENET_TRANS	ENETCONN_MCT1	40
			ENET_TRANS	ENETCONN_MCT0	40
		ENET_COMP_PHY	COMP_ENET	ENET_RDAC	39
SD					
SD_DATA	SD_DATA	SD_PHY	SD	ENET_CR_DATA<7..0> NO_TEST=TRUE	39
SD_DATA	SD_DATA	SD_PHY	SD	SDCONN_DATA<7..0>	39 41
SD_DATA	SD_DATA	SD_PHY	SD	SDCONN_DATA_R<7..0> NO_TEST=TRUE	41
SD_CMD	SD_CMD	SD_PHY	SD	ENET_SD_CMD NO_TEST=TRUE	39
SD_CMD	SD_CMD	SD_PHY	SD	SDCONN_CMD NO_TEST=TRUE	39 41
SD_CMD	SD_CMD	SD_PHY	SD	SDCONN_CMD_R NO_TEST=TRUE	41
SD_CLK	SD_CLK	SD_PHY	SD	ENET_SD_CLK NO_TEST=TRUE	39
SD_CLK	SD_CLK	SD_PHY	SD	SDCONN_CLK NO_TEST=TRUE	39 41
SD_CLK	SD_CLK	SD_PHY	SD	SDCONN_CLK_R NO_TEST=TRUE	41
		SD_PHY	SD	ENET_MEDIA SENSE	15 39
		SD_PHY	SD	ENET_SD_DETECT_L	39 41
		SD_PHY	SD	SDCONN_WP	39 41
CIV SPI					
SPI_50S	SPI_50S	SPT		ENET_SCLK	39
SPI_50S	SPI_50S	SPT		ENET_MISO	39
SPI_50S	SPI_50S	SPT		ENET_MOSI	39
SPI_50S	SPI_50S	SPT		ENET_CS_L	39

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DCBABA

BOOST_*		
	Spacing	Netname
1608	GENERIC ISO	BOOST_BP
1608	GENERIC ISO	BOOST_BYPASS
1608	GENERIC ISO	BOOST_BYPASS_GATE
1608	PM	BOOST_EN_GATE
1608	PM	BOOST_EN_L
1608	GENERIC ISO	BOOST_RC
1608	GENERIC ISO	BOOST_SS

DP_*			
	Physical	Spacing	Netname
LE00		PM	DP_AUXIO_EN 15 77 79
LE01		PM	DP_GPU_MUX_EN 75
LE02		GENERIC ISO	DP_INTPNL_HPD 75 78
LE03		GENERIC ISO	DP_INT_EG_HPD 75 98
LE04	TBT_GEN_55S	TBT_GEN	DP_TBTPA_DDC_CLK 76 77
LE05	TBT_GEN_55S	TBT_GEN	DP_TBTPA_DDC_DATA 76 77
LE06		GENERIC ISO	DP_TBTPA_HPD 36 77
LE07	TBT_GEN_55S	TBT_GEN	DP_TBTPB_DDC_CLK 76 79
LE08	TBT_GEN_55S	TBT_GEN	DP_TBTPB_DDC_DATA 76 79
LE09		GENERIC ISO	DP_TBTPB_HPD 36 79

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ENETCONN_*

Spacing	Netname
1E4	GENERIC ISO ENETCONN_TCT 40

ENET_*

Physical	Spacing	Netname
1E2	GENERIC ISO	ENET_ACT 40
1E3	GENERIC ISO	ENET_ASF_GPIO 47 48
1E4	GENERIC ISO	ENET_CLKREQ_L 15 18 40
1E5	PM	ENET_CLKREQ_L_Q 39 40
1E6	PM	ENET_CR_1V8_EN 39 40
1E7	PM	ENET_CR_1V8_EN_R 39 40
1E8	PM	ENET_CR_3V3_EN_L 39 40
1E9	PM	ENET_CR_3V3_EN_L_R 39 40
1E10	PM	ENET_CR_PWREN 39 41
1E11	XDP_PHY	ENET_LOW_PWR 26 39 41
1E12	PM	ENET_PWR_EN_L 40
1E13	PM	ENET_PWR_EN_L_R 40
1E14	PM	ENET_RESET_L 39 41
1E15	PM	ENET_SD_RESET_L 26 41
1E16	GENERIC ISO	ENET_SR_DISABLE 39
1E17	GENERIC ISO	ENET_SR_LX 39 40
1E18	PM	ENET_WAKE_L 40

FAN_*

Spacing	Netname
1E19	GENERIC ISO FAN_0_PWM_FET 54
1E20	GENERIC ISO FAN_0_PWM_FILT 54
1E21	GENERIC ISO FAN_0_TACH_FET 54
1E22	GENERIC ISO FAN_0_TACH_FILT 54

FBVDD_*

Spacing	Netname
1E23	GENERIC ISO FBVDD_ALTVO 91 99

FB_*

Spacing	Netname
1E24	GENERIC ISO FB_A0_VREFC 86
1E25	GENERIC ISO FB_A0_VREFD 86
1E26	GENERIC ISO FB_A1_VREFC 86
1E27	GENERIC ISO FB_A1_VREFD 86
1E28	GENERIC ISO FB_B0_VREFC 87
1E29	GENERIC ISO FB_B0_VREFD 87
1E30	GENERIC ISO FB_B1_VREFC 87
1E31	GENERIC ISO FB_B1_VREFD 87
1E32	GENERIC ISO FB_C0_VREFC 88
1E33	GENERIC ISO FB_C0_VREFD 88
1E34	GENERIC ISO FB_C1_VREFC 88
1E35	GENERIC ISO FB_C1_VREFD 88
1E36	GENERIC ISO FB_D0_VREFC 89
1E37	GENERIC ISO FB_D0_VREFD 89
1E38	GENERIC ISO FB_D1_VREFC 89
1E39	GENERIC ISO FB_D1_VREFD 89
1E40	GENERIC ISO FB_SW_LEG_A 84 86

FB_*

Spacing	Netname
1E41	GENERIC ISO FB_SW_LEG_B 84 87
1E42	GENERIC ISO FB_SW_LEG_C 85 88
1E43	GENERIC ISO FB_SW_LEG_D 85 89
1E44	GENERIC ISO FB_VREF_GPU 94

FET_*

Physical	Spacing	Voltage	Netname
1E45	GENERIC ISO		FET_EN_P12V_S0 74
1E46	GENERIC ISO		FET_EN_P12V_S0_BLC 74
1E47	GENERIC ISO		FET_EN_P12V_S0_BLC_R 74
1E48	GENERIC ISO		FET_EN_P12V_S0_R 74
1E49	GENERIC ISO		FET_EN_P12V_S5 74
1E50	GENERIC ISO		FET_EN_P12V_S5_R 74
1E51	GENERIC ISO		FET_EN_VDDQ_S0 74
1E52	GENERIC ISO		FET_HDD_SLGSW 52
1E53	POWER_PHY	POWER 12V	FET_VCC_U7950 74
1E54	POWER_PHY	POWER 12V	FET_VCC_U7970 74
1E55	POWER_PHY	POWER 12V	FET_VCC_U7980 74

FLAG_*

Spacing	Netname
1E56	GENERIC ISO FLAG_V 88 91 92

G3_*

Spacing	Netname
1E57	GENERIC ISO G3_POWERON_L 47 48

GND_*

Spacing	Netname
1E58	SENSE GND_SMC_AVSS 47 48 51 55

GPU_*

Physical	Spacing	Netname
1E59	GENERIC ISO	GPU_ALT_VREF 84 85 91
1E60	GENERIC ISO	GPU_BUFRSTN 91
1E61	GENERIC ISO	GPU_IFPAB_PLLVDD 90
1E62	GENERIC ISO	GPU_IFPA_IOVDD 90
1E63	GENERIC ISO	GPU_IFPB_IOVDD 90
1E64	GENERIC ISO	GPU_IFPC_IOVDD 90
1E65	GENERIC ISO	GPU_IFPC_PLLVDD 90
1E66	XDP_PHY	GPU_JTAG_TDI 91
1E67	XDP_PHY	GPU_JTAG_TDO 91
1E68	XDP_PHY	GPU_JTAG_TMS 91
1E69	XDP_PHY	GPU_JTAG_TRST_L 91
1E70	GENERIC ISO	GPU_MLS_STRAP0 91
1E71	GENERIC ISO	GPU_MLS_STRAP1 91

GPU_*

Physical	Spacing	Netname
1E72	GENERIC ISO	GPU_MLS_STRAP2 91
1E73	GENERIC ISO	GPU_MLS_STRAP3 91
1E74	GENERIC ISO	GPU_MLS_STRAP4 91
1E75	GENERIC ISO	GPU_RESET_L 26 82 91
1E76	SPI_50S	SPI GPU_ROM_CS_L 91
1E77	SPI_50S	SPI GPU_ROM_CS_L_R 91
1E78	GENERIC ISO	GPU_ROM_HOLD_L 91
1E79	SPI_50S	SPI GPU_ROM_SI 91
1E80	SPI_50S	SPI GPU_ROM_SI_R 91
1E81	SPI_50S	SPI GPU_ROM_S0 91
1E82	SPI_50S	SPI GPU_ROM_S0_R 91
1E83	SPI_50S	SPI GPU_ROM_WP_L 91

HDD_*

Spacing	Netname
1E84	GENERIC ISO HDD_12V_S0_GATE 52
1E85	GENERIC ISO HDD_00B_1V00_REF 52
1E86	PM HDD_PWR_EN 21 52
1E87	GENERIC ISO HDD_PWR_EN_L 52
1E88	GENERIC ISO HDD_PWR_EN_R 52

I2C_*

Physical	Spacing	Netname
1E89	SMB_PHY	SMB I2C_TCON_MAS_SCL 50 78
1E90	SMB_PHY	SMB I2C_TCON_MAS_SDA 50 78

IFPD_*

Spacing	Netname
1E91	GENERIC ISO IFPD_RSET 99

IFPEF_*

Spacing	Netname
1E92	GENERIC ISO IFPEF_RSET 99

ISNSA_*

Electrical	Physical	Spacing	Netname
1E93	SNS_CURRENT	SNS_DIFF_PHY	SENSE ISNSA_P12VG3H_N 91
1E94	SNS_CURRENT	SNS_DIFF_PHY	SENSE ISNSA_P12VG3H_P 91
1E95	SNS_CURRENT	SNS_DIFF_PHY	SENSE ISNSA_P12VS0_CPU_P1V05_N 95
1E96	SNS_CURRENT	SNS_DIFF_PHY	SENSE ISNSA_P12VS0_CPU_P1V05_P 95
1E97	SNS_CURRENT	SNS_DIFF_PHY	SENSE ISNSA_P12VS0_CPU_VCCSA_N 95
1E98	SNS_CURRENT	SNS_DIFF_PHY	SENSE ISNSA_P12VS0_CPU_VCCSA_P 95
1E99	SNS_CURRENT	SNS_DIFF_PHY	SENSE ISNSA_P12VS0_FBVDDQ_N 91
1E100	SNS_CURRENT	SNS_DIFF_PHY	SENSE ISNSA_P12VS0_FBVDDQ_P 91
1E101	SNS_CURRENT	SNS_DIFF_PHY	SENSE ISNSA_P12VS0_HDD_N 91
1E102	SNS_CURRENT	SNS_DIFF_PHY	SENSE ISNSA_P12VS0_HDD_P 91

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PCH_*

Physical	Spacing	Netname	
	PM	PCH_BLC_EXT_BOOT	21 80
	PM	PCH_BLC_EXT_BOOT_R	15 21
	PM	PCH_BLC_MCU_RESET	21 80
	PM	PCH_BLC_MCU_RESET_R	15 21
	PM	PCH_CAM_RESET	15 21
	PM	PCH_CAM_RESET_R	21 43
	GENERIC_ISO	PCH_DSMVRMEN	19
CPU_PHY	CPU	PCH_PECT	21
CPU_PHY	CPU	PCH_PROCPWRGD	21
	GENERIC_ISO	PCH_RCIN_L	21
	GENERIC_ISO	PCH_RI_L	19
	GENERIC_ISO	PCH_SMBALERT_L	19 18
	GENERIC_ISO	PCH_SRTCST_L	18
	GENERIC_ISO	PCH_STRP_TOPBLK_SWP_L	20
	GENERIC_ISO	PCH_SUSWLN_L	15 19

PCIE_*

Spacing	Netname	
GENERIC_ISO	PCIE_CLKREQ05_GPIO44_L	18
PM	PCIE_WAKE_L	19 35 48

PEG_*

Spacing	Netname	
GENERIC_ISO	PEG_CLKREQ_L	15 18 83

PGOOD_*

Spacing	Netname	
PM	PGOOD_P1V5_S0_DLY	28

PLT_*

Spacing	Netname	
PCH	PLT_RESET_L	20 26
PCH	PLT_RST_BUF_L	26

PM_*

Spacing	Netname	
PM	PM_CLKRUN_L	47 48 49
PM	PM_DSW_PWRGD	47 48 65
PM	PM_EN_FET_P12V_S0	64 74
PM	PM_EN_FET_P12V_S0_R	74
PM	PM_EN_FET_P3V3_S0	64 74
PM	PM_EN_FET_P3V3_S4	64 74
PM	PM_EN_FET_P5V_S0	64 74
PM	PM_EN_FET_VDDQ_S0	64 74
PM	PM_EN_LDO_DDRVTT_S0	64 72
PM	PM_EN_REG_CPUCORE_S0	64 66

PM_*

Spacing	Netname	
PM	PM_EN_REG_CPU_P1V05_S0	64 69
PM	PM_EN_REG_FBVDDQ_S0	64 99
PM	PM_EN_REG_GPUCORE_S0_R	96
PM	PM_EN_REG_P1V05_S0	64 99
PM	PM_EN_REG_P1V8_S0	64 72
PM	PM_EN_REG_P3V3_S5	64 71 74
PM	PM_EN_REG_P5V_S4	64 71
PM	PM_EN_REG_VCCSA_S0	64 70
PM	PM_EN_REG_VDDQ_S3	64 72
PM	PM_EN_S0	64
PM	PM_EN_S4	64
PM	PM_EN_USB_PWR	45 46 64
PM	PM_LED_A_ALL_SYS_PWRGD	5
PM	PM_LED_A_BLC_G00D	5
PM	PM_LED_A_CPUAXG_PG00D	5
PM	PM_LED_A_GPU_G00D	5
PM	PM_LED_A_PG00D_CPUCORE_S0	5
PM	PM_LED_A_PG00D_CPU_P1V05_S0	5
PM	PM_LED_A_PG00D_REG_FBVDDQ_S0	5
PM	PM_LED_A_PG00D_REG_GPUCORE_S0	5
PM	PM_LED_A_PG00D_REG_P1V05	5
PM	PM_LED_A_PG00D_REG_VDDQ_S3	5
PM	PM_LED_A_S4	5
PM	PM_LED_A_S5	5
PM	PM_LED_A_SLP_S3	5
PM	PM_LED_A_VIDEO_ON	5
PM	PM_LED_K_ALL_SYS_PWRGD	5
PM	PM_LED_K_BLC_G00D	5
PM	PM_LED_K_CPUAXG_PG00D	5
PM	PM_LED_K_GPU_G00D	5
PM	PM_LED_K_PG00D_CPUCORE_S0	5
PM	PM_LED_K_PG00D_CPU_P1V05_S0	5
PM	PM_LED_K_PG00D_REG_FBVDDQ_S0	5
PM	PM_LED_K_PG00D_REG_GPUCORE_S0	5
PM	PM_LED_K_PG00D_REG_P1V05	5
PM	PM_LED_K_PG00D_REG_VDDQ_S3	5
PM	PM_LED_K_SLP_S3	5
PM	PM_MEM_PWRGD_L	28
PM	PM_PCH_APWROK	19 65
PM	PM_PCH_PWROK	15 19 26 35 43 65 88
PM	PM_PCH_PWROK_APWROK	65
PM	PM_PCH_SYS_PWROK	19 48 65
PM	PM_PG00D_FBVDDQ_VDDQ_S0	64
PM	PM_PG00D_FET_P12V_S0	74
PM	PM_PG00D_FET_P12V_S0_BLC	74 82
PM	PM_PG00D_FET_P12V_S5	74
PM	PM_PG00D_FET_P3V3_S0	64 74
PM	PM_PG00D_FET_P5V_S0	64 74
PM	PM_PG00D_FET_VDDQ_S0	28 64 74
PM	PM_PG00D_P3V3_S4_FET	27 35 64 74
PM	PM_PG00D_REG_ALL_P1V05_S0	64 65
PM	PM_PG00D_REG_ALL_P1V05_S0_R	28 64
PM	PM_PG00D_REG_CPUCORE_S0	5 25 65 66
PM	PM_PG00D_REG_CPU_P1V05_S0	5 64 69
PM	PM_PG00D_REG_FBVDDQ_S0	5 64 99
PM	PM_PG00D_REG_P1V05_S0	5 64 99
PM	PM_PG00D_REG_P1V8_S0	64 72
PM	PM_PG00D_REG_P3V3_S5	65 71
PM	PM_PG00D_REG_P5V_S4	64 71
PM	PM_PG00D_REG_VCCSA_S0	64 65 70
PM	PM_PG00D_REG_VDDQ_S3	5 64 72
PM	PM_PWRBTN_L	15 19 25 47
PM	PM_RSMRST_PCH_L	19 65
PM	PM_RSMRST_PCH_L_R	65
PM	PM_SLP_S3_L	5 15 19 28 48 47 48 64
PM	PM_SLP_S4_L	15 19 47 64
PM	PM_SLP_S5_L	15 19 47 64
PCH	PM_SYSRST_L	19 25 26 47
PCH	PM_THRMTRIP_L	21 48
PM	PG00D_P12V_S0_R	65
PM	PG00D_P12V_S0	64 65 74

PP12V_*

Physical	Spacing	Voltage	Netname	
POWER_PHY	POWER	12V	PP12V_LCD	78
POWER_PHY	POWER	12V	PP12V_LCD_EXT	78
POWER_PHY	POWER	12V	PP12V_S0_FAN_0_FILT	54
POWER_PHY	POWER	12V	PP12V_S0_HDD_FET	44 52

PP1V05_*

Physical	Spacing	Voltage	Netname	
POWER_PHY	POWER	1.05V	PP1V05_S0_PCH_VCCADPLLA_F	17 22
POWER_PHY	POWER	1.05V	PP1V05_S0_PCH_VCCADPLL_B_F	17 22

PP1V2_*

Physical	Spacing	Voltage	Netname	
POWER_PHY	POWER	1.2V	PP1V2_ENET_INTREG	49
POWER_PHY	POWER	1.2V	PP1V2_G3H_SMC_VDDC	47
POWER_PHY	POWER	1.2V	PP1V2_S4_ENET_PHY_AVDDL	39
POWER_PHY	POWER	1.2V	PP1V2_S4_ENET_PHY_GPHYPLL	39
POWER_PHY	POWER	1.2V	PP1V2_S4_ENET_PHY_PCIEPLL	39
POWER_PHY	POWER	1.2V	PP1V2_USB_HUB_CRFLT	27
POWER_PHY	POWER	1.2V	PP1V2_USB_HUB_PLIFILT	27

PP1V5_*

Physical	Spacing	Voltage	Netname	
POWER_PHY	POWER	1.5V	PP1V5_S0_DP_BIAS	75

PP1V8_*

Physical	Spacing	Voltage	Netname	
POWER_PHY	POWER	1.8V	PP1V8_S0_PCH_VCCVRM_F	22 24

PP3V3R1V8_*

Physical	Spacing	Netname	
POWER_PHY	POWER	PP3V3R1V8_ENET_LR_OUT_REG	39

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CBA

S5_*

SATALED_*SDCONN_*SD_*SLG_*SMBUS_*

	Physical	Spacing	Netname	
	SMB_PHY	SMB	SMBUS_PCH_CLK	18 50
	SMB_PHY	SMB	SMBUS_PCH_CLK_R	50
	SMB_PHY	SMB	SMBUS_PCH_DATA	18 50
	SMB_PHY	SMB	SMBUS_PCH_DATA_R	50
	SMB_PHY	SMB	SMBUS_SMC_0_S0_SCL	47 50
	SMB_PHY	SMB	SMBUS_SMC_0_S0_SDA	47 50
	SMB_PHY	SMB	SMBUS_SMC_1_S0_SCL	47 50
	SMB_PHY	SMB	SMBUS_SMC_1_S0_SDA	47 50
	SMB_PHY	SMB	SMBUS_SMC_2_S4_SCL	47 50
	SMB_PHY	SMB	SMBUS_SMC_2_S4_SDA	47 50
	SMB_PHY	SMB	SMBUS_SMC_3_SCL	47 50
	SMB_PHY	SMB	SMBUS_SMC_3_SDA	47 50
	SMB_PHY	SMB	SMBUS_SMC_4_ASF_SCL	47 48 50
	SMB_PHY	SMB	SMBUS_SMC_4_ASF_SDA	47 48 50
	SMB_PHY	SMB	SMBUS_SMC_5_G3H_SCL	47 48 50
	SMB_PHY	SMB	SMBUS_SMC_5_G3H_SDA	47 48 50

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DCBA

	Physical	Spacing	Netname	
E425		GENERIC ISO	SMC ACDC ID	6 48
E426		GENERIC ISO	SMC ACDC ID R	
E427		PM	SMC ASSERT RTCRST	6
E428		GENERIC ISO	SMC BLC_MUX_RX_L	48 86
E429		GENERIC ISO	SMC BLC_MUX_TX_L	48 86
E430		GENERIC ISO	SMC CPU_CATERR_L	47 48
E431	CPU_PHY	CPU	SMC CPU_PECI	48
E432		PM	SMC DELAYED_PWRGD	38 47 48 65
E433		GENERIC ISO	SMC_DP_HPD_L	47 48
E434	CLK_XTAL	XTAL	SMC_EXTAL	47 48
E435		GENERIC ISO	SMC_FAN_0_CTL	47 54
E436		GENERIC ISO	SMC_FAN_0_TACH	47 54
E437		GENERIC ISO	SMC_GFX_OVERTEMP	47 48 91
E438		GENERIC ISO	SMC_GFX_OVERTEMP_Q	91
E439		GENERIC ISO	SMC_GFX_OVERTEMP_R_L	91
E440		PM	SMC_GFX_THROTTLE_L	47 91
E441		PM	SMC_GFX_THROTTLE_R_L	91
E442		PM	SMC_LRESET_L	26 47
E443		PM	SMC_MANUAL_RST_L	48
E444		PM	SMC_ONOFF_L	47 48
E445		SENSE	SMC_00B1_RX_CN	44
E446		SENSE	SMC_00B1_RX_FILT	44 52
E447		SENSE	SMC_00B1_RX_L	47 52
E448		SENSE	SMC_00B1_RX_R	52
E449		SENSE	SMC_00B1_TX_L	47 48
E450		SENSE	SMC_00B2_RX_L	44 48
E451		SENSE	SMC_00B2_TX_L	44 48
E452	CPU_PHY	CPU	SMC_PECI_L	47 48
E453	CPU_PHY	CPU	SMC_PECI_L_R	48
E454		PM	SMC_PME_S4_WAKE_L	35 47 48
E455		GENERIC ISO	SMC_PM_G2_EN	47 48 74
E456		PM	SMC_PM_PCH_SYS_PWROK	47 48
E457		CPU	SMC_PROCHOT	47 48
E458		PM	SMC_RESET_L	47 48 49

I61

I62

178

	Spacing	Netname
179	GENERIC ISO	SPI_DESCRIPTOR_OVERRIDE L 15 47

Spacing	Netname
GENERIC ISO	SSD_CLKREQ L

180 15 18

	Spacing	Netname
E81	PM	SYS PWROK_R

I84

I85

Y10.

	Electrical	Physical	Spacing	Netname	
TSNS	SNS TEMP	SNS DIFF PHY	SENSE	TSNS 1 1 N	53
TSNS	SNS TEMP	SNS DIFF PHY	SENSE	TSNS 1 1 P	53
TSNS	SNS TEMP	SNS DIFF PHY	SENSE	TSNS 1 2 N	53
TSNS	SNS TEMP	SNS DIFF PHY	SENSE	TSNS 1 2 P	53
TSNS	SNS TEMP	SNS DIFF PHY	SENSE	TSNS 1 3 N	53
TSNS	SNS TEMP	SNS DIFF PHY	SENSE	TSNS 1 3 P	53
TSNS	SNS TEMP	SNS DIFF PHY	SENSE	TSNS 2 1 N	53
TSNS	SNS TEMP	SNS DIFF PHY	SENSE	TSNS 2 1 P	53
TSNS	SNS TEMP	SNS DIFF PHY	SENSE	TSNS 2 2 N	53
TSNS	SNS TEMP	SNS DIFF PHY	SENSE	TSNS 2 2 P	53
TSNS	SNS TEMP	SNS DIFF PHY	SENSE	TSNS 2 3 N	53
TSNS	SNS TEMP	SNS DIFF PHY	SENSE	TSNS 2 3 P	53
TSNS	SNS TEMP	SNS DIFF PHY	SENSE	TSNS 2 4 N	53
TSNS	SNS TEMP	SNS DIFF PHY	SENSE	TSNS 2 4 P	53
TSNS	SNS TEMP	SNS DIFF PHY	SENSE	TSNS 2 5 N	53
TSNS	SNS TEMP	SNS DIFF PHY	SENSE	TSNS 2 5 P	53
TSNS	SNS TEMP	SNS DIFF PHY	SENSE	TSNS 2 6 N	53
TSNS	SNS TEMP	SNS DIFF PHY	SENSE	TSNS 2 6 P	53
TSNS	SNS TEMP	SNS DIFF PHY	SENSE	TSNS 2 7 N	53
TSNS	SNS TEMP	SNS DIFF PHY	SENSE	TSNS 2 7 P	53
TSNS	SNS TEMP	SNS DIFF PHY	SENSE	TSNS ACDC N	6
TSNS	SNS TEMP	SNS DIFF PHY	SENSE	TSNS ACDC P	6
TSNS	SNS TEMP	SNS DIFF PHY	SENSE	TSNS SKIN N	53
TSNS	SNS TEMP	SNS DIFF PHY	SENSE	TSNS SKIN P	53

	Spacing	Netname	
	PM	UVP IN 1	02
	PM	UVP IN 1 REF	02
	PM	UVP IN 2	02
	PM	UVP IN 3	02
	PM	UVP IN 4	02
	PM	UVP REF	02

